

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 2

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\163-D10peRev 02.pvdb

Date of Analysis : Apr 20, 2026 3:04pm

PV Elite 28, January 2026



Vessel Design Summary:

ASME Code, Section VIII Division 2, 2025

Diameter Spec : 1700.000 x 1706.000 mm ID
 Vessel Design Length, Tangent to Tangent 18200.00 mm
 Distance of Bottom Tangent above Grade 6300.00 mm
 Distance of Base above Grade 300.00 mm
 Specified Datum Line Distance 6000.00 mm
 Internal Design Temperature 130 °C
 Internal Design Pressure 4.100 MPa
 External Design Temperature 130 °C
 External Design Pressure 0.103 MPa
 Maximum Allowable Working Pressure 4 MPa
 External Max. Allowable Working Pressure 1 MPa
 Shop Test Pressure 5.369 MPa
 Required Minimum Design Metal Temperature -12.0 °C
 Warmest Computed Minimum Design Metal Temperature -12.0 °C
 Wind Design Code ASCE/SEI 7-16
 Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	+IT
Head	SA-516 70	...	...	K02700	+IT
Skirt	SA-516 70	...	...	K02700	+IT
Nozzle	SA-105	...	...	K03504	+IT
Nozzle Flg	SA-105	...	...	K03504	+IT
Basering	SA-516 70	...	...	K02700	
Base Bolting	SA-193 B7	...	<= 64	G41400	+IT

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (MPa & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Skirt-3	...	...	...	1.6000	N/A	No
Skirt 2	...	...	...	1.6000	N/A	No
Skirt 1	...	...	...	1.6000	N/A	No
Bottom head	4.142	0.10	5.000	6.0000	No	No
Shell-1	4.137	0.10	4.000	6.0000	N/A	No
Shell-2	4.107	0.10	4.000	6.0000	N/A	No
Shell-3	4.100	0.10	4.000	6.0000	N/A	No
Shell-4	4.100	0.10	4.000	6.0000	N/A	No
Shell-5	4.100	0.10	4.000	1.0000	N/A	No
Shell-7	4.100	0.10	4.000	1.0000	N/A	No
Top head	4.100	0.10	4.000	1.0000	No	No

Liquid Level: 4175.00 mm Dens.: 0.001 kg/cm³ Sp. Gr.: 1.023

Element Types and Properties:

Element	"To" Elev	Element Length	Nominal Thickness	Finished Thickness	Reqd Thk Internal	Reqd Thk External	Long Eff	Circ Eff
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Type	mm	mm	mm	mm	mm	mm		
Skirt	-4000.0	2000.0	14.0	14.0	...	...	0.70	0.70
Skirt	-1000.0	3000.0	14.0	14.0	...	...	0.70	0.70
Skirt	0.0	1000.0	14.0	14.0	...	...	0.70	0.50
Ellipse	50.0	50.0	38.0	32.0	29.2	10.1	1.00	1.00
Cylinder	3050.0	3000.0	34.0	34.0	32.1	20.3	1.00	1.00
Cylinder	6050.0	3000.0	34.0	34.0	31.9	20.3	1.00	1.00
Cylinder	9050.0	3000.0	34.0	34.0	31.8	20.3	1.00	1.00
Cylinder	12855.0	3805.0	34.0	34.0	31.8	20.3	1.00	1.00
Cylinder	15050.0	2195.0	28.0	28.0	26.8	15.2	1.00	1.00
Cylinder	18150.0	3100.0	28.0	28.0	26.8	15.2	1.00	1.00
Ellipse	18200.0	50.0	32.0	26.0	24.0	5.1	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
D_150	WRC-107/537	0.936	n/a	Passed
B_150	WRC-297	0.964	Shell	Passed
A_150	WRC-297	1.000	Shell	Passed
E_40	WRC-297	1.000	Shell	Passed
C_100	WRC-107/537	0.964	n/a	Passed

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm	Cumulative Wind Shear N	Earthquake Shear N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	1000	47279.3	60729.5	654152512	1043027904
20	30	3500	45207.1	60654.6	561628544	921594432
30	40	5500	40668.5	60293.1	432763040	740099456
40	50	6025	39585.1	60133.9	392619904	679861568
50	60	7550	37010.3	58711.4	390704256	676889152
60	70	10550	33593.7	53845.6	284755392	507985184
70	80	13550	27765.5	48434.7	192679296	354502496
80	90	16952.5	21729.6	42117.7	118406400	218618832
90	100	19952.5	14853.9	26838.7	48777912	87376072
100	110	22600	9721.24	17832.6	21795728	38329520
110	120	24175	3734.43	5932.4	930985	1478931

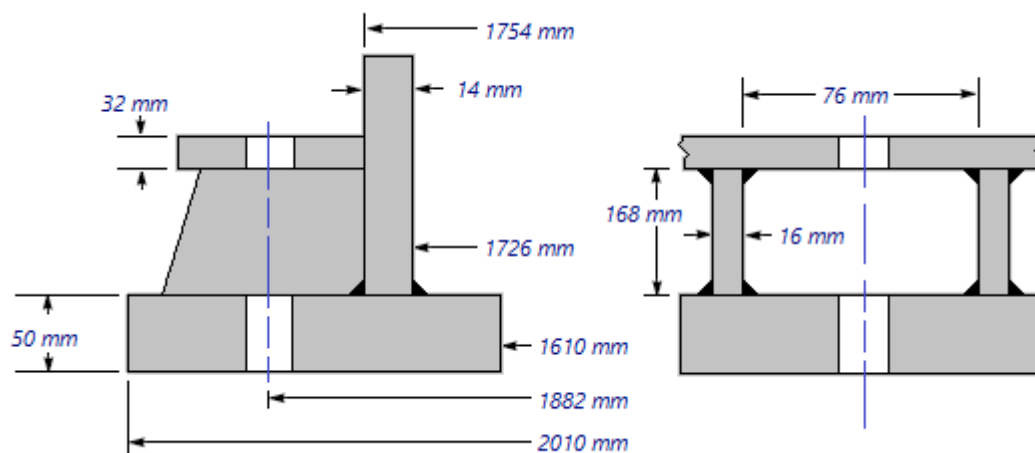
Abs Max of the all of the Stress Ratio's : 0.8147

Basing Data : Continuous Top Ring W/Gussets

Thickness of Basing 50.0000 mm
 Inside Diameter of Basing 1610.0000 mm
 Outside Diameter of Basing 2010.0000 mm
 Nominal Diameter of Bolts 36.0000 mm
 Diameter of Bolt Circle 1882.0000 mm
 Number of Bolts 20

Thickness of Gusset Plates 16.0000 mm
 Average Width of Gusset Plates 118.0000 mm
 Height of Gussets 168.0000 mm
 Distance between Gussets 76.0000 mm
 Thickness of Top Plate or Ring 32.0000 mm
 Radial Width of the Top Plate 128.0000 mm

Basing Sketch for Cylindrical Skirts:



Basing Cross Section View

Loads for Foundation/Support Design:

Total Wind Shear on Support	55054.	N
Total Earthquake Shear on Support	68505.	N
Wind Moment on Support	916363136.	N-mm
Earthquake Moment on Support	1305238528.	N-mm
Maximum Initial Bolt Load	98421.	N

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	45803.8	kgm
Shop Test - Fabricated + water (full)	88378.5	kgm
Shipping - Fab. + rem. intls.+ shipping app.	68387.3	kgm
Erected - Fab. + rem. intls.+ insul. (etc)	68387.3	kgm
Empty - Fab. + intls. + details + wghts.	68387.3	kgm
Operating - Empty + operating liquid (no ca)	83840.2	kgm
Field Test - Empty weight + test medium (full)	106709.0	kgm

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Nozzle Calculation Summary (MPa & mm):

Description	MAWP	Ext	MAPNC	Tbl 4.5.2	Weld	Pass/Fail
D_150	5	OK	...	OK	OK	Passed
MH4_600	4	OK	...	...	OK	Passed
B_150	4	OK	...	OK	OK	Passed
L1_50	4	OK	...	OK	OK	Passed
L2_50	4	OK	...	OK	OK	Passed
L3_50	4	OK	...	OK	OK	Passed
L4_50	4	OK	...	OK	OK	Passed
L5_50	4	OK	...	OK	OK	Passed
L6_50	4	OK	...	OK	OK	Passed
DP1_50	4	OK	...	OK	OK	Passed
UC2_600	4	OK	...	...	OK	Passed
MH3_600	4	OK	...	...	OK	Passed
DP2_50	4	OK	...	OK	OK	Passed
UC1_600	4	OK	...	...	OK	Passed
A_150	4	OK	...	OK	OK	Passed
MH2_600	4	OK	...	...	OK	Passed
E_40	4	OK	...	OK	OK	Passed
MH1_600	4	OK	...	...	OK	Passed
C_100	4	OK	...	OK	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 4 Nozzle : MH1600
 Minimum MAWP Shells/Flanges : 4 Element : Shell-7
 Minimum MAPnc Shells/Flanges : 4 Element : Shell-7

Computed Vessel M.A.W.P. : 4 MPa

VIII-2 Nozzle to Gross Structural Discontinuity Check:

No Nozzles were too close to any gross structural discontinuities per 4.5.13.1

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm	Layout Angle deg	Inside Radius mm
40	D_150	0.000	0.000	77.647
50	MH4_600	849.999	270.000	290.000
50	B_150	2849.999	0.000	77.647
50	L1_50	2550.000	180.000	31.400
50	L2_50	249.999	180.000	31.400
50	L3_50	2049.999	180.000	31.400
50	L4_50	1049.999	180.000	31.400
50	L5_50	2049.999	0.000	31.400
50	L6_50	1049.999	0.000	31.400
60	DP1_50	3699.999	180.000	31.400
60	UC2_600	4049.999	0.000	290.000
70	MH3_600	8349.999	180.000	290.000
70	DP2_50	8549.999	0.000	31.400
80	UC1_600	10049.999	0.000	290.000
90	A_150	14975.000	0.000	77.647
90	MH2_600	14750.000	90.000	290.000
100	E_40	15950.000	0.000	26.000
100	MH1_600	16600.002	180.000	290.000
110	C_100	0.000	0.000	49.663

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.



Ic - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia mm	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
E_40	1 in	LWN	WNF	60.0	10.000	...	...	330	900
L1_50	2 in	LWN	WNF	74.8	12.000	...	...	287	300
L2_50	2 in	LWN	WNF	74.8	12.000	...	...	287	300
L3_50	2 in	LWN	WNF	74.8	12.000	...	...	287	300
L4_50	2 in	LWN	WNF	74.8	12.000	...	...	287	300
L5_50	2 in	LWN	WNF	74.8	12.000	...	...	287	300
L6_50	2 in	LWN	WNF	74.8	12.000	...	...	287	300
DP1_50	2 in	LWN	WNF	74.8	12.000	...	...	287	300
DP2_50	50 mm	Actual	WNF	74.8	12.000	...	...	287	300
C_100	100 mm	160	WNF	114.3	13.487	...	...	328	600
D_150	150 mm	120	WNF	168.3	14.275	...	...	336	600
B_150	150 mm	120	WNF	168.3	14.275	...	...	342	600
A_150	150 mm	120	WNF	168.3	14.275	...	...	336	600
MH4_600	600 mm	Actual	WNF	600.0	16.000	...	...	615	300
UC2_600	600 mm	Actual	WNF	600.0	16.000	...	...	615	300
MH3_600	600 mm	Actual	WNF	600.0	16.000	...	...	615	300
UC1_600	600 mm	Actual	WNF	600.0	16.000	...	...	615	300
MH2_600	600 mm	Actual	WNF	600.0	16.000	...	...	609	300
MH1_600	600 mm	Actual	WNF	600.0	16.000	...	...	609	300

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
E_40	SA-105	28.000	10.000	...	...	...
L1_50	SA-105	30.000	10.000	...	...	...
L2_50	SA-105	30.000	10.000	...	...	...
L3_50	SA-105	30.000	10.000	...	...	...
L4_50	SA-105	30.000	10.000	...	...	...
L5_50	SA-105	30.000	10.000	...	...	...
L6_50	SA-105	30.000	10.000	...	...	...
DP1_50	SA-105	30.000	10.000	...	...	...
DP2_50	SA-105	30.000	10.000	...	...	...
C_100	SA-105	26.000	10.000	...	...	...
D_150	SA-105	30.000	10.000	...	...	...
B_150	SA-105	30.000	10.000	...	...	...
A_150	SA-105	28.000	10.000	...	...	...
MH4_600	SA-105	32.000	10.000	...	...	...
UC2_600	SA-105	32.000	10.000	...	...	...
MH3_600	SA-105	32.000	10.000	...	...	...
UC1_600	SA-105	32.000	10.000	...	...	...
MH2_600	SA-105	28.000	10.000	...	...	...
MH1_600	SA-105	28.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
-------------	-----------------------------	------------------	-----------------	----------------	------------------------

E_40	15949.999	0.0	300.00	0.00	Shell-7
L1_50	2550.000	180.0	250.00	0.00	Shell-1
L2_50	249.999	180.0	250.00	0.00	Shell-1
L3_50	2049.999	180.0	250.00	0.00	Shell-1
L4_50	1049.999	180.0	250.00	0.00	Shell-1
L5_50	2049.999	0.0	250.00	0.00	Shell-1
L6_50	1049.999	0.0	250.00	0.00	Shell-1
DP1_50	3699.999	180.0	250.00	0.00	Shell-2
DP2_50	8549.999	0.0	250.00	0.00	Shell-3
C_100	...	0.0	300.00	0.00	Top head
D_150	...	0.0	300.00	0.00	Bottom head
B_150	2849.999	0.0	300.00	0.00	Shell-1
A_150	14975.000	0.0	300.00	0.00	Shell-5
MH4_600	849.999	270.0	500.00	0.00	Shell-1
UC2_600	4049.999	0.0	500.00	0.00	Shell-2
MH3_600	8349.998	180.0	500.00	0.00	Shell-3
UC1_600	10049.999	0.0	500.00	0.00	Shell-4
MH2_600	14750.000	90.0	500.00	0.00	Shell-5
MH1_600	16600.002	180.0	500.00	0.00	Shell-7



Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	ELLIPTICAL HEAD: 2.0 X 1, 38.0mm THK X 1700.0mm ID X 50.0mm	SA-516 70
1	ELLIPTICAL HEAD: 2.0 X 1, 32.0mm THK X 1706.0mm ID X 50.0mm	SA-516 70
7	PLATFORM: 1000mm X 1500mm WIDE	
5	LADDER: 4000.0mm	...
1	INSULATION: 500mm X 80mm THK	
2	LADDER: 3000.0mm	...
2	INSULATION: 475mm X 80mm THK	
3	INSULATION: 3000mm X 80mm THK	
1	PACKING: 2240.000mm	...
1	PACKING: 2035.000mm	...
1	PACKING: 3275.000mm	...
1	INSULATION: 3804mm X 80mm THK	
1	PACKING: 805.000mm	...
1	INSULATION: 2195mm X 80mm THK	
1	CLADDING: 2195.000mm X 3.000mm THK	...
1	INSULATION: 3100mm X 80mm THK	
1	CLADDING: 3100.000mm X 3.000mm THK	...
3	TRAYS # 3 SPACING 12. mm	
1	PLATFORM: 1000mm X 3764mm WIDE	
1	LADDER: 2500.0mm	...
1	CLADDING: 476.500mm X 3.000mm THK	...
6	CLASS 300 GR 1.1, 600.0mm BLIND FLANGE(S)	SA-105
1	BASERING: 2010.0mm OD X 1610.0 ID X 50.0mm	SA-516 70
20	BASERING BOLTS: 36.0mm DIA	SA-193 B7
20	BASERING BOLT NUTS: 36.0mm DIA	...
1	BASERING TOP RING: 2010.0mm OD X 1754.0 ID X 32.0mm	...
40	BASERING GUSSET PLATES: 118.0mm X 118.0mm X 168.0mm X 16.0mm	...
1	CYLINDRICAL SKIRT SEGMENT 14.0mm THK X 3094.6mm X 2000.0mm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 14.0mm THK X 2327.7mm X 2000.0mm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 14.0mm THK X 3094.6mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 14.0mm THK X 2327.7mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 14.0mm THK X 3094.6mm X 1000.0mm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 14.0mm THK X 2327.7mm X 1000.0mm	SA-516 70



2	CYLINDRICAL SEGMENT 34.0mm THK X 2121.4mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 34.0mm THK X 1097.8mm X 3000.0mm	SA-516 70
2	CYLINDRICAL SEGMENT 34.0mm THK X 3041.2mm X 3000.0mm	SA-516 70
2	CYLINDRICAL SEGMENT 34.0mm THK X 2299.4mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 34.0mm THK X 3041.2mm X 3805.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 34.0mm THK X 2299.4mm X 3805.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 28.0mm THK X 2307.5mm X 2195.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 28.0mm THK X 3051.9mm X 2195.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 28.0mm THK X 2307.5mm X 3100.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 28.0mm THK X 3051.9mm X 3100.0mm	SA-516 70
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	PWHT	Thickness ratio	Gov Thk mm	E*
Notes							
Bottom head	[10]	!	-12	No			
Bottom head	[7]	!	-12	No			
Shell-1	[8]	!	-12	Yes			
Shell-2	[8]	!	-12	Yes			
Shell-3	[8]	!	-12	Yes			
Shell-4	[8]	!	-12	Yes			
Shell-5	[8]	!	-12	No			
Shell-7	[8]	!	-12	No			
Top head	[10]	!	-12	No			
Top head	[7]	!	-12	No			
D150	[1]	!	-12	No			
Nozzle Flg	[5]	!	-18	-63	No		
MH4600	[1]	!	-12	Yes			
Nozzle Flg	[5]	!	-18	-26	Yes		
B150	[1]	!	-12	Yes			
Nozzle Flg	[5]	!	-18	-63	Yes		
L150	[1]	!	-12	Yes			
Nozzle Flg	[5]	!	-17	-17	Yes		
L250	[1]	!	-12	Yes			
Nozzle Flg	[5]	!	-17	-17	Yes		
L350	[1]	!	-12	Yes			
Nozzle Flg	[5]	!	-17	-17	Yes		
L450	[1]	!	-12	Yes			
Nozzle Flg	[5]	!	-17	-17	Yes		
L550	[1]	!	-12	Yes			
Nozzle Flg	[5]	!	-17	-17	Yes		
L650	[1]	!	-12	Yes			
Nozzle Flg	[5]	!	-17	-17	Yes		
DP150	[1]	!	-12	Yes			
Nozzle Flg	[5]	!	-17	-17	Yes		
UC2600	[1]	!	-12	Yes			
Nozzle Flg	[5]	!	-18	-27	Yes		
MH3600	[1]	!	-12	Yes			
Nozzle Flg	[5]	!	-18	-27	Yes		
DP250	[1]	!	-12	Yes			
Nozzle Flg	[5]	!	-18	-27	Yes		
UC1600	[1]	!	-12	Yes			
Nozzle Flg	[5]	!	-18	-27	Yes		
A150	[1]	!	-12	No			
Nozzle Flg	[5]	!	-18	-63	No		
MH2600	[1]	!	-12	No			
Nozzle Flg	[5]	!	-18	-27	No		
E40	[1]	!	-12	No			
Nozzle Flg	[5]	!	-17	-17	No		
MH1600	[1]	!	-12	No			
Nozzle Flg	[5]	!	-18	-27	No		
C100	[1]	!	-12	No			
Nozzle Flg	[5]	!	-18	-63	No		

Warmest MDMT: -12 -17

Required Minimum Design Metal Temperature -12.0 °C
 Warmest Computed Minimum Design Metal Temperature -12.0 °C

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.1.
- [5] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.3.
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.



- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per 3.11.2.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

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Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.

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Units used in this Analysis (SIASME):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	4.4480	N
Mass	Pounds	0.4536	kgm
Area	sq. inches	645.1600	mm ²
Moment	ft. lbs.	1356.3000	N-mm
Stress	lbs./sq.in.	0.0069	MPa
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0069	MPa
Elast. Modulus	lbs./sq.in.	0.0069	MPa
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kg/cm ²
Inertia	in.4	416231.0000	mm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	0.0479	kPa
Elevation	Feet	304.8000	mm
Volume	in.3	0.0164	ltr
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	4.1	MPa
Design Internal Temperature	130.0	°C
Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Minimum Design Metal Temperature	-12.0	°C
Type of Construction	Welded	
Special Service	None	
Use Higher Longitudinal Stresses (Flag)	N	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	Hydrostatic
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	Not Applied
Pressure Test Position	Vertical
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+1.1WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+1.1WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+1.1WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+1.1WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW



Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16	
Wind Load Reduction Scale Factor	1.100	
Basic Wind Speed	[V] 97.2	km/hr
Surface Roughness Category	C: Open Terrain	
Type of Surface	Moderately Smooth	
Base Elevation	300	mm
Percent Wind for Hydrotest	50.0	
Using User defined Wind Press. Vs Elev.	N	
Height of Hill or Escarpment	H or Hh	0 mm
Distance Upwind of Crest	Lh	0 mm
Distance from Crest to the Vessel	x	0 mm
Type of Terrain (Hill, Escarpment)	Flat	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	
Site Elevation above Sea Level	0	mm

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration	[Ss] 0.147
Max. Eff. Ground Acceleration	[S] 0.065
Force Modification Factor R	2.000
Site Class	D
Component Elevation Ratio	z/h 0.000
Amplification Factor	Ap 2.500
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

M.A.W.P + Static Head to Nozzle	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Using Metric Material Databases, ASME II D	Yes
Using ASME VIII-1 Appendix 46	No
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/11
Element From Node	10
Element To Node	20
Element Type	Skirt Support
Description	Skirt-3
Distance "FROM" to "TO"	2000 mm
Skirt Outside Diameter	1754 mm
Diameter of Skirt at Base	1754 mm
Skirt Thickness	14 mm
Internal Corrosion Allowance	0.8 mm



Nominal Thickness	14	mm
External Corrosion Allowance	0.8	mm
Design Temperature Internal Pressure	130	°C
Design Temperature External Pressure	130	°C
Effective Diameter Multiplier	1.3	
Material Name	SA-516 70	+IT
Allowable Stress, Ambient	138	MPa
Allowable Stress, Operating	138	MPa
Allowable Stress, Hydrotest	234	MPa
Material Density	0.00775	kg/cm³
P Number Thickness	0	mm
Yield Stress, Operating	234.4	MPa
MDMT Curve	Impact Tested	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.7	
Weld is pre-Heated	No	
Element From Node	10	
Detail Type	Weight	
Detail ID	Tailing lug	
Dist. from "FROM" Node / Offset dist	0	mm
Miscellaneous Weight	4903	N
Offset from Element Centerline	0	mm

Element#	2/11	
Element From Node	20	
Element To Node	30	
Element Type	Skirt Support	
Description	Skirt 2	
Distance "FROM" to "TO"	3000	mm
Skirt Outside Diameter	1754	mm
Diameter of Skirt at Base	1754	mm
Skirt Thickness	14	mm
Internal Corrosion Allowance	0.8	mm
Nominal Thickness	14	mm
External Corrosion Allowance	0.8	mm
Design Temperature Internal Pressure	130	°C
Design Temperature External Pressure	130	°C
Effective Diameter Multiplier	1.3	
Material Name	SA-516 70	+IT
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.7	
Weld is pre-Heated	No	
Element From Node	20	
Detail Type	Platform	
Detail ID	PL-8	
Dist. from "FROM" Node / Offset dist	2000	mm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	90.0	
Platform Wind Area	1499999	mm²
Platform Weight	8262.9	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	85	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	4000	mm
Unit Weight of Ladder	0.37	N/mm
Platform Length (top head platform)	0	mm



Element#	3/11
Element From Node	30
Element To Node	40
Element Type	Skirt Support
Description	Skirt 1
Distance "FROM" to "TO"	1000 mm
Skirt Outside Diameter	1754 mm
Diameter of Skirt at Base	1754 mm
Skirt Thickness	14 mm
Internal Corrosion Allowance	0.8 mm
Nominal Thickness	14 mm
External Corrosion Allowance	0.8 mm
Design Temperature Internal Pressure	130 °C
Design Temperature External Pressure	130 °C
Effective Diameter Multiplier	1.3
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	0.7
Efficiency, Head-to-Skirt or Circ. Seam	0.5
Weld is pre-Heated	No

Element From Node	30
Detail Type	Insulation
Detail ID	Ins
Dist. from "FROM" Node / Offset dist	500 mm
Height/Length of Insulation	500 mm
Thickness of Insulation	80 mm
Density	0.000352 kg/cm³

Element#	4/11
Element From Node	40
Element To Node	50
Element Type	Elliptical
Description	Bottom head
Distance "FROM" to "TO"	50 mm
Inside Diameter	1700 mm
Element Thickness	32 mm
Internal Corrosion Allowance	6 mm
Nominal Thickness	38 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	4.1 MPa
Design Temperature Internal Pressure	130 °C
Design External Pressure	0.1034 MPa
Design Temperature External Pressure	130 °C
Effective Diameter Multiplier	1.3
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No

Element From Node	40
Detail Type	Platform
Detail ID	PL-1
Dist. from "FROM" Node / Offset dist	0 mm
Platform Start Angle (degrees)	0.0
Platform End Angle (degrees)	360.0
Platform Wind Area	2999999 mm²
Platform Weight	28572 N
Platform Railing Weight	0 N/mm
Platform Grating Weight	0.017 kg/cm²
Platform Width	1500 mm
Platform Height	1000 mm
Platform Clearance or End Offset	115 mm
Platform Force Coefficient	1.0
Ladder Layout Angle	0.0
Ladder Start Elevation	0 mm
Ladder End Elevation	3000 mm



Unit Weight of Ladder	0.37	N/mm
Platform Length (top head platform)	0	mm
Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid 40	
Dist. from "FROM" Node / Offset dist	-425	mm
Height/Length of Liquid	475	mm
Liquid Density	0.001021	kg/cm ³
Element From Node	40	
Detail Type	Insulation	
Detail ID	Ins: 10	
Dist. from "FROM" Node / Offset dist	-425	mm
Height/Length of Insulation	475	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm ³
Element From Node	40	
Detail Type	Nozzle	
Detail ID	D150	
Dist. from "FROM" Node / Offset dist	0	mm
Nozzle Diameter	150	mm
Nozzle Schedule	120	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	2317.6	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	
Element From Node	40	
Detail Type	Weight	
Detail ID	D FY	
Dist. from "FROM" Node / Offset dist	0	mm
Miscellaneous Weight	1843.5	N
Offset from Element Centerline	2118	mm
Element From Node	40	
Detail Type	Weight	
Detail ID	D anti-swirl ba	
Dist. from "FROM" Node / Offset dist	-400	mm
Miscellaneous Weight	98.06	N
Offset from Element Centerline	0	mm

Element#	5/11	
Element From Node	50	
Element To Node	60	
Element Type	Cylinder	
Description	Shell-1	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	1700	mm
Element Thickness	34	mm
Internal Corrosion Allowance	6	mm
Nominal Thickness	34	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	4.1	MPa
Design Temperature Internal Pressure	130	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	130	°C
Effective Diameter Multiplier	1.3	
Material Name	SA-516 70 +IT	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	50	
Detail Type	Liquid	



Detail ID	Liquid 50	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	3000	mm
Liquid Density	0.001021	kg/cm ³

Element From Node	50	
Detail Type	Insulation	
Detail ID	Ins: 20	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm ³

Element From Node	50	
Detail Type	Nozzle	
Detail ID	MH4600	
Dist. from "FROM" Node / Offset dist	800	mm
Nozzle Diameter	600	mm
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	270.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	

Element From Node	50	
Detail Type	Nozzle	
Detail ID	B150	
Dist. from "FROM" Node / Offset dist	2800	mm
Nozzle Diameter	150	mm
Nozzle Schedule	120	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	2317.6	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	

Element From Node	50	
Detail Type	Nozzle	
Detail ID	L150	
Dist. from "FROM" Node / Offset dist	2500	mm
Nozzle Diameter	50.8	mm
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	

Element From Node	50	
Detail Type	Nozzle	
Detail ID	L250	
Dist. from "FROM" Node / Offset dist	200	mm
Nozzle Diameter	50.8	mm
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	

Element From Node	50	
Detail Type	Nozzle	
Detail ID	L350	
Dist. from "FROM" Node / Offset dist	2000	mm
Nozzle Diameter	50.8	mm



Nozzle Schedule	None
Nozzle Class	300
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105 +IT

Element From Node	50
Detail Type	Nozzle
Detail ID	L450
Dist. from "FROM" Node / Offset dist	1000 mm
Nozzle Diameter	50.8 mm
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105 +IT

Element From Node	50
Detail Type	Nozzle
Detail ID	L550
Dist. from "FROM" Node / Offset dist	2000 mm
Nozzle Diameter	50.8 mm
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105 +IT

Element From Node	50
Detail Type	Nozzle
Detail ID	L650
Dist. from "FROM" Node / Offset dist	1000 mm
Nozzle Diameter	50.8 mm
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105 +IT

Element From Node	50
Detail Type	Weight
Detail ID	MH4 Davit
Dist. from "FROM" Node / Offset dist	800 mm
Miscellaneous Weight	980.6 N
Offset from Element Centerline	0 mm

Element From Node	50
Detail Type	Weight
Detail ID	B FY
Dist. from "FROM" Node / Offset dist	2800 mm
Miscellaneous Weight	7864.4 N
Offset from Element Centerline	2118 mm

Element From Node	50
Detail Type	Weight
Detail ID	Bed2 sup
Dist. from "FROM" Node / Offset dist	2950 mm
Miscellaneous Weight	98.06 N
Offset from Element Centerline	0 mm

Element From Node	50
Detail Type	Weight



Detail ID	Bed 2 sup grid	
Dist. from "FROM" Node / Offset dist	2950	mm
Miscellaneous Weight	1765.1	N
Offset from Element Centerline	0	mm

Element#	6/11	
Element From Node	60	
Element To Node	70	
Element Type	Cylinder	
Description	Shell-2	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	1700	mm
Element Thickness	34	mm
Internal Corrosion Allowance	6	mm
Nominal Thickness	34	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	4.1	MPa
Design Temperature Internal Pressure	130	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	130	°C
Effective Diameter Multiplier	1.3	
Material Name	SA-516 70	+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	60	
Detail Type	Platform	
Detail ID	PL-2	
Dist. from "FROM" Node / Offset dist	1000	mm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	180.0	
Platform Wind Area	2999999	mm ²
Platform Weight	15211	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	115	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	4000	mm
Unit Weight of Ladder	0.37	N/mm
Platform Length (top head platform)	0	mm

Element From Node	60	
Detail Type	Packing	
Detail ID	Bed 1	
Dist. from "FROM" Node / Offset dist	760	mm
Height of Packed Section	2240	mm
Density	0.000224	kg/cm ³
Percent Volume Holdup	10.0	
Specific Gravity of Packing Liquid	1.023	

Element From Node	60	
Detail Type	Liquid	
Detail ID	Liquid: 60	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	700	mm
Liquid Density	0.001021	kg/cm ³

Element From Node	60	
Detail Type	Insulation	
Detail ID	Ins: 30	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	80	mm



Density	0.000352	kg/cm ³
Element From Node	60	
Detail Type	Nozzle	
Detail ID	DP150	
Dist. from "FROM" Node / Offset dist	650	mm
Nozzle Diameter	50.8	mm
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	+IT
Element From Node	60	
Detail Type	Nozzle	
Detail ID	UC2600	
Dist. from "FROM" Node / Offset dist	1000	mm
Nozzle Diameter	600	mm
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	+IT

Element#	7/11	
Element From Node	70	
Element To Node	80	
Element Type	Cylinder	
Description	Shell-3	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	1700	mm
Element Thickness	34	mm
Internal Corrosion Allowance	6	mm
Nominal Thickness	34	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	4.1	MPa
Design Temperature Internal Pressure	130	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	130	°C
Effective Diameter Multiplier	1.3	
Material Name	SA-516 70	+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	70	
Detail Type	Platform	
Detail ID	PL-3	
Dist. from "FROM" Node / Offset dist	2000	mm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	180.0	
Platform Wind Area	2999999	mm ²
Platform Weight	15211	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	115	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	4000	mm
Unit Weight of Ladder	0.37	N/mm
Platform Length (top head platform)	0	mm



Element From Node	70	
Detail Type	Packing	
Detail ID	Bed 1	
Dist. from "FROM" Node / Offset dist	0	mm
Height of Packed Section	2035	mm
Density	0.000224	kg/cm ³
Percent Volume Holdup	10.0	
Specific Gravity of Packing Liquid	1.023	

Element From Node	70	
Detail Type	Insulation	
Detail ID	Ins: 40	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm ³

Element From Node	70	
Detail Type	Nozzle	
Detail ID	MH3600	
Dist. from "FROM" Node / Offset dist	2300	mm
Nozzle Diameter	600	mm
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	

Element From Node	70	
Detail Type	Nozzle	
Detail ID	DP250	
Dist. from "FROM" Node / Offset dist	2500	mm
Nozzle Diameter	50.8	mm
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	

Element From Node	70	
Detail Type	Weight	
Detail ID	Distributor 2	
Dist. from "FROM" Node / Offset dist	2500	mm
Miscellaneous Weight	1961.2	N
Offset from Element Centerline	0	mm

Element From Node	70	
Detail Type	Weight	
Detail ID	Dist2 sup	
Dist. from "FROM" Node / Offset dist	2450	mm
Miscellaneous Weight	98.06	N
Offset from Element Centerline	0	mm

Element From Node	70	
Detail Type	Weight	
Detail ID	MH3 Davit	
Dist. from "FROM" Node / Offset dist	2300	mm
Miscellaneous Weight	980.6	N
Offset from Element Centerline	0	mm

Element From Node	70	
Detail Type	Weight	
Detail ID	Bed2 lim sup	
Dist. from "FROM" Node / Offset dist	1300	mm
Miscellaneous Weight	98.06	N



Offset from Element Centerline	0	mm
Element From Node	70	
Detail Type	Weight	
Detail ID	Bed2 limiter	
Dist. from "FROM" Node / Offset dist	0	mm
Miscellaneous Weight	490.3	N
Offset from Element Centerline	0	mm

Element#	8/11
Element From Node	80
Element To Node	90
Element Type	Cylinder
Description	Shell-4
Distance "FROM" to "TO"	3805 mm
Inside Diameter	1700 mm
Element Thickness	34 mm
Internal Corrosion Allowance	6 mm
Nominal Thickness	34 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	4.1 MPa
Design Temperature Internal Pressure	130 °C
Design External Pressure	0.1034 MPa
Design Temperature External Pressure	130 °C
Effective Diameter Multiplier	1.3
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	80
Detail Type	Platform
Detail ID	PL-4
Dist. from "FROM" Node / Offset dist	2000 mm
Platform Start Angle (degrees)	0.0
Platform End Angle (degrees)	180.0
Platform Wind Area	2999999 mm ²
Platform Weight	15227 N
Platform Railing Weight	0 N/mm
Platform Grating Weight	0.017 kg/cm ²
Platform Width	1500 mm
Platform Height	1000 mm
Platform Clearance or End Offset	115 mm
Platform Force Coefficient	1.0
Ladder Layout Angle	0.0
Ladder Start Elevation	0 mm
Ladder End Elevation	4000 mm
Unit Weight of Ladder	0.37 N/mm
Platform Length (top head platform)	0 mm

Element From Node	80
Detail Type	Packing
Detail ID	Bed-1
Dist. from "FROM" Node / Offset dist	530 mm
Height of Packed Section	3275 mm
Density	0.000224 kg/cm ³
Percent Volume Holdup	10.0
Specific Gravity of Packing Liquid	1.023

Element From Node	80
Detail Type	Insulation
Detail ID	Ins: 50
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	3805 mm
Thickness of Insulation	80 mm
Density	0.000352 kg/cm ³

Element From Node	80
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Detail Type	Nozzle	
Detail ID	UC1600	
Dist. from "FROM" Node / Offset dist	1000	mm
Nozzle Diameter	600	mm
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	

Element From Node	80	
Detail Type	Weight	
Detail ID	Bed1 sup	
Dist. from "FROM" Node / Offset dist	450	mm
Miscellaneous Weight	98.06	N
Offset from Element Centerline	0	mm

Element From Node	80	
Detail Type	Weight	
Detail ID	Bed 1 sup grid	
Dist. from "FROM" Node / Offset dist	500	mm
Miscellaneous Weight	1765.1	N
Offset from Element Centerline	0	mm

Element From Node	80	
Detail Type	Weight	
Detail ID	Clip wt	
Dist. from "FROM" Node / Offset dist	2000	mm
Miscellaneous Weight	27457	N
Offset from Element Centerline	0	mm

Element From Node	80	
Detail Type	Weight	
Detail ID	Misc wt 7%	
Dist. from "FROM" Node / Offset dist	1950	mm
Miscellaneous Weight	29418	N
Offset from Element Centerline	0	mm

Element#	9/11	
Element From Node	90	
Element To Node	100	
Element Type	Cylinder	
Description	Shell-5	
Distance "FROM" to "TO"	2195	mm
Inside Diameter	1706	mm
Element Thickness	28	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	28	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	4.1	MPa
Design Temperature Internal Pressure	130	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	130	°C
Effective Diameter Multiplier	1.3	
Material Name	SA-516 70 +IT	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	90	
Detail Type	Platform	
Detail ID	PL-5	
Dist. from "FROM" Node / Offset dist	2000	mm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	180.0	
Platform Wind Area	2999999	mm ²



Platform Weight	15203	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	115	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	4000	mm
Unit Weight of Ladder	0.37	N/mm
Platform Length (top head platform)	0	mm

Element From Node	90	
Detail Type	Packing	
Detail ID	Bed-1	
Dist. from "FROM" Node / Offset dist	0.001161	mm
Height of Packed Section	805	mm
Density	0.000224	kg/cm ³
Percent Volume Holdup	10.0	
Specific Gravity of Packing Liquid	1.023	

Element From Node	90	
Detail Type	Insulation	
Detail ID	Ins: 60	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	2195	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm ³

Element From Node	90	
Detail Type	Lining	
Detail ID	Clad	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	2195	mm
Thickness of Lining	3	mm
Density	0.008027	kg/cm ³

Element From Node	90	
Detail Type	Nozzle	
Detail ID	A150	
Dist. from "FROM" Node / Offset dist	2120	mm
Nozzle Diameter	150	mm
Nozzle Schedule	120	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	

Element From Node	90	
Detail Type	Nozzle	
Detail ID	MH2600	
Dist. from "FROM" Node / Offset dist	1895	mm
Nozzle Diameter	600	mm
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	

Element From Node	90	
Detail Type	Weight	
Detail ID	Distributor	
Dist. from "FROM" Node / Offset dist	2000	mm
Miscellaneous Weight	1961.2	N
Offset from Element Centerline	0	mm



Element From Node	90
Detail Type	Weight
Detail ID	Distributor sup
Dist. from "FROM" Node / Offset dist	1950 mm
Miscellaneous Weight	98.06 N
Offset from Element Centerline	0 mm

Element From Node	90
Detail Type	Weight
Detail ID	Bed limiter sup
Dist. from "FROM" Node / Offset dist	1850 mm
Miscellaneous Weight	98.06 N
Offset from Element Centerline	0 mm

Element From Node	90
Detail Type	Weight
Detail ID	Bed limiter
Dist. from "FROM" Node / Offset dist	1900 mm
Miscellaneous Weight	490.3 N
Offset from Element Centerline	0 mm

Element From Node	90
Detail Type	Weight
Detail ID	A FY
Dist. from "FROM" Node / Offset dist	2120 mm
Miscellaneous Weight	15748 N
Offset from Element Centerline	2118 mm

Element#	10/11
Element From Node	100
Element To Node	110
Element Type	Cylinder
Description	Shell-7
Distance "FROM" to "TO"	3100 mm
Inside Diameter	1706 mm
Element Thickness	28 mm
Internal Corrosion Allowance	1 mm
Nominal Thickness	28 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	4.1 MPa
Design Temperature Internal Pressure	130 °C
Design External Pressure	0.1034 MPa
Design Temperature External Pressure	130 °C
Effective Diameter Multiplier	1.3
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	100
Detail Type	Platform
Detail ID	PL-6
Dist. from "FROM" Node / Offset dist	1000 mm
Platform Start Angle (degrees)	0.0
Platform End Angle (degrees)	180.0
Platform Wind Area	2999999 mm ²
Platform Weight	14833 N
Platform Railing Weight	0 N/mm
Platform Grating Weight	0.017 kg/cm ²
Platform Width	1500 mm
Platform Height	1000 mm
Platform Clearance or End Offset	115 mm
Platform Force Coefficient	1.0
Ladder Layout Angle	0.0
Ladder Start Elevation	0 mm
Ladder End Elevation	3000 mm
Unit Weight of Ladder	0.37 N/mm



Platform Length (top head platform)	0	mm
Element From Node	100	
Detail Type	Insulation	
Detail ID	Ins: 70	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3100	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm ³
Element From Node	100	
Detail Type	Lining	
Detail ID	Clad	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	3100	mm
Thickness of Lining	3	mm
Density	0.008027	kg/cm ³
Element From Node	100	
Detail Type	Nozzle	
Detail ID	E40	
Dist. from "FROM" Node / Offset dist	900	mm
Nozzle Diameter	40	mm
Nozzle Schedule	None	
Nozzle Class	900	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	
Element From Node	100	
Detail Type	Nozzle	
Detail ID	MH1600	
Dist. from "FROM" Node / Offset dist	1550	mm
Nozzle Diameter	600	mm
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	
Element From Node	100	
Detail Type	Tray	
Detail ID	TRAY 1-3	
Dist. from "FROM" Node / Offset dist	240	mm
Tray Spacing	304.8	mm
Number of Trays	3	
Tray Weight per Unit Area	0.015	kg/cm ²
Height of Liquid on Tray	0	mm
Density of Liquid on Tray	0	kg/cm ³
Support Ring and Bolting Bar Weight	0	N
Element From Node	100	
Detail Type	Weight	
Detail ID	Demister	
Dist. from "FROM" Node / Offset dist	2850	mm
Miscellaneous Weight	1176.7	N
Offset from Element Centerline	0	mm
Element From Node	100	
Detail Type	Weight	
Detail ID	Demister sup	
Dist. from "FROM" Node / Offset dist	2800	mm
Miscellaneous Weight	490.3	N
Offset from Element Centerline	0	mm
Element From Node	100	



Detail Type	Weight	
Detail ID	TSR bolting bar	
Dist. from "FROM" Node / Offset dist	200	mm
Miscellaneous Weight	784.48	N
Offset from Element Centerline	0	mm
Element From Node	100	
Detail Type	Weight	
Detail ID	MH1 Davit	
Dist. from "FROM" Node / Offset dist	1550	mm
Miscellaneous Weight	980.6	N
Offset from Element Centerline	0	mm
Element From Node	100	
Detail Type	Weight	
Detail ID	Trunnion 1	
Dist. from "FROM" Node / Offset dist	2100	mm
Miscellaneous Weight	4903	N
Offset from Element Centerline	400	mm
Element From Node	100	
Detail Type	Weight	
Detail ID	Trunnion 2	
Dist. from "FROM" Node / Offset dist	2100	mm
Miscellaneous Weight	4903	N
Offset from Element Centerline	400	mm

Element#	11/11	
Element From Node	110	
Element To Node	120	
Element Type	Elliptical	
Description	Top head	
Distance "FROM" to "TO"	50	mm
Inside Diameter	1706	mm
Element Thickness	26	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	32	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	4.1	MPa
Design Temperature Internal Pressure	130	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	130	°C
Effective Diameter Multiplier	1.3	
Material Name	SA-516 70 +IT	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	

Element From Node	110	
Detail Type	Platform	
Detail ID	PL-7	
Dist. from "FROM" Node / Offset dist	500	mm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	0.0	
Platform Wind Area	3764000	mm ²
Platform Weight	24555	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	3764	mm
Platform Height	1000	mm
Platform Clearance or End Offset	1882	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	2500	mm
Unit Weight of Ladder	0.37	N/mm
Platform Length (top head platform)	3764	mm



Element From Node	110	
Detail Type	Insulation	
Detail ID	Ins: 90	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	475	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm ³

Element From Node	110	
Detail Type	Lining	
Detail ID	Clad	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	476.5	mm
Thickness of Lining	3	mm
Density	0.008027	kg/cm ³

Element From Node	110	
Detail Type	Nozzle	
Detail ID	C100	
Dist. from "FROM" Node / Offset dist	0	mm
Nozzle Diameter	100	mm
Nozzle Schedule	160	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	

Element From Node	110	
Detail Type	Weight	
Detail ID	C FY	
Dist. from "FROM" Node / Offset dist	0	mm
Miscellaneous Weight	5540.4	N
Offset from Element Centerline	2093	mm

Element From Node	110	
Detail Type	For./Mom.	
Detail ID	C F/M	
Dist. from "FROM" Node / Offset dist	350	mm
Force in X Direction	5497.9	N
Force in Y Direction	-8246.8	N
Force in Z Direction	5497.9	N
Moment about X Axis	1540080	N-mm
Moment about Y Axis	0	N-mm
Moment about Z Axis	1540080	N-mm
Force/Moment Combination Method	SRSS	



XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
Skirt-3	...	...	2000	...	2000
Skirt 2	...	...	5000	...	3000
Skirt 1	...	...	6000	...	1000
Bottom head	...	...	6050	...	50
Shell-1	...	...	9050	...	3000
Shell-2	...	...	12050	...	3000
Shell-3	...	...	15050	...	3000
Shell-4	...	...	18855	...	3805
Shell-5	...	...	21050	...	2195
Shell-7	...	...	24150	...	3100
Top head	...	...	24200	...	50

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd MPa	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) MPa	Element#
Skirt-3	...		14	1.6	1726	...	1
Skirt 2	...		14	1.6	1726	...	2
Skirt 1	...		14	1.6	1726	...	3
Bottom head	4.1419		38	6	1700	138	4
Shell-1	4.1371		34	6	1700	138	5
Shell-2	4.107		34	6	1700	138	6
Shell-3	4.1		34	6	1700	138	7
Shell-4	4.1		34	6	1700	138	8
Shell-5	4.1		28	1	1706	138	9
Shell-7	4.1		28	1	1706	138	10
Top head	4.1		32	1	1706	138	11

Element Required Thickness and MAWP :

From	To	Design Pressure MPa	M.A.W.P. Corroded MPa	M.A.P. New & Cold MPa	Minimum Thickness mm	Required Thickness mm	Element#
Skirt-3	...	...	No Calc	No Calc	14	No Calc	1
Skirt 2	...	...	No Calc	No Calc	14	No Calc	2
Skirt 1	...	...	No Calc	No Calc	14	No Calc	3
Bottom head	4.1	4.5907	5.71268	32	29.2252	4	
Shell-1	4.1	4.40457	5.41234	34	32.0511	5	
Shell-2	4.1	4.43465	5.41234	34	31.8589	6	
Shell-3	4.1	4.44167	5.41234	34	31.8141	7	
Shell-4	4.1	4.44167	5.41234	34	31.8141	8	
Shell-5	4.1	4.29536	4.45704	28	26.7537	9	
Shell-7	4.1	4.29536	4.45704	28	26.7537	10	
Top head	4.1	4.45468	4.63436	26	23.9946	11	

Minimum 4 4

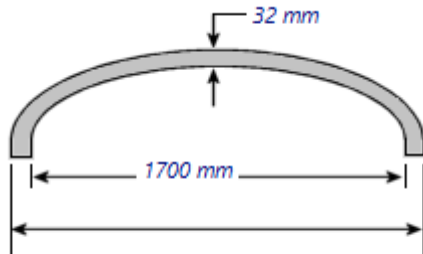
MAWP: 4 MPa, limited by: Shell-7.

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Bottom head

Elliptical Head Design Information:					
Design Pressure	P	4.142 MPa	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		130 °C
Length		50.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	1700.000 mm	Actual Stress		123.4 MPa
Spec. Min. Thk t _e , or s _a		32.0000 mm	Spec. Nominal Thk.		38.000 mm
Surface Area		3698054.2 mm ²	SG of Contents		1.023
Element Volume		756.7 ltr	Weight of Contents		7586.6 N
Empty Weight		11229.5 N	Operating Weight		18816.1 N
			Comp. All. Ope. Stress		115.3 MPa
			Aspect Ratio		2.00

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 21.250 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 10.625 mm

Material treatment options: +IT

Head Calculation - Section 4.3.6 (and Section 4.3.7 for Ellipsoidal Head)

Computed Minimum Required Thickness [t]:

$t_r = 23.2252 \text{ mm}$

$t = t_r + c_i + c_o$
 $= 23.2252 + 6.0 + 0.0$
 $= 29.2252 \text{ mm}$ - see below for the derivation

For Elliptical heads: Compute [h, k, r,] per 4.3.7.1

$h = D / (2 \cdot \text{Aspect Ratio})$
 $= 1700.0 / (2 \cdot 2.0)$
 $= 425.000 \text{ mm}$

$k = D / (2 \cdot h)$
 $= 1700.0 / (2 \cdot 425.0)$
 $= 2.000$

$r = D (0.5/k - 0.08) + c_a$



$$= 1700.0(0.5/2.0 - 0.08) + 6.0$$

$$= 295.000 \text{ mm}$$

$$L = D(0.44 \cdot k + 0.02) + ca$$

$$= 1700.0(0.44 \cdot 2.0 + 0.02) + 6.0$$

$$= 1536.000 \text{ mm}$$

Step 3: Geometric Constants [βTh , ϕTh , Rth]:

Equations 4.3.8 to 4.3.11 as appropriate

t (corroded) and P for the calc.: $t = 23.2252 \text{ mm}$, $P = 4.1419 \text{ MPa}$

Angle [βTh]:

$$= \arccos[(0.5 \cdot D - r)/(L - r)]$$

$$= \arccos[(0.5 \cdot 1712.0 - 295.0)/(1536.0 - 295.0)]$$

$$= 1.1017 \text{ radians - Equation 4.3.8}$$

$$\phi Th = \sqrt{(L \cdot t) / r}$$

$$= \sqrt{(1536.0 \cdot 23.225) / 295.0}$$

$$= 0.6403 \text{ radians - Equation 4.3.9}$$

$\phi Th < \beta Th$, Thus:

$$Rth = (0.5 \cdot D - r) / \cos(\beta Th - \phi Th) + r$$

$$= (0.5 \cdot 1712.0 - 295.0) / \cos(1.102 - 0.64) + 295.0$$

$$= 921.5375 \text{ mm - Equation 4.3.10}$$

Step 4: Coefficients: Equations 4.3.12 to 4.3.15 [$C1$ and $C2$]:

$r/D > 0.08$, Thus:

$$C1 = 0.692(r/D) + 0.605$$

$$= 0.692(295.0/1712.0) + 0.605$$

$$= 0.7242 \text{ - Equation 4.3.13}$$

$$C2 = 1.46 - 2.6(r/D) = 1.46 - 2.6(295.0/1712.0)$$

$$= 1.012, \text{ Equation 4.3.15}$$

Step 5: Pressure to produce knuckle elastic buckling [$Peth$]:

$$= C1 \cdot Et \cdot t^2 / (C2 \cdot Rth(0.5 \cdot Rth - r))$$

$$= 0.724 \cdot 196301.391 \cdot 23.225^2 / (1.012 \cdot 921.538(0.5 \cdot 921.538 - 295.0))$$

$$= 496.06 \text{ MPa, Equation 4.3.16}$$

Step 6: Internal Pressure for max stress in knuckle equals yield [Py]:

$C3 = Sy = 234.397 \text{ MPa}$, Not in the creep range - 4.3.6.1(f)

$$Py = C3 \cdot t / (C2 \cdot Rth(0.5 \cdot Rth / r - 1))$$

$$= 234.397 \cdot 23.225 / (1.012 \cdot 921.538(0.5 \cdot 921.538 / 295.0 - 1))$$

$$= 10.388 \text{ MPa, Equation 4.3.17}$$

Step 7: Value of Internal Press. resulting in buckling of knuckle [G and Pck]:

$$G = Peth/Py = 496.059/10.388 = 47.752$$

$G > 1$, Thus:

Buckling Pressure [Pck]:

$$= ((0.77508 \cdot G - 0.20354 \cdot G^2 + 0.019274 \cdot G^3) /$$

$$(1 + 0.19014 \cdot G - 0.089534 \cdot G^2 + 0.0093965 \cdot G^3)) Py$$

$$= ((0.77508 \cdot 47.752 - 0.20354 \cdot 47.752^2 + 0.019274 \cdot 47.752^3) /$$

$$(1 + 0.19014 \cdot 47.752 - 0.089534 \cdot 47.752^2 + 0.0093965 \cdot 47.752^3)) \cdot 10.388$$

$$= 20.94 \text{ MPa, Equation 4.3.19}$$

Step 8: Compute Allowable Pressure base on buckling of knuckle [Pak]:

$$= Pck/1.5 = 20.94/1.5 = 13.96 \text{ MPa - Equation 4.3.21}$$

Step 9: Compute allowable pressure based on rupture of crown [Pac]:

$$= 2 \cdot S \cdot E / (L/t + 0.5)$$

$$= 2 \cdot 137.997 \cdot 1.0 / (1536.0/23.225 + 0.5)$$

$$= 4.14 \text{ MPa, Equation 4.3.22}$$

Step 10: Compute final pressure [Pa]:

$$= \min(Pak, Pac) = \min(13.963, 4.142)$$

$$= 4.14 \text{ MPa, Equation 4.3.23: (same as original design pressure)}$$



Allowable pressure in Crown at given thickness [Pac]:

$$\begin{aligned} &= 2 \cdot S \cdot E / (L / t + 0.5) \\ &= 2 \cdot 137.997 \cdot 1.0 / (1536.0 / 26.0 + 0.5) \\ &= 4.633 \text{ MPa} \end{aligned}$$

Computed Maximum Allowable Pressure [MAWP]:

$$= \text{MaxPressure} - \text{Pliq} = 4.6326 - 0.042 = 4.5907 \text{ MPa}$$

Computed Maximum Allowable Pressure New and Cold [MAPnc]:

$$= 5.7127 \text{ MPa}$$

Computed Membrane Stress Intensity at Design Pressure [Sm]:

$$= 123.4 \text{ MPa}$$

Required Thickness of Straight Flange = 32.082 mm

Percent Fiber Elongation per Table 6.1

$$\begin{aligned} \text{For Heads formed from Segments } (75 \cdot t_{\text{nom}} / R_f) \cdot (1 - R_f / R_o) & \quad 9.253 \% \\ \text{For one Piece doubly Curved Heads } 100 \cdot \ln(D_b / (D_f - 2t)) & \quad 24.406 \% \end{aligned}$$

Where D_b is the blank diameter for one piece doubly curved surfaces [D_b]:

$$\begin{aligned} &= \text{square root}(4 \cdot \text{Head Surface Area} / 3.14159) \\ &= \text{square root}(4 \cdot 3698054 / 3.14159) \\ &= 2169.910 \text{ mm} \end{aligned}$$

MDMT Calculations in the Knuckle Portion:

Note: This Element/Detail was specified as being Impact Tested.

MDMT Calculations in the Head Straight Flange:

Note: This Element/Detail was specified as being Impact Tested.

Tolerance for Formed Heads per 4.3.2.2:

$$\begin{aligned} \text{Head inner surface maximum deviation outside the specified shape, } 1.25\% \text{ of } D: & \quad 21.250 \text{ mm} \\ \text{Head inner surface maximum deviation inside the specified shape, } 0.625\% \text{ of } D: & \quad 10.625 \text{ mm} \end{aligned}$$

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: Bottom head

Elliptical Head From 40 to 50 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$\begin{aligned} &= 0.075 \cdot E_y (t / R_o) \\ &= 0.075 \cdot 196301 (26.0 / 1533.685) \\ &= 249.587 \text{ MPa} \end{aligned}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$\begin{aligned} &= F_{he} / S_y \\ &= 249.6 / 234.4 \\ &= 1.065 \end{aligned}$$

Engineering Yield to Tensile Stress ratio [R]:

$$\begin{aligned} &= \sigma_y / \sigma_{uts} \\ &= 234.4 / 482.6 \\ &= 0.486 \end{aligned}$$

Ratio [AE]:

$$\begin{aligned} &= F_{he} / E \\ &= 249.6 / 196301.4 \\ &= 0.001271 \end{aligned}$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / (\ln(\ln(1 + \epsilon_{psp})) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / (\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys}) m_1$$

$$= 234.4 (1 + 0.002) / \ln(1 + 0.002) 0.157304$$

$$= 624.376 \text{ MPa}$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m_2 f(m_2^2 - R)$$

$$= 0.6 (1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m_2)) / (m_2 m_2)$$

$$= (482.6 \cdot \exp(0.309)) / (0.309 0.309)$$

$$= 944.570 \text{ MPa}$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R_{1.5} - 0.5 \cdot R_{2.5} - R_{3.5}$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-04	0.271E-04	0.807E-04	-0.632E-04	0.124E+05	0.182E-01
2	120	-4.651	0.763E-06	0.763E-06	0.171E-04	-0.171E-04	0.151E+06	0.472E-03
3	180	-3.245	0.670E-05	0.665E-05	0.423E-04	-0.419E-04	0.529E+05	0.215E-02
4	150	-3.948	0.252E-05	0.252E-05	0.281E-04	-0.281E-04	0.980E+05	0.267E-03
5	135	-4.299	0.143E-05	0.143E-05	0.222E-04	-0.222E-04	0.125E+06	0.188E-03
6	143	-4.124	0.192E-05	0.191E-05	0.251E-04	-0.250E-04	0.112E+06	0.129E-04
7	139	-4.211	0.166E-05	0.166E-05	0.236E-04	-0.236E-04	0.118E+06	0.932E-04
8	141	-4.167	0.178E-05	0.178E-05	0.244E-04	-0.243E-04	0.115E+06	0.417E-04
9	142	-4.146	0.185E-05	0.185E-05	0.247E-04	-0.247E-04	0.113E+06	0.148E-04
10	142	-4.135	0.188E-05	0.188E-05	0.249E-04	-0.249E-04	0.112E+06	0.106E-05
11	143	-4.129	0.190E-05	0.190E-05	0.250E-04	-0.250E-04	0.112E+06	0.589E-05
12	142	-4.132	0.189E-05	0.189E-05	0.249E-04	-0.249E-04	0.112E+06	0.241E-05
13	142	-4.133	0.189E-05	0.188E-05	0.249E-04	-0.249E-04	0.112E+06	0.671E-06
14	142	-4.134	0.188E-05	0.188E-05	0.249E-04	-0.249E-04	0.112E+06	0.197E-06
15	142	-4.133	0.189E-05	0.188E-05	0.249E-04	-0.249E-04	0.112E+06	0.237E-06
16	142	-4.134	0.188E-05	0.188E-05	0.249E-04	-0.249E-04	0.112E+06	0.201E-07
17	142	-4.134	0.188E-05	0.188E-05	0.249E-04	-0.249E-04	0.112E+06	0.883E-07

Factor of Safety [FS]:

Since $Fic > 0.55 F_y$ and $Fic < F_y$

$$= 2.407 - 0.741 \cdot Fic / F_y$$

$$= 2.407 - 0.741 \cdot 142.85 / 234.4$$

$$= 1.955$$

Allowable Circumferential Compressive Stress [Fha]:

$$= Fic / FS$$

$$= 142.849 / 1.955$$

$$= 73.053 \text{ MPa}$$

Allowable External Pressure at the given Thickness [Pa]:

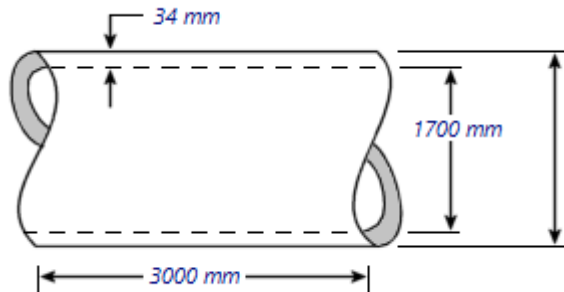
$$= 2 \cdot Fha [t / Ro]$$

$$= 2 \cdot 73.053 [26.0 / 1533.685]$$

$$= 2.477 \text{ MPa}$$

Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell-1

Cylindrical Shell Design Information:					
Design Pressure	P	4.137 MPa	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		130 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	1700.000 mm	Actual Stress		126.5 MPa
Spec. Min. Thk t,e, or sa		34.0000 mm	Spec. Nominal Thk.		34.000 mm
Surface Area		16663008.0 mm ²	SG of Contents		1.023
Element Volume		6810.6 ltr	Weight of Contents		68279.1 N
Empty Weight		42229.9 N	Operating Weight		110509.1 N
			Comp. All. Ope. Stress		116.4 MPa

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 1712.0 \cdot (\exp(4.137/(137.997 \cdot 1.0)) - 1) + 6.0 + 0.0$
 $= 26.0511 + 6.0 + 0.0 = 32.0511 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 137.997 \cdot 1.0 \cdot \ln((2 \cdot 28.0 + 1712.0) / 1712.0) - 0.037$
 $= 4.442 - 0.037 = 4.4046 \text{ MPa}$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 137.997 \cdot 1.0 \cdot \ln((2 \cdot 34.0 + 1700.0) / 1700.0)$
 $= 5.4123 \text{ MPa}$

Computed Stress Intensity at Design Pressure [Si]:
 $= P \cdot D / (E (D_o - D))$
 $= 4.137 \cdot 1712.0 / (1.0 (1768.0 - 1712.0))$
 $= 126.4775 \text{ MPa}$

Computed Membrane Stress at Design Pressure [Sm]:
 $= P / (E \cdot \ln((2 \cdot t + D) / D))$



$$= 4.137 / (1.0 \cdot \ln((2 \cdot 28.0 + 1712.0) / 1712.0))$$
$$= 128.5345 \text{ MPa}$$

Percent Fiber Elongation per Table 6.1

$$\text{Cylinders formed from Plate} \quad (50 \cdot t_{\text{nom}} / R_f) \cdot (1 - R_f / R_o) \quad 1.961 \%$$

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: Shell-1

Cylindrical Shell From 50 to 60 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$
$$= 18483.834 / \sqrt{(884.0 \cdot 28.0)}$$
$$= 117.486$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 2 \cdot (D_o/t)^{0.94}, \text{ Ch} = 0.55 / (D_o/t)$$
$$= 0.55 / (1768.0/28.0)$$
$$= 0.0087$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot \text{Ch} \cdot E_y(t / D_o)$$
$$= 1.6 \cdot 0.009 \cdot 196301(28.0/1768.0)$$
$$= 43.327 \text{ MPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= F_{he} / F_y$$
$$= 43.3/234.4$$
$$= 0.185$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$
$$= 234.4/482.6$$
$$= 0.486$$

Ratio [AE]:

$$= F_{he} / E$$
$$= 43.3/196301.4$$
$$= 0.000221$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$
$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$
$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$
$$= 234.4(1 + 0.002) / \ln(1 + 0.002)^{0.157304}$$
$$= 624.376 \text{ MPa}$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$
$$= 0.6(1.0 - 0.486)$$
$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m_2)) / (m_2 m_2)$$

$$= (482.6 \cdot \exp(0.309)) / (0.309 \cdot 0.309)$$

$$= 944.570 \text{ MPa}$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-04	0.271E-04	0.807E-04	-0.632E-04	0.124E+05	0.193E-01
2	120	-4.651	0.763E-06	0.763E-06	0.171E-04	-0.171E-04	0.151E+06	0.579E-03
3	60	-6.057	0.186E-07	0.186E-07	0.361E-05	-0.361E-05	0.195E+06	0.889E-04
4	30	-6.760	0.454E-09	0.454E-09	0.765E-06	-0.765E-06	0.196E+06	0.670E-04
5	45	-6.409	0.399E-08	0.399E-08	0.190E-05	-0.190E-05	0.196E+06	0.102E-04
6	37	-6.584	0.150E-08	0.150E-08	0.126E-05	-0.126E-05	0.196E+06	0.285E-04
7	41	-6.496	0.250E-08	0.250E-08	0.156E-05	-0.156E-05	0.196E+06	0.921E-05
8	43	-6.452	0.317E-08	0.317E-08	0.172E-05	-0.172E-05	0.196E+06	0.460E-06
9	42	-6.474	0.282E-08	0.282E-08	0.164E-05	-0.164E-05	0.196E+06	0.438E-05
10	42	-6.463	0.299E-08	0.299E-08	0.168E-05	-0.168E-05	0.196E+06	0.196E-05
11	43	-6.458	0.308E-08	0.308E-08	0.170E-05	-0.170E-05	0.196E+06	0.750E-06
12	43	-6.455	0.313E-08	0.313E-08	0.171E-05	-0.171E-05	0.196E+06	0.145E-06
13	43	-6.454	0.315E-08	0.315E-08	0.172E-05	-0.172E-05	0.196E+06	0.158E-06
14	43	-6.455	0.314E-08	0.314E-08	0.172E-05	-0.172E-05	0.196E+06	0.637E-08

Factor of Safety [FS]:

$$\text{Since } F_{ic} [43.27] < 0.55 \cdot 234.4, FS = 2$$

Allowable Circumferential Compressive Stress [Fha]:

$$= F_{ic} / FS$$

$$= 43.275 / 2.0$$

$$= 21.637 \text{ MPa}$$

Allowable External Pressure at the given Thickness [Pa]:

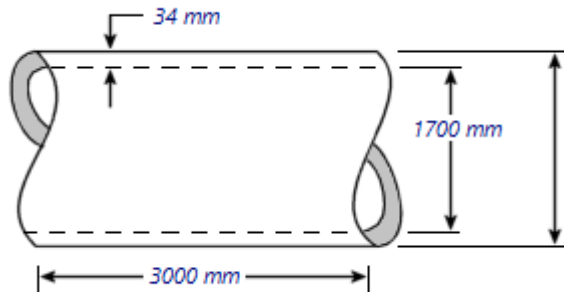
$$= 2 \cdot F_{ha} (t / D_o)$$

$$= 2 \cdot 21.637 (28.0 / 1768.0)$$

$$= 0.685 \text{ MPa}$$

Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell-2

Cylindrical Shell Design Information:					
Design Pressure	P	4.107 MPa	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		130 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	1700.000 mm	Actual Stress		125.6 MPa
Spec. Min. Thk t _e , or s _a		34.0000 mm	Spec. Nominal Thk.		34.000 mm
Surface Area		16663008.0 mm ²	SG of Contents		1.023
Element Volume		6810.6 ltr	Weight of Contents		15931.8 N
Empty Weight		42229.9 N	Operating Weight		58161.7 N
			Comp. All. Ope. Stress		116.4 MPa

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 1712.0 \cdot (\exp(4.107/(137.997 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 25.8589 + 6.0 + 0.0 = 31.8589 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 137.997 \cdot 1.0 \cdot \ln((2 \cdot 28.0 + 1712.0) / 1712.0) - 0.007$$

$$= 4.442 - 0.007 = 4.4347 \text{ MPa}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 137.997 \cdot 1.0 \cdot \ln((2 \cdot 34.0 + 1700.0) / 1700.0)$$

$$= 5.4123 \text{ MPa}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 4.107 \cdot 1712.0 / (1.0 (1768.0 - 1712.0))$$

$$= 125.5577 \text{ MPa}$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 4.107 / (1.0 \cdot \ln((2 \cdot 28.0 + 1712.0) / 1712.0))$$
$$= 127.5998 \text{ MPa}$$

Percent Fiber Elongation per Table 6.1

$$\text{Cylinders formed from Plate} \quad (50 \cdot t_{\text{nom}} / R_f) \cdot (1 - R_f / R_o) \quad 1.961 \%$$

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: Shell-2

Cylindrical Shell From 60 to 70 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$
$$= 18483.834 / \sqrt{(884.0 \cdot 28.0)}$$
$$= 117.486$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 2 \cdot (D_o/t)^{0.94}, \text{ Ch} = 0.55 / (D_o/t)$$
$$= 0.55 / (1768.0/28.0)$$
$$= 0.0087$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot \text{Ch} \cdot E_y(t / D_o)$$
$$= 1.6 \cdot 0.009 \cdot 196301(28.0/1768.0)$$
$$= 43.327 \text{ MPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= F_{he} / F_y$$
$$= 43.3/234.4$$
$$= 0.185$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$
$$= 234.4/482.6$$
$$= 0.486$$

Ratio [AE]:

$$= F_{he} / E$$
$$= 43.3/196301.4$$
$$= 0.000221$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / (\ln(\ln(1 + \epsilon_{psp})) / \ln(1 + \epsilon_{sys}))$$
$$= (\ln(0.486) + (0.00002 - 0.002)) / (\ln(\ln(1 + 0.00002)) / \ln(1 + 0.002))$$
$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$
$$= 234.4(1 + 0.002) / \ln(1 + 0.002)^{0.157304}$$
$$= 624.376 \text{ MPa}$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$
$$= 0.6(1.0 - 0.486)$$
$$= 0.309$$

Curve Fitting Constant [A2]:
 $= (\sigma_{uts} \cdot \exp(m_2)) / (m_2 m_2)$
 $= (482.6 \cdot \exp(0.309)) / (0.309 \cdot 0.309)$
 $= 944.570 \text{ MPa}$

Material Parameter for Stress-Strain Curve Model [K]:
 $= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$
 $= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$
 $= 0.346$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-04	0.271E-04	0.807E-04	-0.632E-04	0.124E+05	0.193E-01
2	120	-4.651	0.763E-06	0.763E-06	0.171E-04	-0.171E-04	0.151E+06	0.579E-03
3	60	-6.057	0.186E-07	0.186E-07	0.361E-05	-0.361E-05	0.195E+06	0.889E-04
4	30	-6.760	0.454E-09	0.454E-09	0.765E-06	-0.765E-06	0.196E+06	0.670E-04
5	45	-6.409	0.399E-08	0.399E-08	0.190E-05	-0.190E-05	0.196E+06	0.102E-04
6	37	-6.584	0.150E-08	0.150E-08	0.126E-05	-0.126E-05	0.196E+06	0.285E-04
7	41	-6.496	0.250E-08	0.250E-08	0.156E-05	-0.156E-05	0.196E+06	0.921E-05
8	43	-6.452	0.317E-08	0.317E-08	0.172E-05	-0.172E-05	0.196E+06	0.460E-06
9	42	-6.474	0.282E-08	0.282E-08	0.164E-05	-0.164E-05	0.196E+06	0.438E-05
10	42	-6.463	0.299E-08	0.299E-08	0.168E-05	-0.168E-05	0.196E+06	0.196E-05
11	43	-6.458	0.308E-08	0.308E-08	0.170E-05	-0.170E-05	0.196E+06	0.750E-06
12	43	-6.455	0.313E-08	0.313E-08	0.171E-05	-0.171E-05	0.196E+06	0.145E-06
13	43	-6.454	0.315E-08	0.315E-08	0.172E-05	-0.172E-05	0.196E+06	0.158E-06
14	43	-6.455	0.314E-08	0.314E-08	0.172E-05	-0.172E-05	0.196E+06	0.637E-08

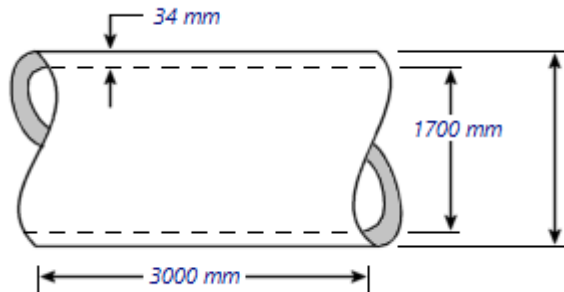
Factor of Safety [FS]:
 Since $Fic [43.27] < 0.55 \cdot 234.4$, $FS = 2$

Allowable Circumferential Compressive Stress [Fha]:
 $= Fic/FS$
 $= 43.275/2.0$
 $= 21.637 \text{ MPa}$

Allowable External Pressure at the given Thickness [Pa]:
 $= 2 \cdot Fha(t / Do)$
 $= 2 \cdot 21.637(28.0/1768.0)$
 $= 0.685 \text{ MPa}$

Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell-3

Cylindrical Shell Design Information:					
Design Pressure	P	4.100 MPa	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		130 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	1700.000 mm	Actual Stress		125.3 MPa
Spec. Min. Thk t,e, or sa		34.0000 mm	Spec. Nominal Thk.		34.000 mm
Surface Area		16663008.0 mm²	SG of Contents		0.000
Element Volume		6810.6 ltr	Weight of Contents		0.0 N
Empty Weight		42229.9 N	Operating Weight		42229.9 N
			Comp. All. Ope. Stress		116.4 MPa

Radiography Information:			
...		...	
Circ. Joint Efficiency		Long. Joint Eff.	E or v
	100 %		100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 1712.0 \cdot (\exp(4.1/(137.997 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 25.8141 + 6.0 + 0.0 = 31.8141 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 137.997 \cdot 1.0 \cdot \ln((2 \cdot 28.0 + 1712.0) / 1712.0) - 0.0$$

$$= 4.442 - 0.0 = 4.4417 \text{ MPa}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 137.997 \cdot 1.0 \cdot \ln((2 \cdot 34.0 + 1700.0) / 1700.0)$$

$$= 5.4123 \text{ MPa}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 4.1 \cdot 1712.0 / (1.0 (1768.0 - 1712.0))$$

$$= 125.3431 \text{ MPa}$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 4.1 / (1.0 \cdot \ln((2 \cdot 28.0 + 1712.0) / 1712.0))$$
$$= 127.3817 \text{ MPa}$$

Percent Fiber Elongation per Table 6.1

$$\text{Cylinders formed from Plate} \quad (50 \cdot t_{\text{nom}} / R_f) \cdot (1 - R_f / R_o) \quad 1.961 \%$$

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: Shell-3

Cylindrical Shell From 70 to 80 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$
$$= 18483.834 / \sqrt{(884.0 \cdot 28.0)}$$
$$= 117.486$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 2 \cdot (D_o/t)^{0.94}, \text{ Ch} = 0.55 / (D_o/t)$$
$$= 0.55 / (1768.0/28.0)$$
$$= 0.0087$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot \text{Ch} \cdot E_y(t / D_o)$$
$$= 1.6 \cdot 0.009 \cdot 196301(28.0/1768.0)$$
$$= 43.327 \text{ MPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= F_{he} / F_y$$
$$= 43.3/234.4$$
$$= 0.185$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$
$$= 234.4/482.6$$
$$= 0.486$$

Ratio [AE]:

$$= F_{he} / E$$
$$= 43.3/196301.4$$
$$= 0.000221$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / (\ln(\ln(1 + \epsilon_{psp})) / \ln(1 + \epsilon_{sys}))$$
$$= (\ln(0.486) + (0.00002 - 0.002)) / (\ln(\ln(1 + 0.00002)) / \ln(1 + 0.002))$$
$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$
$$= 234.4(1 + 0.002) / \ln(1 + 0.002)^{0.157304}$$
$$= 624.376 \text{ MPa}$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$
$$= 0.6(1.0 - 0.486)$$
$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m_2)) / (m_2 m_2)$$

$$= (482.6 \cdot \exp(0.309)) / (0.309 \cdot 0.309)$$

$$= 944.570 \text{ MPa}$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-04	0.271E-04	0.807E-04	-0.632E-04	0.124E+05	0.193E-01
2	120	-4.651	0.763E-06	0.763E-06	0.171E-04	-0.171E-04	0.151E+06	0.579E-03
3	60	-6.057	0.186E-07	0.186E-07	0.361E-05	-0.361E-05	0.195E+06	0.889E-04
4	30	-6.760	0.454E-09	0.454E-09	0.765E-06	-0.765E-06	0.196E+06	0.670E-04
5	45	-6.409	0.399E-08	0.399E-08	0.190E-05	-0.190E-05	0.196E+06	0.102E-04
6	37	-6.584	0.150E-08	0.150E-08	0.126E-05	-0.126E-05	0.196E+06	0.285E-04
7	41	-6.496	0.250E-08	0.250E-08	0.156E-05	-0.156E-05	0.196E+06	0.921E-05
8	43	-6.452	0.317E-08	0.317E-08	0.172E-05	-0.172E-05	0.196E+06	0.460E-06
9	42	-6.474	0.282E-08	0.282E-08	0.164E-05	-0.164E-05	0.196E+06	0.438E-05
10	42	-6.463	0.299E-08	0.299E-08	0.168E-05	-0.168E-05	0.196E+06	0.196E-05
11	43	-6.458	0.308E-08	0.308E-08	0.170E-05	-0.170E-05	0.196E+06	0.750E-06
12	43	-6.455	0.313E-08	0.313E-08	0.171E-05	-0.171E-05	0.196E+06	0.145E-06
13	43	-6.454	0.315E-08	0.315E-08	0.172E-05	-0.172E-05	0.196E+06	0.158E-06
14	43	-6.455	0.314E-08	0.314E-08	0.172E-05	-0.172E-05	0.196E+06	0.637E-08

Factor of Safety [FS]:

$$\text{Since } F_{ic} [43.27] < 0.55 \cdot 234.4, FS = 2$$

Allowable Circumferential Compressive Stress [Fha]:

$$= F_{ic}/FS$$

$$= 43.275/2.0$$

$$= 21.637 \text{ MPa}$$

Allowable External Pressure at the given Thickness [Pa]:

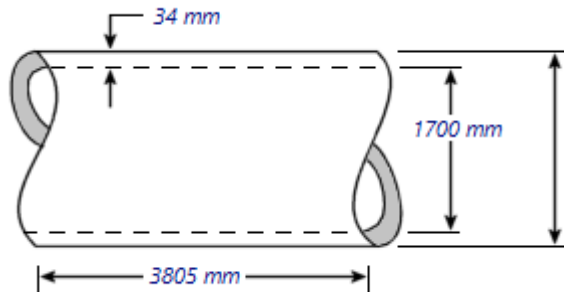
$$= 2 \cdot F_{ha} (t / D_o)$$

$$= 2 \cdot 21.637 (28.0/1768.0)$$

$$= 0.685 \text{ MPa}$$

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Shell-4

Cylindrical Shell Design Information:					
Design Pressure	P	4.100 MPa	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		130 °C
Length		3805.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	1700.000 mm	Actual Stress		125.3 MPa
Spec. Min. Thk t _e , or s _a		34.0000 mm	Spec. Nominal Thk.		34.000 mm
Surface Area		21134248.0 mm ²	SG of Contents		0.000
Element Volume		8638.1 ltr	Weight of Contents		0.0 N
Empty Weight		53561.6 N	Operating Weight		53561.6 N
			Comp. All. Ope. Stress		116.4 MPa

Radiography Information:			
...		...	
Circ. Joint Efficiency		Long. Joint Eff.	E or v
	100 %		100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 1712.0 \cdot (\exp(4.1/(137.997 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 25.8141 + 6.0 + 0.0 = 31.8141 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 137.997 \cdot 1.0 \cdot \ln((2 \cdot 28.0 + 1712.0) / 1712.0) - 0.0$$

$$= 4.442 - 0.0 = 4.4417 \text{ MPa}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 137.997 \cdot 1.0 \cdot \ln((2 \cdot 34.0 + 1700.0) / 1700.0)$$

$$= 5.4123 \text{ MPa}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 4.1 \cdot 1712.0 / (1.0 (1768.0 - 1712.0))$$

$$= 125.3431 \text{ MPa}$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 4.1 / (1.0 \cdot \ln((2 \cdot 28.0 + 1712.0) / 1712.0))$$
$$= 127.3817 \text{ MPa}$$

Percent Fiber Elongation per Table 6.1

$$\text{Cylinders formed from Plate} \quad (50 \cdot t_{\text{nom}} / R_f) \cdot (1 - R_f / R_o) \quad 1.961 \%$$

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: Shell-4

Cylindrical Shell From 80 to 90 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$
$$= 18483.834 / \sqrt{(884.0 \cdot 28.0)}$$
$$= 117.486$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 2 \cdot (D_o/t)^{0.94}, \text{ Ch} = 0.55 / (D_o/t)$$
$$= 0.55 / (1768.0/28.0)$$
$$= 0.0087$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot \text{Ch} \cdot E_y(t / D_o)$$
$$= 1.6 \cdot 0.009 \cdot 196301(28.0/1768.0)$$
$$= 43.327 \text{ MPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= F_{he} / F_y$$
$$= 43.3/234.4$$
$$= 0.185$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$
$$= 234.4/482.6$$
$$= 0.486$$

Ratio [AE]:

$$= F_{he} / E$$
$$= 43.3/196301.4$$
$$= 0.000221$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / (\ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys})))$$
$$= (\ln(0.486) + (0.00002 - 0.002)) / (\ln(\ln(1 + 0.00002) / \ln(1 + 0.002)))$$
$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$
$$= 234.4(1 + 0.002) / \ln(1 + 0.002)^{0.157304}$$
$$= 624.376 \text{ MPa}$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$
$$= 0.6(1.0 - 0.486)$$
$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m_2)) / (m_2 m_2)$$

$$= (482.6 \cdot \exp(0.309)) / (0.309 \cdot 0.309)$$

$$= 944.570 \text{ MPa}$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-04	0.271E-04	0.807E-04	-0.632E-04	0.124E+05	0.193E-01
2	120	-4.651	0.763E-06	0.763E-06	0.171E-04	-0.171E-04	0.151E+06	0.579E-03
3	60	-6.057	0.186E-07	0.186E-07	0.361E-05	-0.361E-05	0.195E+06	0.889E-04
4	30	-6.760	0.454E-09	0.454E-09	0.765E-06	-0.765E-06	0.196E+06	0.670E-04
5	45	-6.409	0.399E-08	0.399E-08	0.190E-05	-0.190E-05	0.196E+06	0.102E-04
6	37	-6.584	0.150E-08	0.150E-08	0.126E-05	-0.126E-05	0.196E+06	0.285E-04
7	41	-6.496	0.250E-08	0.250E-08	0.156E-05	-0.156E-05	0.196E+06	0.921E-05
8	43	-6.452	0.317E-08	0.317E-08	0.172E-05	-0.172E-05	0.196E+06	0.460E-06
9	42	-6.474	0.282E-08	0.282E-08	0.164E-05	-0.164E-05	0.196E+06	0.438E-05
10	42	-6.463	0.299E-08	0.299E-08	0.168E-05	-0.168E-05	0.196E+06	0.196E-05
11	43	-6.458	0.308E-08	0.308E-08	0.170E-05	-0.170E-05	0.196E+06	0.750E-06
12	43	-6.455	0.313E-08	0.313E-08	0.171E-05	-0.171E-05	0.196E+06	0.145E-06
13	43	-6.454	0.315E-08	0.315E-08	0.172E-05	-0.172E-05	0.196E+06	0.158E-06
14	43	-6.455	0.314E-08	0.314E-08	0.172E-05	-0.172E-05	0.196E+06	0.637E-08

Factor of Safety [FS]:

$$\text{Since } F_{ic} [43.27] < 0.55 \cdot 234.4, FS = 2$$

Allowable Circumferential Compressive Stress [Fha]:

$$= F_{ic} / FS$$

$$= 43.275 / 2.0$$

$$= 21.637 \text{ MPa}$$

Allowable External Pressure at the given Thickness [Pa]:

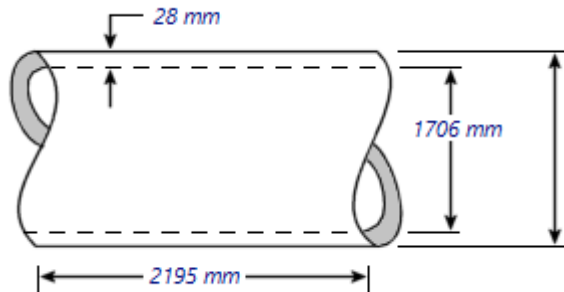
$$= 2 \cdot F_{ha} (t / D_o)$$

$$= 2 \cdot 21.637 (28.0 / 1768.0)$$

$$= 0.685 \text{ MPa}$$

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Shell-5

Cylindrical Shell Design Information:					
Design Pressure	P	4.100 MPa	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		130 °C
Length		2195.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	1706.000 mm	Actual Stress		129.7 MPa
Spec. Min. Thk t _e , or s _a		28.0000 mm	Spec. Nominal Thk.		28.000 mm
Surface Area		12150394.0 mm ²	SG of Contents		0.000
Element Volume		5018.3 ltr	Weight of Contents		0.0 N
Empty Weight		25445.7 N	Operating Weight		25445.7 N
			Comp. All. Ope. Stress		115.9 MPa

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 1708.0 \cdot (\exp(4.1/(137.997 \cdot 1.0)) - 1) + 1.0 + 0.0$$

$$= 25.7537 + 1.0 + 0.0 = 26.7537 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 137.997 \cdot 1.0 \cdot \ln((2 \cdot 27.0 + 1708.0) / 1708.0) - 0.0$$

$$= 4.295 - 0.0 = 4.2954 \text{ MPa}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 137.997 \cdot 1.0 \cdot \ln((2 \cdot 28.0 + 1706.0) / 1706.0)$$

$$= 4.457 \text{ MPa}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 4.1 \cdot 1708.0 / (1.0 (1762.0 - 1708.0))$$

$$= 129.6815 \text{ MPa}$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 4.1 / (1.0 \cdot \ln((2 \cdot 27.0 + 1708.0) / 1708.0))$$
$$= 131.7207 \text{ MPa}$$

Percent Fiber Elongation per Table 6.1

$$\text{Cylinders formed from Plate} \quad (50 \cdot t_{\text{nom}} / R_f) \cdot (1 - R_f / R_o) \quad 1.615 \%$$

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: Shell-5

Cylindrical Shell From 90 to 100 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$
$$= 18483.836 / \sqrt{(881.0 \cdot 27.0)}$$
$$= 119.846$$

Compute the Factor [Ch]:

Since $M_x > 2 \cdot (D_o/t) \cdot 0.94$, $Ch = 0.55 / (D_o/t)$

$$= 0.55 \cdot (1762.0 / 27.0)$$
$$= 0.0084$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y (t / D_o)$$
$$= 1.6 \cdot 0.008 \cdot 196301 (27.0 / 1762.0)$$
$$= 40.562 \text{ MPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / F_y$$
$$= 40.6 / 234.4$$
$$= 0.173$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$
$$= 234.4 / 482.6$$
$$= 0.486$$

Ratio [AE]:

$$= Fhe / E$$
$$= 40.6 / 196301.4$$
$$= 0.000207$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{ps} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{ps}) / \ln(1 + \epsilon_{sys}))$$
$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$
$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$
$$= 234.4 (1 + 0.002) / \ln(1 + 0.002)^{0.157304}$$
$$= 624.376 \text{ MPa}$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$
$$= 0.6 (1.0 - 0.486)$$
$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$
$$= (482.6 \cdot \exp(0.309)) / (0.309^{0.309})$$



= 944.570 MPa

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-04	0.271E-04	0.807E-04	-0.632E-04	0.124E+05	0.193E-01
2	120	-4.651	0.763E-06	0.763E-06	0.171E-04	-0.171E-04	0.151E+06	0.593E-03
3	60	-6.057	0.186E-07	0.186E-07	0.361E-05	-0.361E-05	0.195E+06	0.103E-03
4	30	-6.760	0.454E-09	0.454E-09	0.765E-06	-0.765E-06	0.196E+06	0.529E-04
5	45	-6.409	0.399E-08	0.399E-08	0.190E-05	-0.190E-05	0.196E+06	0.242E-04
6	37	-6.584	0.150E-08	0.150E-08	0.126E-05	-0.126E-05	0.196E+06	0.144E-04
7	41	-6.496	0.250E-08	0.250E-08	0.156E-05	-0.156E-05	0.196E+06	0.487E-05
8	39	-6.540	0.195E-08	0.195E-08	0.141E-05	-0.141E-05	0.196E+06	0.479E-05
9	40	-6.518	0.221E-08	0.221E-08	0.148E-05	-0.148E-05	0.196E+06	0.412E-07
10	40	-6.529	0.208E-08	0.208E-08	0.144E-05	-0.144E-05	0.196E+06	0.237E-05
11	40	-6.524	0.214E-08	0.214E-08	0.146E-05	-0.146E-05	0.196E+06	0.117E-05
12	40	-6.521	0.218E-08	0.218E-08	0.147E-05	-0.147E-05	0.196E+06	0.562E-06
13	40	-6.520	0.219E-08	0.219E-08	0.148E-05	-0.148E-05	0.196E+06	0.261E-06
14	40	-6.519	0.220E-08	0.220E-08	0.148E-05	-0.148E-05	0.196E+06	0.110E-06
15	40	-6.519	0.221E-08	0.221E-08	0.148E-05	-0.148E-05	0.196E+06	0.342E-07
16	40	-6.519	0.221E-08	0.221E-08	0.148E-05	-0.148E-05	0.196E+06	0.351E-08

Factor of Safety [FS]:

$$\text{Since } F_{ic} [40.53] < 0.55 \cdot 234.4, FS = 2$$

Allowable Circumferential Compressive Stress [Fha]:

$$= F_{ic}/FS$$

$$= 40.528/2.0$$

$$= 20.264 \text{ MPa}$$

Allowable External Pressure at the given Thickness [Pa]:

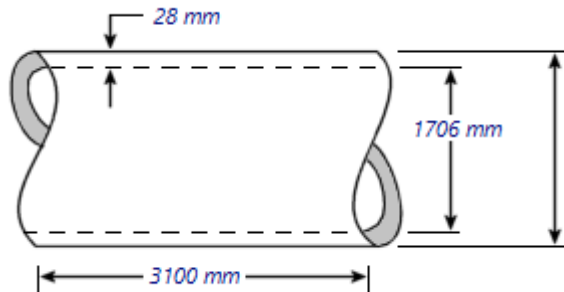
$$= 2 \cdot F_{ha} (t / D_o)$$

$$= 2 \cdot 20.264 (27.0/1762.0)$$

$$= 0.621 \text{ MPa}$$

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Shell-7

Cylindrical Shell Design Information:					
Design Pressure	P	4.100 MPa	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		130 °C
Length		3100.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	1706.000 mm	Actual Stress		129.7 MPa
Spec. Min. Thk t _e , or s _a		28.0000 mm	Spec. Nominal Thk.		28.000 mm
Surface Area		17160006.0 mm ²	SG of Contents		0.000
Element Volume		7087.4 ltr	Weight of Contents		0.0 N
Empty Weight		35936.9 N	Operating Weight		35936.9 N
			Comp. All. Ope. Stress		115.9 MPa

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 1708.0 \cdot (\exp(4.1/(137.997 \cdot 1.0)) - 1) + 1.0 + 0.0$
 $= 25.7537 + 1.0 + 0.0 = 26.7537 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 137.997 \cdot 1.0 \cdot \ln((2 \cdot 27.0 + 1708.0) / 1708.0) - 0.0$
 $= 4.295 - 0.0 = 4.2954 \text{ MPa}$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 137.997 \cdot 1.0 \cdot \ln((2 \cdot 28.0 + 1706.0) / 1706.0)$
 $= 4.457 \text{ MPa}$

Computed Stress Intensity at Design Pressure [Si]:
 $= P \cdot D / (E (D_o - D))$
 $= 4.1 \cdot 1708.0 / (1.0 (1762.0 - 1708.0))$
 $= 129.6815 \text{ MPa}$

Computed Membrane Stress at Design Pressure [Sm]:
 $= P / (E \cdot \ln((2 \cdot t + D) / D))$



$$= 4.1 / (1.0 \cdot \ln((2 \cdot 27.0 + 1708.0) / 1708.0))$$
$$= 131.7207 \text{ MPa}$$

Percent Fiber Elongation per Table 6.1

$$\text{Cylinders formed from Plate} \quad (50 \cdot t_{\text{nom}} / R_f) \cdot (1 - R_f / R_o) \quad 1.615 \%$$

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: Shell-7

Cylindrical Shell From 100 to 110 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$
$$= 18483.836 / \sqrt{(881.0 \cdot 27.0)}$$
$$= 119.846$$

Compute the Factor [Ch]:

Since $M_x > 2 \cdot (D_o/t) \cdot 0.94$, $Ch = 0.55 / (D_o/t)$

$$= 0.55 \cdot (1762.0 / 27.0)$$
$$= 0.0084$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y (t / D_o)$$
$$= 1.6 \cdot 0.008 \cdot 196301 (27.0 / 1762.0)$$
$$= 40.562 \text{ MPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / F_y$$
$$= 40.6 / 234.4$$
$$= 0.173$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$
$$= 234.4 / 482.6$$
$$= 0.486$$

Ratio [AE]:

$$= Fhe / E$$
$$= 40.6 / 196301.4$$
$$= 0.000207$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{ps} - \epsilon_{sys})) / (\ln(1 + \epsilon_{ps}) / \ln(1 + \epsilon_{sys}))$$
$$= (\ln(0.486) + (0.00002 - 0.002)) / (\ln(1 + 0.00002) / \ln(1 + 0.002))$$
$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$
$$= 234.4 (1 + 0.002) / \ln(1 + 0.002)^{0.157304}$$
$$= 624.376 \text{ MPa}$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$
$$= 0.6 (1.0 - 0.486)$$
$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$
$$= (482.6 \cdot \exp(0.309)) / (0.309^{0.309})$$

= 944.570 MPa

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-04	0.271E-04	0.807E-04	-0.632E-04	0.124E+05	0.193E-01
2	120	-4.651	0.763E-06	0.763E-06	0.171E-04	-0.171E-04	0.151E+06	0.593E-03
3	60	-6.057	0.186E-07	0.186E-07	0.361E-05	-0.361E-05	0.195E+06	0.103E-03
4	30	-6.760	0.454E-09	0.454E-09	0.765E-06	-0.765E-06	0.196E+06	0.529E-04
5	45	-6.409	0.399E-08	0.399E-08	0.190E-05	-0.190E-05	0.196E+06	0.242E-04
6	37	-6.584	0.150E-08	0.150E-08	0.126E-05	-0.126E-05	0.196E+06	0.144E-04
7	41	-6.496	0.250E-08	0.250E-08	0.156E-05	-0.156E-05	0.196E+06	0.487E-05
8	39	-6.540	0.195E-08	0.195E-08	0.141E-05	-0.141E-05	0.196E+06	0.479E-05
9	40	-6.518	0.221E-08	0.221E-08	0.148E-05	-0.148E-05	0.196E+06	0.412E-07
10	40	-6.529	0.208E-08	0.208E-08	0.144E-05	-0.144E-05	0.196E+06	0.237E-05
11	40	-6.524	0.214E-08	0.214E-08	0.146E-05	-0.146E-05	0.196E+06	0.117E-05
12	40	-6.521	0.218E-08	0.218E-08	0.147E-05	-0.147E-05	0.196E+06	0.562E-06
13	40	-6.520	0.219E-08	0.219E-08	0.148E-05	-0.148E-05	0.196E+06	0.261E-06
14	40	-6.519	0.220E-08	0.220E-08	0.148E-05	-0.148E-05	0.196E+06	0.110E-06
15	40	-6.519	0.221E-08	0.221E-08	0.148E-05	-0.148E-05	0.196E+06	0.342E-07
16	40	-6.519	0.221E-08	0.221E-08	0.148E-05	-0.148E-05	0.196E+06	0.351E-08

Factor of Safety [FS]:

$$\text{Since } Fic [40.53] < 0.55 \cdot 234.4, FS = 2$$

Allowable Circumferential Compressive Stress [Fha]:

$$= Fic/FS$$

$$= 40.528/2.0$$

$$= 20.264 \text{ MPa}$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot Fha(t / Do)$$

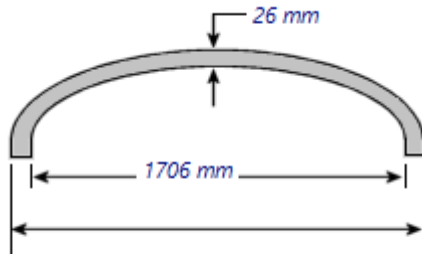
$$= 2 \cdot 20.264(27.0/1762.0)$$

$$= 0.621 \text{ MPa}$$



Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Top head

Elliptical Head Design Information:					
Design Pressure	P	4.100 MPa	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		130 °C
Length		50.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	1706.000 mm	Actual Stress		127.0 MPa
Spec. Min. Thk t _e , or s _a		26.0000 mm	Spec. Nominal Thk.		32.000 mm
Surface Area		3674048.5 mm ²	SG of Contents		0.000
Element Volume		764.4 ltr	Weight of Contents		0.0 N
Empty Weight		9456.4 N	Operating Weight		9456.4 N
			Comp. All. Ope. Stress		114.7 MPa
			Aspect Ratio		2.00

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 21.325 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 10.662 mm

Material treatment options: +IT

Head Calculation - Section 4.3.6 (and Section 4.3.7 for Ellipsoidal Head)

Computed Minimum Required Thickness [t]:
 $t_r = 22.9946 \text{ mm}$

$$\begin{aligned}
 t &= t_r + c_i + c_o \\
 &= 22.9946 + 1.0 + 0.0 \\
 &= 23.9946 \text{ mm} - \text{see below for the derivation}
 \end{aligned}$$

For Elliptical heads: Compute [h, k, r] per 4.3.7.1

$$\begin{aligned}
 h &= D / (2 \cdot \text{Aspect Ratio}) \\
 &= 1706.0 / (2 \cdot 2.0) \\
 &= 426.500 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 k &= D / (2 \cdot h) \\
 &= 1706.0 / (2 \cdot 426.5) \\
 &= 2.000
 \end{aligned}$$

$$r = D (0.5/k - 0.08) + c_a$$



$$= 1706.0(0.5/2.0 - 0.08) + 1.0$$

$$= 291.020 \text{ mm}$$

$$L = D(0.44 \cdot k + 0.02) + ca$$

$$= 1706.0(0.44 \cdot 2.0 + 0.02) + 1.0$$

$$= 1536.400 \text{ mm}$$

Step 3: Geometric Constants [βTh , ϕTh , Rth]:

Equations 4.3.8 to 4.3.11 as appropriate

t (corroded) and P for the calc.: $t = 22.9946 \text{ mm}$, $P = 4.1 \text{ MPa}$

Angle [βTh]:

$$= \arccos[(0.5 \cdot D - r)/(L - r)]$$

$$= \arccos[(0.5 \cdot 1708.0 - 291.02)/(1536.4 - 291.02)]$$

$$= 1.1017 \text{ radians - Equation 4.3.8}$$

$$\phi Th = \sqrt{(L \cdot t) / r}$$

$$= \sqrt{(1536.4 \cdot 22.995) / 291.02}$$

$$= 0.6459 \text{ radians - Equation 4.3.9}$$

$\phi Th < \beta Th$, Thus:

$$Rth = (0.5 \cdot D - r) / \cos(\beta Th - \phi Th) + r$$

$$= (0.5 \cdot 1708.0 - 291.02) / \cos(1.102 - 0.646) + 291.02$$

$$= 918.0291 \text{ mm - Equation 4.3.10}$$

Step 4: Coefficients: Equations 4.3.12 to 4.3.15 [$C1$ and $C2$]:

$r/D > 0.08$, Thus:

$$C1 = 0.692(r/D) + 0.605$$

$$= 0.692(291.02/1708.0) + 0.605$$

$$= 0.7229 \text{ - Equation 4.3.13}$$

$$C2 = 1.46 - 2.6(r/D) = 1.46 - 2.6(291.02/1708.0)$$

$$= 1.017, \text{ Equation 4.3.15}$$

Step 5: Pressure to produce knuckle elastic buckling [$Peth$]:

$$= C1 \cdot Et \cdot t^2 / (C2 \cdot Rth(0.5 \cdot Rth - r))$$

$$= 0.723 \cdot 196301.391 \cdot 22.995^2 / (1.017 \cdot 918.029(0.5 \cdot 918.029 - 291.02))$$

$$= 478.4 \text{ MPa, Equation 4.3.16}$$

Step 6: Internal Pressure for max stress in knuckle equals yield [Py]:

$C3 = Sy = 234.397 \text{ MPa}$, Not in the creep range - 4.3.6.1(f)

$$Py = C3 \cdot t / (C2 \cdot Rth(0.5 \cdot Rth / r - 1))$$

$$= 234.397 \cdot 22.995 / (1.017 \cdot 918.029(0.5 \cdot 918.029 / 291.02 - 1))$$

$$= 10.001 \text{ MPa, Equation 4.3.17}$$

Step 7: Value of Internal Press. resulting in buckling of knuckle [G and Pck]:

$$G = Peth/Py = 478.397/10.001 = 47.836$$

$G > 1$, Thus:

Buckling Pressure [Pck]:

$$= ((0.77508 \cdot G - 0.20354 \cdot G^2 + 0.019274 \cdot G^3) /$$

$$(1 + 0.19014 \cdot G - 0.089534 \cdot G^2 + 0.0093965 \cdot G^3)) Py$$

$$= ((0.77508 \cdot 47.836 - 0.20354 \cdot 47.836^2 + 0.019274 \cdot 47.836^3) /$$

$$(1 + 0.19014 \cdot 47.836 - 0.089534 \cdot 47.836^2 + 0.0093965 \cdot 47.836^3)) \cdot 10.001$$

$$= 20.16 \text{ MPa, Equation 4.3.19}$$

Step 8: Compute Allowable Pressure base on buckling of knuckle [Pak]:

$$= Pck/1.5 = 20.16/1.5 = 13.44 \text{ MPa - Equation 4.3.21}$$

Step 9: Compute allowable pressure based on rupture of crown [Pac]:

$$= 2 \cdot S \cdot E / (L/t + 0.5)$$

$$= 2 \cdot 137.997 \cdot 1.0 / (1536.4/22.995 + 0.5)$$

$$= 4.1 \text{ MPa, Equation 4.3.22}$$

Step 10: Compute final pressure [Pa]:

$$= \min(Pak, Pac) = \min(13.443, 4.1)$$

$$= 4.1 \text{ MPa, Equation 4.3.23: (same as original design pressure)}$$



Allowable pressure in Crown at given thickness [Pac]:

$$\begin{aligned} &= 2 \cdot S \cdot E / (L / t + 0.5) \\ &= 2 \cdot 137.997 \cdot 1.0 / (1536.4 / 25.0 + 0.5) \\ &= 4.455 \text{ MPa} \end{aligned}$$

Computed Maximum Allowable Pressure [MAWP]:

$$= \text{MaxPressure} - \text{Pliq} = 4.4547 - 0.0 = 4.4547 \text{ MPa}$$

Computed Maximum Allowable Pressure New and Cold [MAPnc]:

$$= 4.6344 \text{ MPa}$$

Computed Membrane Stress Intensity at Design Pressure [Sm]:

$$= 127.0 \text{ MPa}$$

Required Thickness of Straight Flange = 26.754 mm

Percent Fiber Elongation per Table 6.1

$$\begin{aligned} \text{For Heads formed from Segments } (75 \cdot t_{\text{nom}} / R_f) \cdot (1 - R_f / R_o) & \quad 7.843 \% \\ \text{For one Piece doubly Curved Heads } 100 \cdot \ln(D_b / (D_f - 2t)) & \quad 23.728 \% \end{aligned}$$

Where D_b is the blank diameter for one piece doubly curved surfaces [D_b]:

$$\begin{aligned} &= \text{square root}(4 \cdot \text{Head Surface Area} / 3.14159) \\ &= \text{square root}(4 \cdot 3674049 / 3.14159) \\ &= 2162.855 \text{ mm} \end{aligned}$$

MDMT Calculations in the Knuckle Portion:

Note: This Element/Detail was specified as being Impact Tested.

MDMT Calculations in the Head Straight Flange:

Note: This Element/Detail was specified as being Impact Tested.

Tolerance for Formed Heads per 4.3.2.2:

$$\begin{aligned} \text{Head inner surface maximum deviation outside the specified shape, } 1.25\% \text{ of } D &: 21.325 \text{ mm} \\ \text{Head inner surface maximum deviation inside the specified shape, } 0.625\% \text{ of } D &: 10.662 \text{ mm} \end{aligned}$$

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: Top head

Elliptical Head From 110 to 120 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$\begin{aligned} &= 0.075 \cdot E_y(t / R_o) \\ &= 0.075 \cdot 196301(25.0 / 1538.227) \\ &= 239.279 \text{ MPa} \end{aligned}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$\begin{aligned} &= F_{he} / S_y \\ &= 239.3 / 234.4 \\ &= 1.021 \end{aligned}$$

Engineering Yield to Tensile Stress ratio [R]:

$$\begin{aligned} &= \sigma_y / \sigma_{uts} \\ &= 234.4 / 482.6 \\ &= 0.486 \end{aligned}$$

Ratio [AE]:

$$\begin{aligned} &= F_{he} / E \\ &= 239.3 / 196301.4 \\ &= 0.001219 \end{aligned}$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / (\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / (\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys}) m_1$$

$$= 234.4 (1 + 0.002) / \ln(1 + 0.002) 0.157304$$

$$= 624.376 \text{ MPa}$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m_2 f(m_2^2 - R)$$

$$= 0.6 (1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m_2)) / (m_2 m_2)$$

$$= (482.6 \cdot \exp(0.309)) / (0.309 0.309)$$

$$= 944.570 \text{ MPa}$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R_{1.5} - 0.5 \cdot R_{2.5} - R_{3.5}$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-04	0.271E-04	0.807E-04	-0.632E-04	0.124E+05	0.183E-01
2	120	-4.651	0.763E-06	0.763E-06	0.171E-04	-0.171E-04	0.151E+06	0.419E-03
3	180	-3.245	0.670E-05	0.665E-05	0.423E-04	-0.419E-04	0.529E+05	0.220E-02
4	150	-3.948	0.252E-05	0.252E-05	0.281E-04	-0.281E-04	0.980E+05	0.319E-03
5	135	-4.299	0.143E-05	0.143E-05	0.222E-04	-0.222E-04	0.125E+06	0.135E-03
6	143	-4.124	0.192E-05	0.191E-05	0.251E-04	-0.250E-04	0.112E+06	0.654E-04
7	139	-4.211	0.166E-05	0.166E-05	0.236E-04	-0.236E-04	0.118E+06	0.407E-04
8	141	-4.167	0.178E-05	0.178E-05	0.244E-04	-0.243E-04	0.115E+06	0.108E-04
9	140	-4.189	0.172E-05	0.172E-05	0.240E-04	-0.240E-04	0.117E+06	0.154E-04
10	140	-4.178	0.175E-05	0.175E-05	0.242E-04	-0.241E-04	0.116E+06	0.239E-05
11	141	-4.173	0.177E-05	0.177E-05	0.243E-04	-0.242E-04	0.115E+06	0.417E-05
12	141	-4.176	0.176E-05	0.176E-05	0.242E-04	-0.242E-04	0.116E+06	0.886E-06
13	140	-4.177	0.176E-05	0.175E-05	0.242E-04	-0.242E-04	0.116E+06	0.751E-06
14	141	-4.176	0.176E-05	0.176E-05	0.242E-04	-0.242E-04	0.116E+06	0.670E-07
15	141	-4.177	0.176E-05	0.176E-05	0.242E-04	-0.242E-04	0.116E+06	0.342E-06
16	141	-4.177	0.176E-05	0.176E-05	0.242E-04	-0.242E-04	0.116E+06	0.138E-06
17	141	-4.176	0.176E-05	0.176E-05	0.242E-04	-0.242E-04	0.116E+06	0.353E-07

Factor of Safety [FS]:

Since $Fic > 0.55 F_y$ and $Fic < F_y$

$$= 2.407 - 0.741 \cdot Fic / F_y$$

$$= 2.407 - 0.741 \cdot 141.01 / 234.4$$

$$= 1.961$$

Allowable Circumferential Compressive Stress [Fha]:

$$= Fic / FS$$

$$= 141.015 / 1.961$$

$$= 71.902 \text{ MPa}$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot Fha [t / Ro]$$

$$= 2 \cdot 71.902 [25.0 / 1538.227]$$

$$= 2.337 \text{ MPa}$$

Hydrostatic Test Pressure Results:

Test Pressure at high point in vessel per Equation 8.1
 $= \gamma_{St/s} \cdot MAWP(St/S)$
 $= 1.25 \cdot 4.2954(1.0)$
 $= 5.369 \text{ MPa}$

Horizontal Test performed per: $\gamma_{St/s} \cdot MAWP(St/S)$

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
Bottom head	130.1	234.0	0.556	5.39
Shell-1	137.3	234.0	0.587	5.39
Shell-2	137.3	234.0	0.587	5.39
Shell-3	137.3	234.0	0.587	5.39
Shell-4	137.3	234.0	0.587	5.39
Shell-5	166.8	234.0	0.713	5.39
Shell-7	166.8	234.0	0.713	5.39
Top head	160.4	234.0	0.685	5.39

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
D_150	Nozzle	165.00	147.60	1.118
MH4_600	Nozzle	165.00	147.60	1.118
B_150	Nozzle	165.00	147.60	1.118
L1_50	Nozzle	165.00	147.60	1.118
L2_50	Nozzle	165.00	147.60	1.118
L3_50	Nozzle	165.00	147.60	1.118
L4_50	Nozzle	165.00	147.60	1.118
L5_50	Nozzle	165.00	147.60	1.118
L6_50	Nozzle	165.00	147.60	1.118
DP1_50	Nozzle	165.00	147.60	1.118
UC2_600	Nozzle	165.00	147.60	1.118
MH3_600	Nozzle	165.00	147.60	1.118
DP2_50	Nozzle	165.00	147.60	1.118
UC1_600	Nozzle	165.00	147.60	1.118
A_150	Nozzle	165.00	147.60	1.118
MH2_600	Nozzle	165.00	147.60	1.118
E_40	Nozzle	165.00	147.60	1.118
MH1_600	Nozzle	165.00	147.60	1.118
C_100	Nozzle	165.00	147.60	1.118

Minimum 1.118

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
Bottom head	138.00	138.00	1.000
Shell-1	138.00	138.00	1.000
Shell-2	138.00	138.00	1.000
Shell-3	138.00	138.00	1.000
Shell-4	138.00	138.00	1.000
Shell-5	138.00	138.00	1.000
Shell-7	138.00	138.00	1.000
Top head	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (MPa):



Description	Ambient	Operating	Ratio
D_150	33.51	237.50	0.141
MH4_600	98.27	237.50	0.414
B_150	33.51	237.50	0.141
L1_50	13.92	237.50	0.059
L2_50	13.92	237.50	0.059
L3_50	13.92	237.50	0.059
L4_50	13.92	237.50	0.059
L5_50	13.92	237.50	0.059
L6_50	13.92	237.50	0.059
DP1_50	13.92	237.50	0.059
UC2_600	98.27	237.50	0.414
MH3_600	98.27	237.50	0.414
DP2_50	13.92	237.50	0.059
UC1_600	98.27	237.50	0.414
A_150	33.52	237.50	0.141
MH2_600	98.27	237.50	0.414
E_40	13.28	237.50	0.056
MH1_600	98.27	237.50	0.414
C_100	20.01	237.50	0.084

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 2, 2025

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B MPa
10	20	6000	...	...	No Calc	No Calc
20	30	6000	...	...	No Calc	No Calc
30	40	6000	...	...	No Calc	No Calc
40	50	No Calc	1764	26	No Calc	No Calc
50	60	18483.8	1768	28	No Calc	No Calc
60	70	18483.8	1768	28	No Calc	No Calc
70	80	18483.8	1768	28	No Calc	No Calc
80	90	18483.8	1768	28	No Calc	No Calc
90	100	18483.8	1762	27	No Calc	No Calc
100	110	18483.8	1762	27	No Calc	No Calc
110	120	No Calc	1758	25	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure MPa	External M.A.W.P. MPa
10	20	...	No Calc	...	No Calc
20	30	...	No Calc	...	No Calc
30	40	...	No Calc	...	No Calc
40	50	32	10.0665	0.10343	2.47688
50	60	34	20.2612	0.10343	0.68535
60	70	34	20.2612	0.10343	0.68535
70	80	34	20.2612	0.10343	0.68535
80	90	34	20.2612	0.10343	0.68535
90	100	28	15.2334	0.10343	0.62103
100	110	28	15.2334	0.10343	0.62103
110	120	26	5.07849	0.10343	2.33717

Minimum

1

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required mm^4	Ring Inertia Available mm^4
10	20	6000	No Calc	No Calc	No Calc
20	30	6000	No Calc	No Calc	No Calc
30	40	6000	No Calc	No Calc	No Calc
40	50	No Calc	No Calc	No Calc	No Calc
50	60	18483.8	25399974	No Calc	No Calc
60	70	18483.8	25399974	No Calc	No Calc
70	80	18483.8	25399974	No Calc	No Calc
80	90	18483.8	25399974	No Calc	No Calc
90	100	18483.8	25399974	No Calc	No Calc
100	110	18483.8	25399974	No Calc	No Calc
110	120	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kgm	Element ID Volume ltr	Corroded Metal Wgt. kgm	Corroded ID Volume ltr	Extra due Misc % kgm
10	20	1913.03	...	1777.44	...	...
20	30	1779.42	...	1576.05	...	...
30	40	593.139	...	525.349	...	...
40	50	1145.17	756.736	964.354	772.062	...
50	60	4306.54	6810.62	3558.83	6907.11	...
60	70	4306.54	6810.62	3558.83	6907.11	...
70	80	4306.54	6810.62	3558.83	6907.11	...
80	90	5462.13	8638.14	4513.79	8760.52	...
90	100	2594.91	5018.34	2503.68	5030.12	...
100	110	3664.79	7087.41	3535.95	7104.03	...
110	120	964.354	764.373	934.218	766.93	...
Total		31036	42696.87	27007	43155.00	0

Weight of Details:

From	Type	Weight of Detail kgm	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
10	Wght	500	...	...	...	Tailing lug
20	Plat	842.638	1151.28	2000	-1151.28	PL-8
30	Insl	81.2194	...	750	...	Ins
40	Plat	2913.7	0.16154E-04	...	...	PL-1
40	Liqd	773.666	...	-212.5	...	Liquid 40
40	Insl	138.594	...	-187.5	...	Ins: 10
40	Nozl	236.344	...	-609	...	D_150
40	Wght	188	2118	...	...	D FY
40	Wght	10	...	-400	...	D anti-swirl ba
50	Liqd	6963	...	1500	...	Liquid 50
50	Insl	491.037	...	1500	...	Ins: 20
50	Nozl	899.148	...	800	1100	MH4_600
50	Nozl	236.344	1000	2800	...	B_150
50	Nozl	24.8193	-975.006	2500	...	L1_50
50	Nozl	24.8193	-975.006	200	...	L2_50
50	Nozl	24.8193	-975.006	2000	...	L3_50
50	Nozl	24.8193	-975.006	1000	...	L4_50
50	Nozl	24.8193	975.006	2000	...	L5_50
50	Nozl	24.8193	975.006	1000	...	L6_50
50	Wght	100	...	800	...	MH4 Davit
50	Wght	802	2118	2800	...	B FY
50	Wght	10	...	2950	...	Bed2 sup
50	Wght	180	...	2950	...	Bed 2 sup grid
60	Plat	1551.18	...	1000	-1161.4	PL-2
60	Pack	1140.23	...	1880	...	Bed 1
60	Liqd	1624.7	...	350	...	Liquid: 60
60	Insl	491.037	...	1500	...	Ins: 30
60	Nozl	24.8193	-975.006	650	...	DP1_50
60	Nozl	899.148	1100	1000	...	UC2_600
70	Plat	1551.18	...	2000	-1161.4	PL-3
70	Pack	1035.88	...	1017.5	...	Bed 1
70	Insl	491.037	...	1500	...	Ins: 40
70	Nozl	899.148	-1100	2300	...	MH3_600
70	Nozl	24.8193	975.006	2500	...	DP2_50
70	Wght	200	...	2500	...	Distributor 2
70	Wght	10	...	2450	...	Dist2 sup
70	Wght	100	...	2300	...	MH3 Davit
70	Wght	10	...	1300	...	Bed2 lim sup
70	Wght	50	...	...	...	Bed2 limiter
80	Plat	1552.78	...	2000	-1161.4	PL-4
80	Pack	1667.08	...	2167.5	...	Bed-1
80	Insl	622.798	...	1902.5	...	Ins: 50
80	Nozl	899.148	1100	999.999	...	UC1_600



80	Wght	10	...	449.999	...
80	Wght	180	...	499.999	...
80	Wght	2800	...	2000	...
80	Wght	3000	...	1950	...
90	Plat	1550.38	...	2000	-1163.19
90	Pack	409.77	...	402.501	...
90	Insl	358.109	...	1097.5	...
90	Lini	283.129	...	1097.5	...
90	Nozl	80.0376	1003	2120	...
90	Nozl	891.575	...	1895	-1103
90	Wght	200	...	2000	...
90	Wght	10	...	1950	...
90	Wght	10	...	1850	...
90	Wght	50	...	1900	...
90	Wght	1606	2118	2120	...
100	Plat	1512.65	...	1000	-1163.19
100	Insl	505.757	...	1550	...
100	Lini	399.863	...	1550	...
100	Nozl	24.7014	1003	900	...
100	Nozl	891.575	-1103	1550	...
100	Tray	1029.16	...	544.8	...
100	Wght	120	...	2850	...
100	Wght	50	...	2800	...
100	Wght	80	...	199.999	...
100	Wght	100	...	1550	...
100	Wght	500	400	2100	...
100	Wght	500	400	2100	...
110	Plat	2504.08	...	500	...
110	Insl	143.176	...	237.5	...
110	Lini	90.1353	...	238.25	...
110	Nozl	48.3931	...	613.5	...
110	Wght	565	2093	...	...
110	Forc	...	...	350	...
60	Pliq	519.91	...	1880.000	...
70	Pliq	472.32	...	1017.500	...
80	Pliq	760.13	...	2167.499	...
90	Pliq	188.16	...	402.501	...

Bed1 sup
 Bed 1 sup grid
 Clip wt
 Misc wt 7%
 PL-5
 Bed-1
 Ins: 60
 Clad
 A_150
 MH2_600
 Distributor
 Distributor sup
 Bed limiter sup
 Bed limiter
 A FY
 PL-6
 Ins: 70
 Clad
 E_40
 MH1_600
 TRAY 1-3
 Demister
 Demister sup
 TSR bolting bar
 MH1 Davit
 Trunnion 1
 Trunnion 2
 PL-7
 Ins: 90
 Clad
 C_100
 C FY
 C F/M

Note: The individual tray liquid weights are listed below, but these weights are included in the tray weights above.

From	Tliq	Weight	X Offset	Y Offset
100	Tliq	0.00	0.000	544.800

Total Weight of Each Detail Type:

Platforms	13978.6
Packing	4253.0
Packing Liquid	1940.5
Liquid	9361.4
Insulation	3322.8
Cladding & Overlay	773.1
Nozzles	6204.1
Trays	1029.2
Weights	11941.0

Sum of the Detail Weights 52803.6 kgm

Weight Summation Results: (kgm)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	31036.6	31036.6	31036.6	31036.6	31036.6	31036.6
Cladding	773.1	773.1	773.1	773.1	773.1	773.1
Nozzles	6204.1	6204.1	6204.1	6204.1	6204.1	6204.1
Wld Weights	7790.0	7790.0	7790.0	7790.0	7790.0	7790.0
Platforms	...	...	13978.6	13978.6	13978.6	13978.6
Packing	...	...	4253.0	4253.0	4253.0	4253.0

Insulation	...	...	3322.8	3322.8	3322.8	3322.8
Trays	...	...	1029.2	1029.2	1029.2	1029.2
Ope Weights	...	...	...	...	...	4151.0
Ope. Liquid	...	...	...	...	...	9361.4
Packing Liq	...	...	...	...	...	1940.5
Test Liquid	...	42574.6	...	...	...	...
Totals	45803.8	88378.5	68387.3	68387.3	68387.3	83840.2

Field Installation Options:

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	45803.8 kgm
Shop Test Wt.	- Fabricated weight + water (full)	88378.5 kgm
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	68387.3 kgm
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	68387.3 kgm
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	68387.3 kgm
Operating Wt.	- Empty weight + operating liq. uncorroded	83840.2 kgm
Field Test Wt.	- Empty weight + test medium (full)	106709.0 kgm

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	34722.4 kgm
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	47497.3 kgm
Mass of the Upper 1/3 of the Vertical Vessel (empty)	32280.1 kgm

Surface Areas of Elements:

From	To	Outside Surface Area mm ²	Inside Surface Area mm ²
10	20	11020707	10844778
20	30	16531061	16267167
30	40	5510353	5422389
40	50	3698054	3399753
50	60	16663008	16022123
60	70	16663008	16022123
70	80	16663008	16022123
80	90	21134248	20321392
90	100	12150394	11764227
100	110	17160006	16614626
110	120	3674048	3422847

Total	140867904.0	136123552.0 mm ²
Total	140.9	136.1 m ²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kgm	Total. Ele. Oper. Wgt. kgm	Total. Ele. Hydro. Wgt. kgm	Total Dtl. Offset Mom. N-mm	Oper. Wgt. No Liquid kgm
10	20	2413.03	2413.03	2413.03	...	2413.03
20	30	2622.06	2622.06	2622.06	9516765	2622.06
30	40	674.359	674.359	674.359	...	674.359
40	50	4443.8	5405.47	5200.08	...	4631.8
50	60	6191.99	14137	12998.5	9702671	7173.99
60	70	8412.96	10557.6	14079.2	17673166	8412.96
70	80	8428.61	9150.93	14199.2	17673166	8678.61
80	90	16013.9	16954.1	22979.7	17691396	16193.9
90	100	6177.91	8232.07	10783.4	27338384	8043.91
100	110	9258.5	9378.5	16341.6	17260638	9378.5
110	120	3750.14	4315.14	4514.04	...	4315.14

Empty Support Force + the Sum of the Y forces	659823.50	N
Operating Support Force + the Sum of the Y forces	830383.06	N
Hydro Support Force + the Sum of the Y forces	1055577.62	N



Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kgm	Cumulative Oper. Wgt. kgm	Cumulative Hydro. Wgt. kgm
10	20	72538.3	83840.2	106805
20	30	70125.2	81427.1	104392
30	40	67503.2	78805.1	101770
40	50	66828.8	78130.7	101096
50	60	62197	72725.2	95895.6
60	70	55023	58588.3	82897.2
70	80	46610.1	48030.7	68818
80	90	37931.5	38879.8	54618.8
90	100	21737.5	21925.7	31639
100	110	13693.6	13693.6	20855.6
110	120	4315.14	4315.14	4514.04

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. N-mm	Cumulative Oper. Mom. N-mm	Cumulative Hydro. Mom. N-mm
10	20	116856184	116856184	116856184
20	30	116856184	116856184	116856184
30	40	107339416	107339416	107339416
40	50	107339416	107339416	107339416
50	60	107339416	107339416	107339416
60	70	97636744	97636744	97636744
70	80	79963592	79963592	79963592
80	90	62290420	62290420	62290420
90	100	44599024	44599024	44599024
100	110	17260638	17260638	17260638
110	120	...	...	...

Nozzle Flange MAWP Results: (MPa & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
D_150	9.14	10.20	130	600	GR 1.1	3.585	9.143	9.143	9.143
MH4_600	4.57	5.10	130	300	GR 1.1	...	...	...	...
B_150	9.14	10.20	130	600	GR 1.1	3.585	9.143	9.143	9.143
L1_50	4.57	5.10	130	300	GR 1.1	...	...	...	...
L2_50	4.57	5.10	130	300	GR 1.1	...	...	...	...
L3_50	4.57	5.10	130	300	GR 1.1	...	...	...	...
L4_50	4.57	5.10	130	300	GR 1.1	...	...	...	...
L5_50	4.57	5.10	130	300	GR 1.1	...	...	...	...
L6_50	4.57	5.10	130	300	GR 1.1	...	...	...	...
DP1_50	4.57	5.10	130	300	GR 1.1	...	...	...	...
UC2_600	4.57	5.10	130	300	GR 1.1	...	...	...	...
MH3_600	4.57	5.10	130	300	GR 1.1	...	...	...	...
DP2_50	4.57	5.10	130	300	GR 1.1	...	...	...	...
UC1_600	4.57	5.10	130	300	GR 1.1	...	...	...	...
A_150	9.14	10.20	130	600	GR 1.1	3.585	9.143	9.143	9.143
MH2_600	4.57	5.10	130	300	GR 1.1	...	...	...	...
E_40	13.72	15.31	130	900	GR 1.1	13.777	6.799	13.717	6.799
MH1_600	4.57	5.10	130	300	GR 1.1	...	...	...	...
C_100	9.14	10.20	130	600	GR 1.1	4.621	9.093	9.143	9.093
Min Rating	4.574	5.100, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME 4.16.12

The 4.16.12 Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the 4.16.12 method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME N-mm	Axial Force FE N	MAWP PD MPa	Maximum Pressure Pmax MPa	Pass Fail
D_150	192.151	0.50	4454994	11246	4.295	9.143	Pass
B_150	192.151	0.50	4454994	11246	4.295	9.143	Pass
A_150	192.151	0.50	4454994	11246	4.295	9.143	Pass
E_40	58.801	0.50	495000	3749	4.295	6.799	Pass
C_100	135.001	0.50	1979995	7497	4.295	9.093	Pass

Evaluating Flange on Nozzle [D150]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(4454994) + 4 \cdot \text{abs}(11246)192.151 \leq \pi \cdot 192.151^3 \cdot [(9.143 - 4.295) + 500 \cdot 9.143]$$

$$= 0.79927E+08 \leq 0.20999E+09 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle D150 [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(9.143, 0.500 \cdot 9.143 + 9.143 - 79926872.0 / (\pi \cdot 192.151^3))$$

$$= \min(9.143, 10.129)$$

$$= 9.143 \text{ MPa}$$

Evaluating Flange on Nozzle [B150]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(4454994) + 4 \cdot \text{abs}(11246)192.151 \leq \pi \cdot 192.151^3 \cdot [(9.143 - 4.295) + 500 \cdot 9.143]$$

$$= 0.79927E+08 \leq 0.20999E+09 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle B150 [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$



$= \min(9.143, 0.500 \cdot 9.143 + 9.143 - 79926872.0/(\pi \cdot 192.151^3))$
 $= \min(9.143, 10.129)$
 $= 9.143 \text{ MPa}$

Evaluating Flange on Nozzle [A150]:

$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$
 $= 16(4454994) + 4 \cdot \text{abs}(11246)192.151 \leq \pi \cdot 192.151^3 [(9.143 - 4.295) + .500 \cdot 9.143]$
 $= 0.79927E+08 \leq 0.20999E+09 \text{ Check Passes}$

Maximum Pressure for Flange on Nozzle A150 [Pmax]:

$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$
 $= \min(9.143, 0.500 \cdot 9.143 + 9.143 - 79926872.0/(\pi \cdot 192.151^3))$
 $= \min(9.143, 10.129)$
 $= 9.143 \text{ MPa}$

Evaluating Flange on Nozzle [E40]:

$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$
 $= 16(495000) + 4 \cdot \text{abs}(3749)58.801 \leq \pi \cdot 58.801^3 [(13.717 - 4.295) + .500 \cdot 13.717]$
 $= 0.88020E+07 \leq 0.10402E+08 \text{ Check Passes}$

Maximum Pressure for Flange on Nozzle E40 [Pmax]:

$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$
 $= \min(13.717, 0.500 \cdot 13.717 + 13.717 - 8802025.0/(\pi \cdot 58.801^3))$
 $= \min(13.717, 6.799)$
 $= 6.799 \text{ MPa}$

Evaluating Flange on Nozzle [C100]:

$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$
 $= 16(1979995) + 4 \cdot \text{abs}(7497)135.001 \leq \pi \cdot 135.001^3 [(9.143 - 4.295) + .500 \cdot 9.143]$
 $= 0.35730E+08 \leq 0.72826E+08 \text{ Check Passes}$

Maximum Pressure for Flange on Nozzle C100 [Pmax]:

$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$
 $= \min(9.143, 0.500 \cdot 9.143 + 9.143 - 35730028.0/(\pi \cdot 135.001^3))$
 $= \min(9.143, 9.093)$
 $= 9.093 \text{ MPa}$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 1.1578 Hz.

The Natural Frequency for the Vessel (Ope...) is 1.13643 Hz.

The Natural Frequency for the Vessel (Filled) is 0.96919 Hz.

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Forces/Moments Applied to Vessel (Combined w/Wind Loads)

From	To	X and Z Dir Force Res. N	X,Z Moment and For Res N-mm
10	20	...	...
20	30	...	...
30	40	...	...
40	50	...	...
50	60	...	...
60	70	...	...
70	80	...	...
80	90	...	...
90	100	...	...
100	110	...	...
110	120	7775.19	4900422

Forces/Moments Applied to Vessel (Combined w/Seismic Loads)

From	To	X and Z Dir Force Res. N	X,Z Moment and For Res N-mm
10	20	...	...
20	30	...	...
30	40	...	...
40	50	...	...
50	60	...	...
60	70	...	...
70	80	...	...
80	90	...	...
90	100	...	...
100	110	...	...
110	120	7775.19	4900422

User Input Forces and Moments:

From Node	Distance From	- - - - Fx	Forces - - - - Fy	- - - - Fz	- - - - Mx	Moments - - - - My	- - - - Mz
110	350.00	5498.	-8247.	5498.	0.154E+07	0.154E+07	

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	97.2 km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
= $\min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$
= $\min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.188 \cdot 0.914) / (1 + 1.7 \cdot 3.4 \cdot 0.188)))$
= $\min(0.85, 0.884)$
= 0.850

Natural Frequency of Vessel (Operating)	1.136 Hz
Natural Frequency of Vessel (Empty)	1.158 Hz
Natural Frequency of Vessel (Test)	0.969 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.634
Structure Height to Diameter ratio	13.192
Height to top of Structure	24732.498 mm

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:
Exposure Category: C from Table 26.11-1
 $\alpha = 9.5$; $Zg = 274320.0$ mm

Effective Height $[z]$:
= Centroid Height + Vessel Base Elevation
= $1000.0 + 300.0 = 1300.0$ mm
= 4.265 ft. Imperial Units

Velocity Pressure coefficient evaluated at height z $[Kz]$:
Because z (4.265 ft.) < 15 ft.
= $2.01(15 / Zg)^{2/\alpha}$
= $2.01(15/900.0)^{2/9.5}$
= 0.849

Type of Hill: No Hill

Wind Directionality Factor $[Kd]$:
= 1, per Table 26.6-1

As there is No Hill Present: $[Kzt]$:



K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$$

$$= 1.0$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 0.849 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 60.399^2)$$

$$= 16.0 \text{ psf } [0.766] \text{ kPa}$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 16.0 \cdot 0.85 \cdot 0.634 \cdot 49.088$$

$$= 423.5 \text{ lbs. } [1883.8] \text{ N}$$

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz kPa
Skirt-3	1300.0	0.000	0.000	0.000	0.849	1.000	0.766
Skirt 2	3800.0	0.000	0.000	0.000	0.849	1.000	0.766
Skirt 1	5800.0	0.000	0.000	0.000	0.892	1.000	0.766
Bottom head	6325.0	0.000	0.000	0.000	0.909	1.000	0.766
Shell-1	7850.0	0.000	0.000	0.000	0.951	1.000	0.766
Shell-2	10850.0	0.000	0.000	0.000	1.018	1.000	0.766
Shell-3	13850.0	0.000	0.000	0.000	1.072	1.000	0.766
Shell-4	17252.5	0.000	0.000	0.000	1.123	1.000	0.766
Shell-5	20252.5	0.000	0.000	0.000	1.161	1.000	0.766
Shell-7	22900.0	0.000	0.000	0.000	1.192	1.000	0.766
Top head	24699.2	0.000	0.000	0.000	1.211	1.000	0.766

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

- Cf - Correction factor for natural frequency
- D - Average outer insulated diameter of vessel, mm
- Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
- Dr - Average outer insulated of top half of vessel, mm
- f - Natural frequency of vibration (Hertz)
- f1 - Natural frequency of bare vessel based on a unit value of $(D/L^2)(10^4)$
- L - Total height of structure mm
- Lc - Total length of conical section(s) of vessel, mm
- tb - Uncorroded plate thickness at bottom of vessel, mm
- V30 - Design Wind Speed provided by user, km/hr
- Vc - Critical wind velocity, km/hr
- Vw - Maximum wind speed at top of structure, km/hr
- W - Total corroded weight of structure, N
- Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. N
- Z - Maximum amplitude of vibration at top of vessel, mm
- Dl - Logarithmic decrement (taken as 0.03 for Welded Structures)
- Vp - Vib. Chance, <= 0.000003 (High); 0.000003 < 0.000004 (Probable)
- P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5
 :: 0.0/24199.998 = 0.0

:: $[D / L^2] \cdot 10^4 < 8$ (English Units, ft.)
 :: $[6.19/79.4^2] \cdot 10^4 = 9.82$ [Geometry Violation]

Compute the vibration possibility. If Vp > 0.000004 no chance. [Vp]:

$$= W / (L \cdot Dr^2)$$

$$= 782625 / (24200.0 \cdot 1872.0^2)$$

$$= 0.92284E-05$$



Since Vp is > 0.000004 no further vibration analysis is required !

Platform Load Calculations

ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N	Cf
PL-8	1499999.88	4300.00	0.77	1149.07	1.00
PL-1	2999999.75	6300.00	0.77	2298.14	1.00
PL-2	2999999.75	10350.00	0.77	2298.14	1.00
PL-3	2999999.75	14350.00	0.77	2298.14	1.00
PL-4	2999999.75	17350.00	0.77	2298.14	1.00
PL-5	2999999.75	21155.00	0.77	2298.14	1.00
PL-6	2999999.75	22350.00	0.77	2298.14	1.00
PL-7	3764000.00	24950.00	0.77	2883.40	1.00

Wind Loads on Masses/Equipment/Piping

ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N
Tailing lug	0.00	300.00	0.77	0.00
D FY	0.00	6300.00	0.77	0.00
D anti-swirl	0.00	5900.00	0.77	0.00
MH4 Davit	0.00	7150.00	0.77	0.00
B FY	0.00	9150.00	0.77	0.00
Bed2 sup	0.00	9300.00	0.77	0.00
Bed 2 sup gri	0.00	9300.00	0.77	0.00
Distributor 2	0.00	14850.00	0.77	0.00
Dist2 sup	0.00	14800.00	0.77	0.00
MH3 Davit	0.00	14650.00	0.77	0.00
Bed2 lim sup	0.00	13650.00	0.77	0.00
Bed2 limiter	0.00	12350.00	0.77	0.00
Bed1 sup	0.00	15800.00	0.77	0.00
Bed 1 sup gri	0.00	15850.00	0.77	0.00
Clip wt	0.00	17350.00	0.77	0.00
Misc wt 7%	0.00	17300.00	0.77	0.00
Distributor	0.00	21155.00	0.77	0.00
Distributor s	0.00	21105.00	0.77	0.00
Bed limiter s	0.00	21005.00	0.77	0.00
Bed limiter	0.00	21055.00	0.77	0.00
A FY	0.00	21275.00	0.77	0.00
Demister	0.00	24200.00	0.77	0.00
Demister sup	0.00	24150.00	0.77	0.00
TSR bolting b	0.00	21550.00	0.77	0.00
MH1 Davit	0.00	22900.00	0.77	0.00
Trunnion 1	0.00	23450.00	0.77	0.00
Trunnion 2	0.00	23450.00	0.77	0.00
C FY	0.00	24450.00	0.77	0.00

The Natural Frequency for the Vessel (Ope...) is 1.13643 Hz.

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area mm ²	Wind Pressure kPa	Element Wind Load N
10	20	1300	2280.2	4560399	0.76606	2072.21
20	30	3800	2280.2	6840599	0.76606	4538.61
30	40	5800	2384.2	2384199	0.76606	1083.36
40	50	6325	2501.2	125060	0.76606	2574.79
50	60	7850	2506.4	7519201	0.76606	3416.67
60	70	10850	2506.4	7519201	0.76606	5828.13
70	80	13850	2506.4	7519201	0.76606	6035.88
80	90	17252.5	2506.4	9536852	0.76606	6875.72
90	100	20252.5	2498.6	5484427	0.76606	5132.69
100	110	22900	2498.6	7745659	0.76606	5986.81



110	120	24699.2	2493.4	1167470	0.76606	3734.43
-----	-----	---------	--------	---------	---------	---------

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	2.000
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	2.500
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned}SMS &= Fa * Ss = 1.6 * 0.147 = 0.235 \\SM1 &= Fv * S1 = 2.4 * 0.065 = 0.156 \\SDS &= 2/3 * Sms = 2/3 * 0.235 = 0.157 \\SD1 &= 2/3 * Sm1 = 2/3 * 0.156 = 0.104\end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
= $C_t \cdot h_n^x$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
= $0.020(80.7956^{0.75})$
= 0.539 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
= (1/Natural Frequency) = (1/1.136)
= 0.880

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
= Minimum Value of ($1.7 \cdot 0.539$, 0.88) per 12.8.2
= 0.880

Seismic Response Coefficient per equation 12.8-2 [Cs]:
= $Sds / (R / I)$
= $0.157 / (2.0 / 1.25)$
= 0.098

Maximum value of C_s per equation 12.8-3 [Cs]:
= $Sd1 / (T(R / I_e))$
= $0.104 / (0.88(2.0 / 1.25))$
= 0.074

Minimum value of C_s per equation 15.4-1 [Cs]:
= $\max((0.044 \cdot SDS \cdot I_e), 0.030)$
= $\max((0.044 \cdot 0.1568 \cdot 1.25), 0.030)$
= 0.030

Total Base Shear [V]:
= $C_s \cdot W$ (Equation 12.8-1):
= $0.0739 \cdot 822136$
= 60729.531 N

Vertical load per 12.4-4, [Ev]:
= $0.2 \cdot Sds \cdot W$
= $0.2 \cdot 0.157 \cdot 822136$



= 25782.2 N

Distribute the Base shear force to each element according to the equations

$F_x = C_{vx} * V$ (eqn. 12.8-11) and the vertical distribution factor

$C_{vx} = W_x * h_x^k / (\text{Sum of } W_i * h_i^k)$ and k is an exponent which is related to the period of Vibration.

In this case, the value of k was 1.19

The Natural Frequency for the Vessel (Ope...) is 1.13643 Hz.

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight N	Element Ope Load N	Element Emp Load N	Element Vertical Load N
10	20	1000	23662.1	74.9251	74.2247	17535.4
20	30	3500	25711.9	361.522	354.517	16793.3
30	40	5500	6612.76	159.209	155.553	15987
40	50	6025	53006	1422.42	1141.66	15779.6
50	60	7550	138627	4865.82	2076.93	14117.4
60	70	10550	103527	5410.94	4190.54	9770.02
70	80	13550	89734	6316.98	5643.25	6523.4
80	90	16952.5	166251	15279	13972	3709.34
90	100	19952.5	80723.6	9006.12	6534.84	-1504.3
100	110	22600	91965.5	11900.2	11347.1	-4035.8
110	120	24175	42314.2	5932.4	4977.04	-6919.84



Bending Moments due to user defined forces and moments.

User Force/Moment Shear and Bending

From	To	Distance to Support mm	Cumulative Shr Wind Cas N	Cumulative Shr Eqk Cas N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	1000	7775.19	7775.19	192747424	192747424
20	30	3500	7775.19	7775.19	177190752	177190752
30	40	5500	7775.19	7775.19	153855696	153855696
40	50	6025	7775.19	7775.19	146077360	146077360
50	60	7550	7775.19	7775.19	145688448	145688448
60	70	10550	7775.19	7775.19	122353416	122353416
70	80	13550	7775.19	7775.19	99018392	99018392
80	90	16952.5	7775.19	7775.19	75683352	75683352
90	100	19952.5	7775.19	7775.19	46086740	46086740
100	110	22600	7775.19	7775.19	29013282	29013282
110	120	24175	7775.19	7775.19	4900422	4900422

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

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The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm	Cumulative Wind Shear N	Earthquake Shear N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	1000	47279.3	60729.5	654152512	1043027904
20	30	3500	45207.1	60654.6	561628544	921594432
30	40	5500	40668.5	60293.1	432763040	740099456
40	50	6025	39585.1	60133.9	392619904	679861568
50	60	7550	37010.3	58711.4	390704256	676889152
60	70	10550	33593.7	53845.6	284755392	507985184
70	80	13550	27765.5	48434.7	192679296	354502496
80	90	16952.5	21729.6	42117.7	118406400	218618832
90	100	19952.5	14853.9	26838.7	48777912	87376072
100	110	22600	9721.24	17832.6	21795728	38329520
110	120	24175	3734.43	5932.4	930985	1478931

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Applied Forces Case.

Wind Load Deflection:

From	To	Cumulative Shear N	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	7775.19	0.01725	0.068061	0.67120E-04
20	30	7775.19	0.20412	0.40774	0.00015722
30	40	7775.19	0.48981	0.57868	0.00018443
40	50	7775.19	0.58329	0.58792	0.0001851
50	60	7775.19	0.87921	1.19629	0.00021948
60	70	7775.19	1.53691	1.89883	0.00024788
70	80	7775.19	2.2798	2.67758	0.00027029
80	90	7775.19	3.20275	3.74677	0.0002901
90	100	7775.19	4.06749	4.39225	0.00029747
100	110	7775.19	4.85602	5.32343	0.00030218
110	120	7775.19	5.33099	5.33854	0.00030218

Allowable deflection at the Tower Top (For)(6.000"/100ft. Criteria)
 Allowable deflection : 121.000 Actual deflection : 5.339 mm

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear N	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	47279.3	0.059531	0.23234	0.00022659
20	30	45207.1	0.68537	1.34641	0.00050411
30	40	40668.5	1.60839	1.88961	0.00058105
40	50	39585.1	1.90415	1.91871	0.00058291
50	60	37010.3	2.8299	3.80766	0.00067174
60	70	33593.7	4.8418	5.92279	0.0007343
70	80	27765.5	7.04195	8.1915	0.00077491
80	90	21729.6	9.6825	11.1992	0.00080214
90	100	14853.9	12.082	12.9685	0.00080907
100	110	9721.24	14.2245	15.4825	0.00081187
110	120	3734.43	15.5028	15.5231	0.00081187

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed km/hr	2nd Crit. Wind Speed km/hr
10	20	46.5175	290.735
20	30	46.5175	290.735
30	40	48.6392	303.995
40	50	51.0261	318.913
50	60	51.1322	319.576
60	70	51.1322	319.576
70	80	51.1322	319.576
80	90	51.1322	319.576
90	100	50.973	318.581
100	110	50.973	318.581
110	120	50.8669	317.918

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
 Allowable deflection : 121.000 Actual deflection : 15.523 mm



Total Deflection in the Operating Condition + Applied Forces:

Allowable deflection : 121.000 Actual deflection : 20.862 mm

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Longitudinal Stress Constants:

From	To	Metal Area New mm ²	Metal Area Corroded mm ²	Section Modulus New mm ³	Section Modulus Corroded mm ³
10	20	76529.2	67782.5	33026588	29278256
20	30	76529.2	67782.5	33026588	29278256
30	40	76529.2	67782.5	33026588	29278256
40	50	174119	141962	74051240	60786932
50	60	185215	153058	78777088	65542800
60	70	185215	153058	78777088	65542800
70	80	185215	153058	78777088	65542800
80	90	185215	153058	78777088	65542800
90	100	152531	147168	65088336	62871224
100	110	152531	147168	65088336	62871224
110	120	141472	136109	60364948	58142800



Longitudinal Allowable Stresses:

From	To	Tensile MPa	Hydrotest Tensile MPa	Compressive MPa	Hydrotest Compressive MPa
10	20	96.5979	233.996	-100.904	-100.904
20	30	96.5979	233.996	-100.904	-100.904
30	40	68.9985	233.996	-100.904	-100.904
40	50	137.997	233.996	-115.292	-115.292
50	60	137.997	233.996	-116.357	-116.357
60	70	137.997	233.996	-116.357	-116.357
70	80	137.997	233.996	-116.357	-116.357
80	90	137.997	233.996	-116.357	-116.357
90	100	137.997	233.996	-115.876	-115.876
100	110	137.997	233.996	-115.876	-115.876
110	120	137.997	233.996	-114.738	-114.738

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Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure MPa	Longitudinal Stress External Pressure MPa	Longitudinal Stress Hydrotest Pressure MPa
10	20	...	...	...
20	30	...	...	...
30	40	...	...	...
40	50	66.4828	-1.78049	87.0633
50	60	61.6631	-1.65891	80.7515
60	70	61.6631	-1.65891	80.7515
70	80	61.6631	-1.65891	80.7515
80	90	61.6631	-1.65891	80.7515
90	100	63.8316	-1.71362	83.5914
100	110	63.8316	-1.71362	83.5914
110	120	69.018	-1.84445	90.3833

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty MPa	Wght. Str. Operating MPa	Wght. Str. Hydrotest MPa	Wght. Str. Emp. Mom. MPa	Wght. Str. Opr. Mom. MPa
10	20	-10.4949	-12.1301	-15.4527	3.98995	3.98995
20	30	-10.1458	-11.7809	-15.1035	3.98995	3.98995
30	40	-9.76642	-11.4016	-14.7242	3.66501	3.66501
40	50	-4.61658	-4.75063	-4.61658	1.76526	1.76526
50	60	-3.98513	-4.10946	-3.98513	1.63718	1.63718
60	70	-3.52547	-3.6498	-3.52547	1.48919	1.48919
70	80	-2.98643	-3.07745	-2.98643	1.21963	1.21963
80	90	-2.43037	-2.49113	-2.43037	0.95007	0.95007
90	100	-1.44852	-1.46106	-1.44852	0.70914	0.70914
100	110	-0.9125	-0.9125	-0.9125	0.27445	0.27445
110	120	-0.31091	-0.31091	-0.31091	...	...

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. MPa	Bend. Str. Oper. Wind MPa	Bend. Str. Oper. Equ. MPa	Bend. Str. Hyd. Wind MPa	Bend. Str. Hyd. Equ. MPa
10	20	3.98995	22.3354	35.6132	11.1677	...
20	30	3.98995	19.1763	31.467	9.58815	...
30	40	3.66501	14.7763	25.27	7.38815	...
40	50	1.76526	6.45688	11.1808	3.22844	...
50	60	1.63718	5.95915	10.3241	2.97957	...
60	70	1.48919	4.34318	7.74795	2.17159	...
70	80	1.21963	2.93881	5.40699	1.4694	...
80	90	0.95007	1.80597	3.33444	0.90299	...
90	100	0.70914	0.77559	1.38932	0.38779	...
100	110	0.27445	0.34656	0.60946	0.17328	...
110	120	...	0.016007	0.025428	0.0080035	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case	Vortex Shedding Empty Case	Vortex Shedding Test Case	Earthquake Empty Case
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		MPa	MPa	MPa	MPa
10	20	...	...	...	30.4188
20	30	...	...	...	26.9736
30	40	...	...	...	21.8278
40	50	...	...	...	9.69144
50	60	...	...	...	8.9506
60	70	...	...	...	6.76696
70	80	...	...	...	4.72676
80	90	...	...	...	2.91164
90	100	...	...	...	1.22897
100	110	...	...	...	0.54492
110	120	...	...	...	0.021333

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind MPa	Longitudinal Stress Y Forces Seismic MPa
10	20	-0.12168	0.25872
20	30	-0.12168	0.24777
30	40	-0.12168	0.23588
40	50	-0.058097	0.11116
50	60	-0.053885	0.092243
60	70	-0.053885	0.063838
70	80	-0.053885	0.042624
80	90	-0.053885	0.024237
90	100	-0.056041	-0.010223
100	110	-0.056041	-0.027425
110	120	-0.060595	-0.050845

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded MPa	Earthquake For/Mom Corroded MPa	Wind For/Mom No Corrosion MPa	Earthquake For/Mom No Corrosion MPa
10	20	6.58119	6.58119	5.83426	5.83426
20	30	6.05002	6.05002	5.36338	5.36338
30	40	5.25326	5.25326	4.65705	4.65705
40	50	2.40233	2.40233	1.97202	1.97202
50	60	2.22209	2.22209	1.84878	1.84878
60	70	1.86617	1.86617	1.55266	1.55266
70	80	1.51026	1.51026	1.25654	1.25654
80	90	1.15435	1.15435	0.96042	0.96042
90	100	0.7328	0.7328	0.70784	0.70784
100	110	0.46132	0.46132	0.44561	0.44561
110	120	0.084256	0.084256	0.081154	0.081154

Bending Stresses due to Blast Load:

From	To	Bending Stress MPa
10	20	...
20	30	...
30	40	...
40	50	...
50	60	...
60	70	...
70	80	...
80	90	...
90	100	...
100	110	...
110	120	...





Stress Combination Load Cases for Vertical Vessels:

Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

For Division 2, the overall stress intensity is computed in accordance with the Section VIII Division II Example problem manual PTB-3, example 4.3.6, stress due to combined loads.
 The stresses are combined per equation 4.3.44 to determine the overall stress intensity.
 The unity check is from Section 4.4.12 equations 4.4.85 and 4.4.90.

An asterisk (*) in the final column denotes overstress.

Note: Performing Combined Stress Analysis per VIII-2 paragraph 4.3.10.2.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	43.52	96.60	-43.52	110.83	0.4506	0.3927	0.2862
20	39.48	96.60	-39.48	110.83	0.4087	0.3563	0.2541
30	33.58	69.00	-33.58	110.83	0.4867	0.3030	0.2061
40	15.30	138.00	-15.30	138.64	0.1109	0.1104	0.2285
50	13.86	138.00	-13.86	165.07	0.1004	0.0839	0.0595
60	11.28	138.00	-11.28	165.07	0.0817	0.0683	0.0466
70	8.71	138.00	-8.71	165.07	0.0631	0.0528	0.0343
80	6.39	138.00	-6.39	165.07	0.0463	0.0387	0.0237
90	3.72	138.00	-3.72	163.70	0.0270	0.0227	0.0135
100	2.05	138.00	-2.05	163.70	0.0149	0.0125	0.0087
110	0.47	138.00	-0.47	71.90	0.0034	0.0066	...

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	51.23	96.60	-51.23	110.83	0.5303	0.4622	0.3565



20	46.91	96.60	-46.91	110.83	0.4856	0.4233	0.3219
30	40.28	69.00	-40.28	110.83	0.5837	0.3634	0.2674
40	18.36	138.00	-18.36	138.79	0.1331	0.1323	0.2765
50	16.70	138.00	-16.70	165.07	0.1210	0.1012	0.0776
60	13.58	138.00	-13.58	165.07	0.0984	0.0823	0.0613
70	10.40	138.00	-10.40	165.07	0.0754	0.0630	0.0452
80	7.42	138.00	-7.42	165.07	0.0538	0.0450	0.0304
90	4.13	138.00	-4.13	163.70	0.0299	0.0252	0.0163
100	2.22	138.00	-2.22	163.70	0.0161	0.0136	0.0106
110	0.47	138.00	-0.47	137.01	0.0034	0.0034	0.0142

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	45.16	96.60	-45.16	110.83	0.4675	0.4075	0.2862
20	41.12	96.60	-41.12	110.83	0.4257	0.3710	0.2541
30	35.22	69.00	-35.22	110.83	0.5104	0.3178	0.2061
40	15.43	138.00	-15.43	138.61	0.1118	0.1113	0.2404
50	13.98	138.00	-13.98	165.07	0.1013	0.0847	0.0595
60	11.40	138.00	-11.40	165.07	0.0826	0.0691	0.0466
70	8.80	138.00	-8.80	165.07	0.0638	0.0533	0.0343
80	6.46	138.00	-6.46	165.07	0.0468	0.0391	0.0237
90	3.73	138.00	-3.73	163.70	0.0271	0.0228	0.0135
100	2.05	138.00	-2.05	163.70	0.0149	0.0125	0.0087
110	0.47	138.00	-0.47	71.90	0.0034	0.0066	...

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	58.06	96.60	-58.06	110.83	0.6010	0.5238	0.4017
20	53.04	96.60	-53.04	110.83	0.5491	0.4786	0.3610
30	45.35	69.00	-45.35	110.83	0.6573	0.4092	0.2974
40	19.99	138.00	-19.99	138.82	0.1448	0.1440	0.3130
50	18.20	138.00	-18.20	165.07	0.1319	0.1103	0.0859
60	14.69	138.00	-14.69	165.07	0.1064	0.0890	0.0673
70	11.17	138.00	-11.17	165.07	0.0810	0.0677	0.0493
80	7.91	138.00	-7.91	165.07	0.0573	0.0479	0.0329
90	4.30	138.00	-4.30	163.70	0.0312	0.0263	0.0173
100	2.29	138.00	-2.29	163.70	0.0166	0.0140	0.0106
110	0.47	138.00	-0.47	137.01	0.0034	0.0034	0.0147

Analysis of Load Case 5 : NP+HW+HI

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	30.61	234.00	-30.61	100.90	0.1308	0.3034	0.1713
20	28.68	234.00	-28.68	100.90	0.1226	0.2842	0.1674
30	25.78	234.00	-25.78	100.90	0.1102	0.2555	0.1632
40	9.61	234.00	-9.61	115.29	0.0411	0.0834	...
50	8.60	234.00	-8.60	116.36	0.0368	0.0739	0.0319
60	7.19	234.00	-7.19	116.36	0.0307	0.0618	0.0282
70	5.68	234.00	-5.68	116.36	0.0243	0.0488	0.0239
80	4.28	234.00	-4.28	116.36	0.0183	0.0368	0.0195
90	2.55	234.00	-2.55	115.88	0.0109	0.0220	0.0117
100	1.36	234.00	-1.36	115.88	0.0058	0.0117	0.0087
110	0.32	234.00	-0.32	114.74	0.0014	0.0028	0.0103

Analysis of Load Case 6 : NP+HW+HE

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	19.44	234.00	-19.44	100.90	0.0831	0.1927	0.1713
20	19.09	234.00	-19.09	100.90	0.0816	0.1892	0.1674
30	18.39	234.00	-18.39	100.90	0.0786	0.1822	0.1632
40	6.38	234.00	-6.38	115.29	0.0273	0.0554	0.1837
50	5.62	234.00	-5.62	116.36	0.0240	0.0483	0.0319
60	5.01	234.00	-5.01	116.36	0.0214	0.0431	0.0282
70	4.21	234.00	-4.21	116.36	0.0180	0.0361	0.0239



80	3.38	234.00	-3.38	116.36	0.0144	0.0291	0.0195
90	2.16	234.00	-2.16	115.88	0.0092	0.0186	0.0117
100	1.19	234.00	-1.19	115.88	0.0051	0.0102	0.0074
110	0.31	234.00	-0.31	114.74	0.0013	0.0027	...

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	45.16	96.60	-45.16	110.83	0.4675	0.4075	0.2862
20	41.12	96.60	-41.12	110.83	0.4257	0.3710	0.2541
30	35.22	69.00	-35.22	110.83	0.5104	0.3178	0.2061
40	107.38	138.00		138.61	0.7781		0.2404
50	110.63	138.00		165.07	0.8017		0.0595
60	109.45	138.00		165.07	0.7932		0.0466
70	108.99	138.00		165.07	0.7898		0.0343
80	108.81	138.00		165.07	0.7885		0.0237
90	112.41	138.00		163.70	0.8146		0.0135
100	112.35	138.00		163.70	0.8141		0.0087
110	110.12	138.00		114.74	0.7980		...

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	58.06	96.60	-58.06	110.83	0.6010	0.5238	0.4017
20	53.04	96.60	-53.04	110.83	0.5491	0.4786	0.3610
30	45.35	69.00	-45.35	110.83	0.6573	0.4092	0.2974
40	107.97	138.00		138.82	0.7824		0.3130
50	111.30	138.00		165.07	0.8066		0.0859
60	109.88	138.00		165.07	0.7962		0.0673
70	109.23	138.00		165.07	0.7916		0.0493
80	108.92	138.00		165.07	0.7893		0.0329
90	112.43	138.00		163.70	0.8147		0.0173
100	112.36	138.00		163.70	0.8142		0.0106
110	110.12	138.00		114.74	0.7980		...

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	45.16	96.60	-45.16	110.83	0.4675	0.4075	0.2862
20	41.12	96.60	-41.12	110.83	0.4257	0.3710	0.2541
30	35.22	69.00	-35.22	110.83	0.5104	0.3178	0.2061
40	17.21	138.00	-17.21	138.61	0.1247	0.1242	0.2404
50	15.64	138.00	-15.64	165.07	0.1133	0.0948	0.2356
60	13.06	138.00	-13.06	165.07	0.0946	0.0791	0.2302
70	10.46	138.00	-10.46	165.07	0.0758	0.0634	0.2241
80	8.11	138.00	-8.11	165.07	0.0588	0.0492	0.2174
90	5.45	138.00	-5.45	163.70	0.0395	0.0333	0.2228
100	3.76	138.00	-3.76	163.70	0.0273	0.0230	0.2096
110	2.32	138.00	-2.32	137.04	0.0168	0.0169	...

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	58.06	96.60	-58.06	110.83	0.6010	0.5238	0.4017
20	53.04	96.60	-53.04	110.83	0.5491	0.4786	0.3610
30	45.35	69.00	-45.35	110.83	0.6573	0.4092	0.2974
40	21.77	138.00	-21.77	138.82	0.1577	0.1568	0.3130
50	19.86	138.00	-19.86	165.07	0.1439	0.1203	0.2419
60	16.35	138.00	-16.35	165.07	0.1185	0.0990	0.2337
70	12.83	138.00	-12.83	165.07	0.0930	0.0777	0.2257
80	9.56	138.00	-9.56	165.07	0.0693	0.0579	0.2178
90	6.02	138.00	-6.02	163.70	0.0436	0.0368	0.2223
100	4.00	138.00	-4.00	163.70	0.0290	0.0244	0.2089
110	2.32	138.00	-2.32	137.05	0.0168	0.0169	0.0022

Analysis of Load Case 11 : HP+HW+HI



From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	30.61	234.00	-30.61	100.90	0.1308	0.3034	0.1713
20	28.68	234.00	-28.68	100.90	0.1226	0.2842	0.1674
30	25.78	234.00	-25.78	100.90	0.1102	0.2555	0.1632
40	114.87	234.00		115.29	0.4909		...
50	119.60	234.00		116.36	0.5111		0.0319
60	119.55	234.00		116.36	0.5109		0.0282
70	119.51	234.00		116.36	0.5107		0.0239
80	119.48	234.00		116.36	0.5106		0.0195
90	144.43	234.00		115.88	0.6173		0.0117
100	144.42	234.00		115.88	0.6172		0.0087
110	139.24	234.00		114.74	0.5951		...

Analysis of Load Case 12 : HP+HW+HE

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	19.44	234.00	-19.44	100.90	0.0831	0.1927	0.1713
20	19.09	234.00	-19.09	100.90	0.0816	0.1892	0.1674
30	18.39	234.00	-18.39	100.90	0.0786	0.1822	0.1632
40	114.29	234.00		115.29	0.4884		0.1837
50	119.32	234.00		116.36	0.5099		0.0319
60	119.35	234.00		116.36	0.5100		0.0282
70	119.37	234.00		116.36	0.5101		0.0239
80	119.40	234.00		116.36	0.5103		0.0195
90	144.43	234.00		115.88	0.6172		0.0117
100	144.42	234.00		115.88	0.6172		0.0074
110	139.24	234.00		114.74	0.5951		...

Analysis of Load Case 13 : IP+WE+EW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	14.48	96.60	-14.48	110.83	0.1499	0.1307	0.1170
20	14.14	96.60	-14.14	110.83	0.1463	0.1275	0.1131
30	13.43	69.00	-13.43	110.83	0.1947	0.1212	0.1089
40	106.87	138.00		73.05	0.7744		0.1114
50	109.77	138.00		165.07	0.7954		0.0323
60	108.91	138.00		165.07	0.7892		0.0285
70	108.68	138.00		165.07	0.7875		0.0242
80	108.64	138.00		165.07	0.7873		0.0197
90	112.35	138.00		163.70	0.8142		0.0118
100	112.33	138.00		163.70	0.8140		0.0074
110	110.12	138.00		114.74	0.7980		...

Analysis of Load Case 14 : IP+WF+CW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	10.74	96.60	-10.74	110.83	0.1112	0.0969	0.1198
20	10.43	96.60	-10.43	110.83	0.1080	0.0941	0.1163
30	10.10	69.00	-10.10	110.83	0.1464	0.0911	0.1126
40	106.85	138.00		73.05	0.7743		...
50	109.65	138.00		165.07	0.7945		0.0275
60	108.81	138.00		165.07	0.7885		0.0244
70	108.61	138.00		165.07	0.7870		0.0206
80	108.59	138.00		165.07	0.7869		0.0167
90	112.33	138.00		163.70	0.8140		0.0115
100	112.32	138.00		163.70	0.8140		0.0072
110	110.12	138.00		114.74	0.7980		...

Analysis of Load Case 15 : IP+VO+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	16.12	96.60	-16.12	110.83	0.1669	0.1454	0.1353
20	15.77	96.60	-15.77	110.83	0.1633	0.1423	0.1314
30	15.07	69.00	-15.07	110.83	0.2184	0.1359	0.1271



40	106.87	138.00	73.05	0.7744	0.1362
50	109.78	138.00	165.07	0.7955	0.0333
60	108.92	138.00	165.07	0.7893	0.0296
70	108.68	138.00	165.07	0.7876	0.0249
80	108.64	138.00	165.07	0.7873	0.0202
90	112.35	138.00	163.70	0.8142	0.0119
100	112.33	138.00	163.70	0.8140	0.0074
110	110.12	138.00	114.74	0.7980	...

Analysis of Load Case 16 : IP+VE+EW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	14.48	96.60	-14.48	110.83	0.1499	0.1307	0.1170
20	14.14	96.60	-14.14	110.83	0.1463	0.1275	0.1131
30	13.43	69.00	-13.43	110.83	0.1947	0.1212	0.1089
40	106.87	138.00		73.05	0.7744		0.1114
50	109.77	138.00		165.07	0.7954		0.0323
60	108.91	138.00		165.07	0.7892		0.0285
70	108.68	138.00		165.07	0.7875		0.0242
80	108.64	138.00		165.07	0.7873		0.0197
90	112.35	138.00		163.70	0.8142		0.0118
100	112.33	138.00		163.70	0.8140		0.0074
110	110.12	138.00		114.74	0.7980		...

Analysis of Load Case 17 : NP+VO+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	16.12	96.60	-16.12	110.83	0.1669	0.1454	0.1353
20	15.77	96.60	-15.77	110.83	0.1633	0.1423	0.1314
30	15.07	69.00	-15.07	110.83	0.2184	0.1359	0.1271
40	6.52	138.00	-6.52	73.05	0.0472	0.0892	0.1362
50	5.75	138.00	-5.75	165.07	0.0416	0.0348	0.0333
60	5.14	138.00	-5.14	165.07	0.0372	0.0311	0.0296
70	4.30	138.00	-4.30	165.07	0.0311	0.0260	0.0249
80	3.44	138.00	-3.44	165.07	0.0249	0.0208	0.0202
90	2.17	138.00	-2.17	163.70	0.0157	0.0133	0.0119
100	1.19	138.00	-1.19	163.70	0.0086	0.0073	0.0074
110	0.31	138.00	-0.31	71.90	0.0023	0.0043	...

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	22.44	96.60	-22.44	110.83	0.2323	0.2025	0.1324
20	21.57	96.60	-21.57	110.83	0.2233	0.1946	0.1286
30	20.08	69.00	-20.08	110.83	0.2911	0.1812	0.1245
40	106.94	138.00		73.05	0.7749		...
50	109.94	138.00		165.07	0.7967		0.0325
60	109.03	138.00		165.07	0.7901		0.0290
70	108.76	138.00		165.07	0.7881		0.0246
80	108.69	138.00		165.07	0.7877		0.0200
90	112.38	138.00		163.70	0.8143		0.0120
100	112.34	138.00		163.70	0.8141		0.0077
110	110.12	138.00		114.74	0.7980		...

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	22.44	96.60	-22.44	110.83	0.2323	0.2025	0.1324
20	21.57	96.60	-21.57	110.83	0.2233	0.1946	0.1286
30	20.08	69.00	-20.08	110.83	0.2911	0.1812	0.1245
40	10.59	138.00	-10.59	73.05	0.0767	0.1449	...
50	9.54	138.00	-9.54	165.07	0.0691	0.0578	0.2278
60	8.60	138.00	-8.60	165.07	0.0623	0.0521	0.2248
70	7.42	138.00	-7.42	165.07	0.0538	0.0450	0.2204
80	6.23	138.00	-6.23	165.07	0.0451	0.0377	0.2149
90	4.63	138.00	-4.63	163.70	0.0335	0.0283	0.2211

100	3.39	138.00	-3.39	163.70	0.0246	0.0207	0.2085
110	2.29	138.00	-2.29	137.23	0.0166	0.0167	0.0232

Analysis of Load Case 20 : BL+IP+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	16.12	96.60	-16.12	110.83	0.1669	0.1454	0.1353
20	15.77	96.60	-15.77	110.83	0.1633	0.1423	0.1314
30	15.07	69.00	-15.07	110.83	0.2184	0.1359	0.1271
40	106.87	138.00		73.05	0.7744		0.1362
50	109.78	138.00		165.07	0.7955		0.0333
60	108.92	138.00		165.07	0.7893		0.0296
70	108.68	138.00		165.07	0.7876		0.0249
80	108.64	138.00		165.07	0.7873		0.0202
90	112.35	138.00		163.70	0.8142		0.0119
100	112.33	138.00		163.70	0.8140		0.0074
110	110.12	138.00		114.74	0.7980		...

Absolute Maximum of the all of the Stress Ratio's 0.8147

Governing Element: Shell-5

Governing Load Case 8 : IP+OW+EQ+FS+BS

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
Skirt-3	...	-11.9	4.0	42.2	34.3	+ Tens
Skirt-3	...	-11.9	-4.0	-42.2	-58.1	- Comp
Skirt 2	...	-11.5	4.0	37.5	30.0	+ Tens
Skirt 2	...	-11.5	-4.0	-37.5	-53.0	- Comp
Skirt 1	...	-11.2	3.7	30.5	23.0	+ Tens
Skirt 1	...	-11.2	-3.7	-30.5	-45.4	- Comp
Bottom head	66.5	-4.6	1.8	13.6	77.2	+ Tens
Bottom head	66.5	-4.6	-1.8	-13.6	...	- Comp
Shell-1	61.7	-4.0	1.6	12.5	71.8	+ Tens
Shell-1	61.7	-4.0	-1.6	-12.5	...	- Comp
Shell-2	61.7	-3.6	1.5	9.6	69.2	+ Tens
Shell-2	61.7	-3.6	-1.5	-9.6	...	- Comp
Shell-3	61.7	-3.0	1.2	6.9	66.8	+ Tens
Shell-3	61.7	-3.0	-1.2	-6.9	...	- Comp
Shell-4	61.7	-2.5	1.0	4.5	64.6	+ Tens
Shell-4	61.7	-2.5	-1.0	-4.5	...	- Comp
Shell-5	63.8	-1.5	0.7	2.1	65.2	+ Tens
Shell-5	63.8	-1.5	-0.7	-2.1	...	- Comp
Shell-7	63.8	-0.9	0.3	1.1	64.2	+ Tens
Shell-7	63.8	-0.9	-0.3	-1.1	...	- Comp
Top head	69.0	-0.4	...	0.1	68.8	+ Tens
Top head	69.0	-0.4	...	-0.1	...	- Comp

Additional Results per Section 4.4, see 4.4.15

Nomenclature :

- L - Section Length mm
- Fxa - Allowable comp mem stress due to compression with lamc <= 0.15, MPa
- Fba - Allowable all comp mem stress due to bending moment, MPa
- Fva - Allowable shear stress when only shear is present, MPa
- Fha - Allowable hoop comp memb stress under ext pressure alone, MPa



Fxha - Allowable axial comp w/hoop compression and lamc <= 0.15, MPa
 Fhxa - Allowable hoop comp w/axial compression and lamc <= 0.15, MPa
 Faha - Allowable axial comp w/hoop compression and lamc > 0.15, MPa
 Fbha - Allowable axial comp from bending w/hoop compression, MPa
 Fhba - Allowable hoop comp w/long compression due to moment, MPa

Allowable Stress Results for Case 1 : NP+EW+1.1WI+FW+BW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.834	6000	110	114	69		6895	110	6895	6895	6895
20	0.834	6000	110	114	69		6895	110	6895	6895	6895
30	0.834	6000	110	114	69		6895	110	6895	6895	6895
40						138	6895		6895	6895	6895
50	0.987	18483	165	165	81	21	6895	165	6895	6895	6895
60	0.987	18483	165	165	81	21	6895	165	6895	6895	6895
70	0.987	18483	165	165	81	21	6895	165	6895	6895	6895
80	0.987	18483	165	165	81	21	6895	165	6895	6895	6895
90	0.987	18483	163	163	80	20	6895	163	6895	6895	6895
100	0.987	18483	163	163	80	20	6895	163	6895	6895	6895
110						71	6895		6895	6895	6895

Allowable Stress Results for Case 2 : NP+EW+EE+FS+BS

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.834	6000	110	114	69		6895	110	6895	6895	6895
20	0.834	6000	110	114	69		6895	110	6895	6895	6895
30	0.834	6000	110	114	69		6895	110	6895	6895	6895
40						138	6895		6895	6895	6895
50	0.987	18483	165	165	81	21	6895	165	6895	6895	6895
60	0.987	18483	165	165	81	21	6895	165	6895	6895	6895
70	0.987	18483	165	165	81	21	6895	165	6895	6895	6895
80	0.987	18483	165	165	81	21	6895	165	6895	6895	6895
90	0.987	18483	163	163	80	20	6895	163	6895	6895	6895
100	0.987	18483	163	163	80	20	6895	163	6895	6895	6895
110						137	6895		6895	6895	6895

Allowable Stress Results for Case 3 : NP+OW+1.1WI+FW+BW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.834	6000	110	114	69		6895	110	6895	6895	6895
20	0.834	6000	110	114	69		6895	110	6895	6895	6895
30	0.834	6000	110	114	69		6895	110	6895	6895	6895
40						138	6895		6895	6895	6895
50	0.987	18483	165	165	81	21	6895	165	6895	6895	6895
60	0.987	18483	165	165	81	21	6895	165	6895	6895	6895
70	0.987	18483	165	165	81	21	6895	165	6895	6895	6895
80	0.987	18483	165	165	81	21	6895	165	6895	6895	6895
90	0.987	18483	163	163	80	20	6895	163	6895	6895	6895
100	0.987	18483	163	163	80	20	6895	163	6895	6895	6895
110						71	6895		6895	6895	6895

Allowable Stress Results for Case 4 : NP+OW+EQ+FS+BS

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.834	6000	110	114	69		6895	110	6895	6895	6895
20	0.834	6000	110	114	69		6895	110	6895	6895	6895
30	0.834	6000	110	114	69		6895	110	6895	6895	6895
40						138	6895		6895	6895	6895
50	0.987	18483	165	165	81	21	6895	165	6895	6895	6895
60	0.987	18483	165	165	81	21	6895	165	6895	6895	6895
70	0.987	18483	165	165	81	21	6895	165	6895	6895	6895
80	0.987	18483	165	165	81	21	6895	165	6895	6895	6895
90	0.987	18483	163	163	80	20	6895	163	6895	6895	6895
100	0.987	18483	163	163	80	20	6895	163	6895	6895	6895
110						137	6895		6895	6895	6895

Allowable Stress Results for Case 5 : NP+HW+HI

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.823	6000	110	115	69		6895	110	6895	6895	6895
20	0.823	6000	110	115	69		6895	110	6895	6895	6895
30	0.823	6000	110	115	69		6895	110	6895	6895	6895
40						73	6895		6895	6895	6895
50	0.976	18483	165	165	81	22	6895	165	6895	6895	6895
60	0.976	18483	165	165	81	22	6895	165	6895	6895	6895



70	0.976	18483	165	165	81	22	6895	165	6895	6895	6895
80	0.976	18483	165	165	81	22	6895	165	6895	6895	6895
90	0.976	18483	164	164	80	20	6895	164	6895	6895	6895
100	0.976	18483	164	164	80	20	6895	164	6895	6895	6895
110						72	6895		6895	6895	6895

Allowable Stress Results for Case 6 : NP+HW+HE

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.823	6000	110	115	69		6895	110	6895	6895	6895
20	0.823	6000	110	115	69		6895	110	6895	6895	6895
30	0.823	6000	110	115	69		6895	110	6895	6895	6895
40						73	6895		6895	6895	6895
50	0.976	18483	165	165	81	22	6895	165	6895	6895	6895
60	0.976	18483	165	165	81	22	6895	165	6895	6895	6895
70	0.976	18483	165	165	81	22	6895	165	6895	6895	6895
80	0.976	18483	165	165	81	22	6895	165	6895	6895	6895
90	0.976	18483	164	164	80	20	6895	164	6895	6895	6895
100	0.976	18483	164	164	80	20	6895	164	6895	6895	6895
110						72	6895		6895	6895	6895

Allowable Stress Results for Case 7 : IP+OW+1.1WI+FW+BW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.834	6000	110	114	69		6895	110	6895	6895	6895
20	0.834	6000	110	114	69		6895	110	6895	6895	6895
30	0.834	6000	110	114	69		6895	110	6895	6895	6895
40						138	6895		6895	6895	6895
50	0.987	18483	165	165	81	21	6895	165	6895	6895	6895
60	0.987	18483	165	165	81	21	6895	165	6895	6895	6895
70	0.987	18483	165	165	81	21	6895	165	6895	6895	6895
80	0.987	18483	165	165	81	21	6895	165	6895	6895	6895
90	0.987	18483	163	163	80	20	6895	163	6895	6895	6895
100	0.987	18483	163	163	80	20	6895	163	6895	6895	6895
110							6895		6895	6895	6895

Allowable Stress Results for Case 8 : IP+OW+EQ+FS+BS

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.834	6000	110	114	69		6895	110	6895	6895	6895
20	0.834	6000	110	114	69		6895	110	6895	6895	6895
30	0.834	6000	110	114	69		6895	110	6895	6895	6895
40						138	6895		6895	6895	6895
50	0.987	18483	165	165	81	21	6895	165	6895	6895	6895
60	0.987	18483	165	165	81	21	6895	165	6895	6895	6895
70	0.987	18483	165	165	81	21	6895	165	6895	6895	6895
80	0.987	18483	165	165	81	21	6895	165	6895	6895	6895
90	0.987	18483	163	163	80	20	6895	163	6895	6895	6895
100	0.987	18483	163	163	80	20	6895	163	6895	6895	6895
110							6895		6895	6895	6895

Allowable Stress Results for Case 9 : EP+OW+1.1WI+FW+BW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.834	6000	110	114	69		6895	110	6895	6895	6895
20	0.834	6000	110	114	69		6895	110	6895	6895	6895
30	0.834	6000	110	114	69		6895	110	6895	6895	6895
40						138	6895		6895	6895	6895
50	0.987	18483	165	165	81	21	38	21	28	61	20
60	0.987	18483	165	165	81	21	35	21	26	49	20
70	0.987	18483	165	165	81	21	31	21	23	36	21
80	0.987	18483	165	165	81	21	27	21	20	25	21
90	0.987	18483	163	163	80	20	19	20	14	13	20
100	0.987	18483	163	163	80	20	15	20	11	6	20
110						137	6895		6895	6895	6895

Allowable Stress Results for Case 10 : EP+OW+EQ+FS+BS

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.834	6000	110	114	69		6895	110	6895	6895	6895
20	0.834	6000	110	114	69		6895	110	6895	6895	6895
30	0.834	6000	110	114	69		6895	110	6895	6895	6895
40						138	6895		6895	6895	6895
50	0.987	18483	165	165	81	21	37	21	27	85	19
60	0.987	18483	165	165	81	21	34	21	25	69	20



70	0.987	18483	165	165	81	21	30	21	22	51	20
80	0.987	18483	165	165	81	21	27	21	20	35	21
90	0.987	18483	163	163	80	20	18	20	13	16	20
100	0.987	18483	163	163	80	20	15	20	11	8	20
110						137	6895		6895	6895	6895

Allowable Stress Results for Case 11 : HP+HW+HI

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.823	6000	110	115	69		6895	110	6895	6895	6895
20	0.823	6000	110	115	69		6895	110	6895	6895	6895
30	0.823	6000	110	115	69		6895	110	6895	6895	6895
40						73	6895		6895	6895	6895
50	0.976	18483	165	165	81	22	6895	165	6895	6895	6895
60	0.976	18483	165	165	81	22	6895	165	6895	6895	6895
70	0.976	18483	165	165	81	22	6895	165	6895	6895	6895
80	0.976	18483	165	165	81	22	6895	165	6895	6895	6895
90	0.976	18483	164	164	80	20	6895	164	6895	6895	6895
100	0.976	18483	164	164	80	20	6895	164	6895	6895	6895
110							6895		6895	6895	6895

Allowable Stress Results for Case 12 : HP+HW+HE

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.823	6000	110	115	69		6895	110	6895	6895	6895
20	0.823	6000	110	115	69		6895	110	6895	6895	6895
30	0.823	6000	110	115	69		6895	110	6895	6895	6895
40						73	6895		6895	6895	6895
50	0.976	18483	165	165	81	22	6895	165	6895	6895	6895
60	0.976	18483	165	165	81	22	6895	165	6895	6895	6895
70	0.976	18483	165	165	81	22	6895	165	6895	6895	6895
80	0.976	18483	165	165	81	22	6895	165	6895	6895	6895
90	0.976	18483	164	164	80	20	6895	164	6895	6895	6895
100	0.976	18483	164	164	80	20	6895	164	6895	6895	6895
110							6895		6895	6895	6895

Allowable Stress Results for Case 13 : IP+WE+EW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.834	6000	110	114	69		6895	110	6895	6895	6895
20	0.834	6000	110	114	69		6895	110	6895	6895	6895
30	0.834	6000	110	114	69		6895	110	6895	6895	6895
40						73	6895		6895	6895	6895
50	0.987	18483	165	165	81	21	6895	165	6895	6895	6895
60	0.987	18483	165	165	81	21	6895	165	6895	6895	6895
70	0.987	18483	165	165	81	21	6895	165	6895	6895	6895
80	0.987	18483	165	165	81	21	6895	165	6895	6895	6895
90	0.987	18483	163	163	80	20	6895	163	6895	6895	6895
100	0.987	18483	163	163	80	20	6895	163	6895	6895	6895
110							6895		6895	6895	6895

Allowable Stress Results for Case 14 : IP+WF+CW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.834	6000	110	114	69		6895	110	6895	6895	6895
20	0.834	6000	110	114	69		6895	110	6895	6895	6895
30	0.834	6000	110	114	69		6895	110	6895	6895	6895
40						73	6895		6895	6895	6895
50	0.987	18483	165	165	81	21	6895	165	6895	6895	6895
60	0.987	18483	165	165	81	21	6895	165	6895	6895	6895
70	0.987	18483	165	165	81	21	6895	165	6895	6895	6895
80	0.987	18483	165	165	81	21	6895	165	6895	6895	6895
90	0.987	18483	163	163	80	20	6895	163	6895	6895	6895
100	0.987	18483	163	163	80	20	6895	163	6895	6895	6895
110							6895		6895	6895	6895

Allowable Stress Results for Case 15 : IP+VO+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.834	6000	110	114	69		6895	110	6895	6895	6895
20	0.834	6000	110	114	69		6895	110	6895	6895	6895
30	0.834	6000	110	114	69		6895	110	6895	6895	6895
40						73	6895		6895	6895	6895
50	0.987	18483	165	165	81	21	6895	165	6895	6895	6895
60	0.987	18483	165	165	81	21	6895	165	6895	6895	6895



70	0.987	18483	165	165	81	21	6895	165	6895	6895	6895
80	0.987	18483	165	165	81	21	6895	165	6895	6895	6895
90	0.987	18483	163	163	80	20	6895	163	6895	6895	6895
100	0.987	18483	163	163	80	20	6895	163	6895	6895	6895
110							6895		6895	6895	6895

Allowable Stress Results for Case 16 : IP+VE+EW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.834	6000	110	114	69		6895	110	6895	6895	6895
20	0.834	6000	110	114	69		6895	110	6895	6895	6895
30	0.834	6000	110	114	69		6895	110	6895	6895	6895
40						73	6895		6895	6895	6895
50	0.987	18483	165	165	81	21	6895	165	6895	6895	6895
60	0.987	18483	165	165	81	21	6895	165	6895	6895	6895
70	0.987	18483	165	165	81	21	6895	165	6895	6895	6895
80	0.987	18483	165	165	81	21	6895	165	6895	6895	6895
90	0.987	18483	163	163	80	20	6895	163	6895	6895	6895
100	0.987	18483	163	163	80	20	6895	163	6895	6895	6895
110							6895		6895	6895	6895

Allowable Stress Results for Case 17 : NP+VO+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.834	6000	110	114	69		6895	110	6895	6895	6895
20	0.834	6000	110	114	69		6895	110	6895	6895	6895
30	0.834	6000	110	114	69		6895	110	6895	6895	6895
40						73	6895		6895	6895	6895
50	0.987	18483	165	165	81	21	6895	165	6895	6895	6895
60	0.987	18483	165	165	81	21	6895	165	6895	6895	6895
70	0.987	18483	165	165	81	21	6895	165	6895	6895	6895
80	0.987	18483	165	165	81	21	6895	165	6895	6895	6895
90	0.987	18483	163	163	80	20	6895	163	6895	6895	6895
100	0.987	18483	163	163	80	20	6895	163	6895	6895	6895
110						71	6895		6895	6895	6895

Allowable Stress Results for Case 18 : FS+BS+IP+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.834	6000	110	114	69		6895	110	6895	6895	6895
20	0.834	6000	110	114	69		6895	110	6895	6895	6895
30	0.834	6000	110	114	69		6895	110	6895	6895	6895
40						73	6895		6895	6895	6895
50	0.987	18483	165	165	81	21	6895	165	6895	6895	6895
60	0.987	18483	165	165	81	21	6895	165	6895	6895	6895
70	0.987	18483	165	165	81	21	6895	165	6895	6895	6895
80	0.987	18483	165	165	81	21	6895	165	6895	6895	6895
90	0.987	18483	163	163	80	20	6895	163	6895	6895	6895
100	0.987	18483	163	163	80	20	6895	163	6895	6895	6895
110							6895		6895	6895	6895

Allowable Stress Results for Case 19 : FS+BS+EP+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.834	6000	110	114	69		6895	110	6895	6895	6895
20	0.834	6000	110	114	69		6895	110	6895	6895	6895
30	0.834	6000	110	114	69		6895	110	6895	6895	6895
40						73	6895		6895	6895	6895
50	0.987	18483	165	165	81	21	37	21	27	14	21
60	0.987	18483	165	165	81	21	34	21	25	12	21
70	0.987	18483	165	165	81	21	30	21	22	9	21
80	0.987	18483	165	165	81	21	27	21	20	7	21
90	0.987	18483	163	163	80	20	18	20	13	4	20
100	0.987	18483	163	163	80	20	15	20	11	2	20
110						137	6895		6895	6895	6895

Allowable Stress Results for Case 20 : BL+IP+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.834	6000	110	114	69		6895	110	6895	6895	6895
20	0.834	6000	110	114	69		6895	110	6895	6895	6895
30	0.834	6000	110	114	69		6895	110	6895	6895	6895
40						73	6895		6895	6895	6895
50	0.987	18483	165	165	81	21	6895	165	6895	6895	6895
60	0.987	18483	165	165	81	21	6895	165	6895	6895	6895



70	0.987	18483	165	165	81	21	6895	165	6895	6895	6895
80	0.987	18483	165	165	81	21	6895	165	6895	6895	6895
90	0.987	18483	163	163	80	20	6895	163	6895	6895	6895
100	0.987	18483	163	163	80	20	6895	163	6895	6895	6895
110							6895		6895	6895	6895

Note:

Load cases involving Wind and Seismic are assumed to cause a net section bending moment. These load case components are: EQ, EE, WI, WE, VO, VE, VF, HE, HI, WF. Static bending moments due to the weights of nozzles etc. are considered to be local in nature and will not cause a net section bending moment. This is an important consideration when computing the unity check per section 4.4.12 equations 4.4.86, 4.4.87, 4.4.90, 4.4.91 and 4.4.92.

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Skirt Support Analysis Per EN 13445-3 Section 16.12:

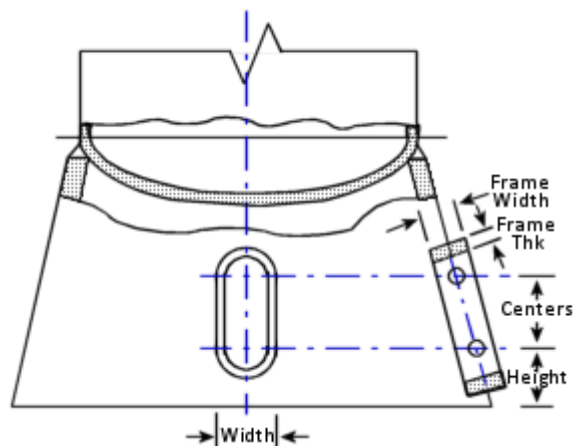
The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

The access opening(s) is/are in section 4-4 only. The other calculations are for areas where no access openings exist. Please be aware of this when you review the results.

The skirt comprises more than one piece. If this is the case, the Openings will only be considered in this Section of the Skirt! Otherwise, make the whole skirt all one piece.

Item: Skirt 2

Total vessel (empty) weight	Fg	688314.438 N
Weight of vessel contents (liquid)	Ff	91797.477 N
Vessel Gross Weight	Fg + Ff	780111.938 N
Skirt Mean Diameter	Dz	1740.000 mm
Skirt Analysis Thickness (Corroded)	ez	12.400 mm
Skirt Height	H	3000.000 mm
Additional Axial Force at section 1-1	F1	-8246.810 N
Moment at section 1-1	M1	893955136.000 N-mm
Applied Moment - Bottom of Skirt	Mb	1098785152.000 N-mm
Weight at Bottom of Skirt	W	798474.188 N



Skirt Access Opening Input Data:

Access #	Diameter mm	Layout Angle deg	Cntr. Dist. From Bottom mm	Obround Cntr-Cntr mm	Frame Width mm	Frame Thickness mm
1	300.0000	0.0	2000.00	0.00	80.000	16.000

Section properties at least metal area [A]:

Number of Openings At Critical Height = 1

Access No	Layout Angle 1 Deg.	Layout Angle 2 Deg.
1	-9.9282	9.9282

The points for integrating Ixx, Iyy are as follows:

FROM THETA1 Deg	TO THETA2 Deg
-350.072	-9.928



Height of Weakest Section	H	2000.009	mm
Rotation of Principle Axes	a	-0.000	deg.
Centroid Position in X direction	g	-15.653	mm
Centroid Position in Y direction	h	0.000	mm
Principle Moment of Inertia	Ixx	25675976704.000	mm ⁴
Principle Moment of Inertia	Iyy	24715915264.000	mm ⁴
Distance to Extreme Fiber	ybar	876.200	mm
Distance to Extreme Fiber	xbar	891.853	mm
Minimum Section Modulus	W4	27713024.000	mm ³
Metal Area at the Cutting Plane	A4	66604.328	mm ²

Half Opening Width Subtended Angle at Weakest Section [δ]:

$$= \sin^{-1}(\text{Largest Opening Width} / \text{Dz})$$

$$= \sin^{-1}(300.0/1740.0)$$

$$= 0.1733 \text{ rad } (9.9282 \text{ deg})$$

Skirt Mean Radius at Section 4-4 [r]:

$$= \text{Dz} / 2$$

$$= 1740.0/2$$

$$= 870.0 \text{ mm}$$

This calculation of ratio below is per an earlier version of EN 13445.

Opening ratio at weakest section: EN 13445 equation 16.12-76 [ratio]:

$$= \delta \cdot \sqrt{(r / e_z)}$$

$$= 0.173 \cdot \sqrt{(870.0/12.4)}$$

$$= 1.451$$

Ok as ratio is < 2 - See EN 13445 16.12.76(a).

Analysis of the Section at Weakest Section Modulus [Section 4-4]:

Skirt Corroded Thickness [ez]:

$$= \text{ex}(\text{new}) - \text{ci} - \text{cext}$$

$$= 14.0 - 0.8 - 0.8$$

$$= 12.4 \text{ mm}$$

Find the moment acting at the Critical Height [M4]:

$$= (\text{Mbottom} - \text{Mtop}) \cdot (\text{H} - \text{CutHeight}) / \text{H} + \text{Mtop}$$

$$= (.10988\text{E}+10 - 0.89396\text{E}+09) \cdot (3000.0 - 2000.009) / 3000.0 + 0.89396\text{E}+09$$

$$= 0.96223\text{E}+09 \text{ N-mm}$$

Compute the section modulus [Zxx and Zyy]:

$$\text{Zxx} = \text{Ixx} / \text{ybar} = 0.25676\text{E}+11 / 876.2 = 29303812 \text{ mm}^3$$

$$\text{Zyy} = \text{Iyy} / \text{xbar} = 0.24716\text{E}+11 / 891.853 = 27713024 \text{ mm}^3$$

$$\text{Z} = \min(\text{Zxx}, \text{Zyy}) = \min(29303812, 27713024) = 27713024 \text{ mm}^3$$

The balance of the calculations are per EN 13445-3, and they may be treated as academic.

Radius of Gyration of Skirt at 4-4 [rg]:

$$= \sqrt{(\min(\text{Ixx}, \text{Iyy}) / \text{Area}(\text{A4}))}$$

$$= \sqrt{(\min(0.25676\text{E}+11, 0.24716\text{E}+11) / 66604.328)}$$

$$= 609.168 \text{ mm}$$

Note: rg is not included in the analysis.

Skirt Slenderness Ratio [rg/L]:

$$= \text{rg} / \text{L}$$

$$= 609.168 / 3000.0$$

$$= 0.203$$

Location of the Skirt Centroid at critical height [e]:

$$= \sqrt{(\text{ex}^2 + \text{ey}^2)}$$

$$= \sqrt{(-15.653^2 + 0.^2)}$$

$$= 0.000 \text{ mm}$$

$$\text{Where } \text{ex} = -15.653 : \text{ey} = 0.$$

Moment Increment at Centroid [ΔM4]:



$$\begin{aligned} &= \text{abs}(F1 - Ff - Fg) \cdot e = F4 \cdot e \\ &= \text{abs}(-8246.81 - 688314 - 91797.48) \cdot 15.653 \\ &= 12345080.000 \text{ N-mm} \end{aligned}$$

Check of the Skirt in Regions with Openings, per 16.12.4.3:

Protrusion, Figure 16.12-5 [ht]:

$$\begin{aligned} &= \text{abs}(lt - ea3) / 2 \\ &= \text{abs}(80.0 - 12.4) / 2 \\ &= 33.800 \text{ mm} \end{aligned}$$

Mean Diameter of Opening, Figure 16.12-5 [d]:

$$\begin{aligned} &= \text{opening width} - et \\ &= 300.0 - 16.0 \\ &= 284.000 \text{ mm} \end{aligned}$$

Angle, 16.12-72 [δ]:

$$\begin{aligned} &= \sin^{-1}(d / D3) \\ &= \sin^{-1}(284.0 / 1740.0) \\ &= 0.164 \text{ radians} \end{aligned}$$

Area in the Frame [At]:

$$\begin{aligned} &= 2 \cdot lt \cdot et \\ &= 2 \cdot 80.0 \cdot 16.0 \\ &= 2560.000 \text{ mm}^2 \end{aligned}$$

Cross Sectional Shell Area at Opening, 16.12-73(b) [AS]:

$$\begin{aligned} &= (\pi - \delta) \cdot D3 \cdot ea3 \\ &= (\pi - 0.164) \cdot 1740.0 \cdot 12.4 \\ &= 64245.582 \text{ mm}^2 \end{aligned}$$

Combined Area at Opening, 16.12-73 [A4]:

$$\begin{aligned} &= AS + AT \\ &= 64245.582 + 2560.0 \\ &= 66805.586 \text{ mm}^2 \end{aligned}$$

Computed Offset, 16.12-74(b) [yt]:

$$\begin{aligned} &= 0.5 \cdot D3 \cdot \cos(\delta) + ht - 0.5 \cdot lt \\ &= 0.5 \cdot 1740.0 \cdot \cos(0.164) + 33.8 - 0.5 \cdot 80.0 \\ &= 852.133 \text{ mm} \end{aligned}$$

Distance to Neutral Axis, 16.12-74 [yG]:

$$\begin{aligned} &= (0.5 \cdot D3 \cdot ea3 \cdot d - 2 \cdot lt \cdot eat \cdot yt) / A4 \\ &= (0.5 \cdot 1740.0 \cdot 12.4 \cdot 284.0 - 2 \cdot 80.0 \cdot 16.0 \cdot 852.133) / 66805.586 \\ &= 13.207 \text{ mm} \end{aligned}$$

Computed Offset [ymax]:

$$\begin{aligned} &= \max[\text{abs}(0.5 \cdot d3 \cdot \cos(\delta) + ht + yG); \text{abs}(0.5 \cdot D3 - yG)] \\ &= \max[\text{abs}(0.5 \cdot 1740.0 \cdot \cos(0.164) + 33.8 + 13.207); \text{abs}(0.5 \cdot 1740.0 - 13.207)] \\ &= 905.341 \text{ mm} \end{aligned}$$

Inertia Value, 16.12-75(b) [IS]:

$$\begin{aligned} &= (\pi - \delta - \sin(\delta) \cdot \cos(\delta)) \cdot ea3 \cdot (D3/2)^3 \\ &= (\pi - 0.164 - \sin(0.164) \cdot \cos(0.164)) \cdot 12.4 \cdot (1740.0/2)^3 \\ &= 22998841344.0 \text{ mm}^4 \end{aligned}$$

Section Modulus [W4]:

$$\begin{aligned} &= (Is + At(yt^2 + lt^2/12) - A4 \cdot yG^2) / ymax \\ &= (0.22999E+11 + 2560.0(852.133^2 + 80.0^2/12) - 66805.586 \cdot 13.207^2) / 905.341 \\ &= 27445438.000 \text{ mm}^3 \end{aligned}$$

Weakening Factor, 16.12-70(b) [psi1]:

$$\begin{aligned} &= \min(1; A4 / (\pi \cdot D3 \cdot ea3)) \\ &= \min(1; 66805.586 / (\pi \cdot 1740.0 \cdot 12.4)) \\ &= 0.986 \end{aligned}$$

Weakening Factor, 16.12-70(c) [psi2]:

$$\begin{aligned} &= \min(1; 4 \cdot W4 / (\pi \cdot D^2 / 3 / (\pi \cdot D3 \cdot ea3))) \\ &= \min(1; 4.0 \cdot 27445438 / (\pi \cdot 1740.0^2 \cdot 12.4)) \\ &= 0.931 \end{aligned}$$



Maximum Bending Moment [Mmax]:

$$\begin{aligned} &= \pi/4 \cdot d^3 \cdot e a^3 \cdot \sigma_{call} \\ &= \pi/4 \cdot 1740.0^2 \cdot 12.4 \cdot 100.9 \\ &= 2976177152.0 \text{ N-mm} \end{aligned}$$

Maximum Compressive Force [Fc,max]:

$$\begin{aligned} &= \pi \cdot d^3 \cdot e a^3 \cdot \sigma_{call} \\ &= \pi \cdot 1740.0 \cdot 12.4 \cdot 100.9 \\ &= 6839013.0 \text{ N} \end{aligned}$$

Unity Check of Weakened Section, 16.12-70 [UC1]:

$$\begin{aligned} &= F4 / (\psi_{i1} \cdot F_{c,max}) + (\text{abs}(M4) + F4 \cdot \text{abs}(y_q)) / (\psi_{i2} \cdot M_{max}) \\ &= 788359 / (0.986 \cdot 6839013) + (\text{abs}(0.96223\text{E}+09) + 788359 \cdot \text{abs}(13.207)) / \\ &\quad (0.931 \cdot 0.29762\text{E}+10) \\ &= 0.468 \text{ must be } \leq \text{ to } 1, \text{ Passed} \end{aligned}$$

The stresses are satisfactory in the region with the opening(s).

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Skirt Support Analysis Per EN 13445-3 Section 16.12:

The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

The access opening(s) is/are in section 4-4 only. The other calculations are for areas where no access openings exist. Please be aware of this when you review the results.

Item: Skirt 1

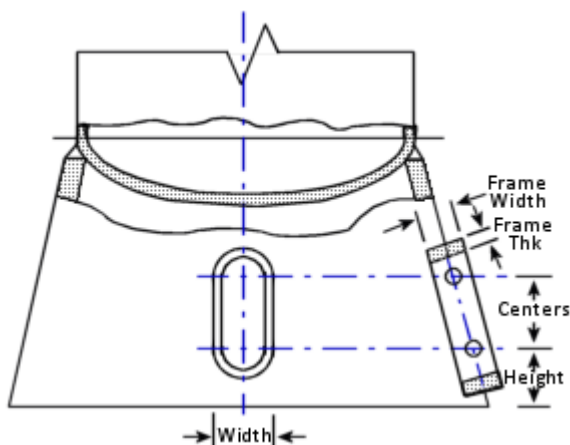
Total vessel (empty) weight	Fg	688314.438 N
Head Weight below section 2-2	DeltaFg	45419.461 N
Weight of vessel contents (liquid)	Ff	91797.477 N
Vessel Gross Weight	Fg + Ff	780111.938 N
Skirt Mean Diameter	Dz	1740.000 mm
Skirt Analysis Thickness (Corroded)	ez	12.400 mm
Skirt Height	H	1000.000 mm
Additional Axial Force at section 1-1	F1	-8246.810 N
Moment at section 1-1	M1	825938944.000 N-mm
Applied Moment - Bottom of Skirt	Mb	893955136.000 N-mm
Weight at Bottom of Skirt	W	772762.312 N

Structural Type B Skirt per EN 13445 Figure 16.12-2:

Type of Head:	Elliptical
Head Mean Diameter at Skirt Top	Db 1738.000 mm
Head Corroded Thickness	eb 26.000 mm

Per EN 13445 Section 7.5.4 [R, r]:

$$\begin{aligned}
 h_i &= 431.0 \text{ mm} \\
 K &= D_i / (2 * h_i) = 1712.0 / (2 * 431.0) = 1.9861 \\
 R &= D_i (0.44 * K + 0.02) = 1712.0 * (0.44 * 1.986 + 0.02) = 1530.3136 \text{ mm} \\
 r &= D_i ((0.5 / K) - 0.08) = 1712.0 * ((0.5 / 1.986) - 0.08) = 294.04 \text{ mm}
 \end{aligned}$$



Skirt Access Opening Input Data:

Access #	Diameter mm	Layout Angle deg	Cntr. Dist. From Bottom mm	Obround Cntr-Cntr mm	Frame Width mm	Frame Thickness mm
1	50.0000	0.0	800.00	0.00	80.000	16.000
2	50.0000	90.0	800.00	0.00	80.000	16.000
3	50.0000	180.0	800.00	0.00	80.000	16.000
4	50.0000	270.0	800.00	0.00	80.000	16.000

Interference check:



Access	Access	Pitch mm	Ligament mm
1	2	1366.5699	1316.5699
2	3	1366.5699	1316.5699
3	4	1366.5699	1316.5699
1	4	1366.5699	1316.5699

Interference Check: No Overlap Detected between Openings

Minimum Pitch between openings is: 1366.5699 mm
 Minimum Ligament between Openings is: 1316.5699 mm

Recommended minimum Ligament [lMin]:
 $= 2 \cdot \sqrt{(D \cdot t)}$
 $= 2 \cdot \sqrt{(1740.0 \cdot 12.4)} \text{ mm}$
 $= 293.775 \text{ mm}$

Carefully evaluate this acceptance criteria.

Section properties at least metal area [A]:

Number of Openings At Critical Height = 4

Access No	Layout Angle 1 Deg.	Layout Angle 2 Deg.
1	-1.6467	1.6467
2	88.3533	91.6467
3	178.3533	181.6467
4	268.3533	271.6467

The points for integrating lxx, lyy are as follows:

FROM THETA1 Deg	TO THETA2 Deg
1.647	88.353
91.647	178.353
181.647	268.353
-88.353	-1.647

Height of Weakest Section	H	799.970	mm
Rotation of Principle Axes	a	-1.077	deg.
Centroid Position in X direction	g	-0.000	mm
Centroid Position in Y direction	h	0.000	mm
Principle Moment of Inertia	Ixx	28590211072.000	mm ⁴
Principle Moment of Inertia	Iyy	28590217216.000	mm ⁴
Distance to Extreme Fiber	ybar	876.200	mm
Distance to Extreme Fiber	xbar	876.200	mm
Minimum Section Modulus	W4	32629808.000	mm ³
Metal Area at the Cutting Plane	A4	75542.664	mm ²

Half Opening Width Subtended Angle at Weakest Section [δ]:
 $= \sin^{-1}(\text{Largest Opening Width} / D_z)$
 $= \sin^{-1}(50.0/1740.0)$
 $= 0.0287 \text{ rad (1.6467 deg)}$

Skirt Mean Radius at Section 4-4 [r]:
 $= D_z / 2$
 $= 1740.0/2$
 $= 870.0 \text{ mm}$

Because the Code uses the words 'may be' in the evaluation below, the results should be considered optional.

Check areas for openings per EN 13445 Equation 16.12-76 [A,check]:

At,i	Adelta,i	Status	lt,i <= 8*eat,i
------	----------	--------	-----------------



2560.0	421.6	...	...
2560.0	421.6	...	...
2560.0	421.6	...	...
2560.0	421.6	...	...

This calculation of ratio below is per an earlier version of EN 13445.

Opening ratio at weakest section: EN 13445 equation 16.12-76 [ratio]:
 $= \delta \cdot \sqrt{(r / e_z)}$
 $= 0.029 \cdot \sqrt{(870.0/12.4)}$
 $= 0.241$

Ok as ratio is < 2 - See EN 13445 16.12.76(a).

Analysis of the Section at Weakest Section Modulus [Section 4-4]:

Skirt Corroded Thickness [ez]:
 $= ex(new) - ci - cext$
 $= 14.0 - 0.8 - 0.8$
 $= 12.4 \text{ mm}$

Find the moment acting at the Critical Height [M4]:
 $= (M_{bottom} - M_{top}) \cdot (H - CutHeight) / H + M_{top}$
 $= (.89396E+09 - 0.82594E+09) \cdot (1000.0 - 799.97) / 1000.0 + 0.82594E+09$
 $= 0.83954E+09 \text{ N-mm}$

Compute the section modulus [Zxx and Zyy]:
 $Z_{xx} = I_{xx} / y_{bar} = 0.28590E+11 / 876.2 = 32629808 \text{ mm}^3$
 $Z_{yy} = I_{yy} / x_{bar} = 0.28590E+11 / 876.2 = 32629816 \text{ mm}^3$
 $Z = \min(Z_{xx}, Z_{yy}) = \min(32629808, 32629816) = 32629808 \text{ mm}^3$

The balance of the calculations are per EN 13445-3, and they may be treated as academic.

Radius of Gyration of Skirt at 4-4 [rg]:
 $= \sqrt{(\min(I_{xx}, I_{yy}) / Area(A4))}$
 $= \sqrt{(\min(0.28590E+11, 0.28590E+11) / 75542.664)}$
 $= 615.195 \text{ mm}$

Note: rg is not included in the analysis.

Skirt Slenderness Ratio [rg/L]:
 $= rg / L$
 $= 615.195 / 1000.0$
 $= 0.615$

Location of the Skirt Centroid at critical height [e]:
 $= \sqrt{(ex^2 + ey^2)}$
 $= \sqrt{(-0.0^2 + 0.0^2)}$
 $= 15.653 \text{ mm}$

Where $ex = -0.0$: $ey = 0.0$

Moment Increment at Centroid [$\Delta M4$]:
 $= \text{abs}(F1 - Ff - Fg) \cdot e = F4 \cdot e$
 $= \text{abs}(-8246.81 - 688314 - 91797.48) \cdot 0.0$
 $= 0.000 \text{ N-mm}$

16.12.3.4 Checking at connection areas:

Force at the joint, 16.12-1 [Fzp]:
 $= F1 - Fg - Ff + 4 \cdot M1 / Dz$
 $= -8246.81 - 688314 - 91797.477 + 4 \cdot 0.82594E+09 / 1740.0$
 $= 1109582 \text{ N}$

Force at the joint, 16.12-2 [Fzq]:
 $= F1 - Fg - Ff - 4 \cdot M1 / Dz$
 $= -8246.81 - 688314 - 91797.477 - 4 \cdot 0.82594E+09 / 1740.0$
 $= -2686299.5 \text{ N}$

16.12.3.4.1 Membrane stresses:



Membrane stress at 1-1, 16.12-3 & 4:

Stress [σ_{1pm}]:

$$= (F_{zp} + \Delta F_g) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b) \\
= (1109582 + 45419.461 + 91797.477) / (\pi \cdot 1738.0 \cdot 26.0) + 4.1 \cdot 1738.0 / (4 \cdot 26.0) \\
= 77.30 \text{ MPa}$$

Stress [σ_{1qm}]:

$$= (F_{zq} + \Delta F_g + F_f) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b) \\
= (-0.3E+07 + 45419.461 + 91797.477) / (\pi \cdot 1738.0 \cdot 26.0) + 4.1 \cdot 1738.0 / (4 \cdot 26.0) \\
= 50.56 \text{ MPa}$$

Membrane stress at 2-2, 16.12-9:

Stress [σ_{2pm} & σ_{2qm}]:

$$= (F_f + \Delta F_g) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b) \\
= (91797.477 + 45419.461) / (\pi \cdot 1738.0 \cdot 26.0) + 4.1 \cdot 1738.0 / (4 \cdot 26.0) \\
= 69.48 \text{ MPa}$$

Membrane stress at 3-3, 16.12 12 & 13:

Stress [σ_{3pm}]:

$$= F_{zp} / (\pi \cdot D_z \cdot e_z) \\
= 1109582 / (\pi \cdot 1740.0 \cdot 12.4) \\
= 16.37 \text{ MPa}$$

Stress [σ_{3qm}]:

$$= F_{zq} / (\pi \cdot D_z \cdot e_z) \\
= -2686299.500 / (\pi \cdot 1740.0 \cdot 12.4) \\
= -39.63 \text{ MPa}$$

16.12.3.4.2 Bending stresses:

Cosine of angle, 16.12-25 [$\cos(\gamma)$]:

$$= 1 - 0.5(D_b + e_b - D_z + e_z) / (r + e_b) \\
= 1 - 0.5(1738.0 + 26.0 - 1740.0 + 12.4) / (294.04 + 26.0) \\
= 0.9431$$

$$\gamma = \arccos(\cos(\gamma)) = \arccos(0.943) = 19.416 \text{ deg}$$

Dimension [a]:

$$= 0.5 \cdot \sqrt{e_b^2 + e_z^2 + 2 \cdot e_b \cdot e_z \cdot \cos(\gamma)} \\
= 0.5 \cdot \sqrt{26.0^2 + 12.4^2 + 2 \cdot 26.0 \cdot 12.4 \cdot 0.943} \\
= 18.96 \text{ mm}$$

Bending moment at verification points p and q [M_p & M_q]:

$$M_p = a \cdot F_{zp} = 18.96 \cdot 1109582.0 = 21.0459 \times 10^6 \text{ N-mm} \\
M_q = a \cdot F_{zq} = 18.96 \cdot -2686299.5 = -50.9523 \times 10^6 \text{ N-mm}$$

Correction factor [C]:

$$= 0.63 - 0.057(e_b/e_z)^2 \\
= 0.63 - 0.057(26.0/12.4)^2 \\
= 0.379$$

Bending stresses in sections 1-1 to 3-3 at the outer surface (o), 16.12-26 to 29:

Stress [$\sigma_{1pb(o)}$ & $\sigma_{2pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot D_b \cdot e_b^2) \\
= 0.379 \cdot 6 \cdot 21045950 / (\pi \cdot 1738.0 \cdot 26.0^2) \\
= 12.98 \text{ MPa}$$

Stress [$\sigma_{1qb(o)}$ & $\sigma_{2qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot D_b \cdot e_b^2) \\
= 0.379 \cdot 6 \cdot -5095230 / (\pi \cdot 1738.0 \cdot 26.0^2) \\
= -31.41 \text{ MPa}$$

Stress [$\sigma_{3pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot D_z \cdot e_z^2) \\
= 0.379 \cdot 6 \cdot 21045950 / (\pi \cdot 1740.0 \cdot 12.4^2)$$



= 56.98 MPa

Stress [$\sigma_{3qb(o)}$]:

$$\begin{aligned} &= C \cdot 6 \cdot M_q / (\pi \cdot D_z \cdot e_z^2) \\ &= 0.379 \cdot 6 \cdot -0.5E+08 / (\pi \cdot 1740.0 \cdot 12.4^2) \\ &= -137.95 \text{ MPa} \end{aligned}$$

Geometry variables, 16.12-32 to 36:

$$\begin{aligned} eb/Db &= 1.024/68.425 = 0.015 \\ y &= 125 \cdot eb/Db = 125 \cdot 26.0/1738.0 = 1.87 \end{aligned}$$

As this an elliptical head, it will be treated a a Korbogen type head

Geometric value [α]:

$$= 4.2 - 0.2 \cdot y = 4.2 - 0.2 \cdot 1.87 = 3.826$$

Bending stress in sections 1-1 and 2-2 Equation 16.12-31 [$\sigma_{1b(p)}$ & $\sigma_{2b(p)}$]:

$$\begin{aligned} &= (P + Ph) \cdot Db ((y / y_A) \cdot \alpha - 1) / (4 \cdot eb) \\ &= (4.1 + 0.0) \cdot 1738.0 ((19.42/40.0) \cdot 3.826 - 1) / (4 \cdot 26.0) \\ &= 58.73 \text{ MPa} \end{aligned}$$

16.12.3.4.3 Total stresses and strength conditions:

Structural shape B and C:

Stress [σ_{1pitot}]:

$$\begin{aligned} &= \sigma_{1pm} - \sigma_{1pb(o)} + \sigma_{1b(p)} \\ &= 77.301 - 12.976 + 58.727 \\ &= 123.05 \text{ MPa} \end{aligned}$$

Stress [σ_{1potot}]:

$$\begin{aligned} &= \sigma_{1pm} + \sigma_{1pb(o)} - \sigma_{1b(p)} \\ &= 77.301 + 12.976 - 58.727 \\ &= 31.55 \text{ MPa} \end{aligned}$$

Stress [σ_{1qitot}]:

$$\begin{aligned} &= \sigma_{1qm} - \sigma_{1qb(o)} + \sigma_{1b(p)} \\ &= 50.56 - -31.414 + 58.727 \\ &= 140.70 \text{ MPa} \end{aligned}$$

Stress [σ_{1qotot}]:

$$\begin{aligned} &= \sigma_{1qm} + \sigma_{1qb(o)} - \sigma_{1b(p)} \\ &= 50.56 + -31.414 - 58.727 \\ &= -39.58 \text{ MPa} \end{aligned}$$

Stress [σ_{2pitot}]:

$$\begin{aligned} &= \sigma_{2pm} + \sigma_{2pb(o)} + \sigma_{2b(p)} \\ &= 69.484 + 12.976 + 58.727 \\ &= 141.19 \text{ MPa} \end{aligned}$$

Stress [σ_{2potot}]:

$$\begin{aligned} &= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)} \\ &= 69.484 - 12.976 - 58.727 \\ &= -2.22 \text{ MPa} \end{aligned}$$

Stress [σ_{2qitot}]:

$$\begin{aligned} &= \sigma_{2qm} + \sigma_{2qb(o)} + \sigma_{2b(p)} \\ &= 69.484 + -31.414 + 58.727 \\ &= 96.80 \text{ MPa} \end{aligned}$$

Stress [σ_{2qotot}]:

$$\begin{aligned} &= \sigma_{2qm} - \sigma_{2qb(o)} - \sigma_{2b(p)} \\ &= 69.484 - -31.414 - 58.727 \\ &= 42.17 \text{ MPa} \end{aligned}$$

Stress [σ_{3pitot}]:

$$\begin{aligned} &= \sigma_{3pm} - \sigma_{3pb} \\ &= 16.371 - 56.982 \\ &= -40.61 \text{ MPa} \end{aligned}$$

HEXAGON

Unity Check of Weakened Section, 16.12-70 [UC1]:



$$= F4 / (\text{psi1} \cdot Fc, \text{max}) + (\text{abs}(M4) + F4 \cdot \text{abs}(yq)) / (\text{psi2} \cdot M\text{max}))$$

$$= 788359 / (1.0 \cdot 6839013) + (\text{abs}(0.83954\text{E}+09) + 788359 \cdot \text{abs}(0.0)) /$$

$$(1.0 \cdot 0.29762\text{E}+10)$$

$$= 0.397 \text{ must be } \leq \text{ to } 1, \text{ Passed}$$

Please note: None of the following stresses in the table below are in the region where there is an access opening.

Summary of Stresses: (MPa)

Stress	Actual	Allowed	Result	Equation
Stot1pi	123.05	385.12	Pass	46
Stot1po	31.55	385.12	Pass	47
Stot1qi	140.70	401.64	Pass	48
Stot1qo	-39.58	401.64	Pass	49
Stot2pi	141.19	390.67	Pass	50
Stot2po	-2.22	390.67	Pass	51
Stot2qi	96.80	390.67	Pass	52
Stot2qo	42.17	390.67	Pass	53
Stot3pi	-40.61	412.70	Pass	54
Stot3po	73.35	412.70	Pass	55
Stot3qi	98.32	406.40	Pass	56
Stot3qo	-177.59	406.40	Pass	57

The stresses are satisfactory in the region without openings.
 The stresses are satisfactory in the region with the opening(s).

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Shop/Field Installation Options :

Platform(s) installed in the Shop.
Packing is installed in the Shop.
Trays are installed in the Shop.
Insulation is installed in the Shop.
Lining is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Platforms	15174.902 mm
Center of Gravity of the Packing	14717.554 mm
Center of Gravity of the Liquid	7725.413 mm
Center of Gravity of the Insulation	14878.841 mm
Center of Gravity of the Lining	21838.938 mm
Center of Gravity of the Nozzles	14455.457 mm
Center of Gravity of the Trays	21594.799 mm
Center of Gravity of the Added Weights (Operating)	16893.195 mm
Center of Gravity of the Added Weights (Empty)	16681.521 mm
Center of Gravity of Bare Shell New and Cold	13051.609 mm
Center of Gravity of Bare Shell Corroded	13240.775 mm
Vessel CG in the Operating Condition	13928.114 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	14446.753 mm
Vessel CG in the Test Condition	14703.299 mm

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Skirt Data :

Skirt Outside Diameter at Base	SOD, D6	1754.0001	mm
Skirt Thickness	STHK	14.0000	mm
Skirt Internal Corrosion Allowance	SCA	0.8000	mm
Skirt External Corrosion Allowance		0.8000	mm
Skirt Material		SA-516 70	[Impact Tested]

Basing Input: Type of Geometry: Continuous Top Ring W/Gussets

Thickness of Basing	TBA, ea4	50.0000	mm
Design Temperature of the Basing		38.00	°C
Basing Material		SA-516 70	
Basing Ope. All. Stress	BASOPE, f4	175.00	MPa
Basing Yield Stress		262.00	MPa
Inside Diameter of Basing	DI, D4	1610.0000	mm
Outside Diameter of Basing	DOU, D5	2010.0000	mm
Nominal Diameter of Bolts	BND, d(b0)	36.0000	mm
Bolt Corrosion Allowance	BCA	1.5000	mm
Bolt Material		SA-193 B7	
Bolt Material Class/Thickness Range		<= 64	
Bolt Operating Allowable Stress	SA, fB	138.00	MPa
Number of Bolts	RN, nb	20	
Diameter of Bolt Circle	DC, DBC	1882.0000	mm
Bolt Allowable Shear Stress		82.800	MPa
Ultimate Comp. Strength of Concrete	FPC	41.0	MPa
Allowable Comp. Strength of Concrete	FC	22.2	MPa
Modular ratio Steel/Concrete		9.833	
Thickness of Gusset Plates	TGA, e7	16.0000	mm
Width of Gussets at Top Plate	TWDT	118.0000	mm
Width of Gussets at Base Plate	BWDT	118.0000	mm
Gusset Plate Elastic Modulus	E	201746.5	MPa
Gusset Plate Yield Stress	SY	262.0	MPa
Height of Gussets	HG, h1	168.0000	mm
Distance between Gussets	RG, b6	76.0000	mm
Dist. from Bolt Center to Gusset (Rg/2)	CG	38.0000	mm
Number of Gussets per bolt	NG	2	
Thickness of Top Plate or Ring	TTA, e5	32.0000	mm
Radial Width of the Top Plate	TOPWTH, b5	128.0000	mm
Anchor Bolt Hole Dia. in Top Plate	dia,H	50.0000	mm
External Corrosion Allowance	CA	6.0000	mm
Dead Weight of Vessel	DW	678852.2	N
Operating Weight of Vessel	ROW	830383.1	N
Earthquake Moment on Basing	EQMOM	1112491008.0	N-mm
Wind Moment on Basing	WIMOM	723615680.0	N-mm
User Moment at the Base [Earthquake Case]		192747424.0	N-mm (UEQMOM)
User Moment at the Base [WindLoad Case]		192747424.0	N-mm (UWIMOM)
Percent Bolt Preload	ppl	100.0	
Use AISC A5.2 Increase in Fc and Bolt Stress		No	
Use Allowable Weld Stress per AISC J2.5		No	
Factor for Increase of Allowables	Fact	1.0000	

Results for Brownell and Young Basing Analysis : Analyze Option

Note: This analysis is based on Neutral Axis shift method for Steel on Concrete (or a material with significantly different Young's modulus).

Calculation of Bolt Loads for Multiple Load Conditions:

Bolt Load, Earthquake Operating Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$



$$= ((4 \cdot 0.13052E+10/1882.0) - 804601) / 20$$

$$= 98421.3 \text{ N}$$

Bolt Load, Windload Empty Case:

$$= ((4 \cdot M/DC) - W) / nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0.91636E+09/1882.0) - 678852) / 20$$

$$= 63399.8 \text{ N}$$

Bolt Load, Windload Vortex Shedding Case:

$$= ((4 \cdot M/DC) - W) / nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0/1882.0) - 830383) / 20$$

$$= -41519.2 \text{ N}$$

Governing Bolt Load Condition, Earthquake + Operating Condition:

Area Available in 1 Bolt (corroded) Abss : 600.6298 mm²
 Area Available in all the Bolts Abss * RN : 12012.5967 mm²

Trial#	k	knew	Cc	Ct	z	j	Ft	Fc	Fs
2	0.299	0.149	1.506	2.445	0.438	0.781	436432.2	1241033.1	60239.2
4	0.224	0.261	1.293	2.608	0.454	0.777	422128.3	1226729.2	54625.3
6	0.280	0.289	1.455	2.485	0.442	0.780	432756.6	1237357.4	58755.9
8	0.294	0.296	1.493	2.455	0.439	0.781	435507.0	1240108.0	59863.0
10	0.298	0.298	1.503	2.447	0.438	0.781	436200.4	1240801.2	60144.8
12	0.298	0.298	1.504	2.447	0.438	0.781	436258.3	1240859.2	60168.4

The Actual Stress in a Single Bolt [Sbolt]:

$$= 2 \cdot Ft / (T1 \cdot Dc \cdot Ct)$$

$$= 2 \cdot 436258.3 / (2.032 \cdot 1882.0 \cdot 2.447)$$

$$= 93.269 \text{ MPa, Should be } \leq 138.0 \text{ MPa}$$

Thickness of the Band of Bolting Steel [T1]

$$= RN \cdot Bolt \text{ Area} / (\pi \cdot Dc)$$

$$= 20 \cdot 600.63 / (\pi \cdot 1882.0)$$

$$= 2.032 \text{ mm}$$

Check the Bearing Stress in the Concrete [fc(max)]

$$= fc[(2kd + t3) / (2kd)]$$

$$= 4.024[(2 \cdot 0.2979 \cdot 1882.0 + 200.0) / (2 \cdot 0.2979 \cdot 1882.0)]$$

$$= 4.742 \text{ MPa, Should be } \leq 22.2 \text{ MPa}$$

Values for table 10.3, l = 128.0, b = 147.812, l/b = 0.865965

Maximum Moment per unit width [Mmax]:

$$= \max(Mx, My) = \max(8345.64, 12619.279) = 12619.279 \text{ N}$$

Reqd Thickness of Basing, Brownell & Young Method [T]:

$$= \sqrt{(6 \cdot Mmax / fallow)} + Ca$$

$$= \sqrt{(6 \cdot 12619.279 / 172.9)} + 6.0$$

$$= 26.926 \text{ mm}$$

Nomenclature:

a = (Dc - Ds) / 2	Skirt Distance to Bolt Circle
P = Sa * Abss	Maximum Load on one Bolt
l	Radial Top Plate Width
lavg	Average Gusset Plate Width
g1 = Gamma 1	Constant Term f(b/l)
g2 = Gamma 2	Constant Term f(b/l)
g = Flat distance / 2	Nut 1/2 Dimension (from Tema)
Fb	Allowable Bending Stress

Values for table 10.6, l = 128.0, b = 76.0, b/l = 0.59375

As b/l (0.594) is less than 1, inverting b/l = 1.684.

Moment Term, based on geometry [Mo]:

$$= P / (4 \cdot \pi) [1.3(\ln((2l \sin(\pi \cdot a/l)) / (\pi \cdot g))) + 1] - [(0.7 - g2)P / (4\pi)]$$

$$= 82879.89 / (4 \cdot \pi) [1.3(\ln((2 \cdot 128.0 \cdot \sin(\pi \cdot 64.0 / 128.0)) / (\pi \cdot 30.0))) + 1] - [(0.7 - 0.051) \cdot 82879.89 / (4 \cdot \pi)]$$

$$= 10882.5586 \text{ N}$$



Required Thickness of Continuous Top Ring [Tc]:

$$= \sqrt{(6 \cdot \text{Abs}(\text{Mo}) / \text{Fb}) + \text{Ca}}$$

$$= \sqrt{(6 \cdot \text{Abs}(10882.56) / 174.66) + 6.0}$$

$$= 25.3357 \text{ mm}$$

Required Gusset Plate Thickness [tg]:

$$= P / (\text{Stress Term} \cdot \text{lavg}) + \text{Ca}$$

$$= 82879.89 / (124.11 \cdot 118.0) + 6.0$$

$$= 15.525 \text{ (not less than } 9.525 + 6.0 \text{) mm}$$

Bolt spacing [m]:

$$= \pi \cdot \text{Bolt Circle Diameter} / \text{number of Bolts}$$

$$= \pi \cdot 1882.0 / 20 = 295.624 \text{ mm}$$

Req. Skirt Thk. to withstand Local Bending, (Brownell and Young) [t]:

$$= 1.76 (P \cdot a / (m \cdot (h + tba) \cdot 1.5 \cdot \text{Sktope}))^{2/3} \cdot r^{1/3} + \text{Ca}$$

$$= 1.76 (82880 \cdot 64.0 / (295.62 \cdot 218.0 \cdot 207))^{2/3} \cdot 877.01^{1/3} + \text{Ca}$$

$$= 9.11 + 1.6 = 10.71 \text{ mm}$$

Shear Stress in a Single Bolt [tb]:

$$= \text{Shear Force} / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 68504.7 / (600.63 \cdot 20)$$

$$= 5.7 \text{ MPa. Must be less than } 82.8 \text{ MPa.}$$

Summary of Basing Thickness Calculations

Required Basing Thickness (tension)	26.9263	mm
Actual Basing Thickness as entered by user	50.0000	mm
Required Thickness of Chair Cap	25.3357	mm
Actual Top Ring Thickness as entered by user	32.0000	mm
Required Gusset thickness, + CA	15.5250	mm
Actual Gusset Thickness as entered by user	16.0000	mm
Required Thickness of Skirt for Local Stress	10.7100	mm
Given Thickness of Skirt	14.0000	mm
Required Gusset Height to meet local stress	214.9106	mm

Weld Size Calculations per Steel Plate Engineering Data - Vol. 2

Compute the Weld load at the Skirt/Base Junction [W]:

$$= \text{SkirtStress} \cdot (\text{SkirtThickness} - \text{CA})$$

$$= 58.056 \cdot (14.0 - 1.6)$$

$$= 719.83 \text{ N/mm}$$

Results for Computed Minimum Basing Weld Size [BWeld]:

$$= W / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 720 / [(0.4 \cdot 234) \cdot 2 \cdot 0.707]$$

$$= 5.43 \text{ mm}$$

Results for Computed Minimum Gusset and Top Plate to Skirt Weld Size

Vertical Plate Load [Wv]:

$$= \text{Bolt Load} / (\text{Cmwth} + 2 \cdot (\text{Hg} + \text{Tta}))$$

$$= 82879.9 / (160.0 + 2 \cdot (168.0 + 32.0))$$

$$= 148.0 \text{ N/mm}$$

Horizontal Plate Load [Wh]:

$$= \text{Bolt Load} \cdot e / (\text{Cmwth} \cdot (\text{Hg} + \text{Tta}) + 0.6667 \cdot (\text{Hg} + \text{Tta})^2)$$

$$= 82879.9 \cdot 64.0 / (160.0 \cdot (200.0) + 0.6667 \cdot (200.0)^2)$$

$$= 90.414 \text{ N/mm}$$

Resultant Weld Load [Wr]:

$$= \sqrt{Wv^2 + Wh^2}$$

$$= \sqrt{148.0^2 + 90.41^2}$$

$$= 173.432 \text{ N/mm}$$

Results for Computed Min Gusset and Top Plate to Skirt Weld Size [GsWeld]:

$$= Wr / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$



$$= 173.43 / [(0.4 \cdot 234) \cdot 2 \cdot 0.707]$$
$$= 1.308 \text{ mm}$$

Results for Computed Minimum Gusset to Top Plate Weld Size

Weld Load [Wv]:

$$= \text{Bolt Load} / (2 \cdot \text{TopWth})$$
$$= 82879.9 / (2 \cdot 118.0)$$
$$= 351.186 \text{ N/mm}$$

Weld Load [Wh]:

$$= \text{Bolt Load} \cdot e / (2 \cdot \text{Hgt} \cdot \text{TopWth})$$
$$= 82879.9 \cdot 64.0 / (2 \cdot 200.0 \cdot 118.0)$$
$$= 112.379 \text{ N/mm}$$

Resultant Weld Load [Wr]:

$$= \sqrt{Wv^2 + Wh^2}$$
$$= \sqrt{351.19^2 + 112.38^2}$$
$$= 368.729 \text{ N/mm}$$

Results for Computed Min Gusset to Top Plate Weld Size [GtpWeld]:

$$= Wr / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$
$$= 368.73 / [(0.4 \cdot 234) \cdot 2 \cdot 0.707]$$
$$= 2.782 \text{ mm}$$

Note: The calculated weld sizes need not exceed the component thickness framing into the weld. At the same time, the weld must meet a minimum size specification which is 3/16 in. (4.76 mm) or 1/4 in. (6.35 mm), depending on the component thickness.

Summary of Required Weld Sizes:

Required Basing to Skirt Double Fillet Weld Size	6.3500	mm
Required Gusset to Skirt Double Fillet Weld Size	6.3500	mm
Required Top Plate to Skirt Weld Size	6.3500	mm
Required Gusset to Top Plate Double Fillet Weld Size	2.7815	mm

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: D150

From: 40

Pressure for Reinforcement Calculations	P	4.3372	MPa
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Elliptical Head	D	1700.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	32.0000	mm
Head Internal Corrosion Allowance	c	6.0000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	0.0000	mm
-------------------------------	----	--------	----

User Entered Minimum Design Metal Temperature		-12.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.60	MPa
Allowable Stress At Ambient	Sna	165.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		150.0000	mm

Size and Thickness Basis		Minimum	
Nominal Thickness		120	

Flange Material		SA-105	+IT
-----------------	--	--------	-----

Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	38.5105	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm

Corrosion Allowance	can	6.0000	mm
---------------------	-----	--------	----

Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgmv	30.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

Flange Class		600	
Flange Grade		GR 1.1	

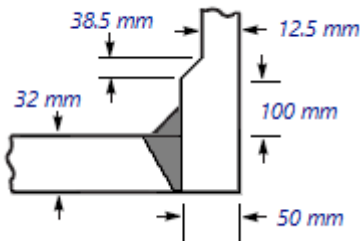
Piping Exit Angle [North Clockwise reference]		90.0	deg
Bend Radius Multiplier		1.50	
Horizontal Run Length		1500.0	mm
Centerline Distance to Tangent		2000.0	mm

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: D150

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		143.294	mm.
Actual Thickness Used in Calculation		12.490	mm



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 155.294 \cdot (\exp(4.337/(147.6 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 2.3155 + 6.0 + 0.0 = 8.3155 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 147.6 \cdot 1.0 \cdot \ln((2 \cdot 6.49 + 155.294) / 155.294) - 0.0$$

$$= 11.849 - 0.0 = 11.8491 \text{ MPa}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	12.220 mm
Min. required shell thickness, internal p + c :	35.488 mm
Min. required shell thickness, external p + c :	16.066 mm

Nozzle neck minimum required thickness : 12.220 mm

Minimum nozzle neck thickness 12.497 mm , is greater than required thickness 12.220 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained :	56.2,	Allowable :	147.6 MPa	Passed
Expansion :	0.0,	Allowable :	334.6 MPa	Passed
Occasional :	24.9,	Allowable :	196.3 MPa	Passed
Shear :	25.0,	Allowable :	103.3 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[147.6/138.0, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 50.0 - 6.0$$

$$= 44.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 12.49 - 6.0$$

$$= 6.490 \text{ mm}$$

Effective Pressure Radius [Reff]:

$$= 0.9/6 \cdot D_i [2 + (D_i/(2h))^2]$$

$$= 0.9/6 \cdot 1712.0 [2 + (1712.0/(2 \cdot 431.0))^2]$$

$$= 1526.550 \text{ mm}$$

Effective Length of Vessel Wall [LR]:



$$\begin{aligned} &= \min(\sqrt{(\text{Reff} \cdot t)}, 2 \cdot Rn) \\ &= \min(\sqrt{(1526.55 \cdot 26.0)}, 2 \cdot 77.647) \\ &= 155.294 \text{ mm} \end{aligned}$$

Intermediate value [Cn]:

$$\begin{aligned} &= \min((t + te)/tn)^{0.35}, 1) \\ &= \min((26.0 + 0.0)/44.0)^{0.35}, 1) \\ &= \min(0.832, 1) \\ &= 0.832 \end{aligned}$$

Intermediate value [Cp]:

$$\begin{aligned} &= \exp((0.35 \cdot di - X_o)/(16t)) \\ &= \exp((0.35 \cdot 1712.0 - 121.647)/(16 \cdot 26.0)) \\ &= 3.152 \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min(t + te + Fp \cdot \sqrt{(Rn \cdot tn)}, Lpr1 + t) \\ &= \min(26.0 + 0.0 + 0.832 \cdot \sqrt{(77.647 \cdot 44.0)}, 300.0 + 26.0) \\ &= 74.621 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot frp) / LR) \\ &= 26.0 + ((0.0 \cdot 1.0) / 155.294) \\ &= 26.000 \text{ mm} \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \\ &= 26.0 \cdot 155.294 \\ &= 4037.647 \text{ mm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= tn \cdot LH \\ &= 44.0 \cdot 74.621 \\ &= 3283.310 \text{ mm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg41}^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 50.000 \text{ mm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 \\ &= 4037.647 + 1.0(3283.31 + 0.0) + 50.0 + 0.0 + 0.0 \\ &= 7370.957 \text{ mm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= tn / \ln[1 + tn/Rn] \\ &= 44.0 / \ln[1 + 44.0/77.647] \\ &= 98.006 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= teff / \ln[1 + teff/Reff] \\ &= 26.0 / \ln[1 + 26.0/1526.55] \\ &= 1539.510 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 137.997 \cdot 1.0 \\ &= 207.0 \text{ MPa} \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot Rxn \cdot LH \\ &= 4.337 \cdot 98.006 \cdot 74.621 \\ &= 31716.8 \text{ N} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot Rxs \cdot (LR + tn)/2 \\ &= 4.337 \cdot 1539.51 \cdot (155.294 + 44.0)/2 \end{aligned}$$



= 665307.2 N

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= (P \cdot Rxs \cdot Rnc)/2 \\ &= (4.337 \cdot 1539.51 \cdot 77.647)/2 \\ &= 259210.6 \text{ N} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN)/P \\ &= (665307 + 259210.6 + 31716.8)/4.337 \\ &= 220490.1 \text{ mm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / (2 \cdot Ap/AT - Rxs/(2 \cdot teff)) \\ &= 206.995 / (2 \cdot 220490.109/7370.957 - 1539.51/(2 \cdot 26.0)) \\ &= 6.8 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= 2 \cdot S[t/Rxs] \\ &= 2 \cdot 137.997[26.0/1539.51] \\ &= 4.7 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(Pmax1, Pmax2) \\ &= \min(6.849, 4.661) \\ &= 4.661 \text{ MPa} \end{aligned}$$

Average Primary Membrane Stress [σAvg]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (31716.77 + 665307 + 259210.562)/7370.957 \\ &= 129.741 \text{ MPa} \end{aligned}$$

General Primary Membrane Stress [σCirc]:

$$\begin{aligned} &= P \cdot Rxs / (2 \cdot teff) \\ &= 4.337 \cdot 1539.51 / (2 \cdot 26.0) \\ &= 128.4 \text{ MPa} \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ) \\ &= \max(2 \cdot 129.741 - 128.408, 128.408) \\ &= 131.1 \text{ MPa} \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	207.00	MPa
Local Primary Membrane Stress	PL	131.07	MPa
Maximum Allowable Working Pressure	Pmax	4.66	MPa

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (Rnc + tn) / Rnc \\ &= (77.647 + 44.0)/77.647 \\ &= 1.567 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned} &= \pi/2 \cdot (Rn + tn) \\ &= \pi/2 \cdot (77.647 + 44.0) \\ &= 191.083 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 44.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2) \\ &= \min(259211 \cdot 1.57, 1.5 \cdot 147.6(3715.825+0.0), \pi/4 \cdot 4.3 \cdot 77.65^2 \cdot 1.57^2) \\ &= 50404.387 \text{ N} \end{aligned}$$



Weld Stress Value [τ]:

$$\begin{aligned} &= fwelds / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot tw_1 + 0.49 \cdot L_{43T})) \\ &= 50404.387 / (191.083(0.49 \cdot 7.071 + 0.6 \cdot 26.0 + 0.49 \cdot 0.0)) \\ &= 13.837 < \text{or} = \text{to } 137.997 \text{ Weld Stress Passed} \end{aligned}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [R_{xn}]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 44.0 / \ln[1 + 44.0/77.647] \\ &= 98.006 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [R_{xs}]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 26.0 / \ln[1 + 26.0/1526.55] \\ &= 1539.510 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [S_{allow}]:

$$\begin{aligned} &= 1.5 \cdot S \\ &= 1.5 \cdot 137.997 \\ &= 207.0 \text{ MPa} \end{aligned}$$

Force acting on the Nozzle [f_N]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot L_H \\ &= 0.103 \cdot 98.006 \cdot 74.621 \\ &= 756.3 \text{ N} \end{aligned}$$

Force acting on the Shell [f_S]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot (L_R + t_n)/2 \\ &= 0.103 \cdot 1539.51 \cdot (155.294 + 44.0)/2 \\ &= 15864.8 \text{ N} \end{aligned}$$

Discontinuity Force from External Pressure [f_Y]:

$$\begin{aligned} &= (P \cdot R_{xs} \cdot R_{nc})/2 \\ &= (0.103 \cdot 1539.51 \cdot 77.647)/2 \\ &= 6181.1 \text{ N} \end{aligned}$$

Area Resisting External Pressure [A_p]:

$$\begin{aligned} &= (f_S + f_Y + f_N)/P \\ &= (15864.8 + 6181.1 + 756.3)/0.103 \\ &= 220490.1 \text{ mm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [P_{max1}]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p/AT - R_{xs}/(2 \cdot t_{eff})) \\ &= 206.995 / (2 \cdot 220490.141/7370.957 - 1539.51/(2 \cdot 26.0)) \\ &= 6.8 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [P_{max2}]:

$$\begin{aligned} &= 2 \cdot S[t/R_{xs}] \\ &= 2 \cdot 137.997[26.0/1539.51] \\ &= 4.7 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure [P_{max}]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(6.849, 4.661) \\ &= 4.661 \text{ MPa} \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (f_N + f_S + f_Y) / AT \\ &= (756.314 + 15864.835 + 6181.104) / 7370.957 \\ &= 3.094 \text{ MPa} \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot R_{xs} / (2 \cdot t_{eff}) \\ &= 0.103 \cdot 1539.51 / (2 \cdot 26.0) \\ &= 3.1 \text{ MPa} \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 3.094 - 3.062, 3.062) \end{aligned}$$



= 3.1 MPa

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	207.00	MPa
Local Primary Membrane Stress	PL	3.13	MPa
Maximum Allowable External Pressure	Pmax	4.66	MPa

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -63 °C

The Drop for this Nozzle is : 4.7637 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 336.7636 mm

Input Echo, WRC107/537 Item 1, Description: D150 :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	6.0000	mm
Vessel Diameter	Dv	3060.000	mm
Vessel Thickness	Tv	32.000	mm
Design Temperature	T1	130.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	MPa
Vessel Hot S.I. Allowable	Smh	138.00	MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	6.0000	mm
Nozzle Diameter	Dn	143.294	mm
Nozzle Thickness	Tn	50.000	mm
Nozzle Material		SA-105	
Nozzle UNS Number		K03504	
Nozzle Cold S.I. Allowable	SNmc	165.00	MPa
Nozzle Hot S.I. Allowable	SNmh	147.60	MPa
Design Internal Pressure	Dp	4.337	MPa
Include Pressure Thrust		No	

External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-11245.7	N
Longitudinal Shear (SUS)	(V1) V1	7497.1	N
Circumferential Shear (SUS)	(Vc) V2	7497.1	N
Circumferential Moment (SUS)	(Mc) M1	3150156.0	N-mm
Longitudinal Moment (SUS)	(Ml) M2	3150156.0	N-mm
Torsional Moment (SUS)	Mt	4500244.5	N-mm

Use Interactive Control No
 WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No

No
No

The diagram shows a vertical shaft passing through a rotating body. The shaft is labeled with P at the top and M_t (torque) around it. The body is labeled with M_1 and M_2 (torques) and V_1 and V_2 (velocities) around it. The shaft is labeled with D (diameter) and B (width). The body is labeled with A and C (points on its surface).

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1



Rad. Bend. P	7.5	-7.5	7.5	-7.5	7.5	-7.5	7.5	-7.5
Rad. Memb. MC	0.0	0.0	0.0	0.0	-2.1	-2.1	2.1	2.1
Rad. Bend. MC	0.0	0.0	0.0	0.0	-24.5	24.5	24.5	-24.5
Rad. Memb. ML	-2.1	-2.1	2.1	2.1	0.0	0.0	0.0	0.0
Rad. Bend. ML	-24.5	24.5	24.5	-24.5	0.0	0.0	0.0	0.0
Tot. Rad. Str.	-18.0	15.9	35.1	-28.8	-18.0	15.9	35.1	-28.8
Tang. Memb. P	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8
Tang. Bend. P	2.4	-2.4	2.4	-2.4	2.4	-2.4	2.4	-2.4
Tang. Memb. MC	0.0	0.0	0.0	0.0	-2.2	-2.2	2.2	2.2
Tang. Bend. MC	0.0	0.0	0.0	0.0	-9.1	9.1	9.1	-9.1
Tang. Memb. ML	-2.2	-2.2	2.2	2.2	0.0	0.0	0.0	0.0
Tang. Bend. ML	-9.1	9.1	9.1	-9.1	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-7.1	6.3	15.5	-7.4	-7.1	6.3	15.5	-7.4
Shear VC	0.8	0.8	-0.8	-0.8	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-0.8	-0.8	0.8	0.8
Shear MT	1.9	1.9	1.9	1.9	1.9	1.9	1.9	1.9
Tot. Shear	2.6	2.6	1.1	1.1	1.1	1.1	2.6	2.6
Str. Int.	18.6	16.6	35.2	28.9	18.1	16.1	35.5	29.1

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		129.2	129.2	129.2	129.2	129.2	129.2	129.2	129.2
Rad. Pl (SUS)		-1.0	-1.0	3.1	3.1	-1.0	-1.0	3.1	3.1
Rad. Q (SUS)		-17.0	17.0	32.0	-32.0	-17.0	17.0	32.0	-32.0
Long. Pm (SUS)		129.2	129.2	129.2	129.2	129.2	129.2	129.2	129.2
Long. Pl (SUS)		-0.4	-0.4	4.0	4.0	-0.4	-0.4	4.0	4.0
Long. Q (SUS)		-6.7	6.7	11.5	-11.5	-6.7	6.7	11.5	-11.5
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.8	0.8	-0.8	-0.8	-0.8	-0.8	0.8	0.8
Shear Q (SUS)		1.9	1.9	1.9	1.9	1.9	1.9	1.9	1.9
Pm (SUS)		129.2	129.2	129.2	129.2	129.2	129.2	129.2	129.2
Pm+Pl (SUS)		129.3	129.3	133.7	133.7	129.3	129.3	133.7	133.7
Pm+Pl+Q (Total)		122.7	145.8	164.4	121.8	122.2	145.3	164.7	122.1

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	129.20	138.00	Passed
Pm+Pl (SUS)	133.65	207.00	Passed
Pm+Pl+Q (TOTAL)	164.66	468.82	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.



Input, Nozzle Desc: MH4600

From: 50

Pressure for Reinforcement Calculations	P	4.3244	MPa
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	1700.00	mm
Design Length of Section	L	18483.8340	mm
Shell Finished (Minimum) Thickness	t	34.0000	mm
Shell Internal Corrosion Allowance	c	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		850.00	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.60	MPa
Allowable Stress At Ambient	Sna	165.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		568.0000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm

Flange Material		SA-105	+IT
-----------------	--	--------	-----

Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	64.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		80.0000	mm

Corrosion Allowance	can	6.0000	mm
---------------------	-----	--------	----

Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	32.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

This is a Manway, Coupling or Access Opening.

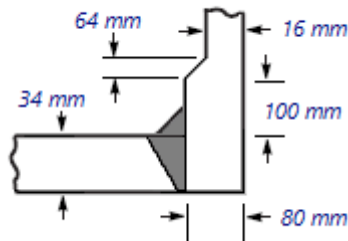
Flange Class		300
Flange Grade		GR 1.1

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: MH4600

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	568.000	mm.
Actual Thickness Used in Calculation	16.000	mm



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 580.0 \cdot (\exp(4.324/(147.6 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 8.6222 + 6.0 + 0.0 = 14.6222 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 147.6 \cdot 1.0 \cdot \ln((2 \cdot 10.0 + 580.0) / 580.0) - 0.0$$

$$= 5.004 - 0.0 = 5.0039 \text{ MPa}$$

This nozzle or its diameter limit intersects weld seam # [3] on this element. The distance between the nozzle and seam is, 239.04 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 518.82 mm

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[147.6/138.0, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 80.0 - 6.0$$

$$= 74.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 16.0 - 6.0$$

$$= 10.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D/t$$

$$= 1712.0/28.0$$

$$= 61.143 \text{ must be less than } 400.$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 1700.0/2 + 6.0$$

$$= 856.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{856.0 \cdot 28.0}, 2 \cdot 290.0)$$



= 154.816 mm

Thickness Limit Candidate [LH1]:

$$\begin{aligned} &= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)} \\ &= \min[1.5 \cdot 28.0, 0.0] + \sqrt{(290.0 \cdot 74.0)} \\ &= 146.492 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 500.0 \\ &= 500.000 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(28.0 + 0.0) \\ &= 224.000 \text{ mm} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles} \\ &= \min[146.492, 500.0, 224.0] + 28.0 \\ &= 174.492 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot frp) / LR) \\ &= 28.0 + ((0.0 \cdot 1.0) / 154.816) \\ &= 28.000 \text{ mm} \end{aligned}$$

Parameter [λ]:

$$\begin{aligned} &= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + teff) \cdot teff})) \\ &= \min(12, (2 \cdot 290.0 + 74.0) / (\sqrt{(1712.0 + 28.0) \cdot 28.0})) \\ &= 2.963 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ &= 28.0 \cdot 154.816 \cdot \max((2.963/5)^{0.85}, 1) \\ &= 4334.849 \text{ mm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= A2A + A2B \\ &= 9472.0 + 2359.664 \\ &= 11831.663 \text{ mm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot Leg^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 50.000 \text{ mm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + frn(A2 + A3) + A41 + A42 + A43 \\ &= 4334.849 + 1.0(11831.663 + 0.0) + 50.0 + 0.0 + 0.0 \\ &= 16216.511 \text{ mm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 74.0 / \ln[1 + 74.0/290.0] \\ &= 325.600 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= teff / \ln[1 + teff/Reff] \\ &= 28.0 / \ln[1 + 28.0/856.0] \\ &= 869.924 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 137.997 \cdot 1.0 \\ &= 207.0 \text{ MPa} \end{aligned}$$

Force acting on the Nozzle [fN]:



$$\begin{aligned} &= P \cdot Rxn \cdot LH \\ &= 4.324 \cdot 325.6 \cdot 174.492 \\ &= 245670.8 \text{ N} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot Rxs(LR + tn) \\ &= 4.324 \cdot 869.924(154.816 + 74.0) \\ &= 860717.9 \text{ N} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot Rxs \cdot Rnc \\ &= 4.324 \cdot 869.924 \cdot 290.0 \\ &= 1090868.6 \text{ N} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN)/P \\ &= (860718 + 1090869 + 245670.8)/4.324 \\ &= 508145.2 \text{ mm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / (2 \cdot Ap/AT - Rxs/teff) \\ &= 206.995 / (2 \cdot 508145/16216.511 - 869.924/28.0) \\ &= 6.6 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t/Rxs] \\ &= 137.997[28.0/869.924] \\ &= 4.4 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(Pmax1, Pmax2) \\ &= \min(6.55, 4.442) \\ &= 4.442 \text{ MPa} \end{aligned}$$

Average Primary Membrane Stress [σ Avg]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (245670.766 + 860718 + 1090869)/16216.511 \\ &= 135.507 \text{ MPa} \end{aligned}$$

General Primary Membrane Stress [σ Circ]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 4.324 \cdot 869.924/28.0 \\ &= 134.4 \text{ MPa} \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ) \\ &= \max(2 \cdot 135.507 - 134.355, 134.355) \\ &= 136.7 \text{ MPa} \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	207.00	MPa
Local Primary Membrane Stress	PL	136.66	MPa
Maximum Allowable Working Pressure	Pmax	4.44	MPa

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (Rnc + tn) / Rnc \\ &= (290.0 + 74.0)/290.0 \\ &= 1.255 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned} &= \pi/2 \cdot (Rn + tn) \\ &= \pi/2 \cdot (290.0 + 74.0) \\ &= 571.770 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :



tc, L41T = 7.071, must be $\geq \min(0.7 * 74.000, 6.000)$ Passed

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2) \\ &= \min(1090869 \cdot 1.26, 1.5 \cdot 147.6(11831.663 + 0.0), \pi/4 \cdot 4.3 \cdot 290.0^2 \cdot 1.26^2) \\ &= 449972.344 \text{ N} \end{aligned}$$

Weld Stress Value [τ]:

$$\begin{aligned} &= fwelds / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T})) \\ &= 449972.344 / (571.77(0.49 \cdot 7.071 + 0.6 \cdot 28.0 + 0.49 \cdot 0.0)) \\ &= 38.838 < \text{or} = \text{to } 137.997 \text{ Weld Stress Passed} \end{aligned}$$

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [R_{xn}]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 74.0 / \ln[1 + 74.0/290.0] \\ &= 325.600 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [R_{xs}]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 28.0 / \ln[1 + 28.0/856.0] \\ &= 869.924 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \\ &= 1.5 \cdot 137.997 \\ &= 207.0 \text{ MPa} \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot L_H \\ &= 0.103 \cdot 325.6 \cdot 174.492 \\ &= 5875.6 \text{ N} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot (L_R + t_n) \\ &= 0.103 \cdot 869.924(154.816 + 74.0) \\ &= 20585.3 \text{ N} \end{aligned}$$

Discontinuity Force from External Pressure [fY]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 0.103 \cdot 869.924 \cdot 290.0 \\ &= 26089.6 \text{ N} \end{aligned}$$

Area Resisting External Pressure [Ap]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (20585.3 + 26089.6 + 5875.6) / 0.103 \\ &= 508145.2 \text{ mm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff}) \\ &= 206.995 / (2 \cdot 508145 / 16216.511 - 869.924 / 28.0) \\ &= 6.6 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S / t / R_{xs} \\ &= 137.997 / (28.0 / 869.924) \\ &= 4.4 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(6.55, 4.442) \\ &= 4.442 \text{ MPa} \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (f_N + f_S + f_Y) / A_T \\ &= (5875.556 + 20585.262 + 26089.631) / 16216.511 \end{aligned}$$



= 3.241 MPa

General Primary Membrane Stress [σ_{Circ}]:

= $P \cdot R_{xs} / t_{eff}$
= $0.103 \cdot 869.924 / 28.0$
= 3.2 MPa

Maximum Local Primary Membrane Stress [PL]:

= $\max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$
= $\max(2 \cdot 3.241 - 3.213, 3.213)$
= 3.3 MPa

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	207.00	MPa
Local Primary Membrane Stress	PL	3.27	MPa
Maximum Allowable External Pressure	Pmax	4.44	MPa

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -26 °C

The Drop for this Nozzle is : 81.8828 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 615.8828 mm

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Input, Nozzle Desc: B150

From: 50

Pressure for Reinforcement Calculations	P	4.3044	MPa
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	1700.00	mm
Design Length of Section	L	18483.8340	mm
Shell Finished (Minimum) Thickness	t	34.0000	mm
Shell Internal Corrosion Allowance	c	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		2850.00	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.60	MPa
Allowable Stress At Ambient	Sna	165.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		150.0000	mm

Size and Thickness Basis		Minimum	
Nominal Thickness		120	

Flange Material		SA-105	+IT
-----------------	--	--------	-----

Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	38.5105	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm

Corrosion Allowance	can	6.0000	mm
---------------------	-----	--------	----

Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

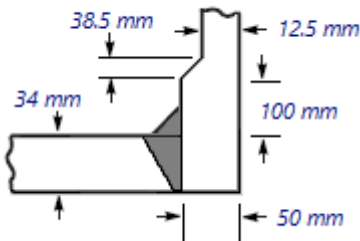
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: B150

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		143.294	mm.
Actual Thickness Used in Calculation		12.490	mm



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 155.294 \cdot (\exp(4.304/(147.6 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 2.2977 + 6.0 + 0.0 = 8.2977 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 147.6 \cdot 1.0 \cdot \ln((2 \cdot 6.49 + 155.294) / 155.294) - 0.0$$

$$= 11.849 - 0.0 = 11.8491 \text{ MPa}$$

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 276.46 mm

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 12.220 mm
 Min. required shell thickness, internal p + c : 33.123 mm
 Min. required shell thickness, external p + c : 26.261 mm

Nozzle neck minimum required thickness : 12.220 mm

Minimum nozzle neck thickness 12.497 mm , is greater than required thickness 12.220 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 56.0,	Allowable	: 147.6 MPa	Passed
Expansion	: 0.0,	Allowable	: 334.8 MPa	Passed
Occasional	: 24.7,	Allowable	: 196.3 MPa	Passed
Shear	: 25.0,	Allowable	: 103.3 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[147.6/138.0, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 50.0 - 6.0$$

$$= 44.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 12.49 - 6.0$$

$$= 6.490 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$



= 1712.0/28.0
= 61.143 must be less than 400.

Effective Pressure Radius [Reff]:

= Di/2 + corrosion allowance
= 1700.0/2 + 6.0
= 856.000 mm

Effective Length of Vessel Wall [LR]:

= $\min(\sqrt{Reff \cdot t}, 2 \cdot Rn)$
= $\min(\sqrt{856.0 \cdot 28.0}, 2 \cdot 77.647)$
= 154.816 mm

Thickness Limit Candidate [LH1]:

= $\min[1.5t, te] + \sqrt{Rn \cdot tn}$
= $\min[1.5 \cdot 28.0, 0.0] + \sqrt{77.647 \cdot 44.0}$
= 58.451 mm

Thickness Limit Candidate [LH2]:

= Lpr1
= 300.0
= 300.000 mm

Thickness Limit Candidate [LH3]:

= 8(t + te)
= 8(28.0 + 0.0)
= 224.000 mm

Effective Nozzle Wall Length Outside the Vessel [LH]:

= $\min[LH1, LH2, LH3] + t$ for set-in nozzles
= $\min[58.451, 300.0, 224.0] + 28.0$
= 86.451 mm

Effective Vessel Thickness [teff]:

= $t + ((A5 \cdot frp) / LR)$
= $28.0 + ((0.0 \cdot 1.0) / 154.816)$
= 28.000 mm

Parameter [λ]:

= $\min(12, (2 \cdot Rn + tn) / (\sqrt{(Di + teff) \cdot teff}))$
= $\min(12, (2 \cdot 77.65 + 44.0) / (\sqrt{(1712.0 + 28.0) \cdot 28.0}))$
= 0.903

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

= $t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$
= $28.0 \cdot 154.816 \cdot \max((0.903/5)^{0.85}, 1)$
= 4334.849 mm²

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

= $tn \cdot LH$
= $44.0 \cdot 86.451$
= 3803.826 mm²

Area Contributed by the Outside Fillet Weld [A41]:

= $0.5 \cdot Leg^{4/3}$
= $0.5 \cdot 10.0^2$
= 50.000 mm²

The total area contributed by A1 through A43 [AT]:

= $A1 + \min(A2 + A3) + A41 + A42 + A43$
= $4334.849 + 1.0(3803.826 + 0.0) + 50.0 + 0.0 + 0.0$
= 8188.675 mm²

Nozzle Radius for Force Calculation [Rxn]:

= $tn / \ln[1 + tn/Rn]$
= $44.0 / \ln[1 + 44.0/77.647]$
= 98.006 mm



Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 28.0 / \ln[1 + 28.0/856.0] \\ &= 869.924 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 137.997 \cdot 1.0 \\ &= 207.0 \text{ MPa} \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot L_H \\ &= 4.304 \cdot 98.006 \cdot 86.451 \\ &= 36466.7 \text{ N} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot R_{xs} (L_R + t_n) \\ &= 4.304 \cdot 869.924 (154.816 + 44.0) \\ &= 744400.9 \text{ N} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 4.304 \cdot 869.924 \cdot 77.647 \\ &= 290723.8 \text{ N} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (744401 + 290723.8 + 36466.7) / 4.304 \\ &= 248974.6 \text{ mm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff}) \\ &= 206.995 / (2 \cdot 248974.594 / 8188.675 - 869.924 / 28.0) \\ &= 7.0 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S / [t / R_{xs}] \\ &= 137.997 [28.0 / 869.924] \\ &= 4.4 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(6.96, 4.442) \\ &= 4.442 \text{ MPa} \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (f_N + f_S + f_Y) / A_T \\ &= (36466.711 + 744401 + 290723.75) / 8188.675 \\ &= 130.874 \text{ MPa} \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot R_{xs} / t_{eff} \\ &= 4.304 \cdot 869.924 / 28.0 \\ &= 133.7 \text{ MPa} \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 130.874 - 133.732, 133.732) \\ &= 133.7 \text{ MPa} \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	207.00	MPa
Local Primary Membrane Stress	PL	133.73	MPa
Maximum Allowable Working Pressure	Pmax	4.44	MPa

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (R_{nc} + t_n) / R_{nc} \\ &= (77.647 + 44.0) / 77.647 \end{aligned}$$



= 1.567 For set-in Nozzles

Weld Length of Nozzle to Shell Weld [L_T]:

$$\begin{aligned} &= \pi/2 \cdot (R_n + t_n) \\ &= \pi/2 \cdot (77.647 + 44.0) \\ &= 191.083 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L_{41T}, L_{42T}, L_{43T}]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$t_c, L_{41T} = 7.071, \text{ must be } \geq \min(0.7 \cdot 44.000, 6.000) \text{ Passed}$$

Weld Load Value [f_{welds}]:

$$\begin{aligned} &= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2) \\ &= \min(290724 \cdot 1.57, 1.5 \cdot 147.6(3803.826 + 0.0), \pi/4 \cdot 4.3 \cdot 77.65^2 \cdot 1.57^2) \\ &= 50022.719 \text{ N} \end{aligned}$$

Weld Stress Value [τ]:

$$\begin{aligned} &= f_{welds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T})) \\ &= 50022.719 / (191.083(0.49 \cdot 7.071 + 0.6 \cdot 28.0 + 0.49 \cdot 0.0)) \\ &= 12.919 < \text{or } = \text{ to } 137.997 \text{ Weld Stress Passed} \end{aligned}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [R_{xn}]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 44.0 / \ln[1 + 44.0/77.647] \\ &= 98.006 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [R_{xs}]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 28.0 / \ln[1 + 28.0/856.0] \\ &= 869.924 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [S_{allow}]:

$$\begin{aligned} &= 1.5 \cdot S \\ &= 1.5 \cdot 137.997 \\ &= 207.0 \text{ MPa} \end{aligned}$$

Force acting on the Nozzle [f_N]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot L_H \\ &= 0.103 \cdot 98.006 \cdot 86.451 \\ &= 876.2 \text{ N} \end{aligned}$$

Force acting on the Shell [f_S]:

$$\begin{aligned} &= P \cdot R_{xs}(L_R + t_n) \\ &= 0.103 \cdot 869.924(154.816 + 44.0) \\ &= 17886.3 \text{ N} \end{aligned}$$

Discontinuity Force from External Pressure [f_Y]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 0.103 \cdot 869.924 \cdot 77.647 \\ &= 6985.5 \text{ N} \end{aligned}$$

Area Resisting External Pressure [A_p]:

$$\begin{aligned} &= (f_S + f_Y + f_N)/P \\ &= (17886.3 + 6985.5 + 876.2)/0.103 \\ &= 248974.6 \text{ mm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [P_{max1}]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff}) \\ &= 206.995 / (2 \cdot 248974.594 / 8188.675 - 869.924 / 28.0) \\ &= 7.0 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [P_{max2}]:

$$\begin{aligned} &= S[t/R_{xs}] \\ &= 137.997[28.0/869.924] \\ &= 4.4 \text{ MPa} \end{aligned}$$



Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(6.96, 4.442)$$

$$= 4.442 \text{ MPa}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (876.216 + 17886.334 + 6985.458) / 8188.675$$

$$= 3.145 \text{ MPa}$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 0.103 \cdot 869.924 / 28.0$$

$$= 3.2 \text{ MPa}$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 3.145 - 3.213, 3.213)$$

$$= 3.2 \text{ MPa}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	207.00	MPa
Local Primary Membrane Stress	PL	3.21	MPa
Maximum Allowable External Pressure	Pmax	4.44	MPa

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -63 °C

The Drop for this Nozzle is : 8.7498 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 342.7497 mm

Input Echo, WRC297 Item 1, Description: B150 :

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	6.0000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	1700.000	mm	
Shell Thickness	34.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-516 70		
Vessel Cold S.I. Allowable	Smc	138.00	MPa
Vessel Hot S.I. Allowable	Smh	138.00	MPa
Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance	6.0000	mm	
Nozzle Diameter	143.294	mm	
Nozzle Thickness	50.0000	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-105		
Nozzle Cold S.I. Allowable	SNmc	165.00	MPa
Nozzle Hot S.I. Allowable	SNmh	147.60	MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.



Design Internal Pressure	Dp	4.30	MPa
Radial Load	P	-11245.70	N
Circumferential Shear	Vc	7497.10	N
Longitudinal Shear	Vl	7497.10	N
Circumferential Moment	Mc	0.315E+07	N-mm
Longitudinal Moment	Ml	0.315E+07	N-mm
Torsional Moment	Mt	0.450E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (4.866) is < 10
 check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 243.294 + 2 \cdot 1.65 \cdot \sqrt{(870.0(34.0 - 6.0))}$
 $= 758.348 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1740.000 mm
Nozzle Outside Diameter	(d) =	243.294 mm
Vessel Thickness used	(T) =	28.000 mm
Nozzle Thickness used	(t) =	44.000 mm
T / t	=	0.636
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.102$$

Nr/P	=	0.105
Mr/P	=	0.201
M0/P	=	0.061
N0/P	=	0.125
MrD/Mc	=	0.298
NrDL/Mc	=	0.107
M0d/Mc	=	0.098
N0DL/Mc	=	0.042
MrD/Ml	=	0.184
NrDL/Ml	=	0.091
M0D/Ml	=	0.062
N0DL/Ml	=	0.124

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-2	-2	2	2
Outplane Bending (Ml)	-6	6	6	-6
Normal Pressure Stress	129	133	129	133
Outplane Stress Summary	128	134	144	126

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	17	-17	17	-17
Inplane Membrane (Mc)	0	0	0	0



Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-18	18	18	-18
Inplane Pressure Stress	64	64	64	64

Inplane Stress Summary	63	65	103	32
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1

Shear Stress Summary	2	2	1	1
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	128	134	144	126
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-9	9	9	-9
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	64	64	64	64

Outplane Stress Summary	61	70	82	52
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	17	-17	17	-17
Inplane Membrane (Mc)	-1	-1	1	1
Inplane Bending (Mc)	-29	29	29	-29
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	129	133	129	133

Inplane Stress Summary	117	145	179	90
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1

Shear Stress Summary	1	1	2	2
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside



Two * Max Shear Stress 117 145 179 90

Nozzle Stresses (MPa):

LONGITUDINAL PLANE
 (Stresses in the
 hoop direction)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-2	-2	2	2
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	7	12	7	12

Hoop Stress Summary 6 12 10 16

Nozzle Stresses (MPa):

LONGITUDINAL PLANE
 (Stresses Normal to
 pipe cross-section)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	5	-5	5	-5
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-2	-2	2	2
Axial Bending (ML)	-5	5	5	-5
Axial Pressure Stress	3	3	3	3

Axial Stress Summary 0 1 17 -4

Nozzle Stresses (MPa):

LONGITUDINAL PLANE
 (Shear stress)

	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1

Shear Stress Summary 1 1 0 0

Nozzle Stresses (MPa):

LONGITUDINAL PLANE
 (Stress Intensities)

Two * Max Shear Stress 7 12 17 20

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Stresses in the
 hoop direction)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	7	12	7	12

Hoop Stress Summary 8 13 9 14

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Stresses Normal to
 pipe cross-section)

Axial Membrane (P) 0 0 0 0

Axial Bending (P)	5	-5	5	-5
Axial Membrane (Mc)	-2	-2	2	2
Axial Bending (Mc)	-9	9	9	-9
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	3	3	3	3

Axial Stress Summary	-3	5	21	-8
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	0	0	1	1

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	11	13	21	23

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	129	133	129	133	129	133	129	133
Circ. Pl (SUS)	-1	-1	3	3	0	0	2	2
Circ. Q (SUS)	-1	1	11	-11	-12	12	46	-46
Long. Pm (SUS)	64	64	64	64	64	64	64	64
Long. Pl (SUS)	0	0	2	2	1	1	1	1
Long. Q (SUS)	-1	1	35	-35	-4	4	14	-14
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	129.0	133.0	129.0	133.0	129.0	133.0	129.0	133.0
Pm+Pl (SUS)	128.0	132.0	132.0	136.0	129.0	133.0	131.0	135.0
Pm+Pl+Q (Total)	127.0	133.0	143.0	125.0	117.0	145.0	177.0	89.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	133.00	138.00	Passed
Pm+Pl (SUS)	136.00	207.00	Passed
Pm+Pl+Q (TOTAL)	177.01	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl



Circ. Pm (SUS)	7	12	7	12	7	12	7	12
Circ. Pl (SUS)	-1	-1	3	3	1	1	1	1
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	3	3	3	3	3	3	3	3
Long. Pl (SUS)	-2	-2	2	2	-2	-2	2	2
Long. Q (SUS)	0	0	10	-10	-4	4	14	-14
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	7.0	12.0	7.0	12.0	7.0	12.0	7.0	12.0
Pm+Pl (SUS)	6.0	11.0	10.0	15.0	8.0	13.0	8.0	13.0
Pm+Pl+Q (Total)	6.2	11.1	15.2	20.1	11.2	13.1	19.1	22.1

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	12.00	147.60	Passed
Pm+Pl (SUS)	15.00	221.40	Passed
Pm+Pl+Q (TOTAL)	22.09	442.80	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: L150

From: 50

Pressure for Reinforcement Calculations	P	4.3074	MPa
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	1700.00	mm
Design Length of Section	L	18483.8340	mm
Shell Finished (Minimum) Thickness	t	34.0000	mm
Shell Internal Corrosion Allowance	c	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		2550.00	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.60	MPa
Allowable Stress At Ambient	Sna	165.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		50.8000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm

Flange Material		SA-105	+IT
Flange Type		N/A	

Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	33.2360	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Style of FVC Nozzle		Long Weld Neck	

Corrosion Allowance	can	6.0000	mm
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Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

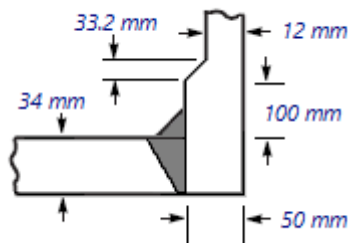
Flange Class		300
Flange Grade		GR 1.1

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: L150

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		50.800	mm.
Actual Thickness Used in Calculation		12.000	mm



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 62.8 \cdot (\exp(4.307/(147.6 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 0.9298 + 6.0 + 0.0 = 6.9298 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 147.6 \cdot 1.0 \cdot \ln((2 \cdot 6.0 + 62.8) / 62.8) - 0.0$$

$$= 25.81 - 0.0 = 25.8098 \text{ MPa}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	10.800 mm
Min. required shell thickness, internal p + c :	33.143 mm
Min. required shell thickness, external p + c :	26.261 mm

Nozzle neck minimum required thickness	: 10.800 mm
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Minimum nozzle neck thickness 11.989 mm , is greater than required thickness 10.800 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[147.6/138.0, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 50.0 - 6.0$$

$$= 44.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 12.0 - 6.0$$

$$= 6.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 1712.0/28.0$$

$$= 61.143 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 1700.0/2 + 6.0$$

$$= 856.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{856.0 \cdot 28.0}, 2 \cdot 31.4)$$

$$= 62.800 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$



$$= \min[1.5 \cdot 28.0, 0.0] + \sqrt{(31.4 \cdot 44.0)}$$
$$= 37.170 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$
$$= 250.0$$
$$= 250.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$
$$= 8(28.0 + 0.0)$$
$$= 224.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles}$$
$$= \min[37.17, 250.0, 224.0] + 28.0$$
$$= 65.170 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$
$$= 28.0 + ((0.0 \cdot 1.0) / 62.8)$$
$$= 28.000 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}}))$$
$$= \min(12, (2 \cdot 31.4 + 44.0) / (\sqrt{(1712.0 + 28.0) \cdot 28.0}))$$
$$= 0.484$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$
$$= 28.0 \cdot 62.8 \cdot \max((0.484/5)^{0.85}, 1)$$
$$= 1758.400 \text{ mm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$
$$= 44.0 \cdot 65.17$$
$$= 2867.475 \text{ mm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$
$$= 0.5 \cdot 10.0^2$$
$$= 50.000 \text{ mm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$
$$= 1758.4 + 1.0(2867.475 + 0.0) + 50.0 + 0.0 + 0.0$$
$$= 4675.875 \text{ mm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$
$$= 44.0 / \ln[1 + 44.0/31.4]$$
$$= 50.228 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$
$$= 28.0 / \ln[1 + 28.0/856.0]$$
$$= 869.924 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$
$$= 1.5 \cdot 137.997 \cdot 1.0$$
$$= 207.0 \text{ MPa}$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$
$$= 4.307 \cdot 50.228 \cdot 65.17$$
$$= 14098.5 \text{ N}$$



Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot Rxs(LR + tn) \\ &= 4.307 \cdot 869.924(62.8 + 44.0) \\ &= 400156.8 \text{ N} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot Rxs \cdot Rnc \\ &= 4.307 \cdot 869.924 \cdot 31.4 \\ &= 117649.1 \text{ N} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN)/P \\ &= (400156.8 + 117649.1 + 14098.5)/4.307 \\ &= 123496.9 \text{ mm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / (2 \cdot Ap/AT - Rxs/teff) \\ &= 206.995 / (2 \cdot 123496.875/4675.875 - 869.924/28.0) \\ &= 9.5 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t/Rxs] \\ &= 137.997[28.0/869.924] \\ &= 4.4 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(Pmax1, Pmax2) \\ &= \min(9.515, 4.442) \\ &= 4.442 \text{ MPa} \end{aligned}$$

Average Primary Membrane Stress [σAvg]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (14098.518 + 400156.812 + 117649.102)/4675.875 \\ &= 113.765 \text{ MPa} \end{aligned}$$

General Primary Membrane Stress [σCirc]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 4.307 \cdot 869.924/28.0 \\ &= 133.8 \text{ MPa} \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ) \\ &= \max(2 \cdot 113.765 - 133.825, 133.825) \\ &= 133.8 \text{ MPa} \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	207.00	MPa
Local Primary Membrane Stress	PL	133.83	MPa
Maximum Allowable Working Pressure	Pmax	4.44	MPa

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (Rnc + tn) / Rnc \\ &= (31.4 + 44.0)/31.4 \\ &= 2.401 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned} &= \pi/2 \cdot (Rn + tn) \\ &= \pi/2 \cdot (31.4 + 44.0) \\ &= 118.438 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 44.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2)$$



$$= \min(117649 \cdot 2.4, 1.5 \cdot 147.6(2867.475 + 0.0), \pi/4 \cdot 4.3 \cdot 31.4^2 \cdot 2.4^2)$$
$$= 19231.371 \text{ N}$$

Weld Stress Value [τ]:

$$= \text{fwelds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$$
$$= 19231.371 / (118.438(0.49 \cdot 7.071 + 0.6 \cdot 28.0 + 0.49 \cdot 0.0))$$
$$= 8.013 < \text{or} = \text{to } 137.997 \text{ Weld Stress Passed}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [R_{xn}]:

$$= t_n / \ln[1 + t_n/R_n]$$
$$= 44.0 / \ln[1 + 44.0/31.4]$$
$$= 50.228 \text{ mm}$$

Shell Radius for Force Calculation [R_{xs}]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$
$$= 28.0 / \ln[1 + 28.0/856.0]$$
$$= 869.924 \text{ mm}$$

Allowable Local Primary Membrane Stress [S_{allow}]:

$$= 1.5 \cdot S$$
$$= 1.5 \cdot 137.997$$
$$= 207.0 \text{ MPa}$$

Force acting on the Nozzle [f_N]:

$$= P \cdot R_{xn} \cdot L_H$$
$$= 0.103 \cdot 50.228 \cdot 65.17$$
$$= 338.5 \text{ N}$$

Force acting on the Shell [f_S]:

$$= P \cdot R_{xs}(L_R + t_n)$$
$$= 0.103 \cdot 869.924(62.8 + 44.0)$$
$$= 9608.2 \text{ N}$$

Discontinuity Force from External Pressure [f_Y]:

$$= P \cdot R_{xs} \cdot R_{nc}$$
$$= 0.103 \cdot 869.924 \cdot 31.4$$
$$= 2824.9 \text{ N}$$

Area Resisting External Pressure [A_p]:

$$= (f_S + f_Y + f_N) / P$$
$$= (9608.2 + 2824.9 + 338.5) / 0.103$$
$$= 123496.9 \text{ mm}^2$$

Maximum Allowable Working Pressure Candidate [P_{max1}]:

$$= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff})$$
$$= 206.995 / (2 \cdot 123496.883 / 4675.875 - 869.924 / 28.0)$$
$$= 9.5 \text{ MPa}$$

Maximum Allowable Working Pressure Candidate [P_{max2}]:

$$= S / (t / R_{xs})$$
$$= 137.997[28.0 / 869.924]$$
$$= 4.4 \text{ MPa}$$

Maximum Allowable Working Pressure [P_{max}]:

$$= \min(P_{max1}, P_{max2})$$
$$= \min(9.515, 4.442)$$
$$= 4.442 \text{ MPa}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / AT$$
$$= (338.52 + 9608.182 + 2824.877) / 4675.875$$
$$= 2.732 \text{ MPa}$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$
$$= 0.103 \cdot 869.924 / 28.0$$
$$= 3.2 \text{ MPa}$$



Maximum Local Primary Membrane Stress [PL]:

= max(2 • σ_{Avg} - σ_{Circ} , σ_{Circ})
= max(2 • 2.732 - 3.213, 3.213)
= 3.2 MPa

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	207.00	MPa
Local Primary Membrane Stress	PL	3.21	MPa
Maximum Allowable External Pressure	Pmax	4.44	MPa

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

The Drop for this Nozzle is : 3.3508 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 287.3507 mm

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Input, Nozzle Desc: L250

From: 50

Pressure for Reinforcement Calculations	P	4.3305	MPa
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	1700.00	mm
Design Length of Section	L	18483.8340	mm
Shell Finished (Minimum) Thickness	t	34.0000	mm
Shell Internal Corrosion Allowance	c	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		250.00	mm
-----------------------------------	--	--------	----

User Entered Minimum Design Metal Temperature		-12.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.60	MPa
Allowable Stress At Ambient	Sna	165.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		50.8000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm

Flange Material		SA-105	+IT
Flange Type		N/A	

Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	33.2360	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Style of FVC Nozzle		Long Weld Neck	

Corrosion Allowance	can	6.0000	mm
---------------------	-----	--------	----

Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

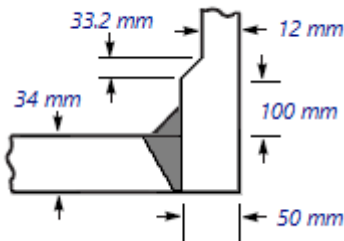
Flange Class		300
Flange Grade		GR 1.1

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: L250

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		50.800	mm.
Actual Thickness Used in Calculation		12.000	mm



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 62.8 \cdot (\exp(4.33/(147.6 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 0.9349 + 6.0 + 0.0 = 6.9349 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 147.6 \cdot 1.0 \cdot \ln((2 \cdot 6.0 + 62.8) / 62.8) - 0.0$$

$$= 25.81 - 0.0 = 25.8098 \text{ MPa}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	10.800 mm
Min. required shell thickness, internal p + c :	33.290 mm
Min. required shell thickness, external p + c :	26.261 mm

Nozzle neck minimum required thickness	: 10.800 mm
--	-------------

Minimum nozzle neck thickness 11.989 mm , is greater than required thickness 10.800 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[147.6/138.0, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 50.0 - 6.0$$

$$= 44.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 12.0 - 6.0$$

$$= 6.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 1712.0/28.0$$

$$= 61.143 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 1700.0/2 + 6.0$$

$$= 856.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{856.0 \cdot 28.0}, 2 \cdot 31.4)$$

$$= 62.800 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$



$$= \min[1.5 \cdot 28.0, 0.0] + \sqrt{(31.4 \cdot 44.0)}$$
$$= 37.170 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$
$$= 250.0$$
$$= 250.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$
$$= 8(28.0 + 0.0)$$
$$= 224.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles}$$
$$= \min[37.17, 250.0, 224.0] + 28.0$$
$$= 65.170 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$
$$= 28.0 + ((0.0 \cdot 1.0) / 62.8)$$
$$= 28.000 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}}))$$
$$= \min(12, (2 \cdot 31.4 + 44.0) / (\sqrt{(1712.0 + 28.0) \cdot 28.0}))$$
$$= 0.484$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$
$$= 28.0 \cdot 62.8 \cdot \max((0.484/5)^{0.85}, 1)$$
$$= 1758.400 \text{ mm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$
$$= 44.0 \cdot 65.17$$
$$= 2867.475 \text{ mm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$
$$= 0.5 \cdot 10.0^2$$
$$= 50.000 \text{ mm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$
$$= 1758.4 + 1.0(2867.475 + 0.0) + 50.0 + 0.0 + 0.0$$
$$= 4675.875 \text{ mm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$
$$= 44.0 / \ln[1 + 44.0/31.4]$$
$$= 50.228 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$
$$= 28.0 / \ln[1 + 28.0/856.0]$$
$$= 869.924 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$
$$= 1.5 \cdot 137.997 \cdot 1.0$$
$$= 207.0 \text{ MPa}$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$
$$= 4.33 \cdot 50.228 \cdot 65.17$$
$$= 14174.0 \text{ N}$$



Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot Rxs(LR + tn) \\ &= 4.33 \cdot 869.924(62.8 + 44.0) \\ &= 402299.5 \text{ N} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot Rxs \cdot Rnc \\ &= 4.33 \cdot 869.924 \cdot 31.4 \\ &= 118279.1 \text{ N} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN)/P \\ &= (402299.5 + 118279.1 + 14174.0)/4.33 \\ &= 123496.9 \text{ mm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / (2 \cdot Ap/AT - Rxs/teff) \\ &= 206.995 / (2 \cdot 123496.867/4675.875 - 869.924/28.0) \\ &= 9.5 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t/Rxs] \\ &= 137.997[28.0/869.924] \\ &= 4.4 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(Pmax1, Pmax2) \\ &= \min(9.515, 4.442) \\ &= 4.442 \text{ MPa} \end{aligned}$$

Average Primary Membrane Stress [σAvg]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (14174.011 + 402299.5 + 118279.07)/4675.875 \\ &= 114.374 \text{ MPa} \end{aligned}$$

General Primary Membrane Stress [σCirc]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 4.33 \cdot 869.924/28.0 \\ &= 134.5 \text{ MPa} \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ) \\ &= \max(2 \cdot 114.374 - 134.542, 134.542) \\ &= 134.5 \text{ MPa} \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	207.00	MPa
Local Primary Membrane Stress	PL	134.54	MPa
Maximum Allowable Working Pressure	Pmax	4.44	MPa

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (Rnc + tn) / Rnc \\ &= (31.4 + 44.0)/31.4 \\ &= 2.401 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned} &= \pi/2 \cdot (Rn + tn) \\ &= \pi/2 \cdot (31.4 + 44.0) \\ &= 118.438 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 44.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2)$$



$$= \min(118279 \cdot 2.4, 1.5 \cdot 147.6(2867.475 + 0.0), \pi/4 \cdot 4.3 \cdot 31.4^2 \cdot 2.4^2)$$
$$= 19334.348 \text{ N}$$

Weld Stress Value [τ]:

$$= \text{fwelds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$$
$$= 19334.348 / (118.438(0.49 \cdot 7.071 + 0.6 \cdot 28.0 + 0.49 \cdot 0.0))$$
$$= 8.056 < \text{or} = \text{to } 137.997 \text{ Weld Stress Passed}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [R_{xn}]:

$$= t_n / \ln[1 + t_n/R_n]$$
$$= 44.0 / \ln[1 + 44.0/31.4]$$
$$= 50.228 \text{ mm}$$

Shell Radius for Force Calculation [R_{xs}]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$
$$= 28.0 / \ln[1 + 28.0/856.0]$$
$$= 869.924 \text{ mm}$$

Allowable Local Primary Membrane Stress [S_{allow}]:

$$= 1.5 \cdot S$$
$$= 1.5 \cdot 137.997$$
$$= 207.0 \text{ MPa}$$

Force acting on the Nozzle [f_N]:

$$= P \cdot R_{xn} \cdot L_H$$
$$= 0.103 \cdot 50.228 \cdot 65.17$$
$$= 338.5 \text{ N}$$

Force acting on the Shell [f_S]:

$$= P \cdot R_{xs}(L_R + t_n)$$
$$= 0.103 \cdot 869.924(62.8 + 44.0)$$
$$= 9608.2 \text{ N}$$

Discontinuity Force from External Pressure [f_Y]:

$$= P \cdot R_{xs} \cdot R_{nc}$$
$$= 0.103 \cdot 869.924 \cdot 31.4$$
$$= 2824.9 \text{ N}$$

Area Resisting External Pressure [A_p]:

$$= (f_S + f_Y + f_N) / P$$
$$= (9608.2 + 2824.9 + 338.5) / 0.103$$
$$= 123496.9 \text{ mm}^2$$

Maximum Allowable Working Pressure Candidate [P_{max1}]:

$$= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff})$$
$$= 206.995 / (2 \cdot 123496.883 / 4675.875 - 869.924 / 28.0)$$
$$= 9.5 \text{ MPa}$$

Maximum Allowable Working Pressure Candidate [P_{max2}]:

$$= S / (t / R_{xs})$$
$$= 137.997[28.0 / 869.924]$$
$$= 4.4 \text{ MPa}$$

Maximum Allowable Working Pressure [P_{max}]:

$$= \min(P_{max1}, P_{max2})$$
$$= \min(9.515, 4.442)$$
$$= 4.442 \text{ MPa}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / AT$$
$$= (338.52 + 9608.182 + 2824.877) / 4675.875$$
$$= 2.732 \text{ MPa}$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$
$$= 0.103 \cdot 869.924 / 28.0$$
$$= 3.2 \text{ MPa}$$



Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}}) \\ &= \max(2 \cdot 2.732 - 3.213, 3.213) \\ &= 3.2 \text{ MPa} \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	207.00	MPa
Local Primary Membrane Stress	PL	3.21	MPa
Maximum Allowable External Pressure	Pmax	4.44	MPa

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

The Drop for this Nozzle is : 3.3508 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 287.3507 mm

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Input, Nozzle Desc: L350

From: 50

Pressure for Reinforcement Calculations	P	4.3124	MPa
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	1700.00	mm
Design Length of Section	L	18483.8340	mm
Shell Finished (Minimum) Thickness	t	34.0000	mm
Shell Internal Corrosion Allowance	c	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		2050.00	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.60	MPa
Allowable Stress At Ambient	Sna	165.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		50.8000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm

Flange Material		SA-105	+IT
Flange Type		N/A	

Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	33.2360	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Style of FVC Nozzle		Long Weld Neck	

Corrosion Allowance	can	6.0000	mm
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Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

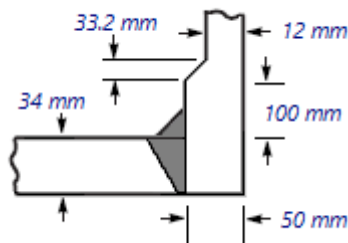
Flange Class		300
Flange Grade		GR 1.1

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: L350

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	50.800	mm.
Actual Thickness Used in Calculation	12.000	mm



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 62.8 \cdot (\exp(4.312/(147.6 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 0.9309 + 6.0 + 0.0 = 6.9309 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 147.6 \cdot 1.0 \cdot \ln((2 \cdot 6.0 + 62.8) / 62.8) - 0.0$$

$$= 25.81 - 0.0 = 25.8098 \text{ MPa}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	10.800 mm
Min. required shell thickness, internal p + c :	33.175 mm
Min. required shell thickness, external p + c :	26.261 mm

Nozzle neck minimum required thickness	: 10.800 mm
--	-------------

Minimum nozzle neck thickness 11.989 mm , is greater than required thickness 10.800 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[147.6/138.0, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 50.0 - 6.0$$

$$= 44.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 12.0 - 6.0$$

$$= 6.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 1712.0/28.0$$

$$= 61.143 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 1700.0/2 + 6.0$$

$$= 856.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{856.0 \cdot 28.0}, 2 \cdot 31.4)$$

$$= 62.800 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$



$$= \min[1.5 \cdot 28.0, 0.0] + \sqrt{(31.4 \cdot 44.0)}$$
$$= 37.170 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$
$$= 250.0$$
$$= 250.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$
$$= 8(28.0 + 0.0)$$
$$= 224.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$
$$= \min[37.17, 250.0, 224.0] + 28.0$$
$$= 65.170 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$
$$= 28.0 + ((0.0 \cdot 1.0) / 62.8)$$
$$= 28.000 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}}))$$
$$= \min(12, (2 \cdot 31.4 + 44.0) / (\sqrt{(1712.0 + 28.0) \cdot 28.0}))$$
$$= 0.484$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$
$$= 28.0 \cdot 62.8 \cdot \max((0.484/5)^{0.85}, 1)$$
$$= 1758.400 \text{ mm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$
$$= 44.0 \cdot 65.17$$
$$= 2867.475 \text{ mm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$
$$= 0.5 \cdot 10.0^2$$
$$= 50.000 \text{ mm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$
$$= 1758.4 + 1.0(2867.475 + 0.0) + 50.0 + 0.0 + 0.0$$
$$= 4675.875 \text{ mm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$
$$= 44.0 / \ln[1 + 44.0/31.4]$$
$$= 50.228 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$
$$= 28.0 / \ln[1 + 28.0/856.0]$$
$$= 869.924 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$
$$= 1.5 \cdot 137.997 \cdot 1.0$$
$$= 207.0 \text{ MPa}$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$
$$= 4.312 \cdot 50.228 \cdot 65.17$$
$$= 14114.9 \text{ N}$$



Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot Rxs(LR + tn) \\ &= 4.312 \cdot 869.924(62.8 + 44.0) \\ &= 400622.6 \text{ N} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot Rxs \cdot Rnc \\ &= 4.312 \cdot 869.924 \cdot 31.4 \\ &= 117786.0 \text{ N} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN)/P \\ &= (400622.6 + 117786.0 + 14114.9)/4.312 \\ &= 123496.9 \text{ mm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / (2 \cdot Ap/AT - Rxs/teff) \\ &= 206.995 / (2 \cdot 123496.867/4675.875 - 869.924/28.0) \\ &= 9.5 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t/Rxs] \\ &= 137.997[28.0/869.924] \\ &= 4.4 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(Pmax1, Pmax2) \\ &= \min(9.515, 4.442) \\ &= 4.442 \text{ MPa} \end{aligned}$$

Average Primary Membrane Stress [σAvg]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (14114.929 + 400622.594 + 117786.047)/4675.875 \\ &= 113.897 \text{ MPa} \end{aligned}$$

General Primary Membrane Stress [σCirc]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 4.312 \cdot 869.924/28.0 \\ &= 134.0 \text{ MPa} \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ) \\ &= \max(2 \cdot 113.897 - 133.981, 133.981) \\ &= 134.0 \text{ MPa} \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	207.00	MPa
Local Primary Membrane Stress	PL	133.98	MPa
Maximum Allowable Working Pressure	Pmax	4.44	MPa

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (Rnc + tn) / Rnc \\ &= (31.4 + 44.0)/31.4 \\ &= 2.401 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned} &= \pi/2 \cdot (Rn + tn) \\ &= \pi/2 \cdot (31.4 + 44.0) \\ &= 118.438 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 44.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2)$$



$$= \min(117786 \cdot 2.4, 1.5 \cdot 147.6(2867.475 + 0.0), \pi/4 \cdot 4.3 \cdot 31.4^2 \cdot 2.4^2)$$
$$= 19253.756 \text{ N}$$

Weld Stress Value [τ]:

$$= \text{fwelds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$$
$$= 19253.756 / (118.438(0.49 \cdot 7.071 + 0.6 \cdot 28.0 + 0.49 \cdot 0.0))$$
$$= 8.023 < \text{or} = \text{to } 137.997 \text{ Weld Stress Passed}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [R_{xn}]:

$$= t_n / \ln[1 + t_n/R_n]$$
$$= 44.0 / \ln[1 + 44.0/31.4]$$
$$= 50.228 \text{ mm}$$

Shell Radius for Force Calculation [R_{xs}]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$
$$= 28.0 / \ln[1 + 28.0/856.0]$$
$$= 869.924 \text{ mm}$$

Allowable Local Primary Membrane Stress [S_{allow}]:

$$= 1.5 \cdot S$$
$$= 1.5 \cdot 137.997$$
$$= 207.0 \text{ MPa}$$

Force acting on the Nozzle [f_N]:

$$= P \cdot R_{xn} \cdot L_H$$
$$= 0.103 \cdot 50.228 \cdot 65.17$$
$$= 338.5 \text{ N}$$

Force acting on the Shell [f_S]:

$$= P \cdot R_{xs} (L_R + t_n)$$
$$= 0.103 \cdot 869.924 (62.8 + 44.0)$$
$$= 9608.2 \text{ N}$$

Discontinuity Force from External Pressure [f_Y]:

$$= P \cdot R_{xs} \cdot R_{nc}$$
$$= 0.103 \cdot 869.924 \cdot 31.4$$
$$= 2824.9 \text{ N}$$

Area Resisting External Pressure [A_p]:

$$= (f_S + f_Y + f_N) / P$$
$$= (9608.2 + 2824.9 + 338.5) / 0.103$$
$$= 123496.9 \text{ mm}^2$$

Maximum Allowable Working Pressure Candidate [P_{max1}]:

$$= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff})$$
$$= 206.995 / (2 \cdot 123496.883 / 4675.875 - 869.924 / 28.0)$$
$$= 9.5 \text{ MPa}$$

Maximum Allowable Working Pressure Candidate [P_{max2}]:

$$= S / (t / R_{xs})$$
$$= 137.997 [28.0 / 869.924]$$
$$= 4.4 \text{ MPa}$$

Maximum Allowable Working Pressure [P_{max}]:

$$= \min(P_{max1}, P_{max2})$$
$$= \min(9.515, 4.442)$$
$$= 4.442 \text{ MPa}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / AT$$
$$= (338.52 + 9608.182 + 2824.877) / 4675.875$$
$$= 2.732 \text{ MPa}$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$
$$= 0.103 \cdot 869.924 / 28.0$$
$$= 3.2 \text{ MPa}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}}) \\ &= \max(2 \cdot 2.732 - 3.213, 3.213) \\ &= 3.2 \text{ MPa} \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	207.00	MPa
Local Primary Membrane Stress	PL	3.21	MPa
Maximum Allowable External Pressure	Pmax	4.44	MPa

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

The Drop for this Nozzle is : 3.3508 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 287.3507 mm

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Input, Nozzle Desc: L450

From: 50

Pressure for Reinforcement Calculations	P	4.3224	MPa
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	1700.00	mm
Design Length of Section	L	18483.8340	mm
Shell Finished (Minimum) Thickness	t	34.0000	mm
Shell Internal Corrosion Allowance	c	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		1050.00	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.60	MPa
Allowable Stress At Ambient	Sna	165.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		50.8000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm

Flange Material		SA-105	+IT
Flange Type		N/A	

Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	33.2360	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Style of FVC Nozzle		Long Weld Neck	

Corrosion Allowance	can	6.0000	mm
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Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

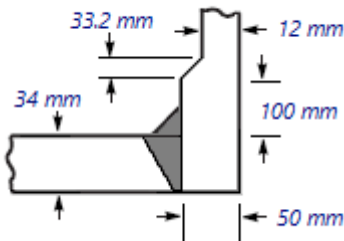
Flange Class		300
Flange Grade		GR 1.1

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: L450

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		50.800	mm.
Actual Thickness Used in Calculation		12.000	mm



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 62.8 \cdot (\exp(4.322/(147.6 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 0.9331 + 6.0 + 0.0 = 6.9331 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 147.6 \cdot 1.0 \cdot \ln((2 \cdot 6.0 + 62.8) / 62.8) - 0.0$$

$$= 25.81 - 0.0 = 25.8098 \text{ MPa}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	10.800 mm
Min. required shell thickness, internal p + c :	33.239 mm
Min. required shell thickness, external p + c :	26.261 mm

Nozzle neck minimum required thickness	: 10.800 mm
--	-------------

Minimum nozzle neck thickness 11.989 mm , is greater than required thickness 10.800 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[147.6/138.0, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 50.0 - 6.0$$

$$= 44.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 12.0 - 6.0$$

$$= 6.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 1712.0/28.0$$

$$= 61.143 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 1700.0/2 + 6.0$$

$$= 856.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{856.0 \cdot 28.0}, 2 \cdot 31.4)$$

$$= 62.800 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$



$$= \min[1.5 \cdot 28.0, 0.0] + \sqrt{(31.4 \cdot 44.0)}$$
$$= 37.170 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$
$$= 250.0$$
$$= 250.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$
$$= 8(28.0 + 0.0)$$
$$= 224.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles}$$
$$= \min[37.17, 250.0, 224.0] + 28.0$$
$$= 65.170 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$
$$= 28.0 + ((0.0 \cdot 1.0) / 62.8)$$
$$= 28.000 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}}))$$
$$= \min(12, (2 \cdot 31.4 + 44.0) / (\sqrt{(1712.0 + 28.0) \cdot 28.0}))$$
$$= 0.484$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$
$$= 28.0 \cdot 62.8 \cdot \max((0.484/5)^{0.85}, 1)$$
$$= 1758.400 \text{ mm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$
$$= 44.0 \cdot 65.17$$
$$= 2867.475 \text{ mm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$
$$= 0.5 \cdot 10.0^2$$
$$= 50.000 \text{ mm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$
$$= 1758.4 + 1.0(2867.475 + 0.0) + 50.0 + 0.0 + 0.0$$
$$= 4675.875 \text{ mm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$
$$= 44.0 / \ln[1 + 44.0/31.4]$$
$$= 50.228 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$
$$= 28.0 / \ln[1 + 28.0/856.0]$$
$$= 869.924 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$
$$= 1.5 \cdot 137.997 \cdot 1.0$$
$$= 207.0 \text{ MPa}$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$
$$= 4.322 \cdot 50.228 \cdot 65.17$$
$$= 14147.8 \text{ N}$$



Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot Rxs(LR + tn) \\ &= 4.322 \cdot 869.924(62.8 + 44.0) \\ &= 401554.2 \text{ N} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot Rxs \cdot Rnc \\ &= 4.322 \cdot 869.924 \cdot 31.4 \\ &= 118060.0 \text{ N} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN)/P \\ &= (401554.2 + 118060.0 + 14147.8)/4.322 \\ &= 123496.9 \text{ mm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / (2 \cdot Ap/AT - Rxs/teff) \\ &= 206.995 / (2 \cdot 123496.875/4675.875 - 869.924/28.0) \\ &= 9.5 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t/Rxs] \\ &= 137.997[28.0/869.924] \\ &= 4.4 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(Pmax1, Pmax2) \\ &= \min(9.515, 4.442) \\ &= 4.442 \text{ MPa} \end{aligned}$$

Average Primary Membrane Stress [σAvg]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (14147.752 + 401554.219 + 118059.953)/4675.875 \\ &= 114.162 \text{ MPa} \end{aligned}$$

General Primary Membrane Stress [σCirc]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 4.322 \cdot 869.924/28.0 \\ &= 134.3 \text{ MPa} \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ) \\ &= \max(2 \cdot 114.162 - 134.293, 134.293) \\ &= 134.3 \text{ MPa} \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	207.00	MPa
Local Primary Membrane Stress	PL	134.29	MPa
Maximum Allowable Working Pressure	Pmax	4.44	MPa

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (Rnc + tn) / Rnc \\ &= (31.4 + 44.0)/31.4 \\ &= 2.401 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned} &= \pi/2 \cdot (Rn + tn) \\ &= \pi/2 \cdot (31.4 + 44.0) \\ &= 118.438 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 44.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2)$$



$$= \min(118060 \cdot 2.4, 1.5 \cdot 147.6(2867.475 + 0.0), \pi/4 \cdot 4.3 \cdot 31.4^2 \cdot 2.4^2) \\ = 19298.529 \text{ N}$$

Weld Stress Value [τ]:

$$= \text{fwelds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T})) \\ = 19298.529 / (118.438(0.49 \cdot 7.071 + 0.6 \cdot 28.0 + 0.49 \cdot 0.0)) \\ = 8.041 < \text{or} = \text{to } 137.997 \text{ Weld Stress Passed}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [R_{xn}]:

$$= t_n / \ln[1 + t_n/R_n] \\ = 44.0 / \ln[1 + 44.0/31.4] \\ = 50.228 \text{ mm}$$

Shell Radius for Force Calculation [R_{xs}]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ = 28.0 / \ln[1 + 28.0/856.0] \\ = 869.924 \text{ mm}$$

Allowable Local Primary Membrane Stress [S_{allow}]:

$$= 1.5 \cdot S \\ = 1.5 \cdot 137.997 \\ = 207.0 \text{ MPa}$$

Force acting on the Nozzle [f_N]:

$$= P \cdot R_{xn} \cdot L_H \\ = 0.103 \cdot 50.228 \cdot 65.17 \\ = 338.5 \text{ N}$$

Force acting on the Shell [f_S]:

$$= P \cdot R_{xs} (L_R + t_n) \\ = 0.103 \cdot 869.924 (62.8 + 44.0) \\ = 9608.2 \text{ N}$$

Discontinuity Force from External Pressure [f_Y]:

$$= P \cdot R_{xs} \cdot R_{nc} \\ = 0.103 \cdot 869.924 \cdot 31.4 \\ = 2824.9 \text{ N}$$

Area Resisting External Pressure [A_p]:

$$= (f_S + f_Y + f_N) / P \\ = (9608.2 + 2824.9 + 338.5) / 0.103 \\ = 123496.9 \text{ mm}^2$$

Maximum Allowable Working Pressure Candidate [P_{max1}]:

$$= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff}) \\ = 206.995 / (2 \cdot 123496.883 / 4675.875 - 869.924 / 28.0) \\ = 9.5 \text{ MPa}$$

Maximum Allowable Working Pressure Candidate [P_{max2}]:

$$= S / (t / R_{xs}) \\ = 137.997 [28.0 / 869.924] \\ = 4.4 \text{ MPa}$$

Maximum Allowable Working Pressure [P_{max}]:

$$= \min(P_{max1}, P_{max2}) \\ = \min(9.515, 4.442) \\ = 4.442 \text{ MPa}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / AT \\ = (338.52 + 9608.182 + 2824.877) / 4675.875 \\ = 2.732 \text{ MPa}$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff} \\ = 0.103 \cdot 869.924 / 28.0 \\ = 3.2 \text{ MPa}$$



Maximum Local Primary Membrane Stress [PL]:

= max(2 • σ_{Avg} - σ_{Circ} , σ_{Circ})
= max(2 • 2.732 - 3.213, 3.213)
= 3.2 MPa

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	207.00	MPa
Local Primary Membrane Stress	PL	3.21	MPa
Maximum Allowable External Pressure	Pmax	4.44	MPa

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

The Drop for this Nozzle is : 3.3508 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 287.3507 mm

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Input, Nozzle Desc: L550

From: 50

Pressure for Reinforcement Calculations	P	4.3124	MPa
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	1700.00	mm
Design Length of Section	L	18483.8340	mm
Shell Finished (Minimum) Thickness	t	34.0000	mm
Shell Internal Corrosion Allowance	c	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		2050.00	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.60	MPa
Allowable Stress At Ambient	Sna	165.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		50.8000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm

Flange Material		SA-105	+IT
Flange Type		N/A	

Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	33.2360	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Style of FVC Nozzle		Long Weld Neck	

Corrosion Allowance	can	6.0000	mm
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Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

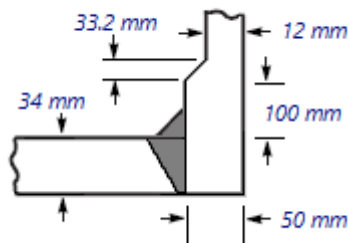
Flange Class		300
Flange Grade		GR 1.1

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: L550

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		50.800	mm.
Actual Thickness Used in Calculation		12.000	mm



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 62.8 \cdot (\exp(4.312/(147.6 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 0.9309 + 6.0 + 0.0 = 6.9309 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 147.6 \cdot 1.0 \cdot \ln((2 \cdot 6.0 + 62.8) / 62.8) - 0.0$$

$$= 25.81 - 0.0 = 25.8098 \text{ MPa}$$

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 138.20 mm

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	10.800 mm
Min. required shell thickness, internal p + c :	33.175 mm
Min. required shell thickness, external p + c :	26.261 mm

Nozzle neck minimum required thickness :	10.800 mm
--	-----------

Minimum nozzle neck thickness 11.989 mm , is greater than required thickness 10.800 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[147.6/138.0, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 50.0 - 6.0$$

$$= 44.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 12.0 - 6.0$$

$$= 6.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 1712.0/28.0$$

$$= 61.143 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 1700.0/2 + 6.0$$

$$= 856.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{856.0 \cdot 28.0}, 2 \cdot 31.4)$$



$$= 62.800 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$\begin{aligned} &= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)} \\ &= \min[1.5 \cdot 28.0, 0.0] + \sqrt{(31.4 \cdot 44.0)} \\ &= 37.170 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 250.0 \\ &= 250.000 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(28.0 + 0.0) \\ &= 224.000 \text{ mm} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles} \\ &= \min[37.17, 250.0, 224.0] + 28.0 \\ &= 65.170 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot frp) / LR) \\ &= 28.0 + ((0.0 \cdot 1.0) / 62.8) \\ &= 28.000 \text{ mm} \end{aligned}$$

Parameter [λ]:

$$\begin{aligned} &= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + teff) \cdot teff})) \\ &= \min(12, (2 \cdot 31.4 + 44.0) / (\sqrt{(1712.0 + 28.0) \cdot 28.0})) \\ &= 0.484 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ &= 28.0 \cdot 62.8 \cdot \max((0.484/5)^{0.85}, 1) \\ &= 1758.400 \text{ mm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot LH \\ &= 44.0 \cdot 65.17 \\ &= 2867.475 \text{ mm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot Leg^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 50.000 \text{ mm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + frn(A2 + A3) + A41 + A42 + A43 \\ &= 1758.4 + 1.0(2867.475 + 0.0) + 50.0 + 0.0 + 0.0 \\ &= 4675.875 \text{ mm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 44.0 / \ln[1 + 44.0/31.4] \\ &= 50.228 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= teff / \ln[1 + teff/Reff] \\ &= 28.0 / \ln[1 + 28.0/856.0] \\ &= 869.924 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 137.997 \cdot 1.0 \\ &= 207.0 \text{ MPa} \end{aligned}$$

Force acting on the Nozzle [fN]:



$$\begin{aligned} &= P \cdot Rxn \cdot LH \\ &= 4.312 \cdot 50.228 \cdot 65.17 \\ &= 14114.9 \text{ N} \end{aligned}$$

$$\begin{aligned} &\text{Force acting on the Shell [fS]:} \\ &= P \cdot Rxs(LR + tn) \\ &= 4.312 \cdot 869.924(62.8 + 44.0) \\ &= 400622.6 \text{ N} \end{aligned}$$

$$\begin{aligned} &\text{Discontinuity Force from Internal Pressure [fY]:} \\ &= P \cdot Rxs \cdot Rnc \\ &= 4.312 \cdot 869.924 \cdot 31.4 \\ &= 117786.0 \text{ N} \end{aligned}$$

$$\begin{aligned} &\text{Area Resisting Internal Pressure [Ap]:} \\ &= (fS + fY + fN)/P \\ &= (400622.6 + 117786.0 + 14114.9)/4.312 \\ &= 123496.9 \text{ mm}^2 \end{aligned}$$

$$\begin{aligned} &\text{Maximum Allowable Working Pressure Candidate [Pmax1]:} \\ &= Sallow / (2 \cdot Ap/AT - Rxs/teff) \\ &= 206.995 / (2 \cdot 123496.867/4675.875 - 869.924/28.0) \\ &= 9.5 \text{ MPa} \end{aligned}$$

$$\begin{aligned} &\text{Maximum Allowable Working Pressure Candidate [Pmax2]:} \\ &= S[t/Rxs] \\ &= 137.997[28.0/869.924] \\ &= 4.4 \text{ MPa} \end{aligned}$$

$$\begin{aligned} &\text{Maximum Allowable Working Pressure [Pmax]:} \\ &= \min(Pmax1, Pmax2) \\ &= \min(9.515, 4.442) \\ &= 4.442 \text{ MPa} \end{aligned}$$

$$\begin{aligned} &\text{Average Primary Membrane Stress [\sigmaAvg]:} \\ &= (fN + fS + fY) / AT \\ &= (14114.929 + 400622.594 + 117786.047)/4675.875 \\ &= 113.897 \text{ MPa} \end{aligned}$$

$$\begin{aligned} &\text{General Primary Membrane Stress [\sigmaCirc]:} \\ &= P \cdot Rxs / teff \\ &= 4.312 \cdot 869.924/28.0 \\ &= 134.0 \text{ MPa} \end{aligned}$$

$$\begin{aligned} &\text{Maximum Local Primary Membrane Stress [PL]:} \\ &= \max(2 \cdot \sigmaAvg - \sigmaCirc, \sigmaCirc) \\ &= \max(2 \cdot 113.897 - 133.981, 133.981) \\ &= 134.0 \text{ MPa} \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	207.00	MPa
Local Primary Membrane Stress	PL	133.98	MPa
Maximum Allowable Working Pressure	Pmax	4.44	MPa

Strength of Nozzle Attachment Welds per 4.5.14

$$\begin{aligned} &\text{Discontinuity Force Factor [ky]:} \\ &= (Rnc + tn) / Rnc \\ &= (31.4 + 44.0)/31.4 \\ &= 2.401 \text{ For set-in Nozzles} \end{aligned}$$

$$\begin{aligned} &\text{Weld Length of Nozzle to Shell Weld [Lt]:} \\ &= \pi/2 \cdot (Rn + tn) \\ &= \pi/2 \cdot (31.4 + 44.0) \\ &= 118.438 \text{ mm} \end{aligned}$$

$$\begin{aligned} &\text{Weld Throat Dimensions, (0.7071 \cdot Leg Dimensions) [L41T, L42T, L43T]:} \\ &= 7.071, 0.000, 0.000, \text{ mm} \end{aligned}$$

Required Weld Throat Dimensions :



tc, L41T = 7.071, must be $\geq \min(0.7 * 44.000, 6.000)$ Passed

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2) \\ &= \min(117786 \cdot 2.4, 1.5 \cdot 147.6(2867.475 + 0.0), \pi/4 \cdot 4.3 \cdot 31.4^2 \cdot 2.4^2) \\ &= 19253.756 \text{ N} \end{aligned}$$

Weld Stress Value [τ]:

$$\begin{aligned} &= fwelds / (L_t(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T})) \\ &= 19253.756 / (118.438(0.49 \cdot 7.071 + 0.6 \cdot 28.0 + 0.49 \cdot 0.0)) \\ &= 8.023 < \text{or} = \text{to } 137.997 \text{ Weld Stress Passed} \end{aligned}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [R_{xn}]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 44.0 / \ln[1 + 44.0/31.4] \\ &= 50.228 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [R_{xs}]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 28.0 / \ln[1 + 28.0/856.0] \\ &= 869.924 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [S_{allow}]:

$$\begin{aligned} &= 1.5 \cdot S \\ &= 1.5 \cdot 137.997 \\ &= 207.0 \text{ MPa} \end{aligned}$$

Force acting on the Nozzle [f_N]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot L_H \\ &= 0.103 \cdot 50.228 \cdot 65.17 \\ &= 338.5 \text{ N} \end{aligned}$$

Force acting on the Shell [f_S]:

$$\begin{aligned} &= P \cdot R_{xs}(L_R + t_n) \\ &= 0.103 \cdot 869.924(62.8 + 44.0) \\ &= 9608.2 \text{ N} \end{aligned}$$

Discontinuity Force from External Pressure [f_Y]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 0.103 \cdot 869.924 \cdot 31.4 \\ &= 2824.9 \text{ N} \end{aligned}$$

Area Resisting External Pressure [A_p]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (9608.2 + 2824.9 + 338.5) / 0.103 \\ &= 123496.9 \text{ mm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [P_{max1}]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff}) \\ &= 206.995 / (2 \cdot 123496.883 / 4675.875 - 869.924 / 28.0) \\ &= 9.5 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [P_{max2}]:

$$\begin{aligned} &= S / t / R_{xs} \\ &= 137.997 / [28.0 / 869.924] \\ &= 4.4 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure [P_{max}]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(9.515, 4.442) \\ &= 4.442 \text{ MPa} \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (f_N + f_S + f_Y) / A_T \\ &= (338.52 + 9608.182 + 2824.877) / 4675.875 \\ &= 2.732 \text{ MPa} \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:



= $P \cdot Rxs / teff$
= $0.103 \cdot 869.924 / 28.0$
= 3.2 MPa

Maximum Local Primary Membrane Stress [PL]:
= $\max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$
= $\max(2 \cdot 2.732 - 3.213, 3.213)$
= 3.2 MPa

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	207.00	MPa
Local Primary Membrane Stress	PL	3.21	MPa
Maximum Allowable External Pressure	Pmax	4.44	MPa

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

The Drop for this Nozzle is : 3.3508 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 287.3507 mm

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Input, Nozzle Desc: L650

From: 50

Pressure for Reinforcement Calculations	P	4.3224	MPa
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	1700.00	mm
Design Length of Section	L	18483.8340	mm
Shell Finished (Minimum) Thickness	t	34.0000	mm
Shell Internal Corrosion Allowance	c	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		1050.00	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.60	MPa
Allowable Stress At Ambient	Sna	165.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		50.8000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm

Flange Material		SA-105	+IT
Flange Type		N/A	

Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	33.2360	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Style of FVC Nozzle		Long Weld Neck	

Corrosion Allowance	can	6.0000	mm
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Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

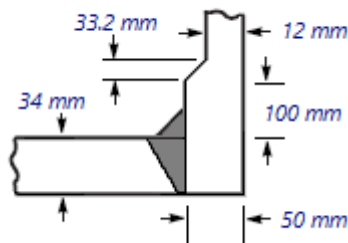
Flange Class		300
Flange Grade		GR 1.1

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: L650

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		50.800	mm.
Actual Thickness Used in Calculation		12.000	mm



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 62.8 \cdot (\exp(4.322/(147.6 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 0.9331 + 6.0 + 0.0 = 6.9331 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 147.6 \cdot 1.0 \cdot \ln((2 \cdot 6.0 + 62.8) / 62.8) - 0.0$$

$$= 25.81 - 0.0 = 25.8098 \text{ MPa}$$

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 138.20 mm

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	10.800 mm
Min. required shell thickness, internal p + c	: 33.239 mm
Min. required shell thickness, external p + c	: 26.261 mm

Nozzle neck minimum required thickness	: 10.800 mm
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Minimum nozzle neck thickness 11.989 mm , is greater than required thickness 10.800 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[147.6/138.0, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 50.0 - 6.0$$

$$= 44.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 12.0 - 6.0$$

$$= 6.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 1712.0/28.0$$

$$= 61.143 \text{ must be less than } 400.$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 1700.0/2 + 6.0$$

$$= 856.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{856.0 \cdot 28.0}, 2 \cdot 31.4)$$



= 62.800 mm

Thickness Limit Candidate [LH1]:

$$\begin{aligned} &= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)} \\ &= \min[1.5 \cdot 28.0, 0.0] + \sqrt{(31.4 \cdot 44.0)} \\ &= 37.170 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 250.0 \\ &= 250.000 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(28.0 + 0.0) \\ &= 224.000 \text{ mm} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles} \\ &= \min[37.17, 250.0, 224.0] + 28.0 \\ &= 65.170 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot frp) / LR) \\ &= 28.0 + ((0.0 \cdot 1.0) / 62.8) \\ &= 28.000 \text{ mm} \end{aligned}$$

Parameter [λ]:

$$\begin{aligned} &= \min\left(12, \left(2 \cdot R_n + t_n\right) / \left(\sqrt{\left(D_i + t_{eff}\right) \cdot t_{eff}}\right)\right) \\ &= \min\left(12, \left(2 \cdot 31.4 + 44.0\right) / \left(\sqrt{\left(1712.0 + 28.0\right) \cdot 28.0}\right)\right) \\ &= 0.484 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max\left((\lambda/5)^{0.85}, 1\right) \\ &= 28.0 \cdot 62.8 \cdot \max\left((0.484/5)^{0.85}, 1\right) \\ &= 1758.400 \text{ mm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot LH \\ &= 44.0 \cdot 65.17 \\ &= 2867.475 \text{ mm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 50.000 \text{ mm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + frn(A2 + A3) + A41 + A42 + A43 \\ &= 1758.4 + 1.0(2867.475 + 0.0) + 50.0 + 0.0 + 0.0 \\ &= 4675.875 \text{ mm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 44.0 / \ln[1 + 44.0/31.4] \\ &= 50.228 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 28.0 / \ln[1 + 28.0/856.0] \\ &= 869.924 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 137.997 \cdot 1.0 \\ &= 207.0 \text{ MPa} \end{aligned}$$

Force acting on the Nozzle [fN]:



$$\begin{aligned} &= P \cdot Rxn \cdot LH \\ &= 4.322 \cdot 50.228 \cdot 65.17 \\ &= 14147.8 \text{ N} \end{aligned}$$

$$\begin{aligned} &\text{Force acting on the Shell [fS]:} \\ &= P \cdot Rxs(LR + tn) \\ &= 4.322 \cdot 869.924(62.8 + 44.0) \\ &= 401554.2 \text{ N} \end{aligned}$$

$$\begin{aligned} &\text{Discontinuity Force from Internal Pressure [fY]:} \\ &= P \cdot Rxs \cdot Rnc \\ &= 4.322 \cdot 869.924 \cdot 31.4 \\ &= 118060.0 \text{ N} \end{aligned}$$

$$\begin{aligned} &\text{Area Resisting Internal Pressure [Ap]:} \\ &= (fS + fY + fN)/P \\ &= (401554.2 + 118060.0 + 14147.8)/4.322 \\ &= 123496.9 \text{ mm}^2 \end{aligned}$$

$$\begin{aligned} &\text{Maximum Allowable Working Pressure Candidate [Pmax1]:} \\ &= Sallow / (2 \cdot Ap/AT - Rxs/teff) \\ &= 206.995 / (2 \cdot 123496.875/4675.875 - 869.924/28.0) \\ &= 9.5 \text{ MPa} \end{aligned}$$

$$\begin{aligned} &\text{Maximum Allowable Working Pressure Candidate [Pmax2]:} \\ &= S[t/Rxs] \\ &= 137.997[28.0/869.924] \\ &= 4.4 \text{ MPa} \end{aligned}$$

$$\begin{aligned} &\text{Maximum Allowable Working Pressure [Pmax]:} \\ &= \min(Pmax1, Pmax2) \\ &= \min(9.515, 4.442) \\ &= 4.442 \text{ MPa} \end{aligned}$$

$$\begin{aligned} &\text{Average Primary Membrane Stress [\sigma Avg]:} \\ &= (fN + fS + fY) / AT \\ &= (14147.752 + 401554.219 + 118059.953)/4675.875 \\ &= 114.162 \text{ MPa} \end{aligned}$$

$$\begin{aligned} &\text{General Primary Membrane Stress [\sigma Circ]:} \\ &= P \cdot Rxs / teff \\ &= 4.322 \cdot 869.924/28.0 \\ &= 134.3 \text{ MPa} \end{aligned}$$

$$\begin{aligned} &\text{Maximum Local Primary Membrane Stress [PL]:} \\ &= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ) \\ &= \max(2 \cdot 114.162 - 134.293, 134.293) \\ &= 134.3 \text{ MPa} \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	207.00	MPa
Local Primary Membrane Stress	PL	134.29	MPa
Maximum Allowable Working Pressure	Pmax	4.44	MPa

Strength of Nozzle Attachment Welds per 4.5.14

$$\begin{aligned} &\text{Discontinuity Force Factor [ky]:} \\ &= (Rnc + tn) / Rnc \\ &= (31.4 + 44.0)/31.4 \\ &= 2.401 \text{ For set-in Nozzles} \end{aligned}$$

$$\begin{aligned} &\text{Weld Length of Nozzle to Shell Weld [Lt]:} \\ &= \pi/2 \cdot (Rn + tn) \\ &= \pi/2 \cdot (31.4 + 44.0) \\ &= 118.438 \text{ mm} \end{aligned}$$

$$\begin{aligned} &\text{Weld Throat Dimensions, (0.7071 \cdot Leg Dimensions) [L41T, L42T, L43T]:} \\ &= 7.071, 0.000, 0.000, \text{ mm} \end{aligned}$$

Required Weld Throat Dimensions :



tc, L41T = 7.071, must be $\geq \min(0.7 \cdot 44.000, 6.000)$ Passed

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2) \\ &= \min(118060 \cdot 2.4, 1.5 \cdot 147.6(2867.475 + 0.0), \pi/4 \cdot 4.3 \cdot 31.4^2 \cdot 2.4^2) \\ &= 19298.529 \text{ N} \end{aligned}$$

Weld Stress Value [τ]:

$$\begin{aligned} &= \text{fwelds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T})) \\ &= 19298.529 / (118.438(0.49 \cdot 7.071 + 0.6 \cdot 28.0 + 0.49 \cdot 0.0)) \\ &= 8.041 < \text{or} = \text{to } 137.997 \text{ Weld Stress Passed} \end{aligned}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 44.0 / \ln[1 + 44.0/31.4] \\ &= 50.228 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 28.0 / \ln[1 + 28.0/856.0] \\ &= 869.924 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \\ &= 1.5 \cdot 137.997 \\ &= 207.0 \text{ MPa} \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot L_H \\ &= 0.103 \cdot 50.228 \cdot 65.17 \\ &= 338.5 \text{ N} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot R_{xs}(L_R + t_n) \\ &= 0.103 \cdot 869.924(62.8 + 44.0) \\ &= 9608.2 \text{ N} \end{aligned}$$

Discontinuity Force from External Pressure [fY]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 0.103 \cdot 869.924 \cdot 31.4 \\ &= 2824.9 \text{ N} \end{aligned}$$

Area Resisting External Pressure [Ap]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (9608.2 + 2824.9 + 338.5) / 0.103 \\ &= 123496.9 \text{ mm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff}) \\ &= 206.995 / (2 \cdot 123496.883 / 4675.875 - 869.924 / 28.0) \\ &= 9.5 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t/R_{xs}] \\ &= 137.997[28.0/869.924] \\ &= 4.4 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(9.515, 4.442) \\ &= 4.442 \text{ MPa} \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (f_N + f_S + f_Y) / A_T \\ &= (338.52 + 9608.182 + 2824.877) / 4675.875 \\ &= 2.732 \text{ MPa} \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:



= $P \cdot Rxs / teff$
= $0.103 \cdot 869.924 / 28.0$
= 3.2 MPa

Maximum Local Primary Membrane Stress [PL]:
= $\max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$
= $\max(2 \cdot 2.732 - 3.213, 3.213)$
= 3.2 MPa

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	207.00	MPa
Local Primary Membrane Stress	PL	3.21	MPa
Maximum Allowable External Pressure	Pmax	4.44	MPa

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

The Drop for this Nozzle is : 3.3508 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 287.3507 mm

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Input, Nozzle Desc: DP150

From: 60

Pressure for Reinforcement Calculations	P	4.2959	MPa
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	1700.00	mm
Design Length of Section	L	18483.8340	mm
Shell Finished (Minimum) Thickness	t	34.0000	mm
Shell Internal Corrosion Allowance	c	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		3700.00	mm
-----------------------------------	--	---------	----

User Entered Minimum Design Metal Temperature		-12.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.60	MPa
Allowable Stress At Ambient	Sna	165.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		50.8000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm

Flange Material		SA-105	+IT
Flange Type		N/A	

Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	33.2360	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Style of FVC Nozzle		Long Weld Neck	

Corrosion Allowance	can	6.0000	mm
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Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

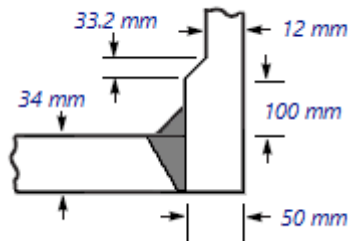
Flange Class		300
Flange Grade		GR 1.1

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: DP150

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	50.800	mm.
Actual Thickness Used in Calculation	12.000	mm



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 62.8 \cdot (\exp(4.296/(147.6 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 0.9273 + 6.0 + 0.0 = 6.9273 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 147.6 \cdot 1.0 \cdot \ln((2 \cdot 6.0 + 62.8) / 62.8) - 0.0$$

$$= 25.81 - 0.0 = 25.8098 \text{ MPa}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	10.800 mm
Min. required shell thickness, internal p + c :	33.069 mm
Min. required shell thickness, external p + c :	26.261 mm

Nozzle neck minimum required thickness :	10.800 mm
--	-----------

Minimum nozzle neck thickness 11.989 mm , is greater than required thickness 10.800 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[147.6/138.0, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 50.0 - 6.0$$

$$= 44.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 12.0 - 6.0$$

$$= 6.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 1712.0/28.0$$

$$= 61.143 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 1700.0/2 + 6.0$$

$$= 856.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{856.0 \cdot 28.0}, 2 \cdot 31.4)$$

$$= 62.800 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$



$$= \min[1.5 \cdot 28.0, 0.0] + \sqrt{(31.4 \cdot 44.0)}$$
$$= 37.170 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$
$$= 250.0$$
$$= 250.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$
$$= 8(28.0 + 0.0)$$
$$= 224.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles}$$
$$= \min[37.17, 250.0, 224.0] + 28.0$$
$$= 65.170 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$
$$= 28.0 + ((0.0 \cdot 1.0) / 62.8)$$
$$= 28.000 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}}))$$
$$= \min(12, (2 \cdot 31.4 + 44.0) / (\sqrt{(1712.0 + 28.0) \cdot 28.0}))$$
$$= 0.484$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$
$$= 28.0 \cdot 62.8 \cdot \max((0.484/5)^{0.85}, 1)$$
$$= 1758.400 \text{ mm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$
$$= 44.0 \cdot 65.17$$
$$= 2867.475 \text{ mm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$
$$= 0.5 \cdot 10.0^2$$
$$= 50.000 \text{ mm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$
$$= 1758.4 + 1.0(2867.475 + 0.0) + 50.0 + 0.0 + 0.0$$
$$= 4675.875 \text{ mm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$
$$= 44.0 / \ln[1 + 44.0/31.4]$$
$$= 50.228 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$
$$= 28.0 / \ln[1 + 28.0/856.0]$$
$$= 869.924 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$
$$= 1.5 \cdot 137.997 \cdot 1.0$$
$$= 207.0 \text{ MPa}$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$
$$= 4.296 \cdot 50.228 \cdot 65.17$$
$$= 14060.8 \text{ N}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot Rxs(LR + tn) \\ &= 4.296 \cdot 869.924(62.8 + 44.0) \\ &= 399085.5 \text{ N} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot Rxs \cdot Rnc \\ &= 4.296 \cdot 869.924 \cdot 31.4 \\ &= 117334.1 \text{ N} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN)/P \\ &= (399085.5 + 117334.1 + 14060.8)/4.296 \\ &= 123496.9 \text{ mm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / (2 \cdot Ap/AT - Rxs/teff) \\ &= 206.995 / (2 \cdot 123496.875/4675.875 - 869.924/28.0) \\ &= 9.5 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t/Rxs] \\ &= 137.997[28.0/869.924] \\ &= 4.4 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(Pmax1, Pmax2) \\ &= \min(9.515, 4.442) \\ &= 4.442 \text{ MPa} \end{aligned}$$

Average Primary Membrane Stress [σAvg]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (14060.772 + 399085.469 + 117334.117)/4675.875 \\ &= 113.460 \text{ MPa} \end{aligned}$$

General Primary Membrane Stress [σCirc]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 4.296 \cdot 869.924/28.0 \\ &= 133.5 \text{ MPa} \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ) \\ &= \max(2 \cdot 113.46 - 133.467, 133.467) \\ &= 133.5 \text{ MPa} \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	207.00	MPa
Local Primary Membrane Stress	PL	133.47	MPa
Maximum Allowable Working Pressure	Pmax	4.44	MPa

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (Rnc + tn) / Rnc \\ &= (31.4 + 44.0)/31.4 \\ &= 2.401 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned} &= \pi/2 \cdot (Rn + tn) \\ &= \pi/2 \cdot (31.4 + 44.0) \\ &= 118.438 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 44.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2)$$



$$= \min(117334 \cdot 2.4, 1.5 \cdot 147.6(2867.475 + 0.0), \pi/4 \cdot 4.3 \cdot 31.4^2 \cdot 2.4^2)$$
$$= 19179.883 \text{ N}$$

Weld Stress Value [τ]:

$$= \text{fwelds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$$
$$= 19179.883 / (118.438(0.49 \cdot 7.071 + 0.6 \cdot 28.0 + 0.49 \cdot 0.0))$$
$$= 7.992 < \text{or} = \text{to } 137.997 \text{ Weld Stress Passed}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [R_{xn}]:

$$= t_n / \ln[1 + t_n/R_n]$$
$$= 44.0 / \ln[1 + 44.0/31.4]$$
$$= 50.228 \text{ mm}$$

Shell Radius for Force Calculation [R_{xs}]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$
$$= 28.0 / \ln[1 + 28.0/856.0]$$
$$= 869.924 \text{ mm}$$

Allowable Local Primary Membrane Stress [S_{allow}]:

$$= 1.5 \cdot S$$
$$= 1.5 \cdot 137.997$$
$$= 207.0 \text{ MPa}$$

Force acting on the Nozzle [f_N]:

$$= P \cdot R_{xn} \cdot L_H$$
$$= 0.103 \cdot 50.228 \cdot 65.17$$
$$= 338.5 \text{ N}$$

Force acting on the Shell [f_S]:

$$= P \cdot R_{xs} (L_R + t_n)$$
$$= 0.103 \cdot 869.924 (62.8 + 44.0)$$
$$= 9608.2 \text{ N}$$

Discontinuity Force from External Pressure [f_Y]:

$$= P \cdot R_{xs} \cdot R_{nc}$$
$$= 0.103 \cdot 869.924 \cdot 31.4$$
$$= 2824.9 \text{ N}$$

Area Resisting External Pressure [A_p]:

$$= (f_S + f_Y + f_N) / P$$
$$= (9608.2 + 2824.9 + 338.5) / 0.103$$
$$= 123496.9 \text{ mm}^2$$

Maximum Allowable Working Pressure Candidate [P_{max1}]:

$$= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff})$$
$$= 206.995 / (2 \cdot 123496.883 / 4675.875 - 869.924 / 28.0)$$
$$= 9.5 \text{ MPa}$$

Maximum Allowable Working Pressure Candidate [P_{max2}]:

$$= S / (t / R_{xs})$$
$$= 137.997 [28.0 / 869.924]$$
$$= 4.4 \text{ MPa}$$

Maximum Allowable Working Pressure [P_{max}]:

$$= \min(P_{max1}, P_{max2})$$
$$= \min(9.515, 4.442)$$
$$= 4.442 \text{ MPa}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / AT$$
$$= (338.52 + 9608.182 + 2824.877) / 4675.875$$
$$= 2.732 \text{ MPa}$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$
$$= 0.103 \cdot 869.924 / 28.0$$
$$= 3.2 \text{ MPa}$$



Maximum Local Primary Membrane Stress [PL]:
= max(2 • σ_{Avg} - σ_{Circ} , σ_{Circ})
= max(2 • 2.732 - 3.213, 3.213)
= 3.2 MPa

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	207.00	MPa
Local Primary Membrane Stress	PL	3.21	MPa
Maximum Allowable External Pressure	Pmax	4.44	MPa

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

The Drop for this Nozzle is : 3.3508 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 287.3507 mm

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Input, Nozzle Desc: UC2600

From: 60

Pressure for Reinforcement Calculations	P	4.2954	MPa
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	1700.00	mm
Design Length of Section	L	18483.8340	mm
Shell Finished (Minimum) Thickness	t	34.0000	mm
Shell Internal Corrosion Allowance	c	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		4050.00	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.60	MPa
Allowable Stress At Ambient	Sna	165.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		568.0000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm

Flange Material		SA-105	+IT
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Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	64.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		80.0000	mm

Corrosion Allowance	can	6.0000	mm
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Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	32.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

This is a Manway, Coupling or Access Opening.

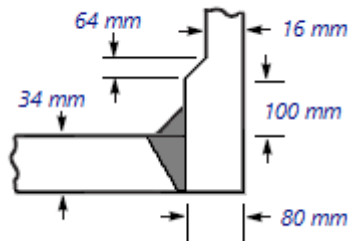
Flange Class		300
Flange Grade		GR 1.1

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: UC2600

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	568.000	mm.
Actual Thickness Used in Calculation	16.000	mm



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 580.0 \cdot (\exp(4.295/(147.6 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 8.5634 + 6.0 + 0.0 = 14.5634 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 147.6 \cdot 1.0 \cdot \ln((2 \cdot 10.0 + 580.0) / 580.0) - 0.0$$

$$= 5.004 - 0.0 = 5.0039 \text{ MPa}$$

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[147.6/138.0, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 80.0 - 6.0$$

$$= 74.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 16.0 - 6.0$$

$$= 10.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 1712.0/28.0$$

$$= 61.143 \text{ must be less than } 400.$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 1700.0/2 + 6.0$$

$$= 856.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{856.0 \cdot 28.0}, 2 \cdot 290.0)$$

$$= 154.816 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$



$$= \min[1.5 \cdot 28.0, 0.0] + \sqrt{(290.0 \cdot 74.0)}$$
$$= 146.492 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$
$$= 500.0$$
$$= 500.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$
$$= 8(28.0 + 0.0)$$
$$= 224.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles}$$
$$= \min[146.492, 500.0, 224.0] + 28.0$$
$$= 174.492 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$
$$= 28.0 + ((0.0 \cdot 1.0) / 154.816)$$
$$= 28.000 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}}))$$
$$= \min(12, (2 \cdot 290.0 + 74.0) / (\sqrt{(1712.0 + 28.0) \cdot 28.0}))$$
$$= 2.963$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$
$$= 28.0 \cdot 154.816 \cdot \max((2.963/5)^{0.85}, 1)$$
$$= 4334.849 \text{ mm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A2A + A2B$$
$$= 9472.0 + 2359.664$$
$$= 11831.663 \text{ mm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$
$$= 0.5 \cdot 10.0^2$$
$$= 50.000 \text{ mm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$
$$= 4334.849 + 1.0(11831.663 + 0.0) + 50.0 + 0.0 + 0.0$$
$$= 16216.511 \text{ mm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$
$$= 74.0 / \ln[1 + 74.0/290.0]$$
$$= 325.600 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$
$$= 28.0 / \ln[1 + 28.0/856.0]$$
$$= 869.924 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$
$$= 1.5 \cdot 137.997 \cdot 1.0$$
$$= 207.0 \text{ MPa}$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$
$$= 4.295 \cdot 325.6 \cdot 174.492$$
$$= 244018.7 \text{ N}$$



Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot Rxs(LR + tn) \\ &= 4.295 \cdot 869.924(154.816 + 74.0) \\ &= 854929.7 \text{ N} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot Rxs \cdot Rnc \\ &= 4.295 \cdot 869.924 \cdot 290.0 \\ &= 1083532.6 \text{ N} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN)/P \\ &= (854930 + 1083533 + 244018.7)/4.295 \\ &= 508145.2 \text{ mm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / (2 \cdot Ap/AT - Rxs/teff) \\ &= 206.995 / (2 \cdot 508145/16216.511 - 869.924/28.0) \\ &= 6.6 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t/Rxs] \\ &= 137.997[28.0/869.924] \\ &= 4.4 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(Pmax1, Pmax2) \\ &= \min(6.55, 4.442) \\ &= 4.442 \text{ MPa} \end{aligned}$$

Average Primary Membrane Stress [σAvg]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (244018.672 + 854930 + 1083533)/16216.511 \\ &= 134.595 \text{ MPa} \end{aligned}$$

General Primary Membrane Stress [σCirc]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 4.295 \cdot 869.924/28.0 \\ &= 133.5 \text{ MPa} \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ) \\ &= \max(2 \cdot 134.595 - 133.451, 133.451) \\ &= 135.7 \text{ MPa} \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	207.00	MPa
Local Primary Membrane Stress	PL	135.74	MPa
Maximum Allowable Working Pressure	Pmax	4.44	MPa

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (Rnc + tn) / Rnc \\ &= (290.0 + 74.0)/290.0 \\ &= 1.255 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned} &= \pi/2 \cdot (Rn + tn) \\ &= \pi/2 \cdot (290.0 + 74.0) \\ &= 571.770 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 74.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2)$$



$$= \min(1083533 \cdot 1.26, 1.5 \cdot 147.6(11831.663 + 0.0), \pi/4 \cdot 4.3 \cdot 290.0^2 \cdot 1.26^2)$$
$$= 446946.344 \text{ N}$$

Weld Stress Value [τ]:

$$= \text{fwelds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$$
$$= 446946.344 / (571.77(0.49 \cdot 7.071 + 0.6 \cdot 28.0 + 0.49 \cdot 0.0))$$
$$= 38.577 < \text{or} = \text{to } 137.997 \text{ Weld Stress Passed}$$

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [R_{xn}]:

$$= t_n / \ln[1 + t_n/R_n]$$
$$= 74.0 / \ln[1 + 74.0/290.0]$$
$$= 325.600 \text{ mm}$$

Shell Radius for Force Calculation [R_{xs}]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$
$$= 28.0 / \ln[1 + 28.0/856.0]$$
$$= 869.924 \text{ mm}$$

Allowable Local Primary Membrane Stress [S_{allow}]:

$$= 1.5 \cdot S$$
$$= 1.5 \cdot 137.997$$
$$= 207.0 \text{ MPa}$$

Force acting on the Nozzle [f_N]:

$$= P \cdot R_{xn} \cdot L_H$$
$$= 0.103 \cdot 325.6 \cdot 174.492$$
$$= 5875.6 \text{ N}$$

Force acting on the Shell [f_S]:

$$= P \cdot R_{xs} (L_R + t_n)$$
$$= 0.103 \cdot 869.924 (154.816 + 74.0)$$
$$= 20585.3 \text{ N}$$

Discontinuity Force from External Pressure [f_Y]:

$$= P \cdot R_{xs} \cdot R_{nc}$$
$$= 0.103 \cdot 869.924 \cdot 290.0$$
$$= 26089.6 \text{ N}$$

Area Resisting External Pressure [A_p]:

$$= (f_S + f_Y + f_N) / P$$
$$= (20585.3 + 26089.6 + 5875.6) / 0.103$$
$$= 508145.2 \text{ mm}^2$$

Maximum Allowable Working Pressure Candidate [P_{max1}]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$
$$= 206.995 / (2 \cdot 508145 / 16216.511 - 869.924 / 28.0)$$
$$= 6.6 \text{ MPa}$$

Maximum Allowable Working Pressure Candidate [P_{max2}]:

$$= S[t / R_{xs}]$$
$$= 137.997[28.0/869.924]$$
$$= 4.4 \text{ MPa}$$

Maximum Allowable Working Pressure [P_{max}]:

$$= \min(P_{max1}, P_{max2})$$
$$= \min(6.55, 4.442)$$
$$= 4.442 \text{ MPa}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / A_T$$
$$= (5875.556 + 20585.262 + 26089.631) / 16216.511$$
$$= 3.241 \text{ MPa}$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$



= $0.103 \cdot 869.924/28.0$
= 3.2 MPa

Maximum Local Primary Membrane Stress [PL]:

= $\max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$
= $\max(2 \cdot 3.241 - 3.213, 3.213)$
= 3.3 MPa

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	207.00	MPa
Local Primary Membrane Stress	PL	3.27	MPa
Maximum Allowable External Pressure	Pmax	4.44	MPa

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

[Note: This Element/Detail was specified as being Impact Tested.](#)

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

[Note: This Element/Detail was specified as being Impact Tested.](#)

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -27 °C

The Drop for this Nozzle is : 81.8828 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 615.8828 mm

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Input, Nozzle Desc: MH3600

From: 70

Pressure for Reinforcement Calculations	P	4.2954	MPa
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	1700.00	mm
Design Length of Section	L	18483.8340	mm
Shell Finished (Minimum) Thickness	t	34.0000	mm
Shell Internal Corrosion Allowance	c	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		8350.00	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.60	MPa
Allowable Stress At Ambient	Sna	165.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		568.0000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm

Flange Material		SA-105	+IT
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Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	64.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		80.0000	mm

Corrosion Allowance	can	6.0000	mm
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Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	32.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

This is a Manway, Coupling or Access Opening.

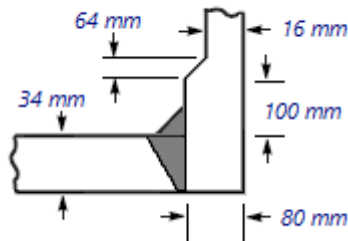
Flange Class		300
Flange Grade		GR 1.1

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: MH3600

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	568.000	mm.
Actual Thickness Used in Calculation	16.000	mm



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 580.0 \cdot (\exp(4.295/(147.6 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 8.5634 + 6.0 + 0.0 = 14.5634 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 147.6 \cdot 1.0 \cdot \ln((2 \cdot 10.0 + 580.0) / 580.0) - 0.0$$

$$= 5.004 - 0.0 = 5.0039 \text{ MPa}$$

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[147.6/138.0, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 80.0 - 6.0$$

$$= 74.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 16.0 - 6.0$$

$$= 10.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 1712.0/28.0$$

$$= 61.143 \text{ must be less than } 400.$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 1700.0/2 + 6.0$$

$$= 856.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{856.0 \cdot 28.0}, 2 \cdot 290.0)$$

$$= 154.816 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$



$$= \min[1.5 \cdot 28.0, 0.0] + \sqrt{(290.0 \cdot 74.0)}$$
$$= 146.492 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$
$$= 500.0$$
$$= 500.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$
$$= 8(28.0 + 0.0)$$
$$= 224.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles}$$
$$= \min[146.492, 500.0, 224.0] + 28.0$$
$$= 174.492 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$
$$= 28.0 + ((0.0 \cdot 1.0) / 154.816)$$
$$= 28.000 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}}))$$
$$= \min(12, (2 \cdot 290.0 + 74.0) / (\sqrt{(1712.0 + 28.0) \cdot 28.0}))$$
$$= 2.963$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$
$$= 28.0 \cdot 154.816 \cdot \max((2.963/5)^{0.85}, 1)$$
$$= 4334.849 \text{ mm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A2A + A2B$$
$$= 9472.0 + 2359.664$$
$$= 11831.663 \text{ mm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$
$$= 0.5 \cdot 10.0^2$$
$$= 50.000 \text{ mm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$
$$= 4334.849 + 1.0(11831.663 + 0.0) + 50.0 + 0.0 + 0.0$$
$$= 16216.511 \text{ mm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$
$$= 74.0 / \ln[1 + 74.0/290.0]$$
$$= 325.600 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$
$$= 28.0 / \ln[1 + 28.0/856.0]$$
$$= 869.924 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$
$$= 1.5 \cdot 137.997 \cdot 1.0$$
$$= 207.0 \text{ MPa}$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$
$$= 4.295 \cdot 325.6 \cdot 174.492$$
$$= 244018.7 \text{ N}$$



Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot Rxs(LR + tn) \\ &= 4.295 \cdot 869.924(154.816 + 74.0) \\ &= 854929.7 \text{ N} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot Rxs \cdot Rnc \\ &= 4.295 \cdot 869.924 \cdot 290.0 \\ &= 1083532.6 \text{ N} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN)/P \\ &= (854930 + 1083533 + 244018.7)/4.295 \\ &= 508145.2 \text{ mm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / (2 \cdot Ap/AT - Rxs/teff) \\ &= 206.995 / (2 \cdot 508145/16216.511 - 869.924/28.0) \\ &= 6.6 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t/Rxs] \\ &= 137.997[28.0/869.924] \\ &= 4.4 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(Pmax1, Pmax2) \\ &= \min(6.55, 4.442) \\ &= 4.442 \text{ MPa} \end{aligned}$$

Average Primary Membrane Stress [σAvg]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (244018.672 + 854930 + 1083533)/16216.511 \\ &= 134.595 \text{ MPa} \end{aligned}$$

General Primary Membrane Stress [σCirc]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 4.295 \cdot 869.924/28.0 \\ &= 133.5 \text{ MPa} \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ) \\ &= \max(2 \cdot 134.595 - 133.451, 133.451) \\ &= 135.7 \text{ MPa} \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	207.00	MPa
Local Primary Membrane Stress	PL	135.74	MPa
Maximum Allowable Working Pressure	Pmax	4.44	MPa

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (Rnc + tn) / Rnc \\ &= (290.0 + 74.0)/290.0 \\ &= 1.255 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned} &= \pi/2 \cdot (Rn + tn) \\ &= \pi/2 \cdot (290.0 + 74.0) \\ &= 571.770 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 74.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2)$$



$$= \min(1083533 \cdot 1.26, 1.5 \cdot 147.6(11831.663 + 0.0), \pi/4 \cdot 4.3 \cdot 290.0^2 \cdot 1.26^2)$$
$$= 446946.344 \text{ N}$$

Weld Stress Value [τ]:

$$= \text{fwelds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$$
$$= 446946.344 / (571.77(0.49 \cdot 7.071 + 0.6 \cdot 28.0 + 0.49 \cdot 0.0))$$
$$= 38.577 < \text{or} = \text{to } 137.997 \text{ Weld Stress Passed}$$

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [R_{xn}]:

$$= t_n / \ln[1 + t_n/R_n]$$
$$= 74.0 / \ln[1 + 74.0/290.0]$$
$$= 325.600 \text{ mm}$$

Shell Radius for Force Calculation [R_{xs}]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$
$$= 28.0 / \ln[1 + 28.0/856.0]$$
$$= 869.924 \text{ mm}$$

Allowable Local Primary Membrane Stress [S_{allow}]:

$$= 1.5 \cdot S$$
$$= 1.5 \cdot 137.997$$
$$= 207.0 \text{ MPa}$$

Force acting on the Nozzle [f_N]:

$$= P \cdot R_{xn} \cdot L_H$$
$$= 0.103 \cdot 325.6 \cdot 174.492$$
$$= 5875.6 \text{ N}$$

Force acting on the Shell [f_S]:

$$= P \cdot R_{xs} (L_R + t_n)$$
$$= 0.103 \cdot 869.924 (154.816 + 74.0)$$
$$= 20585.3 \text{ N}$$

Discontinuity Force from External Pressure [f_Y]:

$$= P \cdot R_{xs} \cdot R_{nc}$$
$$= 0.103 \cdot 869.924 \cdot 290.0$$
$$= 26089.6 \text{ N}$$

Area Resisting External Pressure [A_p]:

$$= (f_S + f_Y + f_N) / P$$
$$= (20585.3 + 26089.6 + 5875.6) / 0.103$$
$$= 508145.2 \text{ mm}^2$$

Maximum Allowable Working Pressure Candidate [P_{max1}]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$
$$= 206.995 / (2 \cdot 508145 / 16216.511 - 869.924 / 28.0)$$
$$= 6.6 \text{ MPa}$$

Maximum Allowable Working Pressure Candidate [P_{max2}]:

$$= S[t / R_{xs}]$$
$$= 137.997[28.0/869.924]$$
$$= 4.4 \text{ MPa}$$

Maximum Allowable Working Pressure [P_{max}]:

$$= \min(P_{max1}, P_{max2})$$
$$= \min(6.55, 4.442)$$
$$= 4.442 \text{ MPa}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / A_T$$
$$= (5875.556 + 20585.262 + 26089.631) / 16216.511$$
$$= 3.241 \text{ MPa}$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$



= $0.103 \cdot 869.924/28.0$
= 3.2 MPa

Maximum Local Primary Membrane Stress [PL]:

= $\max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}})$
= $\max(2 \cdot 3.241 - 3.213, 3.213)$
= 3.3 MPa

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	207.00	MPa
Local Primary Membrane Stress	PL	3.27	MPa
Maximum Allowable External Pressure	Pmax	4.44	MPa

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

[Note: This Element/Detail was specified as being Impact Tested.](#)

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

[Note: This Element/Detail was specified as being Impact Tested.](#)

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -27 °C

The Drop for this Nozzle is : 81.8828 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 615.8828 mm

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Input, Nozzle Desc: DP250

From: 70

Pressure for Reinforcement Calculations	P	4.2954	MPa
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	1700.00	mm
Design Length of Section	L	18483.8340	mm
Shell Finished (Minimum) Thickness	t	34.0000	mm
Shell Internal Corrosion Allowance	c	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		8550.00	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.60	MPa
Allowable Stress At Ambient	Sna	165.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		50.8000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm

Flange Material		SA-105	+IT
-----------------	--	--------	-----

Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	33.2360	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm

Corrosion Allowance	can	6.0000	mm
---------------------	-----	--------	----

Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

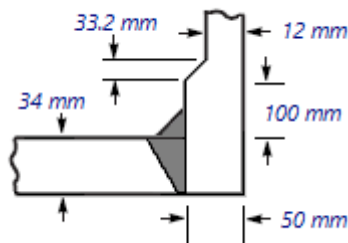
Flange Class		300
Flange Grade		GR 1.1

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: DP250

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	50.800	mm.
Actual Thickness Used in Calculation	12.000	mm



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 62.8 \cdot (\exp(4.295/(147.6 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 0.9272 + 6.0 + 0.0 = 6.9272 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 147.6 \cdot 1.0 \cdot \ln((2 \cdot 6.0 + 62.8) / 62.8) - 0.0$$

$$= 25.81 - 0.0 = 25.8098 \text{ MPa}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	10.800 mm
Min. required shell thickness, internal p + c :	33.065 mm
Min. required shell thickness, external p + c :	26.261 mm

Nozzle neck minimum required thickness	: 10.800 mm
--	-------------

Minimum nozzle neck thickness 11.989 mm , is greater than required thickness 10.800 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[147.6/138.0, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 50.0 - 6.0$$

$$= 44.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 12.0 - 6.0$$

$$= 6.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 1712.0/28.0$$

$$= 61.143 \text{ must be less than } 400.$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 1700.0/2 + 6.0$$

$$= 856.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:



$$\begin{aligned} &= \min(\sqrt{(\text{Reff} \cdot t)}, 2 \cdot R_n) \\ &= \min(\sqrt{(856.0 \cdot 28.0)}, 2 \cdot 31.4) \\ &= 62.800 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH1]:

$$\begin{aligned} &= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)} \\ &= \min[1.5 \cdot 28.0, 0.0] + \sqrt{(31.4 \cdot 44.0)} \\ &= 37.170 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 250.0 \\ &= 250.000 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(28.0 + 0.0) \\ &= 224.000 \text{ mm} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles} \\ &= \min[37.17, 250.0, 224.0] + 28.0 \\ &= 65.170 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot \text{frp}) / LR) \\ &= 28.0 + ((0.0 \cdot 1.0) / 62.8) \\ &= 28.000 \text{ mm} \end{aligned}$$

Parameter [λ]:

$$\begin{aligned} &= \min(12, (2 \cdot R_n + t_n) / (\sqrt{((D_i + t_{eff}) \cdot t_{eff})})) \\ &= \min(12, (2 \cdot 31.4 + 44.0) / (\sqrt{((1712.0 + 28.0) \cdot 28.0)})) \\ &= 0.484 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ &= 28.0 \cdot 62.8 \cdot \max((0.484/5)^{0.85}, 1) \\ &= 1758.400 \text{ mm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot LH \\ &= 44.0 \cdot 65.17 \\ &= 2867.475 \text{ mm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 50.000 \text{ mm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 \\ &= 1758.4 + 1.0(2867.475 + 0.0) + 50.0 + 0.0 + 0.0 \\ &= 4675.875 \text{ mm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 44.0 / \ln[1 + 44.0/31.4] \\ &= 50.228 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/\text{Reff}] \\ &= 28.0 / \ln[1 + 28.0/856.0] \\ &= 869.924 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 137.997 \cdot 1.0 \\ &= 207.0 \text{ MPa} \end{aligned}$$



Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot L_H \\ &= 4.295 \cdot 50.228 \cdot 65.17 \\ &= 14059.1 \text{ N} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot R_{xs} (L_R + t_n) \\ &= 4.295 \cdot 869.924 (62.8 + 44.0) \\ &= 399038.9 \text{ N} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 4.295 \cdot 869.924 \cdot 31.4 \\ &= 117320.4 \text{ N} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (399038.9 + 117320.4 + 14059.1) / 4.295 \\ &= 123496.9 \text{ mm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff}) \\ &= 206.995 / (2 \cdot 123496.875 / 4675.875 - 869.924 / 28.0) \\ &= 9.5 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S / t / R_{xs} \\ &= 137.997 [28.0 / 869.924] \\ &= 4.4 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(9.515, 4.442) \\ &= 4.442 \text{ MPa} \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (f_N + f_S + f_Y) / A_T \\ &= (14059.132 + 399038.906 + 117320.43) / 4675.875 \\ &= 113.447 \text{ MPa} \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot R_{xs} / t_{eff} \\ &= 4.295 \cdot 869.924 / 28.0 \\ &= 133.5 \text{ MPa} \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 113.447 - 133.451, 133.451) \\ &= 133.5 \text{ MPa} \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	207.00	MPa
Local Primary Membrane Stress	PL	133.45	MPa
Maximum Allowable Working Pressure	Pmax	4.44	MPa

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (R_{nc} + t_n) / R_{nc} \\ &= (31.4 + 44.0) / 31.4 \\ &= 2.401 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [LT]:

$$\begin{aligned} &= \pi / 2 \cdot (R_n + t_n) \\ &= \pi / 2 \cdot (31.4 + 44.0) \\ &= 118.438 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$



Required Weld Throat Dimensions :

tc, L41T = 7.071, must be $\geq \min(0.7 * 44.000, 6.000)$ Passed

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2) \\ &= \min(117320 \cdot 2.4, 1.5 \cdot 147.6(2867.475 + 0.0), \pi/4 \cdot 4.3 \cdot 31.4^2 \cdot 2.4^2) \\ &= 19177.645 \text{ N} \end{aligned}$$

Weld Stress Value [τ]:

$$\begin{aligned} &= \text{fwelds} / (L_t(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T})) \\ &= 19177.645 / (118.438(0.49 \cdot 7.071 + 0.6 \cdot 28.0 + 0.49 \cdot 0.0)) \\ &= 7.991 < \text{or} = \text{to } 137.997 \text{ Weld Stress Passed} \end{aligned}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 44.0 / \ln[1 + 44.0/31.4] \\ &= 50.228 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 28.0 / \ln[1 + 28.0/856.0] \\ &= 869.924 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \\ &= 1.5 \cdot 137.997 \\ &= 207.0 \text{ MPa} \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot L_H \\ &= 0.103 \cdot 50.228 \cdot 65.17 \\ &= 338.5 \text{ N} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot R_{xs}(L_R + t_n) \\ &= 0.103 \cdot 869.924(62.8 + 44.0) \\ &= 9608.2 \text{ N} \end{aligned}$$

Discontinuity Force from External Pressure [fY]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 0.103 \cdot 869.924 \cdot 31.4 \\ &= 2824.9 \text{ N} \end{aligned}$$

Area Resisting External Pressure [Ap]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (9608.2 + 2824.9 + 338.5) / 0.103 \\ &= 123496.9 \text{ mm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff}) \\ &= 206.995 / (2 \cdot 123496.883 / 4675.875 - 869.924 / 28.0) \\ &= 9.5 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S / [t / R_{xs}] \\ &= 137.997 [28.0 / 869.924] \\ &= 4.4 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(9.515, 4.442) \\ &= 4.442 \text{ MPa} \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (f_N + f_S + f_Y) / A_T \\ &= (338.52 + 9608.182 + 2824.877) / 4675.875 \\ &= 2.732 \text{ MPa} \end{aligned}$$



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General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 0.103 \cdot 869.924 / 28.0$$

$$= 3.2 \text{ MPa}$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}})$$

$$= \max(2 \cdot 2.732 - 3.213, 3.213)$$

$$= 3.2 \text{ MPa}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	207.00	MPa
Local Primary Membrane Stress	PL	3.21	MPa
Maximum Allowable External Pressure	Pmax	4.44	MPa

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested)

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction	-27 °C
---------------------------------	--------

The Drop for this Nozzle is : 3.3508 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 287.3507 mm

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Input, Nozzle Desc: UC1600

From: 80

Pressure for Reinforcement Calculations	P	4.2954	MPa
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	1700.00	mm
Design Length of Section	L	18483.8340	mm
Shell Finished (Minimum) Thickness	t	34.0000	mm
Shell Internal Corrosion Allowance	c	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		10050.00	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.60	MPa
Allowable Stress At Ambient	Sna	165.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		568.0000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm

Flange Material		SA-105	+IT
-----------------	--	--------	-----

Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	64.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		80.0000	mm

Corrosion Allowance	can	6.0000	mm
---------------------	-----	--------	----

Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	32.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

This is a Manway, Coupling or Access Opening.

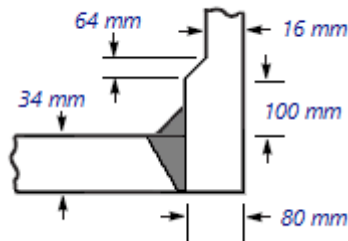
Flange Class		300
Flange Grade		GR 1.1

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: UC1600

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	568.000	mm.
Actual Thickness Used in Calculation	16.000	mm



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 580.0 \cdot (\exp(4.295/(147.6 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 8.5634 + 6.0 + 0.0 = 14.5634 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 147.6 \cdot 1.0 \cdot \ln((2 \cdot 10.0 + 580.0) / 580.0) - 0.0$$

$$= 5.004 - 0.0 = 5.0039 \text{ MPa}$$

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 298.80 mm, while the distance from the center of the nozzle to the edge of the diameter limit is: 518.82 mm

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[147.6/138.0, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 80.0 - 6.0$$

$$= 74.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 16.0 - 6.0$$

$$= 10.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D/t$$

$$= 1712.0/28.0$$

$$= 61.143 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 1700.0/2 + 6.0$$

$$= 856.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{856.0 \cdot 28.0}, 2 \cdot 290.0)$$



= 154.816 mm

Thickness Limit Candidate [LH1]:

$$\begin{aligned} &= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)} \\ &= \min[1.5 \cdot 28.0, 0.0] + \sqrt{(290.0 \cdot 74.0)} \\ &= 146.492 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 500.0 \\ &= 500.000 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(28.0 + 0.0) \\ &= 224.000 \text{ mm} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles} \\ &= \min[146.492, 500.0, 224.0] + 28.0 \\ &= 174.492 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot frp) / LR) \\ &= 28.0 + ((0.0 \cdot 1.0) / 154.816) \\ &= 28.000 \text{ mm} \end{aligned}$$

Parameter [λ]:

$$\begin{aligned} &= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + teff) \cdot teff})) \\ &= \min(12, (2 \cdot 290.0 + 74.0) / (\sqrt{(1712.0 + 28.0) \cdot 28.0})) \\ &= 2.963 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ &= 28.0 \cdot 154.816 \cdot \max((2.963/5)^{0.85}, 1) \\ &= 4334.849 \text{ mm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= A2A + A2B \\ &= 9472.0 + 2359.664 \\ &= 11831.663 \text{ mm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot Leg^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 50.000 \text{ mm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + frn(A2 + A3) + A41 + A42 + A43 \\ &= 4334.849 + 1.0(11831.663 + 0.0) + 50.0 + 0.0 + 0.0 \\ &= 16216.511 \text{ mm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 74.0 / \ln[1 + 74.0/290.0] \\ &= 325.600 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= teff / \ln[1 + teff/Reff] \\ &= 28.0 / \ln[1 + 28.0/856.0] \\ &= 869.924 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 137.997 \cdot 1.0 \\ &= 207.0 \text{ MPa} \end{aligned}$$

Force acting on the Nozzle [fN]:



$$\begin{aligned} &= P \cdot Rxn \cdot LH \\ &= 4.295 \cdot 325.6 \cdot 174.492 \\ &= 244018.7 \text{ N} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot Rxs(LR + tn) \\ &= 4.295 \cdot 869.924(154.816 + 74.0) \\ &= 854929.7 \text{ N} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot Rxs \cdot Rnc \\ &= 4.295 \cdot 869.924 \cdot 290.0 \\ &= 1083532.6 \text{ N} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN)/P \\ &= (854930 + 1083533 + 244018.7)/4.295 \\ &= 508145.2 \text{ mm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / (2 \cdot Ap/AT - Rxs/teff) \\ &= 206.995 / (2 \cdot 508145/16216.511 - 869.924/28.0) \\ &= 6.6 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t/Rxs] \\ &= 137.997[28.0/869.924] \\ &= 4.4 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(Pmax1, Pmax2) \\ &= \min(6.55, 4.442) \\ &= 4.442 \text{ MPa} \end{aligned}$$

Average Primary Membrane Stress [σ Avg]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (244018.672 + 854930 + 1083533)/16216.511 \\ &= 134.595 \text{ MPa} \end{aligned}$$

General Primary Membrane Stress [σ Circ]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 4.295 \cdot 869.924/28.0 \\ &= 133.5 \text{ MPa} \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ) \\ &= \max(2 \cdot 134.595 - 133.451, 133.451) \\ &= 135.7 \text{ MPa} \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	207.00	MPa
Local Primary Membrane Stress	PL	135.74	MPa
Maximum Allowable Working Pressure	Pmax	4.44	MPa

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (Rnc + tn) / Rnc \\ &= (290.0 + 74.0)/290.0 \\ &= 1.255 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned} &= \pi/2 \cdot (Rn + tn) \\ &= \pi/2 \cdot (290.0 + 74.0) \\ &= 571.770 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :



tc, L41T = 7.071, must be $\geq \min(0.7 * 74.000, 6.000)$ Passed

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2) \\ &= \min(1083533 \cdot 1.26, 1.5 \cdot 147.6(11831.663 + 0.0), \pi/4 \cdot 4.3 \cdot 290.0^2 \cdot 1.26^2) \\ &= 446946.344 \text{ N} \end{aligned}$$

Weld Stress Value [τ]:

$$\begin{aligned} &= \text{fwelds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T})) \\ &= 446946.344 / (571.77(0.49 \cdot 7.071 + 0.6 \cdot 28.0 + 0.49 \cdot 0.0)) \\ &= 38.577 < \text{or} = \text{to } 137.997 \text{ Weld Stress Passed} \end{aligned}$$

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [R_{xn}]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 74.0 / \ln[1 + 74.0/290.0] \\ &= 325.600 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [R_{xs}]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 28.0 / \ln[1 + 28.0/856.0] \\ &= 869.924 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \\ &= 1.5 \cdot 137.997 \\ &= 207.0 \text{ MPa} \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot L_H \\ &= 0.103 \cdot 325.6 \cdot 174.492 \\ &= 5875.6 \text{ N} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot (L_R + t_n) \\ &= 0.103 \cdot 869.924(154.816 + 74.0) \\ &= 20585.3 \text{ N} \end{aligned}$$

Discontinuity Force from External Pressure [fY]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 0.103 \cdot 869.924 \cdot 290.0 \\ &= 26089.6 \text{ N} \end{aligned}$$

Area Resisting External Pressure [A_p]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (20585.3 + 26089.6 + 5875.6) / 0.103 \\ &= 508145.2 \text{ mm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [P_{max1}]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff}) \\ &= 206.995 / (2 \cdot 508145 / 16216.511 - 869.924 / 28.0) \\ &= 6.6 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [P_{max2}]:

$$\begin{aligned} &= S / t / R_{xs} \\ &= 137.997 / (28.0 / 869.924) \\ &= 4.4 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure [P_{max}]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(6.55, 4.442) \\ &= 4.442 \text{ MPa} \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (f_N + f_S + f_Y) / A_T \\ &= (5875.556 + 20585.262 + 26089.631) / 16216.511 \end{aligned}$$



= 3.241 MPa

General Primary Membrane Stress [σ_{Circ}]:

= $P \cdot R_{xs} / t_{eff}$
= $0.103 \cdot 869.924 / 28.0$
= 3.2 MPa

Maximum Local Primary Membrane Stress [PL]:

= $\max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$
= $\max(2 \cdot 3.241 - 3.213, 3.213)$
= 3.3 MPa

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	207.00	MPa
Local Primary Membrane Stress	PL	3.27	MPa
Maximum Allowable External Pressure	Pmax	4.44	MPa

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -27 °C

The Drop for this Nozzle is : 81.8828 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 615.8828 mm

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Input, Nozzle Desc: A150

From: 90

Pressure for Reinforcement Calculations	P	4.2954	MPa
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	1706.00	mm
Design Length of Section	L	18483.8359	mm
Shell Finished (Minimum) Thickness	t	28.0000	mm
Shell Internal Corrosion Allowance	c	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		14975.00	mm
-----------------------------------	--	----------	----

User Entered Minimum Design Metal Temperature		-12.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.60	MPa
Allowable Stress At Ambient	Sna	165.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		150.0000	mm

Size and Thickness Basis		Minimum	
Nominal Thickness		120	

Flange Material		SA-105	+IT
-----------------	--	--------	-----

Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	38.5105	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm

Corrosion Allowance	can	6.0000	mm
---------------------	-----	--------	----

Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	28.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

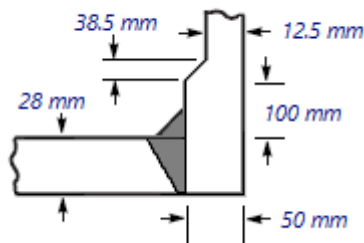
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: A150

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		143.294	mm.
Actual Thickness Used in Calculation		12.490	mm



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 155.294 \cdot (\exp(4.295/(147.6 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 2.2928 + 6.0 + 0.0 = 8.2928 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 147.6 \cdot 1.0 \cdot \ln((2 \cdot 6.49 + 155.294) / 155.294) - 0.0$$

$$= 11.849 - 0.0 = 11.8491 \text{ MPa}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	12.220 mm
Min. required shell thickness, internal p + c :	28.002 mm
Min. required shell thickness, external p + c :	16.233 mm

Nozzle neck minimum required thickness : 12.220 mm

Minimum nozzle neck thickness 12.497 mm , is greater than required thickness 12.220 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained :	55.9,	Allowable :	147.6 MPa	Passed
Expansion :	49.3,	Allowable :	334.8 MPa	Passed
Occasional :	24.7,	Allowable :	196.3 MPa	Passed
Shear :	25.0,	Allowable :	103.3 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[147.6/138.0, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 50.0 - 6.0$$

$$= 44.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 12.49 - 6.0$$

$$= 6.490 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 1708.0/27.0$$

$$= 63.259 \text{ must be less than } 400.$$

Effective Pressure Radius [Reff]:



$$\begin{aligned} &= D_i/2 + \text{corrosion allowance} \\ &= 1706.0/2 + 1.0 \\ &= 854.000 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Effective Length of Vessel Wall [LR]:} \\ &= \min(\sqrt{R_{eff} \cdot t}, 2 \cdot R_n) \\ &= \min(\sqrt{854.0 \cdot 27.0}, 2 \cdot 77.647) \\ &= 151.849 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Thickness Limit Candidate [LH1]:} \\ &= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n} \\ &= \min[1.5 \cdot 27.0, 0.0] + \sqrt{77.647 \cdot 44.0} \\ &= 58.451 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Thickness Limit Candidate [LH2]:} \\ &= L_{pr1} \\ &= 300.0 \\ &= 300.000 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Thickness Limit Candidate [LH3]:} \\ &= 8(t + t_e) \\ &= 8(27.0 + 0.0) \\ &= 216.000 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Effective Nozzle Wall Length Outside the Vessel [LH]:} \\ &= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles} \\ &= \min[58.451, 300.0, 216.0] + 27.0 \\ &= 85.451 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Effective Vessel Thickness [teff]:} \\ &= t + ((A_5 \cdot frp) / LR) \\ &= 27.0 + ((0.0 \cdot 1.0) / 151.849) \\ &= 27.000 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Parameter } [\lambda]: \\ &= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + teff) \cdot teff})) \\ &= \min(12, (2 \cdot 77.65 + 44.0) / (\sqrt{(1708.0 + 27.0) \cdot 27.0})) \\ &= 0.921 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

$$\begin{aligned} \text{Area Contributed by the Vessel Wall [A1]:} \\ &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ &= 27.0 \cdot 151.849 \cdot \max((0.921/5)^{0.85}, 1) \\ &= 4099.913 \text{ mm}^2 \end{aligned}$$

$$\begin{aligned} \text{Area Contributed by the Nozzle Outside the Vessel Wall [A2]:} \\ &= t_n \cdot LH \\ &= 44.0 \cdot 85.451 \\ &= 3759.826 \text{ mm}^2 \end{aligned}$$

$$\begin{aligned} \text{Area Contributed by the Outside Fillet Weld [A41]:} \\ &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 50.000 \text{ mm}^2 \end{aligned}$$

$$\begin{aligned} \text{The total area contributed by A1 through A43 [AT]:} \\ &= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 \\ &= 4099.913 + 1.0(3759.826 + 0.0) + 50.0 + 0.0 + 0.0 \\ &= 7909.738 \text{ mm}^2 \end{aligned}$$

$$\begin{aligned} \text{Nozzle Radius for Force Calculation [Rxn]:} \\ &= t_n / \ln[1 + t_n/R_n] \\ &= 44.0 / \ln[1 + 44.0/77.647] \\ &= 98.006 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Shell Radius for Force Calculation [Rxs]:} \\ &= teff / \ln[1 + teff/R_{eff}] \\ &= 27.0 / \ln[1 + 27.0/854.0] \\ &= 867.429 \text{ mm} \end{aligned}$$



Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 137.997 \cdot 1.0 \\ &= 207.0 \text{ MPa} \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot L_H \\ &= 4.295 \cdot 98.006 \cdot 85.451 \\ &= 35969.3 \text{ N} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot R_{xs} (L_R + t_n) \\ &= 4.295 \cdot 867.429 (151.849 + 44.0) \\ &= 729654.0 \text{ N} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 4.295 \cdot 867.429 \cdot 77.647 \\ &= 289282.0 \text{ N} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (729654 + 289282.0 + 35969.3) / 4.295 \\ &= 245612.7 \text{ mm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff}) \\ &= 206.995 / (2 \cdot 245612.688 / 7909.738 - 867.429 / 27.0) \\ &= 6.9 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S / t / R_{xs} \\ &= 137.997 [27.0 / 867.429] \\ &= 4.3 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(6.905, 4.295) \\ &= 4.295 \text{ MPa} \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (f_N + f_S + f_Y) / A_T \\ &= (35969.316 + 729654 + 289282.031) / 7909.738 \\ &= 133.379 \text{ MPa} \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot R_{xs} / t_{eff} \\ &= 4.295 \cdot 867.429 / 27.0 \\ &= 138.0 \text{ MPa} \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 133.379 - 137.997, 137.997) \\ &= 138.0 \text{ MPa} \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	207.00	MPa
Local Primary Membrane Stress	PL	138.00	MPa
Maximum Allowable Working Pressure	Pmax	4.30	MPa

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (R_{nc} + t_n) / R_{nc} \\ &= (77.647 + 44.0) / 77.647 \\ &= 1.567 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi / 2 \cdot (R_n + t_n)$$



$$= \pi/2 \cdot (77.647 + 44.0) \\ = 191.083 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:
= 7.071, 0.000, 0.000, mm

Required Weld Throat Dimensions :
tc, L41T = 7.071, must be $\geq \min(0.7 \cdot 44.000, 6.000)$ Passed

Weld Load Value [fwelds]:
= $\min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$
= $\min(289282 \cdot 1.57, 1.5 \cdot 147.6(3759.826 + 0.0), \pi/4 \cdot 4.3 \cdot 77.65^2 \cdot 1.57^2)$
= 49917.836 N

Weld Stress Value [τ]:
= $fwelds / (L_t(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$
= $49917.836 / (191.083(0.49 \cdot 7.071 + 0.6 \cdot 27.0 + 0.49 \cdot 0.0))$
= 13.286 < or = to 137.997 Weld Stress Passed

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:
= $t_n / \ln[1 + t_n/R_n]$
= $44.0 / \ln[1 + 44.0/77.647]$
= 98.006 mm

Shell Radius for Force Calculation [Rxs]:
= $t_{eff} / \ln[1 + t_{eff}/R_{eff}]$
= $27.0 / \ln[1 + 27.0/854.0]$
= 867.429 mm

Allowable Local Primary Membrane Stress [Sallow]:
= $1.5 \cdot S$
= $1.5 \cdot 137.997$
= 207.0 MPa

Force acting on the Nozzle [fN]:
= $P \cdot R_{xn} \cdot L_H$
= $0.103 \cdot 98.006 \cdot 85.451$
= 866.1 N

Force acting on the Shell [fS]:
= $P \cdot R_{xs}(L_R + t_n)$
= $0.103 \cdot 867.429(151.849 + 44.0)$
= 17568.8 N

Discontinuity Force from External Pressure [fY]:
= $P \cdot R_{xs} \cdot R_{nc}$
= $0.103 \cdot 867.429 \cdot 77.647$
= 6965.4 N

Area Resisting External Pressure [Ap]:
= $(f_S + f_Y + f_N)/P$
= $(17568.8 + 6965.4 + 866.1)/0.103$
= 245612.7 mm²

Maximum Allowable Working Pressure Candidate [Pmax1]:
= $S_{allow} / (2 \cdot A_p / A_T - R_{xs}/t_{eff})$
= $206.995 / (2 \cdot 245612.703/7909.738 - 867.429/27.0)$
= 6.9 MPa

Maximum Allowable Working Pressure Candidate [Pmax2]:
= $S/t/R_{xs}$
= $137.997[27.0/867.429]$
= 4.3 MPa

Maximum Allowable Working Pressure [Pmax]:
= $\min(P_{max1}, P_{max2})$
= $\min(6.905, 4.295)$
= 4.295 MPa



Average Primary Membrane Stress [σ_{Avg}]:
 $= (fN + fS + fY) / AT$
 $= (866.08 + 17568.834 + 6965.421) / 7909.738$
 $= 3.212 \text{ MPa}$

General Primary Membrane Stress [σ_{Circ}]:
 $= P \cdot Rxs / teff$
 $= 0.103 \cdot 867.429 / 27.0$
 $= 3.3 \text{ MPa}$

Maximum Local Primary Membrane Stress [PL]:
 $= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$
 $= \max(2 \cdot 3.212 - 3.323, 3.323)$
 $= 3.3 \text{ MPa}$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	207.00	MPa
Local Primary Membrane Stress	PL	3.32	MPa
Maximum Allowable External Pressure	Pmax	4.30	MPa

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Flange MDMT including temperature reduction per 3.11:
 Flange MDMT with Temp reduction -63 °C

The Drop for this Nozzle is : 8.7187 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 336.7187 mm

Input Echo, WRC297 Item 1, Description: A150 :

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	1.0000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	1706.000	mm	
Shell Thickness	28.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-516 70		
Vessel Cold S.I. Allowable	Smc	138.00	MPa
Vessel Hot S.I. Allowable	Smh	138.00	MPa
Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance	6.0000	mm	
Nozzle Diameter	143.294	mm	
Nozzle Thickness	50.0000	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-105		
Nozzle Cold S.I. Allowable	SNmc	165.00	MPa
Nozzle Hot S.I. Allowable	SNmh	147.60	MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUStained loads.

Design Internal Pressure	Dp	4.30	MPa
Radial Load	P	-11245.70	N
Circumferential Shear	Vc	7497.10	N



Longitudinal Shear	Vl	7497.10	N
Circumferential Moment	Mc	0.315E+07	N-mm
Longitudinal Moment	Ml	0.315E+07	N-mm
Torsional Moment	Mt	0.450E+07	N-mm

Include Axial Pressure Thrust	Yes
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (4.866) is < 10
 check the Limitations of WRC bulletin 297.*

Note: The radial load includes the effect of the pressure thrust force (Pressure times Area). This may lead to overly conservative results in some cases.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 243.294 + 2 \cdot 1.65 \cdot \sqrt{(867.5(28.0 - 1.0))}$
 $= 748.340 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1735.000 mm
Nozzle Outside Diameter	(d) =	243.294 mm
Vessel Thickness used	(T) =	27.000 mm
Nozzle Thickness used	(t) =	44.000 mm
T / t	=	0.614
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.124$$

Nr/P	=	0.103
Mr/P	=	0.200
M0/P	=	0.061
N0/P	=	0.125
MrD/Mc	=	0.299
NrDL/Mc	=	0.105
M0d/Mc	=	0.097
N0DL/Mc	=	0.043
MrD/Ml	=	0.182
NrDL/Ml	=	0.090
M0D/Ml	=	0.061
N0DL/Ml	=	0.125

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	15	15	15	15
Outplane Bending (P)	46	-46	46	-46
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-2	-2	2	2
Outplane Bending (Ml)	-6	6	6	-6
Normal Pressure Stress	133	138	133	138
Outplane Stress Summary	187	111	204	103

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	13	13	13	13
Inplane Bending (P)	152	-152	152	-152
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0

Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-19	19	19	-19
Inplane Pressure Stress	66	66	66	66
Inplane Stress Summary	211	-54	253	-90

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	1	1

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	211	166	253	193

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	15	15	15	15
Outplane Bending (P)	46	-46	46	-46
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-10	10	10	-10
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	66	66	66	66
Outplane Stress Summary	117	46	140	26

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	13	13	13	13
Inplane Bending (P)	152	-152	152	-152
Inplane Membrane (Mc)	-1	-1	1	1
Inplane Bending (Mc)	-31	31	31	-31
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	133	138	133	138
Inplane Stress Summary	265	28	333	-31

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1
Shear Stress Summary	1	1	2	2

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside



Two * Max Shear Stress 265 46 333 58

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	15	15	15	15
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-2	-2	2	2
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	7	12	7	12
Hoop Stress Summary	20	25	25	30

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	3	3	3	3
Axial Bending (P)	42	-42	42	-42
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-2	-2	2	2
Axial Bending (ML)	-5	5	5	-5
Axial Pressure Stress	3	3	3	3
Axial Stress Summary	41	-32	57	-38

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	1	1	0	0

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	41	58	57	68

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	15	15	15	15
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	7	12	7	12
Hoop Stress Summary	22	27	23	28

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	3	3	3	3
Axial Bending (P)	42	-42	42	-42



Axial Membrane (Mc)	-2	-2	2	2
Axial Bending (Mc)	-9	9	9	-9
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	3	3	3	3

Axial Stress Summary	37	-28	62	-42
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	0	0	1	1

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	37	55	62	71

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	133	138	133	138	133	138	133	138
Circ. Pl (SUS)	13	13	17	17	12	12	14	14
Circ. Q (SUS)	40	-40	52	-52	121	-121	183	-183
Long. Pm (SUS)	66	66	66	66	66	66	66	66
Long. Pl (SUS)	12	12	14	14	15	15	15	15
Long. Q (SUS)	133	-133	171	-171	36	-36	56	-56
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	133.0	138.0	133.0	138.0	133.0	138.0	133.0	138.0
Pm+Pl (SUS)	146.0	151.0	150.0	155.0	145.0	150.0	147.0	152.0
Pm+Pl+Q (Total)	211.0	166.0	251.0	194.0	266.0	45.1	330.0	56.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	138.00	138.00	Failed
Pm+Pl (SUS)	155.00	207.00	Passed
Pm+Pl+Q (TOTAL)	330.01	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl



Circ. Pm (SUS)	7	12	7	12	7	12	7	12
Circ. Pl (SUS)	13	13	17	17	15	15	15	15
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	3	3	3	3	3	3	3	3
Long. Pl (SUS)	1	1	5	5	1	1	5	5
Long. Q (SUS)	37	-37	47	-47	33	-33	51	-51
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	7.0	12.0	7.0	12.0	7.0	12.0	7.0	12.0
Pm+Pl (SUS)	20.0	25.0	24.0	29.0	22.0	27.0	22.0	27.0
Pm+Pl+Q (Total)	41.0	58.0	55.0	68.0	37.1	56.0	59.0	70.0

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	12.00	147.60	Passed
Pm+Pl (SUS)	29.00	221.40	Passed
Pm+Pl+Q (TOTAL)	70.03	442.80	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: MH2600

From: 90

Pressure for Reinforcement Calculations	P	4.2954	MPa
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	1706.00	mm
Design Length of Section	L	18483.8359	mm
Shell Finished (Minimum) Thickness	t	28.0000	mm
Shell Internal Corrosion Allowance	c	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		14750.00	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.60	MPa
Allowable Stress At Ambient	Sna	165.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		568.0000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm

Flange Material		SA-105	+IT
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Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	64.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		80.0000	mm

Corrosion Allowance	can	6.0000	mm
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Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	28.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

This is a Manway, Coupling or Access Opening.

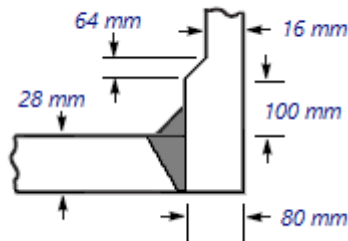
Flange Class		300
Flange Grade		GR 1.1

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: MH2600

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	568.000	mm.
Actual Thickness Used in Calculation	16.000	mm



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 580.0 \cdot (\exp(4.295/(147.6 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 8.5634 + 6.0 + 0.0 = 14.5634 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 147.6 \cdot 1.0 \cdot \ln((2 \cdot 10.0 + 580.0) / 580.0) - 0.0$$

$$= 5.004 - 0.0 = 5.0039 \text{ MPa}$$

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[147.6/138.0, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 80.0 - 6.0$$

$$= 74.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 16.0 - 6.0$$

$$= 10.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 1708.0/27.0$$

$$= 63.259 \text{ must be less than } 400.$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 1706.0/2 + 1.0$$

$$= 854.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{854.0 \cdot 27.0}, 2 \cdot 290.0)$$

$$= 151.849 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$



$$= \min[1.5 \cdot 27.0, 0.0] + \sqrt{(290.0 \cdot 74.0)}$$
$$= 146.492 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$
$$= 500.0$$
$$= 500.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$
$$= 8(27.0 + 0.0)$$
$$= 216.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles}$$
$$= \min[146.492, 500.0, 216.0] + 27.0$$
$$= 173.492 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$
$$= 27.0 + ((0.0 \cdot 1.0) / 151.849)$$
$$= 27.000 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}}))$$
$$= \min(12, (2 \cdot 290.0 + 74.0) / (\sqrt{(1708.0 + 27.0) \cdot 27.0}))$$
$$= 3.022$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$
$$= 27.0 \cdot 151.849 \cdot \max((3.022/5)^{0.85}, 1)$$
$$= 4099.913 \text{ mm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A2A + A2B$$
$$= 9398.0 + 2359.664$$
$$= 11757.663 \text{ mm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$
$$= 0.5 \cdot 10.0^2$$
$$= 50.000 \text{ mm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$
$$= 4099.913 + 1.0(11757.663 + 0.0) + 50.0 + 0.0 + 0.0$$
$$= 15907.575 \text{ mm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$
$$= 74.0 / \ln[1 + 74.0/290.0]$$
$$= 325.600 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$
$$= 27.0 / \ln[1 + 27.0/854.0]$$
$$= 867.429 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$
$$= 1.5 \cdot 137.997 \cdot 1.0$$
$$= 207.0 \text{ MPa}$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$
$$= 4.295 \cdot 325.6 \cdot 173.492$$
$$= 242620.2 \text{ N}$$



Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot Rxs(LR + tn) \\ &= 4.295 \cdot 867.429(151.849 + 74.0) \\ &= 841422.1 \text{ N} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot Rxs \cdot Rnc \\ &= 4.295 \cdot 867.429 \cdot 290.0 \\ &= 1080424.6 \text{ N} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN)/P \\ &= (841422 + 1080425 + 242620.2)/4.295 \\ &= 503951.0 \text{ mm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / (2 \cdot Ap/AT - Rxs/teff) \\ &= 206.995 / (2 \cdot 503951/15907.575 - 867.429/27.0) \\ &= 6.6 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t/Rxs] \\ &= 137.997[27.0/867.429] \\ &= 4.3 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(Pmax1, Pmax2) \\ &= \min(6.627, 4.295) \\ &= 4.295 \text{ MPa} \end{aligned}$$

Average Primary Membrane Stress [σAvg]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (242620.219 + 841422 + 1080425)/15907.575 \\ &= 136.077 \text{ MPa} \end{aligned}$$

General Primary Membrane Stress [σCirc]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 4.295 \cdot 867.429/27.0 \\ &= 138.0 \text{ MPa} \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ) \\ &= \max(2 \cdot 136.077 - 137.997, 137.997) \\ &= 138.0 \text{ MPa} \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	207.00	MPa
Local Primary Membrane Stress	PL	138.00	MPa
Maximum Allowable Working Pressure	Pmax	4.30	MPa

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (Rnc + tn) / Rnc \\ &= (290.0 + 74.0)/290.0 \\ &= 1.255 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned} &= \pi/2 \cdot (Rn + tn) \\ &= \pi/2 \cdot (290.0 + 74.0) \\ &= 571.770 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 74.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2)$$



$$= \min(1080425 \cdot 1.26, 1.5 \cdot 147.6(11757.663 + 0.0), \pi/4 \cdot 4.3 \cdot 290.0^2 \cdot 1.26^2)$$
$$= 446946.344 \text{ N}$$

Weld Stress Value [τ]:

$$= \text{fwelds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$$
$$= 446946.344 / (571.77(0.49 \cdot 7.071 + 0.6 \cdot 27.0 + 0.49 \cdot 0.0))$$
$$= 39.754 < \text{or} = \text{to } 137.997 \text{ Weld Stress Passed}$$

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [R_{xn}]:

$$= t_n / \ln[1 + t_n/R_n]$$
$$= 74.0 / \ln[1 + 74.0/290.0]$$
$$= 325.600 \text{ mm}$$

Shell Radius for Force Calculation [R_{xs}]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$
$$= 27.0 / \ln[1 + 27.0/854.0]$$
$$= 867.429 \text{ mm}$$

Allowable Local Primary Membrane Stress [S_{allow}]:

$$= 1.5 \cdot S$$
$$= 1.5 \cdot 137.997$$
$$= 207.0 \text{ MPa}$$

Force acting on the Nozzle [f_N]:

$$= P \cdot R_{xn} \cdot L_H$$
$$= 0.103 \cdot 325.6 \cdot 173.492$$
$$= 5841.9 \text{ N}$$

Force acting on the Shell [f_S]:

$$= P \cdot R_{xs} (L_R + t_n)$$
$$= 0.103 \cdot 867.429 (151.849 + 74.0)$$
$$= 20260.0 \text{ N}$$

Discontinuity Force from External Pressure [f_Y]:

$$= P \cdot R_{xs} \cdot R_{nc}$$
$$= 0.103 \cdot 867.429 \cdot 290.0$$
$$= 26014.8 \text{ N}$$

Area Resisting External Pressure [A_p]:

$$= (f_S + f_Y + f_N) / P$$
$$= (20260.0 + 26014.8 + 5841.9) / 0.103$$
$$= 503950.9 \text{ mm}^2$$

Maximum Allowable Working Pressure Candidate [P_{max1}]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$
$$= 206.995 / (2 \cdot 503951 / 15907.575 - 867.429 / 27.0)$$
$$= 6.6 \text{ MPa}$$

Maximum Allowable Working Pressure Candidate [P_{max2}]:

$$= S[t / R_{xs}]$$
$$= 137.997 [27.0 / 867.429]$$
$$= 4.3 \text{ MPa}$$

Maximum Allowable Working Pressure [P_{max}]:

$$= \min(P_{max1}, P_{max2})$$
$$= \min(6.627, 4.295)$$
$$= 4.295 \text{ MPa}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / A_T$$
$$= (5841.884 + 20260.02 + 26014.795) / 15907.575$$
$$= 3.276 \text{ MPa}$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$



= $0.103 \cdot 867.429/27.0$
= 3.3 MPa

Maximum Local Primary Membrane Stress [PL]:

= $\max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$
= $\max(2 \cdot 3.276 - 3.323, 3.323)$
= 3.3 MPa

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	207.00	MPa
Local Primary Membrane Stress	PL	3.32	MPa
Maximum Allowable External Pressure	Pmax	4.30	MPa

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

[Note: This Element/Detail was specified as being Impact Tested.](#)

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

[Note: This Element/Detail was specified as being Impact Tested.](#)

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -27 °C

The Drop for this Nozzle is : 81.5644 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 609.5644 mm

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Input, Nozzle Desc: E40

From: 100

Pressure for Reinforcement Calculations	P	4.2954	MPa
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	1706.00	mm
Design Length of Section	L	18483.8359	mm
Shell Finished (Minimum) Thickness	t	28.0000	mm
Shell Internal Corrosion Allowance	c	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		15950.00	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.60	MPa
Allowable Stress At Ambient	Sna	165.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		40.0000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm

Flange Material		SA-105	+IT
Flange Type		N/A	

Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	40.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Style of FVC Nozzle		Long Weld Neck	

Corrosion Allowance	can	6.0000	mm
---------------------	-----	--------	----

Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	28.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

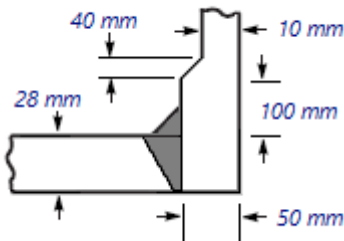
Flange Class		900
Flange Grade		GR 1.1

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: E40

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		40.000	mm.
Actual Thickness Used in Calculation		10.000	mm



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 52.0 \cdot (\exp(4.295/(147.6 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 0.7678 + 6.0 + 0.0 = 6.7678 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 147.6 \cdot 1.0 \cdot \ln((2 \cdot 4.0 + 52.0) / 52.0) - 0.0$$

$$= 21.122 - 0.0 = 21.1217 \text{ MPa}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	9.420 mm
Min. required shell thickness, internal p + c :	28.002 mm
Min. required shell thickness, external p + c :	16.233 mm

Nozzle neck minimum required thickness : 9.420 mm

Minimum nozzle neck thickness 10.008 mm , is greater than required thickness 9.420 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained :	61.2,	Allowable :	147.6 MPa	Passed
Expansion :	0.0,	Allowable :	329.6 MPa	Passed
Occasional :	13.0,	Allowable :	196.3 MPa	Passed
Shear :	40.2,	Allowable :	103.3 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[147.6/138.0, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 50.0 - 6.0$$

$$= 44.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 10.0 - 6.0$$

$$= 4.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 1708.0/27.0$$

$$= 63.259 \text{ must be less than } 400.$$

Effective Pressure Radius [Reff]:



$$\begin{aligned} &= Di/2 + \text{corrosion allowance} \\ &= 1706.0/2 + 1.0 \\ &= 854.000 \text{ mm} \end{aligned}$$

Effective Length of Vessel Wall [LR]:

$$\begin{aligned} &= \min(\sqrt{Reff \cdot t}, 2 \cdot Rn) \\ &= \min(\sqrt{854.0 \cdot 27.0}, 2 \cdot 26.0) \\ &= 52.000 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH1]:

$$\begin{aligned} &= \min[1.5t, te] + \sqrt{Rn \cdot tn} \\ &= \min[1.5 \cdot 27.0, 0.0] + \sqrt{26.0 \cdot 44.0} \\ &= 33.823 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= Lpr1 \\ &= 300.0 \\ &= 300.000 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + te) \\ &= 8(27.0 + 0.0) \\ &= 216.000 \text{ mm} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles} \\ &= \min[33.823, 300.0, 216.0] + 27.0 \\ &= 60.823 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot frp) / LR) \\ &= 27.0 + ((0.0 \cdot 1.0) / 52.0) \\ &= 27.000 \text{ mm} \end{aligned}$$

Parameter [λ]:

$$\begin{aligned} &= \min(12, (2 \cdot Rn + tn) / (\sqrt{(Di + teff) \cdot teff})) \\ &= \min(12, (2 \cdot 26.0 + 44.0) / (\sqrt{(1708.0 + 27.0) \cdot 27.0})) \\ &= 0.444 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ &= 27.0 \cdot 52.0 \cdot \max((0.444/5)^{0.85}, 1) \\ &= 1404.000 \text{ mm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= tn \cdot LH \\ &= 44.0 \cdot 60.823 \\ &= 2676.215 \text{ mm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot Leg^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 50.000 \text{ mm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 \\ &= 1404.0 + 1.0(2676.215 + 0.0) + 50.0 + 0.0 + 0.0 \\ &= 4130.215 \text{ mm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= tn / \ln[1 + tn/Rn] \\ &= 44.0 / \ln[1 + 44.0/26.0] \\ &= 44.427 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= teff / \ln[1 + teff/Reff] \\ &= 27.0 / \ln[1 + 27.0/854.0] \\ &= 867.429 \text{ mm} \end{aligned}$$



Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 137.997 \cdot 1.0 \\ &= 207.0 \text{ MPa} \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot L_H \\ &= 4.295 \cdot 44.427 \cdot 60.823 \\ &= 11605.8 \text{ N} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot R_{xs} (L_R + t_n) \\ &= 4.295 \cdot 867.429 (52.0 + 44.0) \\ &= 357657.8 \text{ N} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 4.295 \cdot 867.429 \cdot 26.0 \\ &= 96865.7 \text{ N} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (357657.8 + 96865.7 + 11605.8) / 4.295 \\ &= 108528.5 \text{ mm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff}) \\ &= 206.995 / (2 \cdot 108528.469 / 4130.215 - 867.429 / 27.0) \\ &= 10.1 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S / t / R_{xs} \\ &= 137.997 [27.0 / 867.429] \\ &= 4.3 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(10.134, 4.295) \\ &= 4.295 \text{ MPa} \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (f_N + f_S + f_Y) / A_T \\ &= (11605.761 + 357657.812 + 96865.656) / 4130.215 \\ &= 112.868 \text{ MPa} \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot R_{xs} / t_{eff} \\ &= 4.295 \cdot 867.429 / 27.0 \\ &= 138.0 \text{ MPa} \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 112.868 - 137.997, 137.997) \\ &= 138.0 \text{ MPa} \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	207.00	MPa
Local Primary Membrane Stress	PL	138.00	MPa
Maximum Allowable Working Pressure	Pmax	4.30	MPa

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (R_{nc} + t_n) / R_{nc} \\ &= (26.0 + 44.0) / 26.0 \\ &= 2.692 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi / 2 \cdot (R_n + t_n)$$



$$= \pi/2 \cdot (26.0 + 44.0) \\ = 109.956 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:
= 7.071, 0.000, 0.000, mm

Required Weld Throat Dimensions :
tc, L41T = 7.071, must be $\geq \min(0.7 \cdot 44.000, 6.000)$ Passed

Weld Load Value [fwelds]:
= $\min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$
= $\min(96866 \cdot 2.69, 1.5 \cdot 147.6(2676.215 + 0.0), \pi/4 \cdot 4.3 \cdot 26.0^2 \cdot 2.69^2)$
= 16529.082 N

Weld Stress Value [τ]:
= $\text{fwelds} / (L_{\tau}(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$
= $16529.082 / (109.956(0.49 \cdot 7.071 + 0.6 \cdot 27.0 + 0.49 \cdot 0.0))$
= 7.645 < or = to 137.997 Weld Stress Passed

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:
= $t_n / \ln[1 + t_n/R_n]$
= $44.0 / \ln[1 + 44.0/26.0]$
= 44.427 mm

Shell Radius for Force Calculation [Rxs]:
= $t_{eff} / \ln[1 + t_{eff}/R_{eff}]$
= $27.0 / \ln[1 + 27.0/854.0]$
= 867.429 mm

Allowable Local Primary Membrane Stress [Sallow]:
= $1.5 \cdot S$
= $1.5 \cdot 137.997$
= 207.0 MPa

Force acting on the Nozzle [fN]:
= $P \cdot R_{xn} \cdot L_H$
= $0.103 \cdot 44.427 \cdot 60.823$
= 279.4 N

Force acting on the Shell [fS]:
= $P \cdot R_{xs}(L_R + t_n)$
= $0.103 \cdot 867.429(52.0 + 44.0)$
= 8611.8 N

Discontinuity Force from External Pressure [fY]:
= $P \cdot R_{xs} \cdot R_{nc}$
= $0.103 \cdot 867.429 \cdot 26.0$
= 2332.4 N

Area Resisting External Pressure [Ap]:
= $(f_S + f_Y + f_N)/P$
= $(8611.8 + 2332.4 + 279.4)/0.103$
= 108528.5 mm²

Maximum Allowable Working Pressure Candidate [Pmax1]:
= $S_{allow} / (2 \cdot A_p / A_T - R_{xs}/t_{eff})$
= $206.995 / (2 \cdot 108528.469 / 4130.215 - 867.429/27.0)$
= 10.1 MPa

Maximum Allowable Working Pressure Candidate [Pmax2]:
= $S/[t/R_{xs}]$
= $137.997[27.0/867.429]$
= 4.3 MPa

Maximum Allowable Working Pressure [Pmax]:
= $\min(P_{max1}, P_{max2})$
= $\min(10.134, 4.295)$
= 4.295 MPa



Average Primary Membrane Stress [σ_{Avg}]:
 $= (fN + fS + fY) / AT$
 $= (279.447 + 8611.794 + 2332.361) / 4130.215$
 $= 2.718 \text{ MPa}$

General Primary Membrane Stress [σ_{Circ}]:
 $= P \cdot Rxs / teff$
 $= 0.103 \cdot 867.429 / 27.0$
 $= 3.3 \text{ MPa}$

Maximum Local Primary Membrane Stress [PL]:
 $= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$
 $= \max(2 \cdot 2.718 - 3.323, 3.323)$
 $= 3.3 \text{ MPa}$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	207.00	MPa
Local Primary Membrane Stress	PL	3.32	MPa
Maximum Allowable External Pressure	Pmax	4.30	MPa

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

The Drop for this Nozzle is : 2.8771 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 330.8771 mm

Input Echo, WRC297 Item 1, Description: E40 :

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	1.0000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	1706.000	mm	
Shell Thickness	28.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-516 70		
Vessel Cold S.I. Allowable	Smc	138.00	MPa
Vessel Hot S.I. Allowable	Smh	138.00	MPa
Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance	6.0000	mm	
Nozzle Diameter	40.000	mm	
Nozzle Thickness	50.0000	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-105		
Nozzle Cold S.I. Allowable	SNmc	165.00	MPa
Nozzle Hot S.I. Allowable	SNmh	147.60	MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUstained loads.

Design Internal Pressure	Dp	4.30	MPa
Radial Load	P	-3748.57	N
Circumferential Shear	Vc	2499.04	N
Longitudinal Shear	Vl	2499.04	N
Circumferential Moment	Mc	350017.62	N-mm
Longitudinal Moment	Ml	350017.62	N-mm
Torsional Moment	Mt	500025.78	N-mm



Include Axial Pressure Thrust No
 Include Pressure Stress Indices per Div. 2 No
 Local Loads applied at end of Nozzle/Attachment No

*Warning - The ratio, Dn/Tn (2.800) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 140.0 + 2 \cdot 1.65 \cdot \sqrt{(867.5(28.0 - 1.0))}$$

$$= 645.046 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1735.000 mm
 Nozzle Outside Diameter (d) = 140.000 mm
 Vessel Thickness used (T) = 27.000 mm
 Nozzle Thickness used (t) = 44.000 mm
 T / t = 0.614
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.647$$

Nr/P = 0.134
 Mr/P = 0.277
 M0/P = 0.085
 N0/P = 0.157
 MrD/Mc = 0.307
 NrDL/Mc = 0.097
 M0d/Mc = 0.103
 N0DL/Mc = -0.035
 MrD/Ml = 0.246
 NrDL/Ml = 0.099
 M0D/Ml = 0.083
 N0DL/Ml = 0.071

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	-1	1	1	-1
Normal Pressure Stress	133	138	133	138
Outplane Stress Summary	135	137	139	134

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	8	-8	8	-8
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-5	5	5	-5
Inplane Pressure Stress	66	66	66	66
Inplane Stress Summary	70	63	81	54

Vessel Stresses (MPa):

LONGITUDINAL PLANE			Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)			Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			0	0	0	0
Outplane Shear (Mt)			0	0	0	0
Shear Stress Summary			1	1	0	0

Vessel Stresses (MPa):

LONGITUDINAL PLANE			Au	Al	Bu	B1
(Stress Intensities)			Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress			135	137	139	134

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Stresses Normal to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)			0	0	0	0
Outplane Bending (P)			2	-2	2	-2
Outplane Membrane (Mc)			0	0	0	0
Outplane Bending (Mc)			-2	2	2	-2
Outplane Membrane (ML)			0	0	0	0
Outplane Bending (ML)			0	0	0	0
Normal Pressure Stress			66	66	66	66
Outplane Stress Summary			68	67	72	62

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Stresses parallel to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)			0	0	0	0
Inplane Bending (P)			8	-8	8	-8
Inplane Membrane (Mc)			0	0	0	0
Inplane Bending (Mc)			-6	6	6	-6
Inplane Membrane (ML)			0	0	0	0
Inplane Bending (ML)			0	0	0	0
Inplane Pressure Stress			133	138	133	138
Inplane Stress Summary			136	136	149	124

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Shear stress normal to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			0	0	0	0
Torsional Shear (Mt)			0	0	0	0
Shear Stress Summary			0	0	1	1

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Stress Intensities)			Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress			136	136	149	124

Nozzle Stresses (MPa):

LONGITUDINAL PLANE			Au	Al	Bu	B1
(Stresses in the hoop direction)			Top Outside	Top Inside	Bottom Outside	Bottom Inside



Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	1	6	1	6

Hoop Stress Summary	2	7	2	7
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	2	-2	2	-2
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-1	-1	1	1
Axial Bending (ML)	-1	1	1	-1
Axial Pressure Stress	0	0	0	0

Axial Stress Summary	0	-1	6	-1
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	0	0	0	0
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	2	8	6	9

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	1	6	1	6

Hoop Stress Summary	2	7	2	7
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	2	-2	2	-2
Axial Membrane (Mc)	-1	-1	1	1
Axial Bending (Mc)	-1	1	1	-1
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	0	0	0	0

Axial Stress Summary	0	0	6	-2
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	2	8	6	9

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	133	138	133	138	133	138	133	138
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	1	-1	3	-3	2	-2	14	-14
Long. Pm (SUS)	66	66	66	66	66	66	66	66
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	3	-3	13	-13	0	0	4	-4
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	133.0	138.0	133.0	138.0	133.0	138.0	133.0	138.0
Pm+Pl (SUS)	133.0	138.0	133.0	138.0	133.0	138.0	133.0	138.0
Pm+Pl+Q (Total)	134.0	137.0	136.0	135.0	135.0	136.0	147.0	124.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	138.00	138.00	Failed
Pm+Pl (SUS)	138.00	207.00	Passed
Pm+Pl+Q (TOTAL)	147.00	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1	6	1	6	1	6	1	6
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	0	0	0	0	0	0	0	0
Long. Pl (SUS)	-1	-1	1	1	-1	-1	1	1
Long. Q (SUS)	1	-1	3	-3	1	-1	3	-3



Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	1.0	6.0	1.0	6.0	1.0	6.0	1.0	6.0
Pm+Pl (SUS)	2.0	7.0	1.0	6.0	2.0	7.0	1.0	6.0
Pm+Pl+Q (Total)	1.0	8.0	4.0	8.0	1.0	8.0	4.0	8.0

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	6.00	147.60	Passed
Pm+Pl (SUS)	7.00	221.40	Passed
Pm+Pl+Q (TOTAL)	8.00	442.80	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: MH1600

From: 100

Pressure for Reinforcement Calculations	P	4.2954	MPa
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	1706.00	mm
Design Length of Section	L	18483.8359	mm
Shell Finished (Minimum) Thickness	t	28.0000	mm
Shell Internal Corrosion Allowance	c	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		16600.00	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.60	MPa
Allowable Stress At Ambient	Sna	165.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		568.0000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm

Flange Material		SA-105	+IT
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Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	64.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		80.0000	mm

Corrosion Allowance	can	6.0000	mm
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Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		28.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

This is a Manway, Coupling or Access Opening.

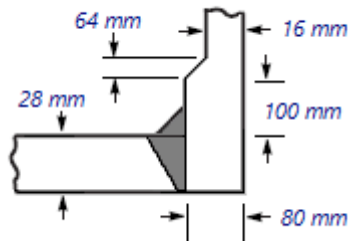
Flange Class		300
Flange Grade		GR 1.1

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: MH1600

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	568.000	mm.
Actual Thickness Used in Calculation	16.000	mm



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 580.0 \cdot (\exp(4.295/(147.6 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 8.5634 + 6.0 + 0.0 = 14.5634 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 147.6 \cdot 1.0 \cdot \ln((2 \cdot 10.0 + 580.0) / 580.0) - 0.0$$

$$= 5.004 - 0.0 = 5.0039 \text{ MPa}$$

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[147.6/138.0, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 80.0 - 6.0$$

$$= 74.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 16.0 - 6.0$$

$$= 10.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 1708.0/27.0$$

$$= 63.259 \text{ must be less than } 400.$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 1706.0/2 + 1.0$$

$$= 854.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{854.0 \cdot 27.0}, 2 \cdot 290.0)$$

$$= 151.849 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$



$$= \min[1.5 \cdot 27.0, 0.0] + \sqrt{(290.0 \cdot 74.0)}$$
$$= 146.492 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$
$$= 500.0$$
$$= 500.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$
$$= 8(27.0 + 0.0)$$
$$= 216.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles}$$
$$= \min[146.492, 500.0, 216.0] + 27.0$$
$$= 173.492 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$
$$= 27.0 + ((0.0 \cdot 1.0) / 151.849)$$
$$= 27.000 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + teff) \cdot teff}))$$
$$= \min(12, (2 \cdot 290.0 + 74.0) / (\sqrt{(1708.0 + 27.0) \cdot 27.0}))$$
$$= 3.022$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$
$$= 27.0 \cdot 151.849 \cdot \max((3.022/5)^{0.85}, 1)$$
$$= 4099.913 \text{ mm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A2A + A2B$$
$$= 9398.0 + 2359.664$$
$$= 11757.663 \text{ mm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot Leg_1^2$$
$$= 0.5 \cdot 10.0^2$$
$$= 50.000 \text{ mm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$
$$= 4099.913 + 1.0(11757.663 + 0.0) + 50.0 + 0.0 + 0.0$$
$$= 15907.575 \text{ mm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$
$$= 74.0 / \ln[1 + 74.0/290.0]$$
$$= 325.600 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= teff / \ln[1 + teff/Reff]$$
$$= 27.0 / \ln[1 + 27.0/854.0]$$
$$= 867.429 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$
$$= 1.5 \cdot 137.997 \cdot 1.0$$
$$= 207.0 \text{ MPa}$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$
$$= 4.295 \cdot 325.6 \cdot 173.492$$
$$= 242620.2 \text{ N}$$



Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot Rxs(LR + tn) \\ &= 4.295 \cdot 867.429(151.849 + 74.0) \\ &= 841422.1 \text{ N} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot Rxs \cdot Rnc \\ &= 4.295 \cdot 867.429 \cdot 290.0 \\ &= 1080424.6 \text{ N} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN)/P \\ &= (841422 + 1080425 + 242620.2)/4.295 \\ &= 503951.0 \text{ mm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / (2 \cdot Ap/AT - Rxs/teff) \\ &= 206.995 / (2 \cdot 503951/15907.575 - 867.429/27.0) \\ &= 6.6 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t/Rxs] \\ &= 137.997[27.0/867.429] \\ &= 4.3 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(Pmax1, Pmax2) \\ &= \min(6.627, 4.295) \\ &= 4.295 \text{ MPa} \end{aligned}$$

Average Primary Membrane Stress [σAvg]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (242620.219 + 841422 + 1080425)/15907.575 \\ &= 136.077 \text{ MPa} \end{aligned}$$

General Primary Membrane Stress [σCirc]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 4.295 \cdot 867.429/27.0 \\ &= 138.0 \text{ MPa} \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ) \\ &= \max(2 \cdot 136.077 - 137.997, 137.997) \\ &= 138.0 \text{ MPa} \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	207.00	MPa
Local Primary Membrane Stress	PL	138.00	MPa
Maximum Allowable Working Pressure	Pmax	4.30	MPa

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (Rnc + tn) / Rnc \\ &= (290.0 + 74.0)/290.0 \\ &= 1.255 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned} &= \pi/2 \cdot (Rn + tn) \\ &= \pi/2 \cdot (290.0 + 74.0) \\ &= 571.770 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 74.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2)$$



$$= \min(1080425 \cdot 1.26, 1.5 \cdot 147.6(11757.663 + 0.0), \pi/4 \cdot 4.3 \cdot 290.0^2 \cdot 1.26^2)$$
$$= 446946.344 \text{ N}$$

Weld Stress Value [τ]:

$$= \text{fwelds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$$
$$= 446946.344 / (571.77(0.49 \cdot 7.071 + 0.6 \cdot 27.0 + 0.49 \cdot 0.0))$$
$$= 39.754 < \text{or} = \text{to } 137.997 \text{ Weld Stress Passed}$$

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [R_{xn}]:

$$= t_n / \ln[1 + t_n/R_n]$$
$$= 74.0 / \ln[1 + 74.0/290.0]$$
$$= 325.600 \text{ mm}$$

Shell Radius for Force Calculation [R_{xs}]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$
$$= 27.0 / \ln[1 + 27.0/854.0]$$
$$= 867.429 \text{ mm}$$

Allowable Local Primary Membrane Stress [S_{allow}]:

$$= 1.5 \cdot S$$
$$= 1.5 \cdot 137.997$$
$$= 207.0 \text{ MPa}$$

Force acting on the Nozzle [f_N]:

$$= P \cdot R_{xn} \cdot L_H$$
$$= 0.103 \cdot 325.6 \cdot 173.492$$
$$= 5841.9 \text{ N}$$

Force acting on the Shell [f_S]:

$$= P \cdot R_{xs} (L_R + t_n)$$
$$= 0.103 \cdot 867.429 (151.849 + 74.0)$$
$$= 20260.0 \text{ N}$$

Discontinuity Force from External Pressure [f_Y]:

$$= P \cdot R_{xs} \cdot R_{nc}$$
$$= 0.103 \cdot 867.429 \cdot 290.0$$
$$= 26014.8 \text{ N}$$

Area Resisting External Pressure [A_p]:

$$= (f_S + f_Y + f_N) / P$$
$$= (20260.0 + 26014.8 + 5841.9) / 0.103$$
$$= 503950.9 \text{ mm}^2$$

Maximum Allowable Working Pressure Candidate [P_{max1}]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$
$$= 206.995 / (2 \cdot 503951 / 15907.575 - 867.429 / 27.0)$$
$$= 6.6 \text{ MPa}$$

Maximum Allowable Working Pressure Candidate [P_{max2}]:

$$= S[t / R_{xs}]$$
$$= 137.997 [27.0 / 867.429]$$
$$= 4.3 \text{ MPa}$$

Maximum Allowable Working Pressure [P_{max}]:

$$= \min(P_{max1}, P_{max2})$$
$$= \min(6.627, 4.295)$$
$$= 4.295 \text{ MPa}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / A_T$$
$$= (5841.884 + 20260.02 + 26014.795) / 15907.575$$
$$= 3.276 \text{ MPa}$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$

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Input, Nozzle Desc: C100

From: 110

Pressure for Reinforcement Calculations	P	4.2954	MPa
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Elliptical Head	D	1706.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	26.0000	mm
Head Internal Corrosion Allowance	c	1.0000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	0.0000	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.60	MPa
Allowable Stress At Ambient	Sna	165.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		100.0000	mm

Size and Thickness Basis		Nominal	
Nominal Thickness		160	

Flange Material		SA-105	+IT
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Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	36.5126	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm

Corrosion Allowance	can	6.0000	mm
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Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	26.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

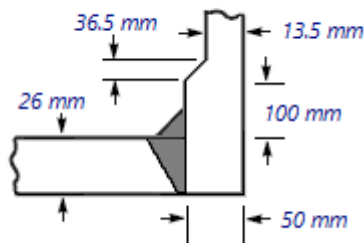
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: C100

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	87.325	mm.
Actual Thickness Used in Calculation	13.487	mm



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 99.325 \cdot (\exp(4.295/(147.6 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 1.4665 + 6.0 + 0.0 = 7.4665 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 147.6 \cdot 1.0 \cdot \ln((2 \cdot 7.487 + 99.325) / 99.325) - 0.0$$

$$= 20.727 - 0.0 = 20.7271 \text{ MPa}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	11.270 mm
Min. required shell thickness, internal p + c :	30.409 mm
Min. required shell thickness, external p + c :	6.078 mm

Nozzle neck minimum required thickness : 11.270 mm

Minimum nozzle neck thickness 11.811 mm , is greater than required thickness 11.270 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained :	54.8,	Allowable :	147.6 MPa	Passed
Expansion :	0.0,	Allowable :	335.9 MPa	Passed
Occasional :	17.4,	Allowable :	196.3 MPa	Passed
Shear :	30.0,	Allowable :	103.3 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[147.6/138.0, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 50.0 - 6.0$$

$$= 44.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 13.487 - 6.0$$

$$= 7.487 \text{ mm}$$

Effective Pressure Radius [Reff]:

$$= 0.9/6 \cdot D_i [2 + (D_i/(2h))^2]$$

$$= 0.9/6 \cdot 1708.0 [2 + (1708.0/(2 \cdot 427.5))^2]$$

$$= 1534.804 \text{ mm}$$

Effective Length of Vessel Wall [LR]:



$$= \min(\sqrt{(\text{Reff} \cdot t)}, 2 \cdot Rn)$$

$$= \min(\sqrt{(1534.804 \cdot 25.0)}, 2 \cdot 49.663)$$

$$= 99.325 \text{ mm}$$

Intermediate value [Cn]:

$$= \min((t + te)/tn)^{0.35}, 1)$$

$$= \min((25.0 + 0.0)/44.0)^{0.35}, 1)$$

$$= \min(0.82, 1)$$

$$= 0.820$$

Intermediate value [Cp]:

$$= \exp((0.35 \cdot di - X_o)/(16t))$$

$$= \exp((0.35 \cdot 1708.0 - 93.663)/(16 \cdot 25.0))$$

$$= 3.527$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min(t + te + Fp \cdot \sqrt{(Rn \cdot tn)}, Lpr1 + t)$$

$$= \min(25.0 + 0.0 + 0.82 \cdot \sqrt{(49.663 \cdot 44.0)}, 300.0 + 25.0)$$

$$= 63.354 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 25.0 + ((0.0 \cdot 1.0) / 99.325)$$

$$= 25.000 \text{ mm}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR$$

$$= 25.0 \cdot 99.325$$

$$= 2483.130 \text{ mm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= tn \cdot LH$$

$$= 44.0 \cdot 63.354$$

$$= 2787.580 \text{ mm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg41}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 50.000 \text{ mm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$

$$= 2483.13 + 1.0(2787.58 + 0.0) + 50.0 + 0.0 + 0.0$$

$$= 5320.710 \text{ mm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= tn / \ln[1 + tn/Rn]$$

$$= 44.0 / \ln[1 + 44.0/49.663]$$

$$= 69.352 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= teff / \ln[1 + teff/Reff]$$

$$= 25.0 / \ln[1 + 25.0/1534.804]$$

$$= 1547.267 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 137.997 \cdot 1.0$$

$$= 207.0 \text{ MPa}$$

Force acting on the Nozzle [fN]:

$$= P \cdot Rxn \cdot LH$$

$$= 4.295 \cdot 69.352 \cdot 63.354$$

$$= 18871.0 \text{ N}$$

Force acting on the Shell [fS]:

$$= P \cdot Rxs \cdot (LR + tn)/2$$

$$= 4.295 \cdot 1547.267 \cdot (99.325 + 44.0)/2$$



= 476234.2 N

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= (P \cdot Rxs \cdot Rnc)/2 \\ &= (4.295 \cdot 1547.267 \cdot 49.663)/2 \\ &= 165016.5 \text{ N} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN)/P \\ &= (476234.2 + 165016.5 + 18871.0)/4.295 \\ &= 153695.6 \text{ mm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / (2 \cdot Ap/AT - Rxs/(2 \cdot teff)) \\ &= 206.995 / (2 \cdot 153695.562/5320.71 - 1547.267/(2 \cdot 25.0)) \\ &= 7.7 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= 2 \cdot S[t/Rxs] \\ &= 2 \cdot 137.997[25.0/1547.267] \\ &= 4.5 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(Pmax1, Pmax2) \\ &= \min(7.716, 4.459) \\ &= 4.459 \text{ MPa} \end{aligned}$$

Average Primary Membrane Stress [σAvg]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (18870.996 + 476234.188 + 165016.547)/5320.71 \\ &= 124.077 \text{ MPa} \end{aligned}$$

General Primary Membrane Stress [σCirc]:

$$\begin{aligned} &= P \cdot Rxs / (2 \cdot teff) \\ &= 4.295 \cdot 1547.267 / (2 \cdot 25.0) \\ &= 132.9 \text{ MPa} \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ) \\ &= \max(2 \cdot 124.077 - 132.921, 132.921) \\ &= 132.9 \text{ MPa} \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	207.00	MPa
Local Primary Membrane Stress	PL	132.92	MPa
Maximum Allowable Working Pressure	Pmax	4.46	MPa

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (Rnc + tn) / Rnc \\ &= (49.663 + 44.0)/49.663 \\ &= 1.886 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned} &= \pi/2 \cdot (Rn + tn) \\ &= \pi/2 \cdot (49.663 + 44.0) \\ &= 147.125 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 44.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2) \\ &= \min(165017 \cdot 1.89, 1.5 \cdot 147.6(3156.808 + 0.0), \pi/4 \cdot 4.3 \cdot 49.66^2 \cdot 1.89^2) \\ &= 29592.730 \text{ N} \end{aligned}$$

Weld Stress Value [τ]:
= $f_{welds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$
= $29592.73 / (147.125(0.49 \cdot 7.071 + 0.6 \cdot 25.0 + 0.49 \cdot 0.0))$
= 10.894 < or = to 137.997 Weld Stress Passed

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [R_{xn}]:
= $t_n / \ln[1 + t_n/R_n]$
= $44.0 / \ln[1 + 44.0/49.663]$
= 69.352 mm

Shell Radius for Force Calculation [R_{xs}]:
= $t_{eff} / \ln[1 + t_{eff}/R_{eff}]$
= $25.0 / \ln[1 + 25.0/1534.804]$
= 1547.267 mm

Allowable Local Primary Membrane Stress [S_{allow}]:
= $1.5 \cdot S$
= $1.5 \cdot 137.997$
= 207.0 MPa

Force acting on the Nozzle [f_N]:
= $P \cdot R_{xn} \cdot L_H$
= $0.103 \cdot 69.352 \cdot 63.354$
= 454.4 N

Force acting on the Shell [f_S]:
= $P \cdot R_{xs} \cdot (L_R + t_n)/2$
= $0.103 \cdot 1547.267 \cdot (99.325 + 44.0)/2$
= 11466.9 N

Discontinuity Force from External Pressure [f_Y]:
= $(P \cdot R_{xs} \cdot R_{nc})/2$
= $(0.103 \cdot 1547.267 \cdot 49.663)/2$
= 3973.3 N

Area Resisting External Pressure [A_p]:
= $(f_S + f_Y + f_N)/P$
= $(11466.9 + 3973.3 + 454.4)/0.103$
= 153695.6 mm²

Maximum Allowable Working Pressure Candidate [P_{max1}]:
= $S_{allow} / (2 \cdot A_p/AT - R_{xs}/(2 \cdot t_{eff}))$
= $206.995 / (2 \cdot 153695.578/5320.71 - 1547.267/(2 \cdot 25.0))$
= 7.7 MPa

Maximum Allowable Working Pressure Candidate [P_{max2}]:
= $2 \cdot S[t/R_{xs}]$
= $2 \cdot 137.997[25.0/1547.267]$
= 4.5 MPa

Maximum Allowable Working Pressure [P_{max}]:
= $\min(P_{max1}, P_{max2})$
= $\min(7.716, 4.459)$
= 4.459 MPa

Average Primary Membrane Stress [σ_{Avg}]:
= $(f_N + f_S + f_Y) / AT$
= $(454.382 + 11466.913 + 3973.319) / 5320.71$
= 2.988 MPa

General Primary Membrane Stress [σ_{Circ}]:
= $P \cdot R_{xs} / (2 \cdot t_{eff})$
= $0.103 \cdot 1547.267 / (2 \cdot 25.0)$
= 3.2 MPa

Maximum Local Primary Membrane Stress [P_L]:
= $\max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$
= $\max(2 \cdot 2.988 - 3.201, 3.201)$



= 3.2 MPa

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	207.00	MPa
Local Primary Membrane Stress	PL	3.20	MPa
Maximum Allowable External Pressure	Pmax	4.46	MPa

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -63 °C

The Drop for this Nozzle is : 2.8211 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 328.8211 mm

Input Echo, WRC107/537 Item 1, Description: C100 :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	1.0000	mm
Vessel Diameter	Dv	3070.800	mm
Vessel Thickness	Tv	26.000	mm
Design Temperature	T1	130.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	MPa
Vessel Hot S.I. Allowable	Smh	138.00	MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	6.0000	mm
Nozzle Diameter	Dn	87.325	mm
Nozzle Thickness	Tn	50.000	mm
Nozzle Material		SA-105	
Nozzle UNS Number		K03504	
Nozzle Cold S.I. Allowable	SNmc	165.00	MPa
Nozzle Hot S.I. Allowable	SNmh	147.60	MPa
Design Internal Pressure	Dp	4.295	MPa
Include Pressure Thrust		No	

External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-7497.1	N
Longitudinal Shear (SUS)	(V1) V1	4998.1	N
Circumferential Shear (SUS)	(Vc) V2	4998.1	N
Circumferential Moment (SUS)	(Mc) M1	1400067.9	N-mm
Longitudinal Moment (SUS)	(Ml) M2	1400067.9	N-mm
Torsional Moment (SUS)	Mt	2000095.1	N-mm

Use Interactive Control No
 WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No

No

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9



Rad. Bend. P	6.9	-6.9	6.9	-6.9	6.9	-6.9	6.9	-6.9
Rad. Memb. MC	0.0	0.0	0.0	0.0	-1.3	-1.3	1.3	1.3
Rad. Bend. MC	0.0	0.0	0.0	0.0	-17.2	17.2	17.2	-17.2
Rad. Memb. ML	-1.3	-1.3	1.3	1.3	0.0	0.0	0.0	0.0
Rad. Bend. ML	-17.2	17.2	17.2	-17.2	0.0	0.0	0.0	0.0
Tot. Rad. Str.	-10.6	9.9	26.3	-21.9	-10.6	9.9	26.3	-21.9
Tang. Memb. P	1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5
Tang. Bend. P	2.2	-2.2	2.2	-2.2	2.2	-2.2	2.2	-2.2
Tang. Memb. MC	0.0	0.0	0.0	0.0	-1.1	-1.1	1.1	1.1
Tang. Bend. MC	0.0	0.0	0.0	0.0	-6.2	6.2	6.2	-6.2
Tang. Memb. ML	-1.1	-1.1	1.1	1.1	0.0	0.0	0.0	0.0
Tang. Bend. ML	-6.2	6.2	6.2	-6.2	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-3.6	4.4	11.0	-5.8	-3.6	4.4	11.0	-5.8
Shear VC	0.7	0.7	-0.7	-0.7	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-0.7	-0.7	0.7	0.7
Shear MT	1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5
Tot. Shear	2.1	2.1	0.8	0.8	0.8	0.8	2.1	2.1
Str. Int.	11.2	10.6	26.3	21.9	10.7	10.0	26.6	22.1

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		133.1	133.1	133.1	133.1	133.1	133.1	133.1	133.1
Rad. Pl (SUS)		-0.4	-0.4	2.2	2.2	-0.4	-0.4	2.2	2.2
Rad. Q (SUS)		-10.3	10.3	24.1	-24.1	-10.3	10.3	24.1	-24.1
Long. Pm (SUS)		133.1	133.1	133.1	133.1	133.1	133.1	133.1	133.1
Long. Pl (SUS)		0.4	0.4	2.6	2.6	0.4	0.4	2.6	2.6
Long. Q (SUS)		-4.0	4.0	8.4	-8.4	-4.0	4.0	8.4	-8.4
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.7	0.7	-0.7	-0.7	-0.7	-0.7	0.7	0.7
Shear Q (SUS)		1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5
Pm (SUS)		133.1	133.1	133.1	133.1	133.1	133.1	133.1	133.1
Pm+Pl (SUS)		133.9	133.9	136.2	136.2	133.9	133.9	136.2	136.2
Pm+Pl+Q (Total)		130.1	143.7	159.4	127.2	129.6	143.1	159.6	127.5

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	133.06	138.00	Passed
Pm+Pl (SUS)	136.17	207.00	Passed
Pm+Pl+Q (TOTAL)	159.63	468.82	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.